A PROOF OF THE REINHARDT CONJECTURE

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Abstract. In 1934 Reinhardt conjectured that, among the centrally symmetric convex disks in the plane, the one that packs worst is the smoothed octagon: a regular octagon whose corners are cut off by arcs of hyperbolas, each asymptotic to the two neighbouring side lines. Its greatest packing density is $\delta_{\mathrm{oct}} = (8 - 4\sqrt{2} - \log 2)/(2\sqrt{2} - 1) = 0.902414\ldots$. We prove the conjecture: every centrally symmetric convex disk has greatest packing density at least δ_{oct}, with equality exactly for the images of the smoothed octagon under invertible affine maps. The proof uses Hales's formulation of the problem as an optimal control problem on $\mathrm{SL}_2(\mathbb{R})$, and Hales and Vajjha's results on its minimizers. The worst disks correspond to the global minimizers of the control problem. A minimizer lifts to a normal Pontryagin extremal whose control is bang-bang with finitely many switches. Its switching pattern has one of five sign types when it has at least six pieces, and each type is excluded by an estimate on the whole period. The patterns with at most five pieces are compared directly, and the cheapest is the octagon's. The proof is formalized in Lean 4 with Mathlib and checked by Lean's kernel.

1. Introduction

A *convex disk* is a compact convex subset of the plane with nonempty interior. For a convex disk K let $\delta(K)$ be its greatest packing density: the supremum of the densities of the packings of the plane by congruent copies of K (Definition 2.1). The circular disk packs with density $\pi/\sqrt{12} = 0.906899\ldots$, and in 1934 Reinhardt [30] asked which centrally symmetric convex disk packs worst, that is, has the least $\delta(K)$. He proposed the *smoothed octagon* K_{oct}: the regular octagon with each corner replaced by an arc of the hyperbola that touches the two sides at the corner and is asymptotic to the two neighbouring side lines (Definition 2.3, Figure 1). Its greatest packing density is

$$\delta_{\mathrm{oct}} = \frac{8 - 4\sqrt{2} - \log 2}{2\sqrt{2} - 1} = 0.902414\ldots.$$

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We prove that the smoothed octagon packs worst, and that it is the only worst disk up to affine maps.

Theorem 1.1. (a) *The smoothed octagon K_{oct} is a centrally symmetric convex disk with $\delta(K_{\mathrm{oct}}) = \delta_{\mathrm{oct}}$.*

(b) *Every centrally symmetric convex disk K satisfies $\delta(K) \geq \delta_{\mathrm{oct}}$.*

(c) *A centrally symmetric convex disk K satisfies $\delta(K) = \delta_{\mathrm{oct}}$ if and only if $K = A(K_{\mathrm{oct}})$ for an invertible affine map A of the plane.*

Part (b) is Reinhardt's conjecture. Part (a) is due to Reinhardt and to Mahler [30, 24]; we prove it again, in the form in which the rest of the proof uses it. Part (c) says that the minimum is attained only at the octagon; since δ is invariant under invertible affine maps, the *if* direction follows from (a).

1.1. **History.** Minkowski [25, 26] studied the densest lattice packings of a convex body and found the critical lattices of a convex disk through its inscribed hexagons; the question of the least dense centrally symmetric disk appears in Blaschke's book [5], where, according to Hales and Vajjha [15, §1.2], it is credited to Courant, who expected the circle to be the answer. Reinhardt [30] showed that a least dense disk exists, that its boundary has no corners, and that every boundary point is the midpoint of a side of a circumscribed hexagon of least area. He described the smoothed octagon, which packs worse than the circle, and left open which disk is the least dense.

Mahler, unaware of Reinhardt's work, reached the same conjecture in three papers of 1947 [22, 23, 24]. He reduced the problem to one in the calculus of variations, proved that it has a solution and that the solution is not an ellipse, and expected that the boundary of an extreme disk consists of line segments and arcs of hyperbolas [23, p. 117]; Hales and Vajjha call this Mahler's first conjecture. Among centrally symmetric octagons he found the regular octagon to be the worst, and he constructed the smoothed octagon and computed its density [24, §12]. Ledermann and Mahler [20] named the smoothed octagon, and found that the smoothed decagon packs better, "a result which seems to support Reinhardt's conjecture".

These authors studied lattice packings. Fejes Tóth [12] and Rogers [31] proved that for a centrally symmetric convex disk no packing by congruent copies, or by translates, is denser than the densest lattice packing (Theorem 2.2), so Reinhardt's lattice problem is the problem for all packings; see the surveys of G. Fejes Tóth [11] and of Bezdek and Kuperberg [4]. Lower bounds for $\delta(K)$ followed: $\sqrt{3}/2 = 0.866\ldots$ by Mahler [21], 0.88 by Ennola [9], and 0.892656 by Tammela [32]. Nazarov [27] proved that the smoothed octagon is a local minimum of δ in the Hausdorff metric.

Hales recast the problem twice. In [16] it is a problem of Bolza in the calculus of variations: the circle is, up to affine maps, the only solution of the Euler–Lagrange equations, it is not a minimizer, and a piecewise analytic minimizer is a smoothed polygon. In [17] it is an optimal control problem on $\mathrm{SL}_2(\mathbb{R})$, the problem this paper works with (Definition 2.6). There the

smoothed octagon, and more generally the smoothed regular $(6k+2)$-gon, is a Pontryagin extremal with bang-bang control, the octagon is a strict local minimum, the optimal solution has no singular arcs, and extremals that avoid the singular locus switch finitely often. Vajjha's thesis [34] and the book of Hales and Vajjha [15] continue this work. The book confines the state of a global minimizer to a compact domain (Theorem 2.14), finds the Fuller system [14] near the singular locus, and proves Mahler's first conjecture: a global minimizer is bang-bang with finitely many switches, so a worst disk is a smoothed polygon. It leaves the Reinhardt conjecture open, and notes that a proof that the smoothed $(6k+2)$-gons are the only extremals would complete it [15, §1.3]; see also [3].

Two related problems have been studied in the same spirit. In three dimensions, Kallus [18] proved that the ball is a local pessimum for packing. For convex disks that need not be centrally symmetric, Fáry [10] proved that triangles have the least lattice packing density, and Kallus [19] proved that the regular heptagon is a local minimum of the double-lattice packing density.

1.2. **The control problem.** Hales's control problem (Definition 2.6) has a state $X(t)$ in the upper half-plane, viewed as a traceless matrix of determinant 1, and a frame $g(t) \in \mathrm{SL}_2(\mathbb{R})$, driven by a measurable control $u(t)$ in a triangle U_T; a trajectory is admissible if it closes up after a rotation by $\pi/3$, and its cost $\mathcal{C}(T, X)$ is an integral of the state, equal to the area of the disk that the trajectory traces when it comes from a disk (Section 2.6). The smoothed octagon has a trajectory of cost $A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}$ (Proposition 2.9). We prove:

Theorem 1.2. *Every admissible trajectory (T, u, g, X) of the control problem satisfies $\mathcal{C}(T, X) \geq A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}$.*

By Theorem 2.12, Theorem 1.2 is equivalent to Theorem 1.1(b). It holds for every terminal time $T > 0$ and every measurable control.

1.3. **Strategy of the proof.** The proof follows Hales's control formulation and uses the results of Hales and Vajjha. It has five steps.

From disks to minimizers (Section 2). A worst disk exists, and after a linear change of coordinates it is traced by an admissible trajectory whose cost is $\sqrt{12}$ times its density. That trajectory is a global minimizer of the control problem, and every global minimizer has the same cost (Theorem 2.12). Its state stays in the compact domain $\mathfrak{h}^{\star\star}$ of Hales and Vajjha (Theorem 2.14).

A finite normal period (Section 3). A global minimizer lifts to a Pontryagin extremal (Theorem 3.2). Its costate stays away from the singular locus (Proposition 3.8), so its control is bang-bang, takes values at the vertices of U_T, and switches finitely often in a period (Proposition 3.12). Rebasing the period at a switch (Lemma 3.14) gives a global minimizer whose lift is normal (Theorem 3.16). The result is a *vertex period*: N pieces, on each of

which the control sits at one vertex of U_T, with explicit one-piece formulas, together with the costate of a normal extremal (Theorem 3.18).

Five classes (Section 4). Each switch moves the control to the next vertex or to the previous one, and we call it forward or reverse accordingly. The cyclic word of switch signs puts every period in one of five classes (Definition 4.7, Proposition 4.8). The costate gives a normal multiplier whose determinant γ^2, a Casimir of the motion, is constant on the period. A period that costs at most A_{oct} has a small Casimir (Proposition 4.17, Corollary 4.18), and the Casimir of a mixed period is positive (Proposition 4.16).

Periods with six or more pieces (Sections 5 to 9). Each class is excluded for $N \geq 6$ by an estimate on the whole period: classes R and M3 by the reverse-four principle and a three-piece log-concavity (Theorems 5.4 and 5.7); class M1, whose sign word is $(- - +)^m$, by an angle gap at each centre and a winding identity (Theorem 6.1); class F by a window at the smallest margin and the Casimir cap (Theorem 7.1); and class M2 by locating the smallest margin at a reverse centre and comparing the windows around it with the cost (Theorems 8.1 and 9.3). Together these give Theorem 4.19: a competitive normal vertex period in $\mathfrak{h}^{\star\star}$ has at most five pieces.

Five or fewer pieces, and equality (Sections 10 and 11). The closure condition leaves few words with at most five pieces: four pieces with every switch forward, as for the octagon, and five pieces with the word 0 1 0 2 0 or with every switch reverse (Proposition 10.3); the last are excluded by Theorem 5.4. The four-piece periods cost at least A_{oct}, with equality only for the octagon, and the five-piece periods cost more (Theorem 10.13). This proves Theorem 1.2 and Theorem 1.1(b). For the equality case, a worst disk is traced by a global minimizer, which by the above is the octagon's trajectory; transporting its six arcs shows that the disk is a linear image of the smoothed octagon (Theorem 11.4). Reinhardt's construction and Hales's trajectory give the same disk up to a linear map (Proposition 11.6), which proves Theorem 1.1(a) and (c).

Most of the work is in the fourth step. The obstruction there is that the classes are not excluded by any local condition at a switch: each exclusion compares the cost of a whole period, or of a window of a few pieces around its smallest margin, with A_{oct}, through scalar inequalities in a few variables. These are collected in Appendices A to F. We prove them by identities, monotonicity and concavity arguments in one variable at a time and tangent-line bounds for the logarithm, and a few polynomial signs on boxes by expansion in the Bernstein basis of the box; no interval arithmetic is used.

1.4. **Formal verification.** Theorems 1.1 and 1.2 are proved in Lean 4 [8] with Mathlib [33], in the repository [37], and Lean's kernel checks every step. The three parts of Theorem 1.1 are stated in Mathlib's vocabulary in a file that imports only Mathlib, and the proof uses only Lean's standard axioms. It builds on formalizations of the book of Hales and Vajjha [38]

and of the Pontryagin maximum principle [36], and it is registered in the Palomar registry [28]. Appendix G describes what is formalized and how it is checked, and lists the Lean declaration that proves each numbered result of the paper. The text of the paper does not depend on the formalization: it is a self-contained mathematical proof.

1.5. **Organization.** Section 2 sets up packings, the smoothed octagon and the control problem, and reduces Theorem 1.1(b) to Theorem 1.2; Table 1 at its end lists the notation. Section 3 reduces a global minimizer to a finite normal vertex period, and Section 4 develops the tools for such periods and states Theorem 4.19. Sections 5 to 9 exclude the five classes for six or more pieces, Section 10 treats at most five pieces and proves Theorem 1.2, and Section 11 proves the equality case. Appendices A to F hold the scalar estimates, and Appendix G the formal verification.

2. Packings, balanced disks and Hales's control problem

This section sets up the problem as the rest of the paper uses it. We recall the packing densities and Fejes Tóth's theorem (Section 2.1), define Reinhardt's smoothed octagon (Section 2.2), describe the balanced disks of Reinhardt, Mahler and Hales (Section 2.3), state Hales's control problem (Section 2.4) and the octagon's trajectory (Section 2.5), prove that the conjecture is equivalent to a lower bound for the cost of every admissible trajectory (Section 2.6), and recall the compact domain of Hales and Vajjha (Section 2.7). Most of this material is due to Reinhardt, Mahler, Fejes Tóth, Hales and Vajjha; we follow the book [15] and refer to it for the proofs that we only sketch. Table 1 at the end of the section lists the notation of the paper.

2.1. **Packing densities.** A *convex disk* is a compact convex set $K \subset \mathbb{R}^2$ with nonempty interior. It is *centrally symmetric* if it has a centre c, so that $2c - x \in K$ for every $x \in K$. A *congruent copy* of K is its image under a Euclidean isometry, which may reverse the orientation. A *packing* is a family of convex disks with pairwise disjoint interiors. Let B_r be the closed disk of radius r about the origin.

Definition 2.1 (Greatest packing density)**.** The *density* of a packing $\mathcal{P}$ exists if the limit

$$\delta(\mathcal{P}) = \lim_{r\to\infty} \frac{1}{\operatorname{area}(B_r)} \sum_{K'\in\mathcal{P}} \operatorname{area}(K' \cap B_r)$$

exists and is finite, that is, if the upper and lower limits are equal and finite. The *greatest packing density* $\delta(K)$ of a convex disk K is the supremum of $\delta(\mathcal{P})$ over the packings $\mathcal{P}$ by congruent copies of K whose density exists [15, Def. 2.1.1].

A *lattice* is a discrete subgroup $L = \mathbb{Z}b_0 + \mathbb{Z}b_1$ of $\mathbb{R}^2$ with b_0, b_1 linearly independent; its determinant is $\det L = |\det(b_0, b_1)|$. If the translates $K + v$,

$v \in L$, form a packing, its density exists and equals $\operatorname{area}(K)/\det L$. The *greatest lattice packing density* $\delta_L(K)$ is the supremum of these values.

Theorem 2.2 (Fejes Tóth [12], Rogers [31])**.** *For every centrally symmetric convex disk K, $\delta(K) = \delta_L(K)$.*

So for centrally symmetric disks the problem is a problem about lattices. A linear map carries lattice packings to lattice packings and multiplies areas and determinants by the same factor, so $\delta(K)$ is invariant under invertible affine maps (Proposition 11.5). The *minimax density* $\delta_{\min}$ is the infimum of $\delta(K)$ over all centrally symmetric convex disks, and a *worst disk* is a centrally symmetric convex disk, centred at the origin, with $\delta(K) = \delta_{\min}$. Worst disks exist, by Blaschke's selection theorem and the continuity of δ_L in the Hausdorff metric (Reinhardt [30], Mahler [23]; [15, §2.2]). In this language the conjecture says that $\delta_{\min} = \delta_{\text{oct}}$ and that the worst disks are the linear images of the smoothed octagon.

2.2. **Reinhardt's smoothed octagon.** For $k \in \mathbb{Z}/8$ let

$$d_k(x) = 1 - \bigl(x_0 \cos(k\pi/4) + x_1 \sin(k\pi/4)\bigr), \qquad x = (x_0, x_1) \in \mathbb{R}^2,$$

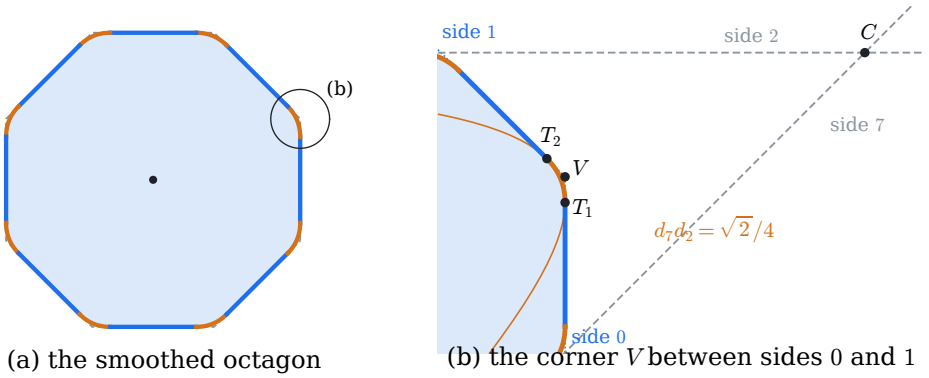
the signed distance from x to the line of the k-th side of the regular octagon of inradius 1, positive on the side of the origin.

Definition 2.3 (The smoothed octagon)**.** Reinhardt's *smoothed octagon* is the set K_{oct} of the points $x \in \mathbb{R}^2$ such that, for every $k \in \mathbb{Z}/8$,

$$d_k(x) \ge 0 \qquad \text{and} \qquad d_k(x) + d_{k+1}(x) \le \tfrac{3}{2} - \sqrt{2} \implies d_{k-1}(x)\, d_{k+2}(x) \ge \tfrac{\sqrt{2}}{4}.$$

The set K_{oct} is the regular octagon $\{d_k \ge 0\}$ with each corner cut off by an arc of a hyperbola (Figure 1). At the corner between sides k and $k+1$ the hyperbola is $d_{k-1} d_{k+2} = \sqrt{2}/4$. Its asymptotes are the lines of the next two sides, $k-1$ and $k+2$, and it touches sides k and $k+1$ at the two points where $d_k + d_{k+1} = \frac{3}{2} - \sqrt{2}$: along either side the product $d_{k-1} d_{k+2}$ is a concave quadratic whose largest value is $\sqrt{2}/4$, attained at the point of tangency. Beyond the chord $d_k + d_{k+1} = \frac{3}{2} - \sqrt{2}$ only the points between the hyperbola and the origin are kept. Each side of the octagon, of length $2(\sqrt{2} - 1)$, keeps a straight middle part of length $2 - \sqrt{2}$. The boundary of K_{oct} is therefore made of eight segments and eight arcs of hyperbolas, and it is C^1, since the arcs leave the segments tangentially. Mahler [24] and Ledermann and Mahler [20] describe the same figure; Hales [16] and Hales and Vajjha [15, §7.5] obtain it, in a different linear position, from the control problem below (Propositions 2.9 and 11.6).

2.3. **Critical hexagons and balanced disks.** Let K be centrally symmetric about the origin. Two translates $K + v$ and $K + w$ have disjoint interiors exactly when $(v - w)/2$ is not an interior point of K, so $K + L$ is a packing exactly when the lattice $\frac{1}{2}L$ has no nonzero point in the interior of K. The densest lattice packings therefore come from the *critical* lattices



(a) the smoothed octagon (b) the corner V between sides 0 and 1

FIGURE 1. (a) Reinhardt's smoothed octagon K_{oct}: the regular octagon of inradius 1 (dashed) with its corners cut off; the straight parts of the sides are blue, the arcs of hyperbolas orange. (b) The corner $V = (1, \sqrt{2}-1)$ between side 0 ($x_0 = 1$) and side 1 ($x_0 + x_1 = \sqrt{2}$). The hyperbola $d_7 d_2 = \sqrt{2}/4$ (thin orange) has the lines of sides 7 and 2 as asymptotes, meeting at $C = (1+\sqrt{2}, 1)$, and touches the two sides at $T_1 = (1, 1-1/\sqrt{2})$ and $T_2 = (\sqrt{2} - \frac{1}{2}, \frac{1}{2})$, where $d_0 + d_1 = \frac{3}{2} - \sqrt{2}$. The arc from T_1 to T_2 replaces the corner. The straight part of each side has length $2 - \sqrt{2}$.

of K, those of least determinant with no nonzero interior point. Reinhardt and Mahler described them through hexagons: a critical lattice has six points $\pm p, \pm q, \pm(q-p)$ on the boundary of K, the vertices of an affinely regular hexagon inscribed in K, and a least such hexagon gives a critical lattice [30, 24]; see [15, §2.2]. For a worst disk, Reinhardt showed that the least hexagons form a one-parameter family whose vertices sweep the whole boundary, and Hales normalized this family as follows.

The *sixth roots of unity* $s_j = (\cos(j\pi/3), \sin(j\pi/3))$, $j \in \mathbb{Z}/6$, satisfy $s_{j+3} = -s_j$, $s_0 + s_2 + s_4 = 0$ and $\det(s_j, s_{j+1}) = \sqrt{3}/2$. A *multi-point* is a map $j \mapsto \sigma_j \in \mathbb{R}^2$ with the same three properties, and six curves $\sigma_j(t)$ that form a multi-point at every time are a *multi-curve*. If the points $\sigma_j(t)$ lie on the boundary of K, the hexagon $\sigma_0(t), \dots, \sigma_5(t)$, of area $3\sqrt{3}/2$, is inscribed in K, and the lattice

$$L(t) = 2\mathbb{Z}\sigma_0(t) + 2\mathbb{Z}\sigma_1(t), \qquad \det L(t) = 4 \cdot \tfrac{\sqrt{3}}{2} = \sqrt{12},$$

contains the six vectors $2\sigma_j(t)$: the translate $K + 2\sigma_j(t)$ touches K at $\sigma_j(t)$.

Definition 2.4 (Balanced pair [15, Def. 2.4.7])**.** A *balanced pair* (K, σ) consists of a convex disk K centred at the origin and a multi-curve σ such that each σ_j is C^1 and regular with Lipschitz derivative, the curvature $\det(\sigma_j', \sigma_j'')$ is nonnegative almost everywhere, each derivative $\sigma_j'(t)$ points into the open cone spanned at $\sigma_j(t)$ by $\sigma_{j+1}(t) - \sigma_j(t)$ and $\sigma_{j+1}(t)$ (the *star*

condition), and the curves have a common period $6t_{\mathrm{f}}$ over which each of them traces the boundary of K once. It is in *circle representation* if $\sigma_j(0) = s_j$ for every j.

Proposition 2.5 ([15, Thm. 2.2.2])**.** *If (K, σ) is a balanced pair, then $\delta(K) = \operatorname{area}(K)/\sqrt{12}$.*

For a balanced disk the lattices $L(t)$ thus have the determinant of a densest lattice packing. For the smoothed octagon the translates $K + L(t)$ are packings, and they form a one-parameter family of densest packings that slide continuously into one another (Figure 2). Baez [2] shows an animation of this family by Greg Egan.

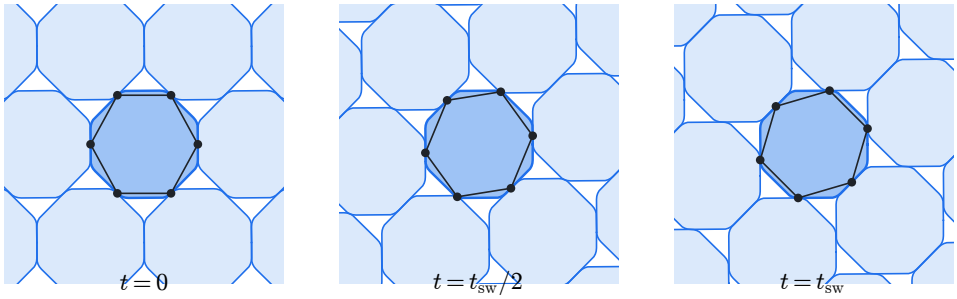


FIGURE 2. Three densest packings of the smoothed octagon from its one-parameter family: the packings $K_{\mathrm{H}} + L(t)$ of the octagon of Proposition 2.9, mapped by the linear map A that carries K_{H} onto K_{oct}. In each, the six neighbours of the dark copy touch it at the six points of a multi-point, the vertices of an inscribed hexagon. Every vector of $L(t)$ other than 0 and the six vectors $2\sigma_j(t)$ is more than twice as long in the norm of K_{H}, so the copies do not overlap. All the lattices have determinant $\sqrt{12}$, and the packings have density δ_{oct}.

2.4. **The control problem.** Let (K, σ) be a balanced pair in circle representation. A multi-point is determined by σ_0 and σ_1, and $\det(\sigma_0, \sigma_1) = \det(s_0, s_1)$, so there is a unique $g(t) \in \mathrm{SL}_2(\mathbb{R})$ with $\sigma_j(t) = g(t)s_j$ for every j. The matrix $X = g^{-1}g'$ is traceless, the star condition says that it lies in an open cone, and after a change of time $\det X = 1$. The six curves trace the boundary of K in turn, $\sigma_j(t + t_{\mathrm{f}}) = \sigma_{j+1}(t)$, which reads $g(t + t_{\mathrm{f}}) = g(t)R$ with R the rotation by $\pi/3$. The convexity of σ_0 becomes a constraint on X', which Hales writes with a control in a triangle, and the area of K becomes an integral of X. Hales derived the problem that results in [16, 17]; [15, Ch. 3–4] gives the derivation in detail.

Write $\langle X, Y \rangle = \operatorname{tr}(XY)$ for the trace form on 2×2 matrices, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \exp(\pi J/3) = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

The *control simplex* is $U_T = \{u = (u_0, u_1, u_2) : u_i \geq 0,\ u_0 + u_1 + u_2 = 1\}$, with vertices $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$. A control $u \in U_T$ has the *control matrix*

$$Z_u = \begin{pmatrix} (u_1 - u_2)/\sqrt{3} & (u_0 - 2u_1 - 2u_2)/3 \\ u_0 & (u_2 - u_1)/\sqrt{3} \end{pmatrix}.$$

For $X = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ the *star functions* are $\rho_0(X) = (c + \sqrt{3}a)/\sqrt{3}$, $\rho_1(X) = (c - \sqrt{3}a)/\sqrt{3}$ and $\rho_2(X) = -(3b + c)/(2\sqrt{3})$. A *star state* is a traceless matrix X with $\det X = 1$ and $\rho_0(X), \rho_1(X), \rho_2(X) > 0$.

Definition 2.6 (The control problem; [15, Problem 3.6.2])**.** Let $T > 0$. A *state trajectory* on $[0, T]$ consists of a measurable control $u : [0, T] \to U_T$, a frame $g : [0, T] \to \mathrm{SL}_2(\mathbb{R})$ and a state X with values in the star states, such that

$$g' = gX, \qquad X' = [P_u, X], \qquad P_u = \frac{Z_u}{\langle Z_u, X\rangle}, \tag{2.1}$$

in the integral sense: the right-hand sides are integrable, and g and X are their indefinite integrals. It is *admissible* if

$$g(0) = I, \qquad g(T) = R, \qquad X(T) = R^{-1}X(0)R.$$

Its *cost* is

$$\mathcal{C}(T, X) = -\frac{3}{2}\int_0^T \langle J, X(t)\rangle \,\mathrm{d}t. \tag{2.2}$$

The terminal time T is free, and no regularity of the control beyond measurability is assumed: this is the class of trajectories in which a minimizer exists [15, §6.1], and the comparison of Section 2.6 must cover all of it. A global minimizer is defined accordingly.

Definition 2.7 (Global minimizer)**.** A *global minimizer* is an admissible trajectory (T, u, g, X) such that $\mathcal{C}(T, X) \leq \mathcal{C}(T', X')$ for every admissible trajectory (T', u', g', X'), with any terminal time and any measurable control.

The geometry of the problem is best seen in three coordinates on the star states. The first is a triple of positive numbers, one for each vertex of the simplex.

Lemma 2.8 (Margins)**.** *For a traceless matrix X let $p_i(X) = -\sqrt{3}\,\langle Z_{\mathbf{e}_i}, X\rangle$, $i \in \mathbb{Z}/3$, be its* margins*. Then $p_i = 2\rho_{2-i}$, and:*

(i) *a traceless X with $\det X = 1$ is a star state if and only if $p_0, p_1, p_2 > 0$;*
(ii) *$p_0p_1 + p_1p_2 + p_2p_0 = 4\det X$, which is 4 on star states;*
(iii) *$-\frac{3}{2}\langle J, X\rangle = \frac{\sqrt{3}}{2}(p_0 + p_1 + p_2)$, which is at least 3 on star states;*
(iv) *$p_i(R^{-1}XR) = p_{i-1}(X)$.*

Proof. For $X = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ the margins are $p_0 = -(3b + c)/\sqrt{3}$, $p_1 = 2(c - \sqrt{3}a)/\sqrt{3}$ and $p_2 = 2(c + \sqrt{3}a)/\sqrt{3}$, which are $2\rho_2$, $2\rho_1$ and $2\rho_0$; this gives (i). Expanding, $p_0p_1 + p_1p_2 + p_2p_0 = 4(-a^2 - bc) = 4\det X$. Since $Z_{\mathbf{e}_0} + Z_{\mathbf{e}_1} + Z_{\mathbf{e}_2} = J$, the sum of the margins is $-\sqrt{3}\langle J, X\rangle$, which is (iii); and

$(p_0+p_1+p_2)^2 \geq 3(p_0p_1+p_1p_2+p_2p_0) = 12$ gives the lower bound $\frac{\sqrt{3}}{2}\cdot 2\sqrt{3} = 3$. Finally (iv) is a direct computation with the entries of $R^{-1}XR$. □

By (iii), every state trajectory on $[0,T]$ costs at least $3T$. The second coordinate is a point of the upper half-plane. For $z = x + iy$ with $y > 0$ put

$$\Phi(z) = \frac{1}{y}\begin{pmatrix} x & -(x^2+y^2) \\ 1 & -x \end{pmatrix}.$$

Then $\Phi(z)$ is traceless with determinant one, and it is a star state exactly when z lies in the *star domain*

$$\mathfrak{h}^\star = \{x+iy :\ y>0,\ |x|<1/\sqrt{3},\ x^2+y^2>1/3\},$$

the ideal triangle with vertices $-1/\sqrt{3}$, $1/\sqrt{3}$ and ∞; every star state is $\Phi(z)$ for exactly one $z \in \mathfrak{h}^\star$ [15, Thm. 4.4.1]. In this coordinate the margins are

$$p_0 = \frac{\sqrt{3}\,(x^2+y^2-1/3)}{y}, \qquad p_1 = \frac{2(1-\sqrt{3}x)}{\sqrt{3}\,y}, \qquad p_2 = \frac{2(1+\sqrt{3}x)}{\sqrt{3}\,y},$$

and each margin vanishes on one side of the ideal triangle. Conjugation by R acts on $\mathfrak{h}^\star$ by the Möbius transformation of R^{-1}, which permutes the three sides. The third coordinate, used from Section 4 on, is the triangular form of the state along a piece of constant control.

On an interval where the control is a vertex $\mathbf{e}_i$, the matrix $P_{\mathbf{e}_i}$ is a constant multiple of $Z_{\mathbf{e}_i}$, the state moves by conjugation by a one-parameter group, and the margin p_i stays constant (Lemma 4.1). On such an interval three of the six contact curves trace segments of the boundary of the disk and the other three trace arcs of hyperbolas. A switch between two vertices is *forward* if it raises the label by one and *reverse* if it lowers it (Figure 3). The words of vertex controls are the subject of Sections 4 to 10.

2.5. **The smoothed octagon's trajectory.** Let

$$y_0 = \sqrt{\frac{2\sqrt{2}-1}{3}}, \qquad t_{\mathrm{sw}} = \frac{\log 2}{2\sqrt{2\sqrt{2}-1}}, \qquad t_{\mathrm{f}} = 4t_{\mathrm{sw}},$$

and let u_{oct} be the control that equals $\mathbf{e}_2, \mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$ on the four intervals $[mt_{\mathrm{sw}}, (m+1)t_{\mathrm{sw}})$, $m = 0,1,2,3$. The *octagon's density* and *cost* are

$$\delta_{\mathrm{oct}} = \frac{8-4\sqrt{2}-\log 2}{2\sqrt{2}-1} = 0.902414\ldots, \qquad A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}} = 3.126054\ldots.$$

Proposition 2.9 (The octagon's trajectory [15, §7.5])**.** *There is exactly one state trajectory $(g_{\mathrm{oct}}, X_{\mathrm{oct}})$ on $[0,t_{\mathrm{f}}]$ with control u_{oct}, $g(0) = I$ and $X(0) = \Phi(iy_0)$. It is admissible, its cost is $A_{\mathrm{oct}} < \pi$, and it is a Pontryagin extremal (Definition 3.1). The convex hull*

$$K_{\mathrm{H}} = \mathrm{conv} \bigcup_{j\in\mathbb{Z}/6} \{g_{\mathrm{oct}}(t)s_j :\ 0 \leq t \leq t_{\mathrm{f}}\}$$

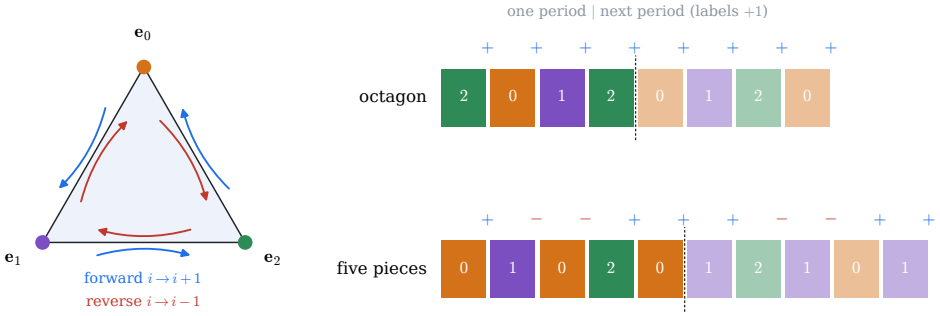


FIGURE 3. Left: the control simplex U_T with its vertices $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$; a switch from $\mathbf{e}_i$ to $\mathbf{e}_{i+1}$ is forward (blue), one from $\mathbf{e}_i$ to $\mathbf{e}_{i-1}$ reverse (red). Right: two words of vertex labels over one period and the next, with the sign of each switch. The terminal condition $X(T) = R^{-1}X(0)R$ raises every label by one from one period to the next, so the switch across the seam (dashed) compares the last label of a period with the first label plus one. The octagon's word 2 0 1 2 has four forward switches; the word 0 1 0 2 0 has the signs $+\ -\ -\ +\ +$. In both cases the signs add up to 1 modulo 3 (Lemma 4.6).

of its six arcs is a convex disk, symmetric about the origin, of area A_{oct} *and greatest packing density* $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$.

Sketch. On a piece of constant vertex control the state equations have explicit solutions (Lemma 4.1). From $z_0 = iy_0$ the control $\mathbf{e}_2$ moves the state along $X(t) = \Phi((W-1)/\sqrt{3} + iy_0W)$, $W = \exp(\sqrt{3}y_0t)$, and the later pieces are conjugates of the first by powers of R. The choice of y_0 makes $\operatorname{tr}(R^{-1}g(t_{\mathrm{sw}})) = \sqrt{2}$, which gives $g(t_{\mathrm{f}}) = R$ and $X(t_{\mathrm{f}}) = R^{-1}X(0)R$; uniqueness holds because the right-hand sides are Lipschitz on each piece. The cost is four times an explicit integral over $[0, t_{\mathrm{sw}}]$, and it equals A_{oct} [15, §7.5]. The curves $\sigma_j(t) = g_{\mathrm{oct}}(t)s_j$, extended by $g(t + t_{\mathrm{f}}) = g(t)R$, form a balanced pair with the radial hull of σ_0, whose area is the cost; so its density is $A_{\mathrm{oct}}/\sqrt{12} = \delta_{\mathrm{oct}}$ by Proposition 2.5. This radial hull is the convex hull of the six arcs, which together trace its boundary. Hales and Vajjha prove that the trajectory is a Pontryagin extremal [15, Thm. 7.5.6]. □

On each of the four pieces the factor $\exp(2t_{\mathrm{sw}}/a)$ of Lemma 4.1, with a the active margin, equals $\sqrt{2}$ (Section 10). Figure 4 shows the six arcs and the contact hexagons, and Figure 5 the state in the upper half-plane. The disk K_{H} is a linear image of K_{oct}, namely $K_{\mathrm{H}} = A^{-1}K_{\mathrm{oct}}$ with $A = \operatorname{diag}(1, (2\sqrt{2}-1)/\sqrt{3})$ (Proposition 11.6): Reinhardt's picture compressed vertically by about five per cent.

2.6. **From disks to trajectories and back.** The reformulation is an equivalence in both directions. One direction realizes a worst disk by an

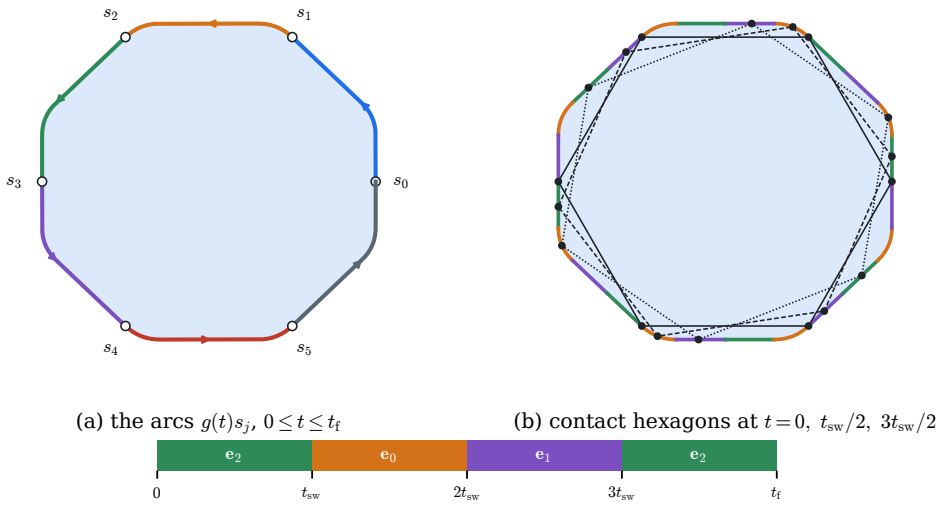


FIGURE 4. The octagon K_H of the trajectory. (a) As t runs over $[0, t_f]$, the frame $g_{oct}(t)$ carries each sixth root s_j (white) along an arc to s_{j+1}; the six arcs close up into the boundary of K_H, and K_H is their convex hull. (b) The boundary coloured by the vertex control that is active when σ_0 passes there (orange $\mathbf{e}_0$, purple $\mathbf{e}_1$, green $\mathbf{e}_2$), and the contact hexagons $\sigma_0(t), \dots, \sigma_5(t)$ at three times. Each hexagon has area $3\sqrt{3}/2$, opposite vertices are antipodal, and consecutive vertices span determinant $\sqrt{3}/2$. Below: the control word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ on $[0, t_f]$.

admissible trajectory of equal cost. The other builds a balanced disk of equal area from every admissible trajectory of cost below π, whatever its control. The second direction is what makes the realization of a worst disk a global minimizer; without it, a cheap trajectory with an irregular control could undercut every disk.

Proposition 2.10 (Realization of a worst disk)**.** *There are a worst disk K, a balanced pair (K, σ) in circle representation with period $6T$, and an admissible trajectory (T, u, g, X) with $\sigma_j(t) = g(t)s_j$ for $t \in [0, T]$, such that*

$$\mathcal{C}(T, X) = \operatorname{area}(K) = \sqrt{12}\,\delta(K) \le A_{oct} < \pi, \qquad T < \pi/3.$$

Sketch. Take a worst disk and move it by a linear map, which does not change its density, into Reinhardt's normal position, in which a critical hexagon has its side midpoints at the sixth roots of unity. Every boundary point is then a contact point of a least hexagon [15, §2.2], and selecting the hexagon whose first contact point has a given polar angle gives six continuous simple closed curves that trace the boundary. The disk has a unique supporting line at each boundary point, so its radial parametrization

is C^1 and regular; reparametrizing by arclength and then by the clock that makes $\det(g^{-1}g') = 1$ gives a balanced pair in circle representation. The contact map $\sigma_j \mapsto \sigma_{j+1}$ is conjugated by this clock to a translation by one sixth of the period, which gives $g(t+T) = g(t)R$. The control is read off from the curvatures; where the three curvatures do not form a nonzero nonnegative vector, which happens only on a null set, any point of the simplex will do. The area formula for balanced disks gives $\mathcal{C} = \operatorname{area}(K)$, and Proposition 2.5 gives $\delta(K) = \operatorname{area}(K)/\sqrt{12}$. Since K_{H} is a centrally symmetric convex disk with $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$, the worst disk has $\operatorname{area}(K) \le A_{\mathrm{oct}} < \pi$, and by Lemma 2.8(iii), $3T \le \mathcal{C} < \pi$. □

Proposition 2.11 (Reverse comparison)**.** *Let (T, u, g, X) be an admissible trajectory with $\mathcal{C}(T, X) < \pi$ and any measurable control. Then there is a balanced pair (K, σ) in circle representation with period $6T$ such that $\operatorname{area}(K) = \mathcal{C}(T, X)$ and $\delta(K) = \mathcal{C}(T, X)/\sqrt{12}$.*

Sketch. The state of an admissible trajectory is Lipschitz, since the control takes values in a compact set and the denominator $\langle Z_u, X\rangle$ stays away from zero on compact sets of star states; the frame is C^1. Extend the trajectory to all of $\mathbb{R}$ by $G(t+T) = G(t)R$ and $Y(t+T) = R^{-1}Y(t)R$; six copies close up because $R^6 = I$, and the integral equations glue across the seams. Let $\sigma_j(t) = G(t)s_j$ and let K be the radial hull of σ_0 over one full period $[0, 6T]$.

The terminal condition $g(T) = R$ alone allows the frame to wind around several times, and this is where the cost bound enters. The reflected first row $(g_{00}, -g_{01})$ of the frame turns with positive angular speed at most $-\langle J, X\rangle$, so a cost below π forces it to turn by exactly $\pi/3$ over $[0, T]$, with no extra multiple of 2π. The determinant-one identity keeps the first column $\sigma_0 = gs_0$ from ever pointing opposite to the reflected first row, so σ_0 also turns exactly once around the origin over $[0, 6T]$, and its positive angular speed makes it simple. The curvature condition of the control makes σ_0 locally convex, and a simple closed locally convex curve bounds a convex disk. The other conditions of a balanced pair follow from the state equations, and the area of the radial hull, $\frac{1}{2}\int_0^{6T} \det(\sigma_0, \sigma_0')$, is the cost. □

Theorem 2.12 (Equivalence)**.** (i) *A worst disk exists, and the trajectory of Proposition 2.10 is a global minimizer of the control problem, of cost $\sqrt{12}\,\delta_{\min}$.*

(ii) *Every global minimizer has cost $\sqrt{12}\,\delta_{\min}$.*

(iii) *$\delta(K) \ge \delta_{\mathrm{oct}}$ for every centrally symmetric convex disk K centred at the origin if and only if $\mathcal{C}(T, X) \ge A_{\mathrm{oct}}$ for every admissible trajectory (T, u, g, X).*

Proof. Let (T, u, g, X) and K be as in Proposition 2.10, and let (T', u', g', X') be any admissible trajectory. If $\mathcal{C}(T', X') \ge \pi$, then $\mathcal{C}(T', X') > \mathcal{C}(T, X)$. Otherwise Proposition 2.11 gives a balanced disk K' with $\mathcal{C}(T', X') = \operatorname{area}(K') = \sqrt{12}\,\delta(K') \ge \sqrt{12}\,\delta(K) = \mathcal{C}(T, X)$. This proves (i). If (T', u', g', X') is another global minimizer, each of the two costs is at most

the other, which gives (ii). For (iii), suppose first that $\delta \geq \delta_{\mathrm{oct}}$ for every such disk. An admissible trajectory of cost at least π costs more than A_{oct}, and one of cost below π costs $\sqrt{12}\,\delta(K') \geq A_{\mathrm{oct}}$ by Proposition 2.11. Conversely, if every admissible trajectory costs at least A_{oct}, then the worst disk of (i) has $\sqrt{12}\,\delta_{\min} = \mathcal{C}(T, X) \geq A_{\mathrm{oct}}$, so $\delta(K) \geq \delta_{\min} \geq \delta_{\mathrm{oct}}$ for every such disk. □

2.7. **The compact domain.** The star domain $\mathfrak{h}^\star$ is not compact, and near its three sides some margin is small. Hales and Vajjha showed that a balanced disk whose area does not exceed the octagon's has its state away from the sides and from the cusps.

Definition 2.13 (The compact domain [15, Def. 5.0.1])**.** $\mathfrak{h}^{\star\star}$ is the set of the points $z = x + iy \in \mathfrak{h}^\star$ with $y \leq 9/2$, $(x \mp 1/\sqrt{3})^2 + y^2 \geq \frac{8}{27}\, y$ for both signs, and $\rho_k(\Phi(z)) \geq 1/15$ for $k = 0, 1, 2$. A star state X lies in $\mathfrak{h}^{\star\star}$ if $X = \Phi(z)$ with $z \in \mathfrak{h}^{\star\star}$.

The three conditions remove the horoball $y > 9/2$ at the cusp ∞, its two images under conjugation by $R^{\pm 1}$ (the open disks of radius $4/27$ tangent to the real line at $\pm 1/\sqrt{3}$), and three neighbourhoods of the sides of $\mathfrak{h}^\star$. By Lemma 2.8, every margin of a state in $\mathfrak{h}^{\star\star}$ is at least $2/15$.

Theorem 2.14 (The compact domain [15, Thm. 5.0.2])**.** *The state of every global minimizer lies in* $\mathfrak{h}^{\star\star}$ *at every time. In particular all its margins are at least* $2/15$*, at every time and for every vertex, active or not.*

Proof. Let (T, u, g, X) be a global minimizer. Its cost is at most that of the octagon's trajectory, which is $A_{\mathrm{oct}} < \pi$ (Proposition 2.9), so Proposition 2.11 applies: the cyclic extension of the trajectory is the boundary trajectory of a balanced disk K of area equal to the cost, with $X = \Phi \circ z$ for a curve z in $\mathfrak{h}^\star$. Hales and Vajjha prove that if $z(t_0)$ lies outside $\mathfrak{h}^{\star\star}$ at some time, then the area of such a disk is strictly larger than the area of the smoothed octagon [15, Thm. 5.0.2]: the area is bounded below by $\sqrt{12}$ times a density function of the single point $z(t_0)$ [15, Thm. 5.0.6], and this function exceeds δ_{oct} off $\mathfrak{h}^{\star\star}$. The cost would then exceed A_{oct}, the cost of an admissible trajectory, which contradicts minimality. □

2.8. **Notation.** Table 1 lists the symbols that are used in more than one section. Symbols that belong to a single case are introduced where they are used.

Table 1: Notation.

Symbol	Meaning	Where
$\delta(K)$, $\delta_L(K)$	greatest packing density, greatest lattice packing density	§2.1

Symbol	Meaning	Where
$\delta_{\min}$	least $\delta(K)$ over centrally symmetric convex disks	§2.1
K_{oct}, K_{H}	Reinhardt's smoothed octagon; the octagon of the trajectory	§2.2, §2.5
$d_k(x)$	distance to the line of the k-th side of the regular octagon	§2.2
s_j, $\sigma_j(t)$	sixth roots of unity; contact curves $g(t)s_j$	§2.3
$\langle X, Y\rangle$	the trace form $\mathrm{tr}(XY)$	§2.4
J, R	rotation generator; rotation by $\pi/3$	§2.4
U_T, $\mathbf{e}_i$	control simplex and its vertices, $i \in \mathbb{Z}/3$	§2.4
Z_u, P_u	control matrix; $Z_u/\langle Z_u, X\rangle$	§2.4
g, X, T	frame in $\mathrm{SL}_2(\mathbb{R})$, star state, terminal time	§2.4
$\mathcal{C}(T, X)$	the cost $-\frac{3}{2}\int_0^T \langle J, X\rangle$	§2.4
ρ_i, p_i	star functions; margins $p_i = 2\rho_{2-i}$	§2.4
$\Phi(z)$, $\mathfrak{h}^\star$, $\mathfrak{h}^{\star\star}$	the state of $z = x+iy$; star domain; compact domain	§2.4, §2.7
δ_{oct}, A_{oct}	$(8-4\sqrt{2}-\log 2)/(2\sqrt{2}-1)$ and $\sqrt{12}\,\delta_{\mathrm{oct}}$	§2.5
y_0, t_{sw}, t_{f}	data of the octagon's trajectory	§2.5
Λ_1, Λ_R, λ	costates and cost multiplier	§3
H, H_i	Hamiltonian; vertex Hamiltonians	§3
N, τ_k, i_k	pieces per period, switching times, labels	§3
a_k, q_k	active margin and factor of piece k	§4
ε_k	sign of the switch after piece k: $+1$ forward, -1 reverse	§4
s_k, K_k	$a_k + a_{k+1}$ and $a_k^2 + 4$	§4
$T_i(q)$	tangent factor of a piece with label i	§4
R, F, M1, M2, M3	the five classes of sign sequences	§4
α_j, β_j, χ_j	costate coordinates; $\chi_j = 2\alpha_j + \beta_j = p_j H_j$	§4
U, μ_j, γ^2, κ	normal multiplier, its margins, the Casimir $\det U$, $\kappa = \sqrt{12\gamma^2}$	§4
y_k, d_k, e_k	switch vectors, switch norms, cross pairings	§4

3. The reduction to finite normal periods

Section 2 produced a global minimizer of the control problem whose state stays in the compact domain $\mathfrak{h}^{\star\star}$. This section shows that, after a change of origin in time, such a minimizer has a control that takes only the three vertex values $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$, switching finitely often in a period, and that it carries a Pontryagin lift with cost multiplier $\lambda = -1$. The conclusion is Theorem 3.18. Every later section starts from it.

The steps are the following. The maximum principle gives a lift $(\Lambda_1, \Lambda_R, \lambda)$ of the minimizer, possibly abnormal ($\lambda = 0$). The danger is the singular

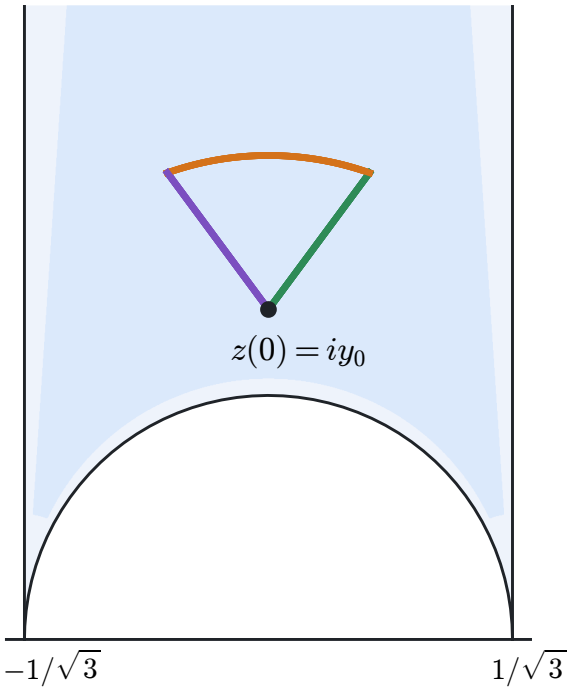


FIGURE 5. The star domain $\mathfrak{h}^\star$ (light), bounded by the lines $x = \pm 1/\sqrt{3}$ and the circle $x^2 + y^2 = 1/3$, and the compact domain $\mathfrak{h}^{\star\star}$ (darker; its cut at $y = 9/2$ lies above the frame). The curve is the state $z(t)$ of the octagon's trajectory, $X(t) = \Phi(z(t))$, over three periods $[0, 3t_{\mathrm{f}}]$, after which it returns to $z(0) = iy_0$; it is coloured by the active vertex control as in Figure 4. The curve stays in $\mathfrak{h}^{\star\star}$, so every margin along it exceeds $2/15$.

locus, where the Hamiltonian does not depend on the control and the maximum condition gives no information. Near it Hales and Vajjha find Fuller's system [15, 14], whose optimal controls switch infinitely often in finite time. A budget inequality, which uses only that the cost is at most $A_{\mathrm{oct}} < \pi$, keeps every lift away from the singular locus in proportion to its size (Proposition 3.8). Away from the singular locus one vertex is excluded on each side of every time (Lemma 3.11). On an interval where one vertex is excluded the control runs along an edge of the simplex, and an edge minimizer has at most four pieces (Proposition 3.10). Compactness of $[0, T]$ then gives finitely many pieces (Proposition 3.12). Merging equal neighbours and restarting the period at a genuine switch gives a vertex representation (Lemma 3.14), and for a lift of a trajectory with a vertex representation the multiplier is nonzero (Theorem 3.16).

The order matters. The clearance and the switch count are homogeneous in $(\lambda, \Lambda_1, \Lambda_R)$, so they hold for abnormal lifts as well as normal ones. The normality argument, on the other hand, reads the costate at the finitely many switching times and uses the twisted closing condition of the period. Finiteness is therefore proved first, for an arbitrary lift, and normality afterwards.

3.1. **Pontryagin extremals.** The maximum principle is applied to the problem of Definition 2.6 with free terminal time and with the closing conditions $g(0) = I$, $g(T) = R$, $X(T) = R^{-1}X(0)R$. Its conclusions, in the form of Hales and Vajjha's Problem 6.7.1 [15], define the following object.

Definition 3.1 (Pontryagin extremal)**.** Let (T, u, g, X) be a state trajectory (Definition 2.6) with $g(0) = I$, $g(T) = R$ and $X(T) = R^{-1}X(0)R$. A *Pontryagin lift* of it consists of two curves $\Lambda_1, \Lambda_R \colon [0, T] \to \mathfrak{sl}_2(\mathbb{R})$ and a constant $\lambda \le 0$, the *cost multiplier*, such that

(1) $\langle \Lambda_R(t), X(t) \rangle = 0$ and $(\lambda, \Lambda_1(t), \Lambda_R(t)) \neq 0$ for every $t \in [0, T]$;

(2) Λ_1 and Λ_R are absolutely continuous and satisfy, for almost every t,

$$\Lambda_1' = [\Lambda_1, X], \qquad \Lambda_R' = [P_u, \Lambda_R] - \langle \Lambda_R, P_u \rangle [P_u, X] + [-\Lambda_1 + \tfrac{3}{2}\lambda J, X], \tag{3.1}$$

with $P_u = Z_u / \langle Z_u, X \rangle$ as in Equation (2.1);

(3) for almost every $t \in (0, T)$ the control $u(t)$ maximizes over U_T the Hamiltonian

$$H(\Lambda_1, \Lambda_R, X; v) = \langle \Lambda_1 - \tfrac{3}{2}\lambda J, X \rangle - \frac{\langle \Lambda_R, Z_v \rangle}{\langle X, Z_v \rangle}, \tag{3.2}$$

evaluated at $(\Lambda_1(t), \Lambda_R(t), X(t))$, and the maximum is 0;

(4) $\Lambda_1(T) = R^{-1}\Lambda_1(0)R$ and $\Lambda_R(T) = R^{-1}\Lambda_R(0)R$.

The tuple $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ is a *Pontryagin extremal*. It is *normal* if $\lambda \neq 0$ and *abnormal* if $\lambda = 0$.

Here Λ_1 is the costate of the frame, read in the left trivialization of $\mathrm{SL}_2(\mathbb{R})$, and $\Lambda_R = [\Lambda_2, X]$ is the part of the costate of the state that the dynamics see. The conditions are positively homogeneous in $(\Lambda_1, \Lambda_R, \lambda)$, so a normal lift can be rescaled to $\lambda = -1$.

Theorem 3.2 (Pontryagin lift)**.** *Every global minimizer (T, u, g, X) of the control problem (Definition 2.7) has a Pontryagin lift $(\Lambda_1, \Lambda_R, \lambda)$.*

Hales and Vajjha use this statement in the proof of their Proposition 8.1.3. It follows from the maximum principle with free terminal time [29, 1], applied on the open subset of $\mathbb{R}^{2\times 2} \times \mathfrak{sl}_2(\mathbb{R})$ cut out by $\det g > 0$ and the star inequalities, with the costates read through the left trivialization. The nontriviality of the reduced costate $(\lambda, \Lambda_1, \Lambda_R)$ needs a separate argument: if $\lambda = 0$ and Λ_1, Λ_R vanish at one time, then $\Lambda_1 \equiv 0$ by the conservation law below, and the equation for Λ_R becomes linear and homogeneous, so $\Lambda_R \equiv 0$.

Two consequences of the costate equations are used throughout. The first is the conservation law of Hales and Vajjha's Corollary 6.5.3,

$$\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)\,g(t) \qquad (0 \le t \le T) \tag{3.3}$$

. In particular $\det \Lambda_1$ is constant. The second concerns the three *vertex Hamiltonians* $H_i(t) = H(\Lambda_1(t), \Lambda_R(t), X(t); \mathbf{e}_i)$. The denominator $-\langle X, Z_v\rangle$ is positive and affine in v, and $-\langle X, Z_v\rangle\big(H(\cdot\,; v) - c\big)$ is affine in v for every constant c. So the maximum of H over the simplex is attained at a vertex, and a vertex carrying positive weight in a maximizing control is itself maximizing.

Lemma 3.3 (Vertex Hamiltonians)**.** *For a Pontryagin extremal the vertex Hamiltonians H_0, H_1, H_2 are continuous on $[0,T]$, and at every $t \in [0,T]$, not only almost every t, all three are ≤ 0 and one of them is 0.*

Proof. Continuity holds because X, Λ_1, Λ_R are continuous and $\langle X, Z_{\mathbf{e}_i}\rangle < 0$ on star states. By the maximum condition, $\max_i H_i(t) = 0$ for almost every t, hence for every t by continuity of $\max_i H_i$. □

Figure 6 shows the three vertex Hamiltonians along the smoothed octagon's period, for its normal lift.

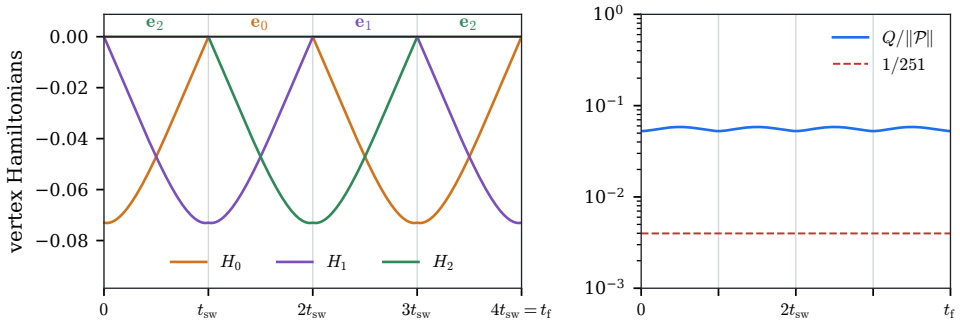


FIGURE 6. The normal Pontryagin lift ($\lambda = -1$) of the smoothed octagon's trajectory over one period $[0, t_{\mathrm{f}}]$, with control word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ and switching time $t_{\mathrm{sw}} = t_{\mathrm{f}}/4$; the lift is that of Hales and Vajjha's Theorem 7.5.1 [15], integrated numerically. Left: the vertex Hamiltonians H_0, H_1, H_2. All three are ≤ 0, the vertex of the current piece has $H_i = 0$, and at each switch the incoming vertex reaches 0. Right: the ratio $Q/\|\mathcal{P}\|$ of Proposition 3.8, on a logarithmic scale. It stays near 0.055, above the bound $1/251$ that the proposition gives for every competitive lift. The computation confirms the closing conditions $\Lambda_1(t_{\mathrm{f}}) = R^{-1}\Lambda_1(0)R$ and $\Lambda_R(t_{\mathrm{f}}) = R^{-1}\Lambda_R(0)R$ and every property stated here.

The singular locus. If the three vertex Hamiltonians are equal at some time, then the maximum condition at that time says nothing about the control. Hales and Vajjha's Lemma 8.2.1 identifies when this happens.

Lemma 3.4 (Singular locus)**.** *Let X be a star state and Λ_R traceless with $\langle \Lambda_R, X\rangle = 0$. Then $H(\Lambda_1, \Lambda_R, X; \mathbf{e}_0) = H(\Lambda_1, \Lambda_R, X; \mathbf{e}_1) = H(\Lambda_1, \Lambda_R, X; \mathbf{e}_2)$ if and only if $\Lambda_R = 0$. If $\Lambda_R \neq 0$, there is a coordinate i such that $v_i = 0$ for every control v maximizing H over U_T.*

So all three vertices tie at a time where $\Lambda_R = 0$; if the Hamiltonian also vanishes there, then $\langle \Lambda_1 - \frac{3}{2}\lambda J, X\rangle = 0$. The *singular locus* is the set of costates with $\Lambda_1 = \frac{3}{2}\lambda J$ and $\Lambda_R = 0$; there $H(\cdot\,; v) = 0$ for every control v. Put

$$A = \Lambda_1 - \tfrac{3}{2}\lambda J, \qquad \|\mathcal{P}\|^2 = \lambda^2 + \|\Lambda_1\|^2 + \|\Lambda_R\|^2, \qquad Q^2 = \|A\|^2 + \|\Lambda_R\|^2,$$

with Frobenius norms. Here $\mathcal{P} = (\lambda, \Lambda_1, \Lambda_R)$ is the extended covector, and Q is its distance from the singular locus.

3.2. **Singular clearance.** The singular locus cannot be excluded by local considerations; the argument uses the whole period and its cost. Write $\omega(Y) = -\frac{1}{2}\langle J, Y\rangle$. For a star state $\omega(Y) \geq 1$, and the cost density is $-\frac{3}{2}\langle J, X\rangle = 3\omega(X)$, so $\mathcal{C}(T, X) = 3\int_0^T \omega(X)$. For a frame h with $\det h = 1$ put $\omega_R(h, Y) = \omega(hYh^{-1})$, the same quantity for the state seen in the fixed frame.

Lemma 3.5 (Winding)**.** *For every admissible trajectory (T, u, h, Y), with $h(0) = I$ and $h(T) = R$,*

$$\int_0^T \big(\omega(Y) + \omega_R(h, Y)\big)\,\mathrm{d}t \;\geq\; \frac{2\pi}{3}$$

.

Proof. Write $\zeta(h) = \frac{1}{2}(h_{00} + h_{11}) + \frac{i}{2}(h_{10} - h_{01}) \in \mathbb{C}$, the coefficient of h on the rotations. Since $\det h = 1$, $|\zeta(h)| \geq 1$, and a direct computation gives $\omega(Y) + \omega_R(h, Y) = 2\,\mathrm{Im}(\overline{\zeta}\,\zeta')$ along $h' = hY$. Let $L(t) = \int_0^t \zeta'/\zeta$, so that $\zeta = e^L$. Then $\mathrm{Im}(\zeta'/\zeta) = (\omega + \omega_R)/(2|\zeta|^2) > 0$, and $\zeta(T) = \zeta(R) = e^{i\pi/3}$ forces $\mathrm{Im}\, L(T) \in \frac{\pi}{3} + 2\pi\mathbb{Z}$; being positive, it is at least $\pi/3$. Since $|\zeta| \geq 1$, $\int_0^T(\omega + \omega_R) = 2\int_0^T |\zeta|^2\,\mathrm{Im}(\zeta'/\zeta) \geq 2\,\mathrm{Im}\, L(T) \geq 2\pi/3$. □

The second ingredient is pointwise: at a star state the control u^* with $P_{u^*} = -X/2$ lies in the simplex and has $H(u^*) = \langle A, X\rangle$, so the maximum condition gives $\langle A, X\rangle \leq 0$. The weights of u^* also bound Λ_R.

Lemma 3.6 (Equilibrium inequality)**.** *Let X be a star state, Λ_R traceless with $\langle \Lambda_R, X\rangle = 0$, and suppose that every vertex Hamiltonian at $(\Lambda_1, \Lambda_R, X)$ is ≤ 0. Then $\langle A, X\rangle \leq 0$. If moreover every margin of X is at least $2/15$, then $\|\Lambda_R\|^2 \leq 2700\,\langle A, X\rangle^2$.*

Proof. With $\varsigma_i = \sqrt{3}\langle \Lambda_R, Z_{\mathbf{e}_i}\rangle$ and the margins p_i of Lemma 2.8, the vertex Hamiltonians are $H_i = \langle A, X\rangle + \varsigma_i/p_i$. The weights $\vartheta_i = p_i(p_j + p_k)/8$, $\{i, j, k\} = \{0, 1, 2\}$, sum to 1 by $p_0p_1 + p_1p_2 + p_2p_0 = 4$, and $\sum_i \vartheta_i\varsigma_i/p_i = \frac{1}{8}\sum_i (p_j + p_k)\varsigma_i$ is a multiple of $\langle \Lambda_R, X\rangle = 0$. Hence $\langle A, X\rangle = \sum_i \vartheta_i H_i \leq 0$. Put $m = -\langle A, X\rangle \geq 0$. From $H_i \leq 0$, $(p_j + p_k)\varsigma_i \leq 8m\vartheta_i \leq 8m$, and

since the three products $(p_j + p_k)\varsigma_i$ sum to zero, each is also $\geq -8m$. With $p_j + p_k \geq 4/15$ this gives $\varsigma_i^2 \leq 900m^2$, and $\|\Lambda_R\|^2 \leq \sum_i \varsigma_i^2$. □

Proposition 3.7 (Budget inequality)**.** *Let (T, u, h, Y) be admissible with $\mathcal{C}(T, Y) \leq A_{\mathrm{oct}}$, let $M \in \mathbb{R}^{2\times 2}$ and $\mu \in \mathbb{R}$, and suppose $\langle h(t)^{-1} M h(t) + \frac{3}{2}\mu J, Y(t)\rangle \leq 0$ for $t \in [0, T]$. Then $\mu \leq 100\,\|M + \frac{3}{2}\mu J\|$. Consequently every Pontryagin extremal with cost at most A_{oct} satisfies*

$$\lambda^2 \leq 10^4 \left\|\Lambda_1(t) - \tfrac{3}{2}\lambda J\right\|^2 \qquad (0 \leq t \leq T)$$

.

Proof. Put $A_0 = M + \frac{3}{2}\mu J$, $a_0 = \|A_0\|$ and $V = hYh^{-1}$. The hypothesis reads $\langle A_0, V\rangle + 3\mu\,\omega_R \leq 3\mu\,\omega$. By the Cauchy–Schwarz inequality and $\|V\|^2 = 4\omega_R^2 - 2$, $\langle A_0, V\rangle \geq -2a_0\omega_R$, so $(3\mu - 2a_0)\omega_R \leq 3\mu\,\omega$ pointwise. Integrate, with $I = \int \omega = \mathcal{C}/3$ and $I_R = \int \omega_R \geq 2\pi/3 - I$ by Lemma 3.5. If $\mu > 100a_0$, then $3\mu - 2a_0 > \frac{149}{50}\mu > 0$ and $\frac{149}{50}(\frac{2\pi}{3} - I) < 3I$, so $I > \frac{149}{50} \cdot \frac{2\pi}{3}/(3 + \frac{149}{50}) > 1.0437$, while $A_{\mathrm{oct}} < 3.127$ gives $I < 1.0424$. For the consequence, rebase the twisted extension of the extremal at t (Section 3.5). The rebased trajectory is admissible with the same cost, its costate at time 0 is $\Lambda_1(t)$ by Equation (3.3), and Lemma 3.6 gives the hypothesis with $M = \Lambda_1(t)$ and $\mu = -\lambda$. □

The gap between $A_{\mathrm{oct}} < 3.127$ and π is what makes the budget effective. It bounds the multiplier by the distance $\|A\|$ from Λ_1 to $\frac{3}{2}\lambda J$; adding $\|\Lambda_R\|$ gives the clearance.

Proposition 3.8 (Singular clearance)**.** *Let $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ be a Pontryagin extremal, normal or abnormal, with $\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$ and $X(t) \in \mathfrak{h}^{\star\star}$ for $t \in [0, T]$. Then*

$$\|\mathcal{P}(t)\|^2 \leq 251^2\, Q(t)^2 \qquad (0 \leq t \leq T)$$

. In particular $Q(t) > 0$ for every t: the costate never meets the singular locus.

Proof. Write $\Lambda_1 = A + \frac{3}{2}\lambda J$, so that $\|\Lambda_1\|^2 = \|A\|^2 + 3\lambda(A_{10} - A_{01}) + \frac{9}{2}\lambda^2$, and use $\lambda^2 \leq 10^4\|A\|^2$ from Proposition 3.7 together with $(A_{10} - A_{01})^2 \leq 2\|A\|^2$. The positivity of Q follows from the nontriviality $\|\mathcal{P}\| > 0$. □

Remark 3.9. The same ingredients give quantitative bounds. Gronwall's inequality for $\|g\|^2$ gives $\|g(t)\|^2 \leq 42$ along a trajectory of cost at most A_{oct}, and with Lemma 3.6 on $\mathfrak{h}^{\star\star}$ it gives the projective bounds $10^{-16}\|\mathcal{P}(0)\|^2 < \|\mathcal{P}(t)\|^2 < 10^{16}\|\mathcal{P}(0)\|^2$. With such bounds, operator norms and a width estimate, the notes [7] (`SINGULAR_COST_BUDGET`, `QUANTITATIVE_REGULAR_SWITCH_BOUND`) bound the number of switches by 10^{56}. Only finiteness is needed below.

3.3. **The edge bound.** Suppose that on a time interval one coordinate of the control vanishes almost everywhere, say $u_0 = 0$. The trajectory then runs along the edge of the simplex opposite $\mathbf{e}_0$, and the control is determined by one number $v = u_1 \in [0, 1]$. A priori v could still alternate between the two remaining vertices infinitely often. The following bound excludes this for a global minimizer.

Proposition 3.10 (Edge bound)**.** *Let (T, u, g, X) be a global minimizer, $0 \le t_0 < t_1 \le T$, and $j \in \{0, 1, 2\}$ with $u_j = 0$ almost everywhere on (t_0, t_1). Then on $[t_0, t_1]$ the control is almost everywhere a concatenation of at most four constant vertex pieces of positive length.*

Proof. Relabeling by a power of R (Section 3.5) moves the missing vertex to $j = 0$ and keeps minimality. On the edge, a star state $X = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ has $b = -(1 + a^2)/c$, and the star inequalities read $c > 0$, $3a^2 < c^2 < 3(1 + a^2)$. With $u = (0, v, 1 - v)$ and $\alpha = (2v - 1)/\sqrt{3}$, the state equation becomes the control system

$$a' = \frac{c}{c - 3\alpha a}, \qquad c' = \frac{3\alpha c}{c - 3\alpha a}, \qquad s' = c,$$

in the state (a, c, s), $s = \int c$, with cost density $\frac{3}{2}(c + (1 + a^2)/c)$. The frame is recovered from $x = a/c$ and s. Replacing the segment $[t_0, t_1]$ of the minimizer by any edge trajectory with the same end states and end frames, and translating the rest of the trajectory in time, gives an admissible trajectory whose cost changes by the difference of the segment costs. Since competitors with any terminal time are allowed, the segment minimizes the edge cost among edge trajectories with its end points and free duration.

The maximum principle with free time applies to this system on the open set given by the star inequalities. It gives a multiplier $\nu \le 0$ and a covector $p = (p_a, p_c, p_s)$ with $(\nu, p(t)) \ne 0$ and $H = 0$, and p_s is constant since the system does not depend on s. The derivative of the Hamiltonian in α is $3c\varsigma/(c - 3\alpha a)^2$ with the *switching function* $\varsigma = ap_a + cp_c$, so $v = 1$ where $\varsigma > 0$ and $v = 0$ where $\varsigma < 0$. The adjoint equation, the homogeneity of the field in (a, c) and $H = 0$ give

$$\varsigma' = -c\,(2p_s + 3\nu(1 + x^2)), \qquad x = a/c, \quad x' = 1/c > 0.$$

Since x is strictly increasing, the quadratic $2p_s + 3\nu(1 + x^2)$ vanishes at most twice in time unless $\nu = p_s = 0$; then ς has at most three zeros by Rolle's theorem, and the control has at most four constant pieces. If $\nu = p_s = 0$, then ς is constant, and it is nonzero, because $\varsigma = 0$ and $H = 0$ would force $p = 0$. □

The edge problem can also be written in the clock x, with a fixed-time maximum principle, as in [7] (`MEASURABLE_EDGE_PMP_BOUND`). With the clock adjoined as a state, that version admits the multiplier $\nu = 0$, $p = (\text{const}, 0, 0)$, which carries no information. In physical time the free-time principle gives $H = 0$, which excludes it.

3.4. **Finitely many switches.**

Lemma 3.11 (Local exclusion of a vertex)**.** *Let $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ be a Pontryagin extremal with $X(t) \in \mathfrak{h}^{\star\star}$ on $[0, T]$, and let $t_0 \in [0, T]$ with $Q(t_0) > 0$. There are $\delta > 0$ and vertices j, k such that $H_j < 0$ on $(t_0, t_0 + \delta) \cap [0, T]$ and $H_k < 0$ on $(t_0 - \delta, t_0) \cap [0, T]$.*

Proof. If $H_i(t_0) < 0$ for some i, continuity gives the claim with $j = k = i$. Otherwise all three vertex Hamiltonians vanish at t_0 (Lemma 3.3), so $\Lambda_R(t_0) = 0$ by Lemma 3.4 and $\langle A, X\rangle = 0$ there. Then $Q(t_0) > 0$ gives $A(t_0) \neq 0$, and $b = [X(t_0), A(t_0)] \neq 0$, since a traceless matrix commuting with X is a multiple of X and $\langle X, X\rangle = -2$. The control-dependent terms of Equation (3.1) are $O(\|\Lambda_R\|)$, so $\Lambda_R'(t_0) = b$, and $h_i = \langle \Lambda_R, Z_{\mathbf{e}_i}\rangle / \langle X, Z_{\mathbf{e}_i}\rangle$ has derivative $c_i = \langle b, Z_{\mathbf{e}_i}\rangle / \langle X, Z_{\mathbf{e}_i}\rangle$ at t_0. As b is traceless and orthogonal to X, Lemma 3.4 applied to b shows that the c_i are not all equal. If $c_k < c_j$, then $H_j - H_k = h_k - h_j$ vanishes at t_0 with negative derivative, so $H_j < H_k \leq 0$ just after t_0 and $H_k < H_j \leq 0$ just before. □

Where $H_j < 0$ on an interval, the coordinate u_j of the control vanishes almost everywhere there, because a vertex with positive weight in a maximizing control is maximizing. So the edge bound applies on both sides of t_0.

Proposition 3.12 (Finitely many switches)**.** *Let $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ be a Pontryagin extremal with $\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$ and $X(t) \in \mathfrak{h}^{\star\star}$ on $[0, T]$, satisfying the clearance of Proposition 3.8, and whose control has the property of Proposition 3.10. Then there is n such that on $[0, T]$ the control is almost everywhere a concatenation of n constant vertex pieces of positive length.*

Proof. By Proposition 3.8, $Q(t_0) > 0$ at every t_0. Lemma 3.11 and the edge bound give, around each t_0, a $\delta > 0$ such that every subinterval of $[0, T]$ inside $(t_0 - \delta, t_0 + \delta)$ has a finite vertex description: on the part after t_0 the vertex j is missing and Proposition 3.10 gives at most four pieces, on the part before t_0 the vertex k is missing, and the two descriptions concatenate at t_0. A Lebesgue number of this cover of the compact interval $[0, T]$ and induction along a uniform partition give a finite description of $[0, T]$. □

A tie of the three vertex Hamiltonians can be treated quantitatively, with three guard lines transported along the flow, which bounds the number of switches explicitly [7]. For finiteness the first-order expansion $\Lambda_R(t) = (t - t_0)[X(t_0), A(t_0)] + o(t - t_0)$ is enough.

3.5. **Vertex representations and rebasing.** The pieces of Proposition 3.12 may have equal neighbouring labels, and the cut at $t = 0$ need not be a switch. Both are repaired by extending the trajectory periodically. The *twisted extension* of an admissible trajectory (T, u, g, X) is defined for $t \geq 0$ by

$$G(t) = g(t - nT)R^n, \qquad Y(t) = R^{-n}X(t - nT)R^n, \qquad n = \lfloor t/T \rfloor,$$

and with the control $v(t)$ equal to $u(t-nT)$ with its coordinates shifted n times, so that the vertex $\mathbf{e}_i$ becomes $\mathbf{e}_{i+n}$ and $Z_{v(t+T)} = R^{-1} Z_{v(t)} R$. The extension satisfies the state equations on every interval $[0, nT]$. The trajectory *rebased* at $s \in [0,T)$ is, for $0 \le t \le T$,

$$\hat{g}(t) = G(s)^{-1} G(s+t), \qquad \hat{X}(t) = Y(s+t), \qquad \hat{u}(t) = v(s+t).$$

It is again admissible, and it has the same cost because J commutes with R; so a rebased global minimizer is a global minimizer. Similarly the *relabeling* $(g, X) \mapsto (R^{-r} g R^r, R^{-r} X R^r)$, with the control coordinates shifted r times, preserves admissibility and cost.

Definition 3.13 (Vertex representation)**.** A *vertex representation* of a control u on $[0,T]$ consists of an integer $N \ge 1$, a strictly increasing sequence $(\tau_k)_{k \in \mathbb{Z}}$ of *switching times* with $\tau_0 = 0$ and $\tau_{k+N} = \tau_k + T$, and *labels* $i_k \in \mathbb{Z}/3$ with $i_{k+N} = i_k + 1$ and $i_{k+1} \ne i_k$ for every $k \in \mathbb{Z}$, such that for $0 \le k < N$ the control equals $\mathbf{e}_{i_k}$ almost everywhere on (τ_k, τ_{k+1}). The interval (τ_k, τ_{k+1}) is the k-th *piece*, and N is the number of pieces of the period.

The condition $i_{k+1} \ne i_k$ for every k, including $k = N-1$, says that equal neighbours have been merged also across the seam $t = T$, where the label after the last piece is $i_N = i_0 + 1$.

Lemma 3.14 (Merging and rebasing)**.** *If on $[0,T]$ the control u is almost everywhere a concatenation of n vertex pieces, there is $s \in [0,T)$ such that the rebased control $\hat{u}$ has a vertex representation with $N \le n$ pieces.*

Proof. Extend the pieces by the twist to a bi-infinite sequence with $\tau_{k+n} = \tau_k + T$ and labels $l_{k+n} = l_k + 1$. Call τ_k a genuine switch if $l_k \ne l_{k-1}$. The increments $l_k - l_{k-1}$ over one period sum to $1 \ne 0$ in $\mathbb{Z}/3$, so some increment is nonzero, and a genuine switch $s = \tau_{k_0}$ lies in $[0,T)$. Merge the pieces of $[s, s+T]$ with equal neighbouring labels, shift by $-s$, and extend the reduced word to $\mathbb{Z}$ by the twist; the seam condition is the genuine switch at s. □

A genuine switch can also be obtained from the fixed-edge obstruction of [7] (`SWITCHING_WORDS`), which uses the closing conditions; the count of increments modulo 3 needs neither those conditions nor optimality.

3.6. **Normality.**

Definition 3.15 (Normal period)**.** A *normal period* is a Pontryagin extremal $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ with $\lambda = -1$, together with a vertex representation of its control u. It is *competitive* if $\mathcal{C}(T, X) \le A_{\mathrm{oct}}$, and it lies in the compact domain if $X(t) \in \mathfrak{h}^{\star\star}$ for $t \in [0,T]$. The *active margin* of the k-th piece is $a_k = p_{i_k}(X(\tau_k))$, with k read modulo N.

Theorem 3.16 (Normality)**.** *Let (T, u, g, X) be a global minimizer with a Pontryagin lift $(\Lambda_1, \Lambda_R, \lambda)$, and suppose that u has a vertex representation. Then $\lambda < 0$, and $(\Lambda_1, \Lambda_R, \lambda)/(-\lambda)$ is a Pontryagin lift with multiplier -1.*

The proof reads an abnormal costate at the switching times, in coordinates in which every piece acts by an explicit linear map, and derives a contradiction with the twisted closing condition.

Coordinates. For $M \in \mathfrak{sl}_2(\mathbb{R})$ put $\mu_j(M) = \sqrt{3}\langle M, Z_{\mathbf{e}_j}\rangle$, so that the margins are $p_j = -\mu_j(X)$. With $\lambda = 0$ put

$$a = -X\Lambda_1 X, \qquad b = \Lambda_R + 2\Lambda_1 + \langle \Lambda_1, X\rangle X.$$

Using $X^2 = -I$ and $X\Lambda X = \langle \Lambda, X\rangle X + \Lambda$ one finds $p_j H_j = 2\mu_j(a) + \mu_j(b)$; this is the abnormal analogue of the switching functions χ_j of Definition 4.9. Moreover $\langle b, X\rangle = \langle \Lambda_R, X\rangle = 0$, and $(a, b) = 0$ only if $(\Lambda_1, \Lambda_R) = 0$. For a label i and $q > 0$ let $L_i(q)$ be the linear map of $\mathbb{R}^3$ that fixes the coordinate i and sends

$$v_{i+1} \mapsto (v_i + v_{i+1})q - v_i, \qquad v_{i-1} \mapsto (v_i + v_{i-1})/q - v_i,$$

the one-arc margin map of Lemma 4.1.

Lemma 3.17 (Abnormal transport)**.** *Let the extremal have $\lambda = 0$, and let $u = \mathbf{e}_i$ almost everywhere on a piece (s_0, s_1), with $q = \exp(2(s_1 - s_0)/p_i(X(s_0)))$. Then $H_i(s_0) = 0$; the margins and the coordinates $\mu(b)$ at s_1 are those at s_0 transformed by $L_i(q)$; and the coordinates $\mu(a)$ are transformed by $L_i(q^2)$.*

Proof. Write $S_j = \sqrt{3}Z_{\mathbf{e}_j}$. On the piece $P_u = -S_i/p_i$ is constant, since the active margin p_i is constant. The state, a and b satisfy $f' = \kappa[P_u, f]$ with $\kappa = 1, 2, 1$ respectively; for b this uses that the active vertex Hamiltonian vanishes on the piece. The relation $[S_i, S_{i+1}] = 2(S_i + S_{i+1})$ and the invariance of the trace form make $\mu_i(f)$ constant and multiply $\mu_i(f) + \mu_{i\pm1}(f)$ by $q^{\pm\kappa}$. □

Proof of Theorem 3.16. Suppose $\lambda = 0$. Continue the lift through the seams by the twist, using the closing conditions $X(T) = R^{-1}X(0)R$, $\Lambda_1(T) = R^{-1}\Lambda_1(0)R$, $\Lambda_R(T) = R^{-1}\Lambda_R(0)R$, which relabel the coordinates. At the switching times τ_k, $k \geq 0$, this gives labels i_k with $i_{k+1} \neq i_k$ and $i_{k+N} = i_k + 1$, factors $q_k > 1$, positive margins, and coordinates $\alpha_k = \mu(a(\tau_k))$, $\beta_k = \mu(b(\tau_k))$, related by Lemma 3.17, with $\sum_{j\neq l} \beta_{k,j} p_l = 0$, the active quantity $\chi_{i_k}(k) = 2\alpha_{k,i_k} + \beta_{k,i_k} = 0$, $\chi_j(k) = 2\alpha_{k,j} + \beta_{k,j} \leq 0$ for all j by Lemma 3.3, and $(\alpha_k, \beta_k) \neq 0$. Four facts about such a sequence follow.

(1) No piece has a triple tie, $\chi_0 = \chi_1 = \chi_2 = 0$, at both of its ends. Indeed, for $j \neq i = i_k$ the pair sums $x = \alpha_i + \alpha_j$ and $y = \beta_i + \beta_j$ are multiplied by ϱ^2 and ϱ along the piece, $\varrho \in \{q_k, q_k^{-1}\}$. The ties $2x + y = 0$ at both ends give $x(\varrho^2 - \varrho) = 0$, so $x = y = 0$. Hence $\beta_i = -2\alpha_i$ and $\beta_j = 2\alpha_i$ for $j \neq i$, and the orthogonality relation becomes $4\alpha_i p_i = 0$. Then $\alpha = \beta = 0$, against nontriviality.

(2) If three consecutive pieces carry labels i, j, i, then χ_j vanishes at the start of the first and at the end of the third.

(3) No three consecutive switches have the same sign: the inequalities $\chi \le 0$ for the missing vertex at the two inner switches, combined with the transports, force triple ties at both ends of the middle piece, against (1).

(4) From the second switch on, neighbouring switches have opposite signs, that is, $i_{k+2} = i_k$ for $k \ge 1$. This follows from (1)–(3).

By (4), $i_{1+2N} = i_1$. By the twist, $i_{1+2N} = i_{1+N} + 1 = i_1 + 2$. Since $2 \neq 0$ in $\mathbb{Z}/3$, this is a contradiction. □

The argument does not use the minimality of (T, u, g, X): an abnormal lift of any trajectory with a vertex representation contradicts the twisted closing condition.

Another argument [7] (`GLOBAL_NORMALITY`) uses needle variations and an endpoint covector pulled back through the endpoint map, and an explicit four-arc matrix computation behind (2) and (3). In the coordinates $\mu(a)$, $\mu(b)$ every piece acts by the one-arc map, the needle inequalities become the maximum condition at the switching times, and both triple-tie implications follow from one forward identity.

3.7. **The finite normal representation.**

Theorem 3.18 (Finite normal representation)**.** *Let (T, u, g, X) be a global minimizer of the control problem. There are a normal period $(T, \hat{u}, \hat{g}, \hat{X}, \Lambda_1, \Lambda_R, -1)$, with a vertex representation of $\hat{u}$, and a time $s \in [0, T)$ such that*

(1) $\hat{g}(t) = G(s)^{-1} G(s + t)$ and $\hat{X}(t) = Y(s + t)$ for $t \in [0, T]$, where (G, Y) is the twisted extension of (g, X);

(2) $\mathcal{C}(T, \hat{X}) = \mathcal{C}(T, X)$, and $(T, \hat{u}, \hat{g}, \hat{X})$ is a global minimizer;

(3) $\hat{X}(t) \in \mathfrak{h}^{\star\star}$ for every $t \in [0, T]$.

In particular the period is competitive, $\mathcal{C}(T, \hat{X}) \le A_{\mathrm{oct}}$.

Proof. The state of the minimizer stays in $\mathfrak{h}^{\star\star}$ (Theorem 2.14), and its cost is at most A_{oct} since the octagon is a competitor. Take a Pontryagin lift (Theorem 3.2); it may be abnormal. It satisfies the clearance (Proposition 3.8), and the minimizer has the edge property (Proposition 3.10), so its control is a finite concatenation of vertex pieces (Proposition 3.12). Merge and rebase at a genuine switch s (Lemma 3.14). The rebased trajectory is a global minimizer with the same cost (Section 3.5), so Theorem 3.2 applies to it directly; Theorem 3.16 makes the new lift normal and rescales it to $\lambda = -1$, and Theorem 2.14 applies to the rebased minimizer as well. □

No costate is transported through the rebasing: the lift of the rebased minimizer is taken afresh, which the minimality of the rebased trajectory allows. The theorem keeps the actual minimizer, with its own frame and state, only restarted at s; the equality case of Section 11 needs this. From Section 4 on, the hatted period is written (T, u, g, X) again, with switching times τ_k, labels i_k and active margins a_k.

4. Vertex periods

Theorem 3.18 turns a global minimizer into a normal vertex period: a normal Pontryagin extremal whose control is a vertex of the simplex on each of finitely many pieces (Definitions 3.13 and 3.15). On a piece of constant vertex control the state, the frame and the costate solve linear differential equations with constant coefficients, so everything that happens along the piece is given by explicit formulas in a few numbers at its start. A period is then described by finitely many numbers, related by algebraic and logarithmic identities, and the maximum principle becomes a list of sign conditions on them. This section collects these formulas and conditions, which the case analysis of Sections 5 to 9 uses, and states the theorem that reduces the problem to periods with at most five pieces.

Throughout, P is a normal vertex period with N pieces, switching times τ_k, labels $i_k \in \mathbb{Z}/3$ and terminal time T, indexed by $k \in \mathbb{Z}$ through the twist $\tau_{k+N} = \tau_k + T$, $i_{k+N} = i_k + 1$ (Definition 3.13). Its state X, frame g, costates Λ_1, Λ_R and cost multiplier $\lambda = -1$ are those of Definition 3.15. We write $m_k(j) = p_j(X(\tau_k))$ for the margins at the switch τ_k ($0 \le k \le N$). Since $p_j(R^{-1}XR) = p_{j-1}(X)$ and $X(T) = R^{-1}X(0)R$, the rule

$$m_{k+N}(j) = m_k(j-1) \tag{4.1}$$

extends them consistently to all $k \in \mathbb{Z}$. The *active margin* of piece k is $a_k = m_k(i_k)$, the margin of its own vertex at its start, and its *factor* is

$$q_k = \exp\bigl(2(\tau_{k+1} - \tau_k)/a_k\bigr) > 1,$$

so that the piece lasts $(a_k/2)\log q_k$. Both sequences are N-periodic. We also write

$$s_k = a_k + a_{k+1}, \qquad K_k = a_k^2 + 4.$$

4.1. **One piece of constant control.** For a label i and $w > 0$ let

$$T_i(w) = \frac{w + w^{-1}}{2}\, I + \frac{\sqrt{3}\,(w - w^{-1})}{2}\, Z_{\mathbf{e}_i}$$

be the *tangent factor* of the vertex $\mathbf{e}_i$. Since $Z_{\mathbf{e}_i}$ is traceless with determinant $-1/3$, $Z_{\mathbf{e}_i}^2 = I/3$, and $T_i(w)$ is the exponential of a multiple of $Z_{\mathbf{e}_i}$: $T_i(w)T_i(v) = T_i(wv)$, $\det T_i(w) = 1$, $T_i(w)^{-1} = T_i(w^{-1})$ and $R^{-1}T_i(w)R = T_{i+1}(w)$.

Lemma 4.1 (One piece)**.** *Let (g, X) be a state trajectory on $[0, T]$ whose control is the vertex $\mathbf{e}_i$ almost everywhere on $(s_0, s_1) \subset [0, T]$, and let $(a, b, c) = (p_i, p_{i+1}, p_{i-1})(X(s_0))$. For $t \in [s_0, s_1]$ put $w = \exp((t - s_0)/a)$ and $q = w^2$. Then*

(a) *$X(t) = T_i(w)^{-1}X(s_0)T_i(w)$, and the tangent frame $V = gX$ satisfies $V(t) = V(s_0)\,T_i(q)$;*

(b) *the margins at time t are*

$$p_i = a, \qquad p_{i+1} = (a + b)q - a, \qquad p_{i-1} = \frac{a + c}{q} - a; \tag{4.2}$$

(c) *with* $q = \exp(2(s_1 - s_0)/a)$*, the cost of the piece is*

$$-\frac{3}{2}\int_{s_0}^{s_1} \langle J, X(t)\rangle \,\mathrm{d}t = \frac{\sqrt{3}}{4}\, a((a+b)(q-1)+(a+c)(1-q^{-1}) - a\log q). \quad (4.3)$$

For a normal vertex period this gives, for every $k \in \mathbb{Z}$*,*

$$m_{k+1}(i_k) = a_k, \qquad m_{k+1}(i_k+1) = \big(a_k + m_k(i_k+1)\big)q_k - a_k,$$
$$m_{k+1}(i_k-1) = \frac{a_k + m_k(i_k-1)}{q_k} - a_k,$$

and the cost $\mathcal{C}(T, X)$ *of the period is the sum of* (4.3) *over the pieces* $0 \le k < N$*.*

Proof. The margin p_i is, up to a factor, the trace pairing with $Z_{\mathbf{e}_i}$, and $\langle Z_{\mathbf{e}_i}, [Z_{\mathbf{e}_i}, X]\rangle = 0$. Along the piece $P_{\mathbf{e}_i}(X) = Z_{\mathbf{e}_i}/\langle Z_{\mathbf{e}_i}, X\rangle$ is a multiple of $Z_{\mathbf{e}_i}$, so p_i is constant and $P_{\mathbf{e}_i}(X) = -(\sqrt{3}/a)Z_{\mathbf{e}_i}$ is a constant matrix. The state equation $X' = [P, X]$ then has the solution $X(t) = e^{(t-s_0)P}X(s_0)e^{-(t-s_0)P}$, and $e^{(t-s_0)P} = T_i(w)^{-1}$. The margins of this conjugate are a direct computation. Writing $g(t) = h(t)e^{-(t-s_0)P}$, the frame equation becomes $h' = h\,(X(s_0)+P)$ with a constant traceless matrix, whose exponential is explicit. Uniqueness of Carathéodory solutions identifies the actual trajectory with this flow, also when the control differs from $\mathbf{e}_i$ on a null set. Integrating $-\frac{3}{2}\langle J, X\rangle = \frac{\sqrt{3}}{2}(p_0+p_1+p_2)$ with (4.2) gives (4.3). □

Each margin stays positive along a piece, so a piece with factor q is admissible exactly when $1 < q < 1 + c/a$: the margin of the label $i-1$ decreases and must not reach 0. We call $i+1$ the *growing* label and $i-1$ the *decreasing* label of the piece.

The margins at a switch lie on the quadric $m_k(0)m_k(1) + m_k(1)m_k(2) + m_k(2)m_k(0) = 4$ (Lemma 2.8), and two of them are active margins: at τ_k the label i_{k-1} has margin a_{k-1} (it was constant along piece $k-1$) and the label i_k has margin a_k. The third margin is therefore $(4 - a_{k-1}a_k)/s_{k-1}$, and the update rule of Lemma 4.1 expresses each factor through the active margins alone.

Lemma 4.2 (Factors from active margins)**.** *For every* k*, according to the signs of the switches before and after piece* k *(Definition 4.5 below),*

forward, forward:	$q_kK_k = s_{k-1}s_k$,	*reverse, forward:*	$q_ks_{k-1} = s_k$,
reverse, reverse:	$q_ks_{k-1}s_k = K_k$,	*forward, reverse:*	$q_ks_k = s_{k-1}$

Proof. Consider two forward switches. The label $i_k - 1 = i_{k-1}$ of the previous piece has margin a_{k-1} at τ_k, so the growing label has margin $b = (4 - a_{k-1}a_k)/s_{k-1}$ and $a_k + b = K_k/s_{k-1}$. The next piece has the growing label, so $a_{k+1} = (a_k + b)q_k - a_k$ and $s_k = K_kq_k/s_{k-1}$. The other three cases are the same computation with the roles of the growing and the decreasing label exchanged. □

Up to the labels, a period is therefore determined by its signs and its active margins, subject to $q_k > 1$ and $a_k a_{k+1} < 4$. At a turn, $q_k > 1$ reads $a_{k+1} > a_{k-1}$ after a reverse switch and $a_{k+1} < a_{k-1}$ after a forward one. The cost takes a symmetric form in these variables. Let

$$G(a,b) = 3ab - 4 - (a^2+b^2)\log(a+b) + \tfrac{1}{2}a^2\log(a^2+4) + \tfrac{1}{2}b^2\log(b^2+4),$$

and $G_a(a,b) = \partial G/\partial a = 3b - 2a\log(a+b) - \frac{a^2+b^2}{a+b} + a\log(a^2+4) + \frac{a^3}{a^2+4}$.

Lemma 4.3 (Signed action)**.** *With the switch signs $\varepsilon_k \in \{\pm 1\}$ of Definition 4.5,*

$$\frac{8\,\mathcal{C}(T,X)}{\sqrt{12}} = \sum_{k=0}^{N-1} \varepsilon_k\, G(a_k, a_{k+1}).$$

Proof. Write the cost (4.3) of piece k, times $4/\sqrt{3}$, with the margins of Lemma 4.2. It splits as $\varepsilon_k F_1(a_k, a_{k+1}) + \varepsilon_{k-1} F_1(a_k, a_{k-1})$, a term for the switch after the piece and one for the switch before it, with

$$F_1(a,n) = a\Big(n - \frac{4-an}{a+n}\Big) - a^2\log(a+n) + \frac{a^2}{2}\log(a^2+4).$$

The two terms that belong to the switch k add up to $\varepsilon_k(F_1(a_k,a_{k+1}) + F_1(a_{k+1},a_k)) = \varepsilon_k G(a_k,a_{k+1})$. □

The derivative of the sum in Lemma 4.3 with respect to one active margin a_k, the others fixed, is $\varepsilon_{k-1}G_a(a_k,a_{k-1}) + \varepsilon_k G_a(a_k,a_{k+1})$. This quantity reappears in the stationarity identity (Lemma 4.13).

4.2. **The tangent closure.** The tangent frame $V = gX$ is multiplied on the right by $T_{i_k}(q_k)$ along piece k (Lemma 4.1(a)). It starts at $V(0) = X(0)$ and ends at $V(T) = R\,R^{-1}X(0)R = X(0)R$, and $X(0)$ is invertible. This gives one matrix equation that every period satisfies.

Lemma 4.4 (Tangent closure)**.** *The factors and labels of a normal vertex period satisfy*

$$T_{i_0}(q_0)\,T_{i_1}(q_1)\cdots T_{i_{N-1}}(q_{N-1}) = R.$$

The closure is simplest in the *triangular coordinates* $Y \mapsto CYC^{-1}$, $C = \left(\begin{smallmatrix}\sqrt{3} & 1\\ \sqrt{3} & -1\end{smallmatrix}\right)$, in which

$$T_0(q) = \begin{pmatrix} q & 0 \\ 0 & q^{-1}\end{pmatrix}, \qquad T_1(q) = \begin{pmatrix} q^{-1} & q-q^{-1} \\ 0 & q\end{pmatrix},$$
$$T_2(q) = \begin{pmatrix} q^{-1} & 0 \\ q^{-1}-q & q\end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \\ -1 & 1\end{pmatrix}.$$

The change of coordinates has $\det C = -2\sqrt{3} < 0$: it reverses orientation, which matters for the sign of ellipticity in Section 4.6. The labels, the factors and the margins at the switches, related by the update rule of Lemma 4.1 and by the closure read from every switch, are all that Section 10 needs to solve the closure for $N \le 5$.

4.3. **Switch signs and the five classes.** Neighbouring labels differ, and labels live in $\mathbb{Z}/3$, so each switch either advances the label by one or moves it back by one.

Definition 4.5 (Signs, runs, turns, centres)**.** The switch k (from piece k to piece $k+1$) is *forward* if $i_{k+1} = i_k + 1$ and *reverse* if $i_{k+1} = i_k - 1$; its sign is $\varepsilon_k = +1$ or $\varepsilon_k = -1$. A *maximal reverse run* of length $L \geq 1$ is a block of reverse switches $k, \ldots, k+L-1$ preceded and followed by forward switches; maximal forward runs are defined in the same way. A forward run of length one is *short*. Piece k is a *turn* if $\varepsilon_{k-1} \neq \varepsilon_k$, a *centre* if $\varepsilon_{k-1} = \varepsilon_k = -1$, and a *forward same-sign piece* if $\varepsilon_{k-1} = \varepsilon_k = +1$.

The margin of the growing label increases along a piece, so a forward switch hands the control to the vertex whose margin has just grown, and a reverse switch hands it to the vertex whose margin has just decreased. Both kinds occur in extremals, but they behave differently: runs of reverse switches carry the first-order obstructions of Section 5, while the forward switches of Section 7 are constrained only through the whole period.

Lemma 4.6 (Sign sum)**.** $\sum_{k=0}^{N-1} \varepsilon_k \equiv 1 \pmod 3$. *In particular an entirely forward period has* $N \equiv 1 \pmod 3$ *and an entirely reverse period has* $N \equiv 2 \pmod 3$.

Proof. The label advances by ε_k at the switch k, and by one over a period: $i_N = i_0 + 1$ (Definition 3.13). Hence $\sum_{k<N} \varepsilon_k \equiv 1$, and if all the signs are equal the sum is $\pm N$. □

The smoothed octagon's period has the word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$: three forward switches inside the period and the seam switch from $\mathbf{e}_2$ to $\mathbf{e}_{2+1} = \mathbf{e}_0$, which is forward as well (Figure 3). So $N = 4$ and the sign sum is $4 \equiv 1$.

Definition 4.7 (The five classes)**.** A vertex period is in class

- R if every switch is reverse;
- F if every switch is forward;
- M3 if it is mixed (both signs occur) and some maximal reverse run has length other than two;
- M1 if it is mixed, every maximal reverse run has length two and every maximal forward run has length one, so that the cyclic sign word is $(--+)^m$;
- M2 if it is mixed, every maximal reverse run has length two and some maximal forward run has length at least two.

Proposition 4.8 (Partition)**.** *Every vertex representation is in one of the classes* R*,* F*,* M3*,* M1*,* M2*.*

Proof. Every switch is forward or reverse. If both signs occur, the period is mixed, and the remaining alternatives are exhaustive by their definitions. □

4.4. **Costate coordinates.** The costate enters the case analysis only through the vertex Hamiltonians, whose signs the maximum principle controls. We use coordinates in which these Hamiltonians are linear and in which the costate equations along a piece decouple into scalar equations.

Definition 4.9 (Costate coordinates)**.** For a star state X and costates Λ_1, Λ_R with $\lambda = -1$, let $L = \Lambda_1 + \frac{3}{2}J$ and

$$\hat{A} = XLX, \qquad \hat{B} = -(\Lambda_R + 2L + \langle L, X\rangle X),$$

and $\alpha_j = p_j(\hat{A})$, $\beta_j = p_j(\hat{B})$, where the margin $p_j(M) = -\sqrt{3}\langle Z_{\mathbf{e}_j}, M\rangle$ is used as a linear function of every matrix M. The *switching function* of the vertex $\mathbf{e}_j$ is $\chi_j = 2\alpha_j + \beta_j$.

The costate is recovered from (X, α, β) by $L = -X\hat{A}X$ and $\Lambda_R = -\hat{B} - 2L - \langle L, X\rangle X$, so these are coordinates.

Lemma 4.10 (The costate along a piece)**.** *Let P be a normal vertex period, and write $\alpha_j(k), \beta_j(k), \chi_j(k)$ for the coordinates at the switch τ_k, extended to $k \in \mathbb{Z}$ by the twist like (4.1).*

(a) *At every time, $p_j(X)\, H_j = \chi_j$ for each vertex Hamiltonian H_j, and $\langle \Lambda_R, X\rangle = 0$ reads $\sum_{j\neq l} \beta_j\, p_l = 0$.*

(b) *Along piece k, with $(a, b, c) = (a_k, m_k(i_k + 1), m_k(i_k - 1))$, $i = i_k$ and factor $q = q_k$, the coordinate α_i increases by $\mathrm{Act}(a, b, c, q)$ and β_i decreases by twice that amount, so that χ_i is constant, and*

$$\begin{aligned}
(\alpha_i + \alpha_{i+1})(k+1) &= q^2(\alpha_i + \alpha_{i+1})(k) + \mathrm{AG}(a, b, q),\\
(\alpha_i + \alpha_{i-1})(k+1) &= q^{-2}(\alpha_i + \alpha_{i-1})(k) + \mathrm{AD}(a, c, q),\\
(\beta_i + \beta_{i+1})(k+1) &= q\,(\beta_i + \beta_{i+1})(k) + \mathrm{BG}(a, b, q),\\
(\beta_i + \beta_{i-1})(k+1) &= q^{-1}(\beta_i + \beta_{i-1})(k) + \mathrm{BD}(a, c, q),
\end{aligned}$$

with the explicit functions $\mathrm{Act}, \mathrm{AG}, \mathrm{AD}, \mathrm{BG}, \mathrm{BD}$ *of Appendix A.1. The same formulas with an intermediate factor $s \in [1, q]$ in place of q give the coordinates inside the piece.*

(c) *$\chi_{i_k}(k) = \chi_{i_k}(k+1) = 0$ for every k; $\chi_j(k) \le 0$ for every j and k; and along piece k the switching functions of the growing and of the decreasing label are ≤ 0 at every intermediate factor $s \in [1, q_k]$.*

Proof. (a) is algebra with the identity $X\Lambda X = \langle \Lambda, X\rangle X + \Lambda$ for traceless Λ. For (b), on a piece the matrix $P = -(\sqrt{3}/a)Z_{\mathbf{e}_i}$ is constant and the costate equations of Definition 3.1 become

$$\hat{A}' = 2[P, \hat{A}] - \tfrac{3}{2}[J, X], \qquad \hat{B}' = [P, \hat{B}] + 3[J, X] - H_i[P, X],$$

with $H_i = 0$ on the piece. In margin coordinates the bracket with $Z_{\mathbf{e}_i}$ annihilates p_i and sends $p_{i\pm1}$ to $\mp(2/\sqrt{3})(p_i + p_{i\pm1})$, while $\frac{3}{2}[J, F]$ has margins $\sqrt{3}\,(p_{j+1}(F) - p_{j-1}(F))$. Each of the six combinations in (b) therefore solves a scalar linear equation $y' = \kappa y + f$ with f explicit through (4.2), and an

integrating factor solves it. (c) is (a) together with the maximum principle: every vertex Hamiltonian is ≤ 0 at every time and the active one vanishes on its piece (Lemma 3.3), and the margins are positive. □

The proofs in Sections 5 to 9 use the costate only through the data listed in Lemma 4.10: the coordinates at the switches, their transport, the vanishing of the active switching functions, the sign conditions at the switches and along the pieces, and $\langle \Lambda_R, X\rangle = 0$. In particular no needle variation of the maximum principle is constructed: each first-order argument becomes an exact linear identity between switching functions at the switches (Section 5).

4.5. **The normal multiplier.** The primary costate is conserved up to conjugation, $\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)g(t)$, so its determinant is a constant of the motion. Read in the moving frame of the state, it gives a matrix whose margins are transported as simply as the margins of the state.

Definition 4.11 (Normal multiplier)**.** Let $\mathcal{U}(t) = \frac{1}{\sqrt{12}}X(t)^{-1}\Lambda_1(t)X(t)$. At the switch k put

$$\mu_j(k) = -\frac{\alpha_j(k)}{2\sqrt{3}} - \frac{\big(m_k(0)+m_k(1)+m_k(2)\big)\,m_k(j)}{4} + \frac{1}{2}$$

; at $k=0$ these are the margins of $\mathcal{U}(0)$. The *Casimir* of P is

$$\gamma^2 = \tfrac{1}{4}(\mu_0\mu_1+\mu_1\mu_2+\mu_2\mu_0)(0) = \det\mathcal{U}(0) = \frac{\det\Lambda_1(0)}{12},$$

and when it is positive, $\kappa = \sqrt{12\gamma^2} = \sqrt{\det\Lambda_1}$. In triangular coordinates the traceless matrix with margins (p_0,p_1,p_2) is

$$Y(p) = \frac{1}{2}\begin{pmatrix} -p_0 & p_0+p_2 \\ -p_0-p_1 & p_0 \end{pmatrix}, \qquad \det Y(p) = \tfrac{1}{4}(p_0p_1+p_1p_2+p_2p_0).$$

The *normal multiplier* read in the frame of piece k is

$$M_k = Y\big(\mu_{i_k}(k), \mu_{i_k+1}(k), \mu_{i_k-1}(k)\big),$$

and $U = M_0$.

Along piece k (label i, factor q) the coordinate μ_i is constant, $\mu_i+\mu_{i+1}$ is multiplied by q^2 and $\mu_i+\mu_{i-1}$ by q^{-2}: the inhomogeneous terms of Lemma 4.10(b) cancel against those of the margins. Hence the quadratic form $\frac{1}{4}(\mu_0\mu_1+\mu_1\mu_2+\mu_2\mu_0)$ takes the value γ^2 at every switch. In the triangular frame the transport is a conjugation, $M_{k+1} = A_k^{-1}M_kA_k$ with

$$A_k = T_0(q_k)\,R^{-\varepsilon_k}.$$

Put $P_n = A_0A_1\cdots A_{n-1}$, so that $M_n = P_n^{-1}UP_n$. Since $R^{-1}T_iR = T_{i+1}$ and $R^3 = -I$, the tangent closure (Lemma 4.4) gives $P_N = \pm I$.

The multiplier enters the estimates through the symmetric matrix $S = -J_0U$, with $J_0 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, whose determinant is γ^2, and through three

sequences of numbers. Let $w_k = -\mu_{i_k}(k)$ and

$$d_k = \frac{w_k + w_{k+1}}{2s_k^2}, \qquad e_k = \frac{w_k}{2K_k}, \qquad j_k = \frac{a_k^2 + a_k a_{k+1} + a_{k+1}^2}{4s_k^2}.$$

The number d_k is the *switch norm* of the switch k and e_k the *cross pairing* at piece k; the number $j_k \geq 3/16$ depends only on the active margins. The names come from the *switch vectors*. In the frame of piece k let the outgoing vector be $o_k = (q_k/s_k, 0)$ if the switch k is forward and $o_k = (0, 1/(q_k s_k))$ if it is reverse, and let the incoming vector be $\iota_k = (0, -1/s_{k-1})$ or $(-1/s_{k-1}, 0)$ according as the switch $k-1$ is forward or reverse. The switch vector of the switch $n \geq 0$ is $y_n = P_n\, o_n$; it is also $P_{n+1}\iota_{n+1}$.

Lemma 4.12 (Switch vectors)**.** (a) $y_n^{\mathsf{T}} S y_n = d_n$ *for every* $n \geq 0$.

(b) *At a same-sign piece* $k \geq 1$, $\iota_k^{\mathsf{T}}(-J_0 M_k)\, o_k = e_k$ *and* $\det(\iota_k, o_k) = \varepsilon_k/K_k$*; hence* $y_{k-1}^{\mathsf{T}} S y_k = e_k$ *and* $\det(y_{k-1}, y_k) = \varepsilon_k/K_k$.

(c) *At a turn* k, $o_k = -\iota_k$*, so* $y_k = -y_{k-1}$.

(d) $y_{N+n} = \pm y_n$*, with the sign of* $P_N = \pm I$.

The entries of $-J_0 M_k$ are $-(\mu_i + \mu_{i+1})/2$, $\mu_i/2$, $\mu_i/2$ and $-(\mu_i + \mu_{i-1})/2$, with $i = i_k$, and (a)–(c) evaluate this quadratic form on the vectors above, using the factors of Lemma 4.2 and the transport of μ.

The maximum principle imposes two kinds of conditions at a switch: the switching function of the new control vanishes there, and the switching function of the old control stays ≤ 0 just after it. The first gives an identity, the second an inequality.

Lemma 4.13 (Stationarity)**.** *For every* k,

$$2\big(\varepsilon_{k-1} s_{k-1} d_{k-1} + \varepsilon_k s_k d_k - (\varepsilon_{k-1} + \varepsilon_k)\, a_k e_k\big) = \frac{1}{8}\big(\varepsilon_{k-1} G_a(a_k, a_{k-1}) + \varepsilon_k G_a(a_k, a_{k+1})\big).$$

At a centre k*, with* $a = a_k$, $p = a_{k-1}$, $n = a_{k+1}$, $r = a/(a+p)$, $t = a/(a+n)$ *and* $\rho = a^2/(a^2+4)$*, this reads*

$$16 e_k = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1 - 8d_{k-1}}{r} - \frac{1 - 8d_k}{t}.$$

The right side is $1/8$ of the derivative of the signed action of Lemma 4.3 with respect to a_k; the left side is the corresponding derivative of the closure condition, paired with the multiplier. The proof is local: by Lemma 4.10, the difference of the two sides is an explicit nonzero multiple of the switching function of the next control at the end of piece k, which vanishes. The identity can also be written as $\partial_{a_k}(\mathcal{C}/\sqrt{12}) = \operatorname{tr}(UV_k)$ with $V_k = (\partial P_N/\partial a_k) P_N^{-1}$, and derived from a boundary identity of the maximum principle ([7], `MIXED_GLOBAL_COSTATE_ELLIPTIC` (8)).

Lemma 4.14 (Switch norms)**.** (a) *At a forward switch* $d_k \geq j_k/2$*; at a reverse switch* $d_k \leq j_k/2$.

(b) *At a turn k, $d_{k-1} = d_k$ and $16(a_{k+1} - a_{k-1})\, d_k = G_a(a_k, a_{k+1}) - G_a(a_k, a_{k-1})$. With $f(x) = x - x^{-1} - \log x$ this is the secant*

$$d_k = \frac{1}{8} \cdot \frac{f(1 + a_{k+1}/a_k) - f(1 + a_{k-1}/a_k)}{a_{k+1}/a_k - a_{k-1}/a_k},$$

and $d_k \geq 3/32$.

Proof. (a) At a reverse switch k the old label $i_k = i_{k+1}+1$ is the growing label of piece $k+1$. Its switching function vanishes at the start of that piece and is ≤ 0 along it. By Lemma 4.10(b) it equals $2s(Y(s-1) + \frac{\sqrt{3}a}{2}F(s))$ at the intermediate factor s, where $a = a_{k+1}$, $Y = (\alpha_i + \alpha_{i+1})(k+1)$ with $i = i_{k+1}$, and F is the function of Appendix A.1, with $F(1) = F'(1) = 0$. Its derivative at $s = 1$ is $2Y$, so $Y \leq 0$, and an exact identity gives $d_k - j_k/2 = Y/(4\sqrt{3}s_k^2)$. A forward switch is the same with the decreasing label. (b) At a turn the transport of μ along the piece and the factor relation of Lemma 4.2 give $d_{k-1} = d_k$, and then $\varepsilon_{k-1} + \varepsilon_k = 0$ reduces Lemma 4.13 to the secant. Since $f'(x) - \frac{3}{4} = (x-2)^2/(4x^2) \geq 0$, every secant slope of f is at least $3/4$. □

The inequalities (a) say that the control just abandoned does not immediately become better again. They can also be derived from a bracket computation in a logarithmic clock ([7], `MIXED_GLOBAL_COSTATE_ELLIPTIC` (17)).

Lemma 4.15 (Gram identity)**.** *At a centre or a forward same-sign piece k, $d_{k-1}d_k - e_k^2 = \gamma^2/K_k^2$.*

This is the Gram determinant of y_{k-1}, y_k for the form S: $\det S = \gamma^2$ and $\det(y_{k-1}, y_k)^2 = 1/K_k^2$ by Lemma 4.12.

4.6. **Ellipticity and the Casimir cap.** The Casimir measures how far the period is from the circle. The circular trajectory, with $X \equiv J$ and $T = \pi/3$, costs π; the notes compute that its multiplier has $\kappa = 3/2$ (`GLOBAL_COST_CASIMIR` (27)). A period of cost below π must have a smaller Casimir, and the inequality of this subsection makes this quantitative. It applies only when Λ_1 is elliptic with the right orientation.

Proposition 4.16 (Ellipticity)**.** *Every mixed normal vertex period has $\gamma^2 > 0$.*

The proof is in Section 5.5. It uses the run structure of mixed periods (Proposition 5.6), which is proved from Lemma 4.10 alone. For entirely forward periods the same argument gives $\gamma^2 > 0$.

When $\gamma^2 > 0$ and $d_0 > 0$, the symmetric form S has positive determinant and is positive on y_0, so it is positive definite. Then $\Lambda_1(0)$ is negative elliptic: $\det \Lambda_1(0) > 0$ and its lower left entry is negative. The positive definite S corresponds to a negative elliptic Λ_1 because the triangular coordinates reverse orientation: if Y is traceless with $\det Y > 0$ and A is invertible, the lower left entries of Y and of $AYA^{-1} \det A$ have the same sign, and $\det C < 0$.

Proposition 4.17 (Cost–Casimir inequality)**.** *Let P be a normal vertex period such that $\det \Lambda_1(0) > 0$ and the lower left entry of $\Lambda_1(0)$ is negative, and let $\kappa = \sqrt{\det \Lambda_1(0)}$. Then*

$$\mathcal{C}(T, X) \geq \frac{4\pi\kappa}{2\kappa + 3}.$$

Proof. A negative elliptic traceless matrix is $\Lambda_1(0) = -\kappa\, hJh^{-1}$ with $\det h = 1$. Put $B(t) = h^{-1}g(t)$, so that $B' = BX$, $\det B = 1$ and $B(T) = B(0)R$. Write $\rho(Y) = -\frac{1}{2}\langle J, Y\rangle$; the cost is $3\int_0^T \rho(X)\,\mathrm{d}t$. By conservation, $\langle \Lambda_1(t), X(t)\rangle = 2\kappa\,\rho(BXB^{-1})$. The maximum principle gives $\langle \Lambda_1 + \frac{3}{2}J, X\rangle \leq 0$ at every time: a positive combination of the three vertex Hamiltonians cancels their Λ_R terms. Hence

$$\rho(X) + \rho(BXB^{-1}) \leq \Big(1 + \frac{3}{2\kappa}\Big)\rho(X).$$

The complex number $z(B) = \frac{1}{2}(B_{11} + B_{22}) + \frac{i}{2}(B_{21} - B_{12})$ has $|z| \geq 1$, its argument increases at the rate $(\rho(X) + \rho(BXB^{-1}))/(2|z|^2)$, and $z(B(0)R) = e^{i\pi/3} z(B(0))$. The argument therefore increases by at least $\pi/3$ over $[0, T]$, and

$$\int_0^T (\rho(X) + \rho(BXB^{-1}))\,\mathrm{d}t \geq \frac{2\pi}{3}.$$

Combining the two displays, $(1 + \frac{3}{2\kappa})\,\mathcal{C}/3 \geq 2\pi/3$. □

The same inequality also follows with a stationary interior comparison control in place of the vertex-wise bound ([7], `GLOBAL_COST_CASIMIR` (4)). The polar-angle argument also gives the winding bound behind the singular clearance (Proposition 3.8).

Corollary 4.18 (Casimir cap)**.** *Let P be a competitive normal vertex period ($\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$) with $\gamma^2 > 0$ and $d_0 > 0$. Then $\gamma^2 \leq 97969/529200$. In particular this holds for every competitive mixed normal vertex period.*

Proof. By the discussion before Proposition 4.17, $\Lambda_1(0)$ is negative elliptic with $\kappa^2 = 12\gamma^2$. With $\mathcal{C} \leq A_{\mathrm{oct}} < 3.127$ and $\pi > 3.14$, the inequality $4\pi\kappa \leq (2\kappa+3)\mathcal{C}$ forces $\kappa < 1.4877$, so $\gamma^2 < 1.4877^2/12 < 97969/529200$. For a mixed period, $\gamma^2 > 0$ by Proposition 4.16, and $d_0 \geq 3/32$ by Corollary 5.8. □

The value $97969/529200 = (313/210)^2/12$ is the notes' cap: $313/210$ solves $4\pi\kappa = (2\kappa + 3)\mathcal{C}$ with the rational bounds $A_{\mathrm{oct}} < 313/100$ and $\pi > 157/50$. The exact threshold is $\kappa_* = 3A_{\mathrm{oct}}/(4\pi - 2A_{\mathrm{oct}}) = 1.4852\ldots$ (Figure 7(b)). For entirely forward periods Section 7 uses the stronger spectral bound $\mathcal{C} \geq \frac{\pi}{3}(\kappa + \frac{3}{2})$; the difference of the two bounds is $\pi(2\kappa - 3)^2/(6(2\kappa + 3)) \geq 0$.

4.7. **Reduction to at most five pieces.**

Theorem 4.19 (At most five pieces)**.** *A competitive normal vertex period whose state stays in the compact domain $\mathfrak{h}^{\star\star}$ has at most five pieces.*

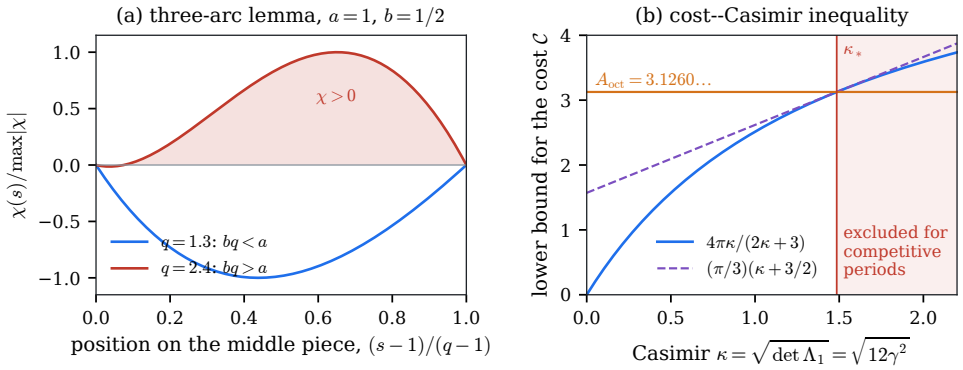


FIGURE 7. (a) The obstruction of the three-arc lemma (Lemma 5.5), growing case, with active margin $a = 1$ of the middle piece and outer margin $b = a_k = 1/2$. The switching function of the outer label vanishes at both ends of the middle piece, and its value inside is fixed by this. For $q = 1.3$ ($bq < a$) it is negative, as the maximum principle requires; for $q = 2.4$ ($bq > a$, that is, $a_k a_{k+2} > a_{k+1}^2$) it becomes positive. The curves are normalized by their maximum modulus. (b) The cost–Casimir inequality of Proposition 4.17 (solid) and the spectral bound of forward periods (dashed), against the octagon's cost A_{oct}. A competitive period has $\kappa < \kappa_* = 3A_{\mathrm{oct}}/(4\pi - 2A_{\mathrm{oct}}) = 1.4852\ldots$; the proof of Corollary 4.18 uses the rational bound $\kappa < 1.4877$. Both bounds equal π at $\kappa = 3/2$.

Proof. Suppose $N \geq 6$. By Proposition 4.8 the period is in one of the five classes, and each class is excluded by a theorem proved in the following sections:

Class	Excluded by	Hypotheses used
R	Theorem 5.4	none
M3	Theorem 5.7	none
M1	Theorem 6.1	competitive
F	Theorem 7.1	competitive, $N \geq 6$
M2	Theorem 8.1	competitive, state in $\mathfrak{h}^{\star\star}$, $N \geq 6$

Only the classes F and M2 use the bound on the number of pieces. □

The classes R and M3 are excluded for every normal vertex period, with no reference to the cost: their obstructions are local in the period and come from the maximum principle alone. The classes M1, F and M2 are excluded only for competitive periods, through the Casimir cap and estimates over the whole period. Section 10 treats the periods with at most five pieces.

5. Classes R and M3

This section excludes the two classes whose obstruction is local: an entirely reverse period (class R) and a mixed period with a maximal reverse run of length other than two (class M3). Both exclusions hold for every normal vertex period, whatever its cost and its number of pieces. They rest on two facts about a few consecutive pieces. Across three reverse switches the active margin of an interior piece cannot be a minimum of the four (Lemma 5.3). Across two switches that return to the same label, the active margins are strictly log-concave (Lemma 5.5). In a run of reverse switches these two facts leave room for exactly one configuration, a run of two switches around a strict minimum (Proposition 5.6). That configuration is the input of the remaining classes. The section ends with the proof of ellipticity (Proposition 4.16), which uses it.

5.1. **The reverse-four principle.** Consider four consecutive pieces $k, k+1, k+2, k+3$ with three reverse switches between them, so that their labels are i, $i-1$, $i+1$, i. Name the margins at the switch τ_{k+1} in the label order $(i, i+1, i-1)$ as (P, Q, Z), and the factors of the two interior pieces as $b = q_{k+1}$ and $c = q_{k+2}$. Then $P = a_k$ and $Z = a_{k+1}$. Along piece $k+1$ (label $i-1$) the label i grows and $i+1$ decreases, so at τ_{k+2}

$$A := m_{k+2}(i+1) = \frac{Z+Q}{b} - Z = a_{k+2}, \qquad B := m_{k+2}(i) = (Z+P)\,b - Z,$$

and along piece $k+2$ (label $i+1$) the label i decreases, so $a_{k+3} = (A+B)/c - A$.

Suppose the third piece is a weak minimum: $a_{k+2} \le a_k, a_{k+1}, a_{k+3}$. In these variables this reads

$$A \le Z, \qquad bA + (b-1)Z \le B, \qquad (2c-1)A \le B. \tag{5.1}$$

The obstruction is a variation of the period ([7], `GLOBAL_REVERSE_PERIOD_EXCLUSION`): insert short pieces of the two inactive controls at the two interior switches (the control $i+1$ at τ_{k+1} and the control i at τ_{k+2}), and change the logarithms of the factors of the four pieces at rates chosen so that the end state and the end frame do not move. To first order the normalized cost $\mathcal{C}/\sqrt{12}$ then changes by $\frac{Z}{4} N(A, B, Z, b, c)$ times the size of the variation, where

$$\begin{aligned} N(A,B,Z,b,c) = {} & b(b-1)A\big(Ac\,\eta(c) - B\,\zeta(c)\big) \\ & + (c-1)\big(bA(B+bZ)\,\eta(b) - B(bA+Z)\,\zeta(b)\big), \end{aligned} \tag{5.2}$$

with $\eta(t) = t - 1 - \log t$ and $\zeta(t) = t\log t - t + 1$, both positive for $t > 1$. Under the minimum hypothesis (5.1) this derivative is negative (Lemma 5.1), so the variation lowers the cost, while the maximum principle says that the inserted controls are not better than the active ones and the variation cannot lower the cost to first order. We do not construct the variation. Its content is an exact linear identity between switching functions at the switches

(Lemma 5.2), and the maximum principle enters only through the signs of these switching functions.

Lemma 5.1 (Sign of the cost derivative)**.** *If $b, c > 1$, $A, Z > 0$ and (5.1) holds, then $N(A, B, Z, b, c) < 0$.*

The proof (Appendix A.3) rests on the inequality $(2t+1)\,\eta(t) < (t+2)\,\zeta(t)$ for $t > 1$, which is the Padé bound $\log t > 3(t^2-1)/(t^2+4t+1)$.

Lemma 5.2 (The reverse-four identities)**.** *In the situation above, with $\chi_j(k)$ the switching functions at the switch τ_k (Definition 4.9),*

$$B(c-1)\,\chi_{i+1}(k+1) + bD\,\chi_i(k+2) = -\sqrt{3}\,Z\,N(A, B, Z, b, c), \tag{5.3}$$

$$(b-1)B\,\chi_{i-1}(k+3) + cD\,\chi_i(k+2) = -\sqrt{3}\,A\,N(Z, B, A, c, b), \tag{5.4}$$

where $D = A(c-1) + Z(b-1)$.

Proof. Express the costate coordinates at τ_{k+2} and τ_{k+3} through those at τ_{k+1} by the transport of Lemma 4.10(b) over the two interior pieces. The left sides are then linear in the six coordinates $\alpha_j(k+1), \beta_j(k+1)$, with coefficients rational in P, Q, Z, b, c and linear in $\log b$, $\log c$. Five linear relations hold among these coordinates: the switching functions of the active labels vanish at τ_{k+1} (labels i and $i-1$), at τ_{k+2} (label $i+1$) and at τ_{k+3} (label i), and $\langle \Lambda_R, X\rangle = 0$ at τ_{k+1}. Each identity is a combination of these five relations with polynomial coefficients, listed in Appendix A.2; after clearing denominators both sides are the same polynomial in the variables and in $\log b$, $\log c$. □

The coefficients $B(c-1)$ and bD in (5.3) are the weights of the two inserted pieces in the variation described above, and N is its cost derivative.

Lemma 5.3 (Reverse-four principle)**.** *Let four consecutive pieces $k, \dots, k+3$ of a normal vertex period have three reverse switches between them. Then*

(a) *it is not the case that $a_{k+2} \le a_k$, $a_{k+2} \le a_{k+1}$ and $a_{k+2} \le a_{k+3}$;*
(b) *it is not the case that $a_{k+1} \le a_k$, $a_{k+1} \le a_{k+2}$ and $a_{k+1} \le a_{k+3}$.*

Proof. (a) The labels $i+1$ at τ_{k+1} and i at τ_{k+2} are inactive on both adjacent pieces, so their switching functions are ≤ 0 (Lemma 4.10(c)). The weights $B(c-1)$ and bD are positive, so the left side of (5.3) is ≤ 0 and $N(A, B, Z, b, c) \ge 0$. The weak minimum at the third piece is (5.1), and Lemma 5.1 gives $N < 0$.

(b) A weak minimum at the second piece reads $Z \le A$, $cZ + (c-1)A \le B$ and $(2b-1)Z \le B$: these are the conditions (5.1) with (A, Z, b, c) replaced by (Z, A, c, b), so $N(Z, B, A, c, b) < 0$. The switching functions $\chi_{i-1}(k+3)$ and $\chi_i(k+2)$ are ≤ 0, and (5.4) gives the contradiction as before. □

Part (b) also follows from (a) by a reflection of the trajectory combined with a reversal of time, which reverses the order of the four pieces and keeps the signs of the switches ([7], `MIXED_GLOBAL_MINIMUM_STRUCTURE`, §§1–4). The second identity (5.4) avoids this transformation of trajectories.

5.2. **Class R.**

Theorem 5.4 (Class R)**.** *No normal vertex period has every switch reverse.*

Proof. The active margins are N-periodic, so one of them, a_m, is a global minimum. Every switch is reverse, so the four pieces $m-2,\dots,m+1$ have three reverse switches between them, and $a_m \le a_{m-2}, a_{m-1}, a_{m+1}$. This contradicts Lemma 5.3(a) for $k = m-2$. □

The window may run through the seam, and it may contain the same piece of the period twice when $N < 4$; the twisted indexing of Section 4 makes no distinction. The argument uses neither the cost nor the number of pieces, and Section 10 uses it also for the entirely reverse words with five pieces.

5.3. **Three pieces with alternating labels.** When the switch after piece k is followed by a switch back to the label of piece k, the middle piece is a turn, and its two outer pieces have the same control. The switching function of that control vanishes at both ends of the middle piece and is ≤ 0 in between. This forces a relation between the three active margins.

Lemma 5.5 (Three-arc log-concavity and turn ordering)**.** *Let $k, k+1, k+2$ be consecutive pieces of a normal vertex period.*

(a) *If $i_{k+2} = i_k$, then $a_k a_{k+2} < a_{k+1}^2$; in particular a_{k+1} is not at most both a_k and a_{k+2}.*

(b) *If the switch k is reverse and the switch $k+1$ forward, then $a_k < a_{k+2}$; if the switch k is forward and the switch $k+1$ reverse, then $a_{k+2} < a_k$.*

Proof. (b) After a reverse switch the label i_k is the growing label of piece $k+1$, so $a_{k+2} = (a_{k+1}+a_k)\,q_{k+1} - a_{k+1} > a_k$ by Lemma 4.1; after a forward switch it is the decreasing label and $a_{k+2} = (a_{k+1}+a_k)/q_{k+1} - a_{k+1} < a_k$.

(a) Write $a = a_{k+1}$ and $q = q_{k+1}$. Suppose first that the switch k is reverse, so that the outer label $i = i_k$ is the growing label of the middle piece, with margin $b = a_k$ at its start, and $a_{k+2} = (a+b)q - a$. Since

$$a^2 - b\big((a+b)q - a\big) = (a+b)(a-bq),$$

the claim is $bq < a$. By Lemma 4.10 and Appendix A.1, the switching function of the label i at the intermediate factor $s \in [1,q]$ is

$$\chi_i(s) = 2s\Big(Y(s-1) + \frac{\sqrt{3}a}{2}F(s)\Big),$$

$$F(s) = (a+b)\log s - (a+b)(s-1) + a\frac{(s-1)^2}{s},$$

for a number Y. It vanishes at $s = 1$ and $s = q$ and is ≤ 0 in between. The vanishing at q fixes Y, and $\chi_i \le 0$ just before q makes the derivative at q nonnegative. Together these say $(q-1)F'(q) \ge F(q)$, that is,

$$(q-1)(2aq - a + bq) \ge (a+b)\,q^2 \log q.$$

If $bq \geq a$, this fails: the right side minus the left side equals $a\big(q^2 \log q - (q-1)(2q-1)\big) + b\big(q^2 \log q - q(q-1)\big)$; the second bracket is positive because $q \log q > q - 1$; if the first bracket is negative, replace a by the larger bq, which leaves $bq^2\big((q+1)\log q - 2(q-1)\big) > 0$. So $bq < a$.

If the switch k is forward, the outer label is the decreasing label of the middle piece, with margin $c = a_k$, and $a_{k+2} = (a+c)/q - a$. Now $a^2 - c\big((a+c)/q - a\big) = (a+c)(a - c/q)$, and the claim is $c < aq$. Here $\chi_i(s) = -(2/s)\big(Y(1-s^{-1}) + \frac{\sqrt{3}a}{2} h(s)\big)$ with $h(s) = (a+c)\log s - a(s-1) - c(1-s^{-1})$, and $h(1) = h'(1) = 0$. The sign at the start of the piece gives $Y \geq 0$. If $aq \leq c$, then

$$h(q) = (c - aq)\big(\log q - (1 - q^{-1})\big) + a\big((q+1)\log q - 2(q-1)\big) > 0,$$

so $\chi_i(q) < 0$, contradicting $\chi_i(q) = 0$. □

The proof uses one end of the middle piece in each case: the right end when the outer label grows, the left end when it decreases. Figure 7(a) shows the switching function of the growing case for an admissible factor and for one with $bq > a$. Another proof of (a) works on the edge of the simplex that contains the two controls: in a clock in which the state moves at unit speed, one transforms the costate accordingly and compares the two switching abscissas ([7], `MIXED_GLOBAL_MINIMUM_STRUCTURE` (12)).

5.4. **Reverse runs.**

Proposition 5.6 (Run structure)**.** *Let P be a mixed normal vertex period, and let the switches $k, \dots, k+L-1$ form a maximal reverse run. Then $L = 2$, and the active margin of the centre piece $k+1$ is strictly smaller than those of the four pieces around it:*

$$a_{k+1} < a_{k-1}, \qquad a_{k+1} < a_k, \qquad a_{k+1} < a_{k+2}, \qquad a_{k+1} < a_{k+3}.$$

Proof. The run is preceded by the forward switch $k-1$ and followed by the forward switch $k+L$.

At the entry, the switches $k-1, k$ are forward then reverse, so $a_{k+1} < a_{k-1}$ by Lemma 5.5(b), and $i_{k+1} = i_{k-1}$. If $a_k \leq a_{k+1}$, then a_k would be at most both a_{k-1} and a_{k+1}, against Lemma 5.5(a). So $a_{k+1} < a_k$.

At the exit, the switches $k+L-1, k+L$ are reverse then forward, so $a_{k+L-1} < a_{k+L+1}$, and Lemma 5.5(a) on the pieces $k+L-1, k+L, k+L+1$ gives $a_{k+L-1} < a_{k+L}$.

If $L = 1$, the entry gives $a_{k+1} < a_k$ and the exit $a_k < a_{k+1}$. If $L \geq 3$, let m be a piece of $k, \dots, k+L$ with the smallest active margin. It is not k, since $a_{k+1} < a_k$, nor $k+L$, since $a_{k+L-1} < a_{k+L}$. If $m = k+1$, the pieces $k, \dots, k+3$ have three reverse switches and the second one is a weak minimum, against Lemma 5.3(b). If $k+2 \leq m \leq k+L-1$, the pieces $m-2, \dots, m+1$ have three reverse switches and the third one is a weak minimum, against Lemma 5.3(a). So $L = 2$. The entry and the exit give the first three inequalities, and Lemma 5.5(b) at the switches $k+1, k+2$ gives $a_{k+1} < a_{k+3}$. □

The proof does not use mixedness beyond the existence of the run: a maximal reverse run is bounded by forward switches by definition, so a period that has one is mixed.

Theorem 5.7 (Class M3)**.** *No normal vertex period is mixed with a maximal reverse run of length other than two.*

Proof. This is the first conclusion of Proposition 5.6. □

In a mixed period every reverse switch therefore belongs to a pattern $+\,-\,-+$ of switch signs, whose middle piece is a centre, and every centre is a strict local minimum of the active margins. The other classes use these facts through the following corollary.

Corollary 5.8 (Mixed periods)**.** *Let P be a mixed normal vertex period.*

(a) *Every reverse switch k is the first or the second switch of a pattern of signs $+\,-\,-+$: either the switches $k-1, k, k+1, k+2$ or the switches $k-2, k-1, k, k+1$ have these signs.*

(b) *At a centre k, $a_k < a_{k-1}$ and $a_k < a_{k+1}$.*

(c) *Every switch norm satisfies $d_k \geq 3/32$.*

Proof. (a) and (b) restate Proposition 5.6. For (c), at a forward switch $d_k \geq j_k/2 \geq 3/32$ by Lemma 4.14(a). By (a), a reverse switch is the outgoing switch of a turn (the piece k after the forward switch $k-1$) or the incoming switch of a turn (the piece $k+1$ before the forward switch $k+1$), and Lemma 4.14(b) gives $d_k \geq 3/32$ in both cases. □

5.5. **Proof of ellipticity.** We prove Proposition 4.16: a mixed normal vertex period has $\gamma^2 > 0$. Suppose $\gamma^2 \leq 0$. The idea is to read the switch vectors y_n of Lemma 4.12 projectively, through a coordinate that changes monotonically from one switch to the next. Because $y_{N+n} = \pm y_n$, the coordinate returns to its value after one period; but at every centre it strictly increases, and nowhere does it decrease.

Lemma 5.9 (Three Gram vectors)**.** *Let S be a symmetric 2×2 matrix with $\det S \leq 0$, and let y, y', y'' be vectors with $y^{\mathsf T}Sy$, $y'^{\mathsf T}Sy'$, $y''^{\mathsf T}Sy''$ all positive. Then $(y^{\mathsf T}Sy')(y^{\mathsf T}Sy'')(y'^{\mathsf T}Sy'') > 0$.*

Proof. For two vectors, $(y^{\mathsf T}Sy)(y'^{\mathsf T}Sy') - (y^{\mathsf T}Sy')^2 = \det S \cdot \det(y,y')^2 \leq 0$. The Gram determinant of three vectors of the plane vanishes: $d\,d'\,d'' - d\,c^2 - d'\,b^2 - d''a^2 + 2abc = 0$ in the obvious notation. With the three two-vector inequalities, $2abc \geq d\,d'\,d'' > 0$. □

Lemma 5.10 (Signs of the cross pairings)**.** *Suppose $\gamma^2 \leq 0$. At a forward same-sign piece k, $e_k > 0$. At a centre k with $a_k < a_{k-1}$, $a_k < a_{k+1}$ and $d_{k-1}, d_k \geq 3/32$, $e_k < 0$.*

Proof. At a same-sign piece, Lemma 4.13 expresses e_k through d_{k-1}, d_k and the active margins. At a forward same-sign piece both switches are forward,

and the lower bounds $d \geq j/2$ of Lemma 4.14(a) give $e_k \geq X_k$ with

$$X_k = \frac{1}{16}\Big(-1 + 2a\Big(\frac{1}{a+p} + \frac{1}{a+n}\Big) - \frac{a^2}{a^2+4} + \log\frac{(a+p)(a+n)}{a^2+4}\Big),$$

$a = a_k$, $p = a_{k-1}$, $n = a_{k+1}$. Here the logarithm is $\log q_k > 0$ by Lemma 4.2, and $a^2/(a^2+4) < 2a/(a+p)$ because $ap < 4$; so $X_k > -1/16$. On the other hand $\gamma^2 \leq 0$ and Lemma 4.15 give $e_k^2 \geq d_{k-1}d_k \geq (3/32)^2$, and $3/32 > 1/16$, so $e_k \geq 3/32 > 0$. At a centre the switches are reverse, the upper bounds $d \leq j/2$ give $e_k \leq X_k$, and now the logarithm is $-\log q_k < 0$ and $2a/(a+p) + 2a/(a+n) < 2$ by the strict minimum, so $X_k < 1/16$ and $e_k \leq -3/32$. □

Proof of Proposition 4.16. Suppose $\gamma^2 \leq 0$. By Corollary 5.8(c) every $d_n = y_n^{\mathsf{T}} S y_n$ is at least $3/32$. Let $h(v) = y_0^{\mathsf{T}} S v$, a nonzero linear form, and let l be the linear form with $h(u)l(v) - h(v)l(u) = c\det(u,v)$ for a constant $c > 0$. By Lemma 5.9 with $y = y_0$, $h(y_n)h(y_m)\,(y_n^{\mathsf{T}} S y_m) > 0$ for all n, m; in particular $h(y_n) \neq 0$, and the projective coordinate $z_n = l(y_n)/h(y_n)$ is defined, with

$$z_{n+1} - z_n = \frac{c\,\det(y_n, y_{n+1})}{h(y_n)\,h(y_{n+1})}.$$

By Lemma 4.12, at a turn $y_{n+1} = -y_n$ and the increment is 0. At a forward same-sign piece $\det(y_n, y_{n+1}) = 1/K_{n+1} > 0$ and $y_n^{\mathsf{T}} S y_{n+1} = e_{n+1} > 0$ (Lemma 5.10), so $h(y_n)h(y_{n+1}) > 0$ and the increment is positive. At a centre $\det(y_n, y_{n+1}) = -1/K_{n+1} < 0$ and $e_{n+1} < 0$, using the strict minimum and the bounds $d \geq 3/32$ of Corollary 5.8; so $h(y_n)h(y_{n+1}) < 0$ and the increment is again positive. A mixed period has a reverse switch, hence a centre (Corollary 5.8(a)), so the increments over one period sum to a positive number. But $y_N = \pm y_0$ and z depends only on the line of the vector, so the sum is $z_N - z_0 = 0$. □

The same telescoping can be organized with a factorization of the indefinite form S in place of Lemma 5.9 ([7], `MIXED_GLOBAL_COSTATE_ELLIPTIC`, §§5–6). For an entirely forward period every piece is a forward same-sign piece and the same telescoping gives $\gamma^2 > 0$.

6. Class M1: every forward run short

A period of class M1 is mixed, every maximal run of reverse switches has length two and every maximal run of forward switches has length one (Definition 4.7). Its cyclic sign word is therefore $(--+)^m$: every reverse run is a pair of reverse switches around a centre piece, and consecutive pairs are separated by single forward switches. This section proves the following theorem.

Theorem 6.1. *No competitive normal vertex period belongs to class* M1*.*

The theorem uses neither the compact domain nor a lower bound on the number of pieces. Its proof compares two ways of counting how far the switch

vectors y_k of Definition 4.11 turn in one period, measured in the positive definite form S. At a turn the switch vector only changes sign, so all the turning happens at the m centres. A whole-period winding identity fixes the total turning at exactly $(m+1)\pi/3$, because the frame of a competitive period turns by exactly $\pi/3$. A local estimate at each centre, which combines the stationarity identity, the Gram identity and the competitive cap on the Casimir γ^2, shows that each centre turns the switch vectors by more than $2\pi/5$. The average $(m+1)\pi/(3m)$ is at most $2\pi/5$ as soon as $m \geq 5$, and the word condition leaves $m = 2$ as the only other case. With two centres the recurrence of the switch vectors closes up only in a rigid configuration whose cost can be computed exactly, and it exceeds A_{oct}.

6.1. **The sign word.**

Lemma 6.2. *Let a vertex representation with N pieces belong to class* M1*. Then $N = 3m$ for an integer $m \geq 1$ with $m \equiv 2 \pmod 3$, and there is $k_0 \in \mathbb{Z}$ such that for every $j \in \mathbb{Z}$ the switches $k_0 + 3j$ and $k_0 + 3j + 1$ are reverse and the switch $k_0 + 3j + 2$ is forward.*

Proof. A mixed representation has a forward switch followed by a reverse one: otherwise every switch after the first forward one would be forward, which the periodicity $\varepsilon_{k+N} = \varepsilon_k$ of the signs forbids. Start at such a pair. The reverse run that begins there has length two, and the forward run that follows it has length one, so the signs continue as $-, -, +, -, -, +, \dots$ with period three. Hence $N = 3m$ and the word is $(--+)^m$. The signs of one period add up to $m - 2m = -m$, and Lemma 4.6 gives $-m \equiv 1 \pmod 3$. □

We fix k_0 and m as in the lemma and call the piece $c_j = k_0 + 3j + 1$, which lies between the two reverse switches of the j-th run, the j-th *centre*. The pieces $c_j + 1$ and $c_j + 2$ are turns: one of their two switches is forward and the other reverse. At a turn the outgoing switch vector is the negative of the incoming one, so the three switch vectors that follow the centre c_j are z_j, $-z_j$, z_j with $z_j = y_{c_j}$. Between two consecutive centres the switch vectors therefore do not turn, and the whole turning of the sequence (y_k) is concentrated at the centres.

6.2. **The data at a centre.**

Fix a centre c, with active margin $a = a_c$ and neighbouring active margins $p = a_{c-1}$ and $n = a_{c+1}$. In this section we write

$$r = \frac{a}{a+p}, \qquad t = \frac{a}{a+n}, \qquad \rho = \frac{a^2}{a^2+4}, \\ u = 8d_{c-1}, \qquad v = 8d_c, \qquad E = 16e_c, \qquad U(s) = 1 - s + s^2. \tag{6.1}$$

Here d_{c-1} and d_c are the switch norms of the two reverse switches at the ends of the centre and e_c is the cross pairing on the centre. The letter ρ is local to this section.

Lemma 6.3. *Let P be a normal vertex period with the sign word of Lemma 6.2, and let γ^2 be its Casimir. At every centre the numbers (6.1) satisfy*

(i) $0 < r < \frac{1}{2}$, $0 < t < \frac{1}{2}$ *and* $0 < \rho < rt$;

(ii) $\frac{3}{4} \le u \le U(r)$ *and* $\frac{3}{4} \le v \le U(t)$;

(iii) *the stationarity identity*

$$E = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1-u}{r} - \frac{1-v}{t}; \tag{6.2}$$

(iv) *the Gram identity* $16(1-\rho)^2\gamma^2 = 4uv - E^2$.

Proof. (i) The centre is the middle piece of a reverse run of length two, so its active margin is strictly smaller than both neighbouring ones (Proposition 5.6). This gives $r, t < 1/2$. The factor of the centre is $q_c = (a^2+4)/((a+p)(a+n))$, and $q_c > 1$ is the inequality $(a+p)(a+n) < a^2+4$, which is $\rho < rt$.

(ii) Both neighbours of the centre are turns. The switch norms at the two ends of a turn are equal and at least $3/32$ (Lemma 4.14); this gives $u, v \ge 3/4$. Both switches of the centre are reverse, so $d_{c-1} \le j_{c-1}/2$ and $d_c \le j_c/2$ by Lemma 4.14, and $4j_{c-1} = (a^2+ap+p^2)/(a+p)^2 = U(r)$, and likewise $4j_c = U(t)$.

(iii) is the stationarity identity of Lemma 4.13 with both incident signs equal to -1, written in the variables (6.1).

(iv) is the Gram identity $d_{c-1}d_c - e_c^2 = \gamma^2/(a^2+4)^2$ at a centre (Lemma 4.15), multiplied by 256, since $1 - \rho = 4/(a^2+4)$. □

Since the period is mixed, $\gamma^2 > 0$ (Proposition 4.16), and the switch norms are positive. The form S, whose determinant is γ^2, is then positive definite, and the number

$$\cos\varphi_c = \frac{e_c}{\sqrt{d_{c-1}d_c}} = \frac{E}{2\sqrt{uv}}$$

is the cosine of the angle $\varphi_c \in (0, \pi)$ between y_{c-1} and y_c in the inner product S; by (iv) its absolute value is less than one. The determinant $\det(y_{c-1}, y_c) = -1/(a^2+4)$ is negative at every centre, so all the centres turn the switch vectors in the same sense. We call φ_c the *centre angle*.

6.3. **The angle gap.** A centre with a large angle cosine needs a large pairing E. The stationarity identity (6.2) allows a large E only when $\log(\rho/(rt))$ is not too negative, that is, when the centre margin is not too small compared with its neighbours. A large centre margin makes the factor $16(1-\rho)^2$ in the Gram identity (iv) small, and then (iv) forces the Casimir γ^2 up, against the competitive cap of Corollary 4.18. The next proposition makes the conflict quantitative.

Proposition 6.4. *Let $\gamma^2, r, t, \rho, u, v, E$ be real numbers satisfying* (i)–(iv) *of Lemma 6.3, and suppose that $\gamma^2 \le 97969/529200$. Then $E < \frac{3}{5}\sqrt{uv}$; that is, $\cos\varphi < 3/10$ for $\cos\varphi = E/(2\sqrt{uv})$.*

The proof, in Appendix B, reduces the seven variables to two in three steps. Write $E = (1-\rho+\log\rho)+\Psi(r,u)+\Psi(t,v)$ with $\Psi(r,u) = r-\log r-(1-u)/r$, and let $\omega(x) = -x-\log(1-x) = \sum_{n\geq 2} x^n/n$ for $0 \leq x < 1$.

(a) One reverse switch at a time: for $s = \sqrt{4u-3}$ one has $u = U((1-s)/2)$, $2r \leq 1-s$, and $\Psi(r,u) \leq \omega(s)+\log 2$, with equality when $r = (1-s)/2$, that is, when the switch bound $u \leq U(r)$ is attained.

(b) The two switches together: define $S \geq 0$ by $S^2 = s^2+s'^2+s^2s'^2/3$, where $s' = \sqrt{4v-3}$. Then $uv = 3(3+S^2)/16$, and $\omega(s)+\omega(s') \leq \omega(S)$ because ω is superadditive in x^2.

(c) The centre: $z = 4\rho/(1-S)$ lies in $(0,1]$, $\rho = (1-S)z/4$, and $E \leq (1-S)(1-z/4)+\log z$.

If $\cos\varphi \geq 3/10$, then $E^2 \geq \frac{9}{25}uv$, and the two-variable inequality of Lemma B.4 gives $16K_*(1-\rho)^2 < 4uv-E^2$ for the cap $K_* = 97969/529200$, which contradicts (iv). Figure 8 shows the region of (S,z) that survives the reductions and the least Casimir compatible with it.

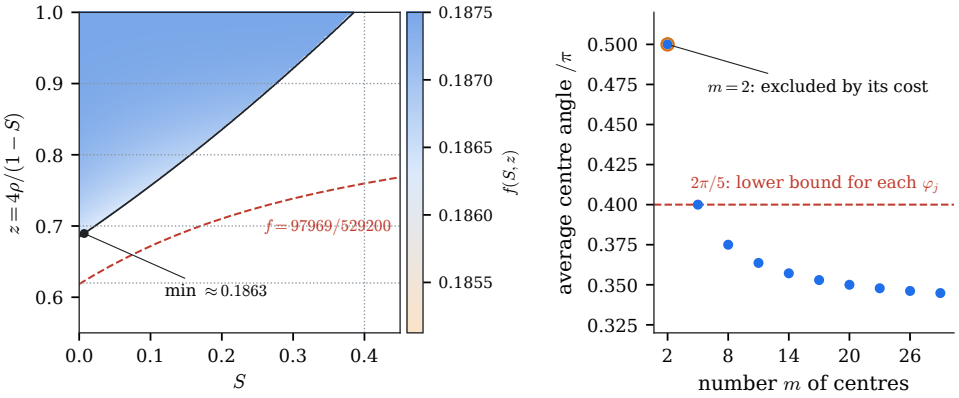


FIGURE 8. Left: the reduced variables of the angle gap. The shaded region is the set of (S,z) with $(1-S)(1-z/4)+\log z \geq \frac{3}{5}\sqrt{3(3+S^2)/16}$, where a centre with $\cos\varphi \geq 3/10$ must lie; on it the colour is the least Casimir $f(S,z) = (4Q-E_{\max}^2)/(16(1-\rho)^2)$ allowed by the Gram identity, with $Q = 3(3+S^2)/16$, $E_{\max} = (1-S)(1-z/4)+\log z$ and $\rho = (1-S)z/4$. Its minimum is about 0.1863, attained near $S = 0$, and the level set $f = 97969/529200 \approx 0.18513$ of the cap (dashed) stays outside the region. The dotted lines are $S = 2/5$, $z = 31/50$ and the ends of the four tangent-line intervals of Lemma B.4. Right: the average centre angle $(m+1)\pi/(3m)$ forced by the winding identity, for $m \equiv 2 \pmod 3$, against the lower bound $2\pi/5$ for every centre angle. The two meet at $m = 5$; only $m = 2$ lies above the line.

Corollary 6.5. *At every centre of a competitive normal vertex period of class* M1*,* $\varphi_c > 2\pi/5$.

Proof. The Casimir and the first switch norm are positive, so the competitive cap $\gamma^2 \le 97969/529200$ of Corollary 4.18 applies. By Lemma 6.3 and Proposition 6.4, $\cos\varphi_c < 3/10$. Since $\cos(2\pi/5) = (\sqrt{5}-1)/4 > 3/10$, which is $\sqrt{5} > 11/5$, and the arccosine decreases, $\varphi_c > 2\pi/5$. □

Remark 6.6. Another argument for Proposition 6.4 is given in [7] (`MIXED_GLOBAL_SHORT_RUN_ANGLE_EXCLUSION`). At a fixed cosine c it solves the stationarity identity for ρ and follows a path along which the Casimir does not increase, to the symmetric configuration $r = t = 1/2$, $u = v = 3/4$; there $1 - \rho_c + \log(4\rho_c) = 3c/2$, and a rational separation at $\rho = 17/100$ beats the cap. Both arguments meet at the same extremal configuration: $S = 0$ is $u = v = 3/4$, and on $S = 0$ the boundary of the region in Figure 8 is the equation for ρ_c, with $c = 3/10$ and $z = 4\rho_c$.

6.4. **The winding identity.**

Proposition 6.7. *Let P be a competitive normal vertex period with the sign word of Lemma 6.2, $N = 3m$, and positive Casimir. Then*

$$\sum_{j=0}^{m-1} \varphi_{c_j} = \frac{(m+1)\pi}{3}, \qquad \varphi_c = \arccos\frac{E_c}{2\sqrt{u_c v_c}},$$

where E_c, u_c, v_c are the numbers (6.1) at the centre c.

The identity says that the switch vectors, which turn only at the centres, turn in total by the angle that the frame of the period prescribes. The proof expresses the switch vectors through the physical frame and follows a continuous angle along the period.

Switch vectors in the physical frame. Let $V(t) = g(t)X(t)$ be the tangent frame. The switch vector at the switch n is, up to a sign and a positive factor, $KV(\tau_n)b$, where $K = R^{i_0}C(-X(0))$ is a fixed matrix with $\det K = -2\sqrt{3}$, C is the triangular change of coordinates, and b is one of two base vectors $b_{\mathrm{in}}(i)$ and $b_{\mathrm{out}}(i) = Rb_{\mathrm{in}}(i)$ that depend only on the label i of the piece. Factor $S = Q^{\mathsf{T}}Q$ with $\det Q < 0$, so that angles for S become Euclidean angles after Q.

A continuous angle. The polar number $z(g) = (g_{00} + g_{11})/2 + i(g_{10} - g_{01})/2$ of a matrix of determinant one has modulus at least one, and for every vector $w \neq 0$ the quotient $gw/(z(g)w)$ lies in the right half-plane, since $\mathrm{Re}(gw \cdot \overline{z(g)w}) = (|gw|^2 + |w|^2)/2$. Let $\Phi(t)$ be a continuous angle of $z(g(t))$. For $w \neq 0$ let $\kappa_t(w)$ be the angle from w to $QKg(t)X(t)w$ obtained by adding $\Phi(t)$ to principal arguments that never leave the slit plane; it depends continuously on t.

Three facts. At a centre c the centre angle is $\varphi_c = \pi/3 + \kappa_{\tau_c}(b_{\mathrm{out}}) - \kappa_{\tau_c}(b_{\mathrm{in}})$. Along a piece the base vector that is being followed is an eigenvector of

the tangent factor of the piece, so $g(t)X(t)b$ keeps its direction and $\kappa_t(b)$ is constant; between consecutive centres the outgoing base vector of one is, up to sign, the incoming base vector of the next. Across the seam, $\kappa_T(w) = \kappa_0(Rw) + \pi/3$, because $g(0) = I$, $g(T) = R$ and $\Phi(T) - \Phi(0) = \pi/3$.

The frame turns by exactly $\pi/3$. The endpoint conditions fix $\Phi(T) - \Phi(0)$ only modulo 2π. For a competitive period the cost is at most $A_{\mathrm{oct}} < \pi$, and the winding theorem behind the reverse comparison (Proposition 2.11) shows that the reflected first row $(g_{00}, -g_{01})$ of the frame turns by exactly $\pi/3$. The real part of the product of that row with the conjugate of $z(g)$ is $(g_{00}^2 + g_{01}^2 + 1)/2 > 0$, so the two angles stay within $\pi/2$ of each other, and $\Phi(T) - \Phi(0) = \pi/3$.

Telescoping. The centre angles are periodic in j with period m, so we may take the m centres of one period in a window that starts at the seam. Summing the centre formula over them, the term $\kappa(b_{\mathrm{out}})$ of each centre cancels the term $\kappa(b_{\mathrm{in}})$ of the next, and the last link crosses the seam, where it gains $\pi/3$. What remains is $m \cdot \pi/3 + \pi/3$.

Remark 6.8. The identity can also be derived from the universal cover of $\mathrm{SL}_2(\mathbb{R})$ ([7], `MIXED_GLOBAL_PROJECTIVE_WINDING`). The tangent frame $X(0)^{-1}g(t)X(t)$ has the relative homotopy class of the canonical path from I to R; the cyclic product of the factors $T_0(q_k)R^{-\varepsilon_k}$ then lifts to the central element $\widetilde{R}^{-3\ell}$, where $\sum \varepsilon_k = 3\ell + 1$. Identify the vectors at the turns, and change the sign of the representative across every centre, so that every centre contributes $\pi - \varphi_c$. The total is $n\pi$ with $n = (N - m - 1)/3 = (2m - 1)/3$, which is the same identity: $m\pi - \sum_j \varphi_{c_j} = (2m - 1)\pi/3$.

6.5. **Five or more centres.**

Proof of Theorem 6.1 for $m \geq 5$. By Corollary 6.5 every centre angle exceeds $2\pi/5$, so the sum in Proposition 6.7 exceeds $2m\pi/5$. For $m \geq 5$, $2m/5 \geq (m+1)/3$, which contradicts Proposition 6.7. Since $m \equiv 2 \pmod 3$, the only other case is $m = 2$. □

6.6. **Two centres.**

For $m = 2$ the average centre angle is $\pi/2 > 2\pi/5$, and the angle count gives nothing. The argument uses instead the recurrence of the switch vectors and the cost identity. For the centre c_j put $K_j = a_{c_j}^2 + 4$, write $b_j = a_{c_j+1}$ and $b'_j = a_{c_j+2}$ for the active margins of the two turns that follow it, and

$$D_j = (a_{c_j} + b_j)^2 + (b'_j + a_{c_{j+1}})^2 - (b_j + b'_j)^2 = s_{c_j}^2 + s_{c_j+2}^2 - s_{c_j+1}^2.$$

Lemma 6.9. *In a normal vertex period with the sign word of Lemma 6.2, the vectors $z_j = y_{c_j}$ satisfy*

$$K_{j+1}z_{j+1} = -K_j z_{j-1} + D_j z_j, \qquad \det(z_{j-1}, z_j) = -1/K_j,$$

and the cyclic product of the period is $\pm I$, so that $z_{j+m} = \pm z_j$ with one sign for all j.

Proof. Work in triangular coordinates and in the frame of the centre $c = c_j$, that is, multiply all vectors by the inverse of the prefix product up to c. Put $A_\pm(q) = T_0(q)R^{\mp 1}$, so that

$$A_+(q) = \begin{pmatrix} q & -q \\ 1/q & 0 \end{pmatrix}, \qquad A_-(q) = \begin{pmatrix} 0 & q \\ -1/q & 1/q \end{pmatrix},$$

and write $K = K_j$ and $s_k = a_k + a_{k+1}$. The factors of the centre and of the two turns after it are $K/(s_{c-1}s_c)$, s_{c+1}/s_c and s_{c+1}/s_{c+2}. In this frame $z_{j-1} = -(1/s_{c-1}, 0)^{\mathsf T}$ and $z_j = (0, s_{c-1}/K)^{\mathsf T}$, which gives the determinant, and

$$\begin{aligned} K_{j+1}z_{j+1} &= s_{c+2}\, A_-\Big(\frac{K}{s_{c-1}s_c}\Big)A_+\Big(\frac{s_{c+1}}{s_c}\Big)A_-\Big(\frac{s_{c+1}}{s_{c+2}}\Big)\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix} K/s_{c-1} \\ (s_{c-1}/K)\, D_j \end{pmatrix} \\ &= -Kz_{j-1} + D_j z_j. \end{aligned}$$

Multiplying back by the prefix product preserves the recurrence and the determinant. The cyclic product of the period is $\pm I$ by the tangent closure (Lemma 4.4). □

Lemma 6.10. *For every normal vertex period,*

$$\frac{2\mathcal{C}}{\sqrt{12}} = 8\sum_{k=0}^{N-1}(\varepsilon_{k-1}+\varepsilon_k)e_k - 2\sum_{k=0}^{N-1}\frac{\varepsilon_{k-1}+\varepsilon_k}{a_k^2+4}.$$

Proof. Multiply the stationarity identity of Lemma 4.13 at the piece k by a_k and sum over a period. The switch norms and the cross pairings are formed from one weight w_k per piece, $d_k = (w_k + w_{k+1})/(2s_k^2)$ and $e_k = w_k/(2(a_k^2+4))$. Regrouped by switches, the left-hand sides add up to $\sum_k(\varepsilon_{k-1}+\varepsilon_k)w_k(1 - a_k^2/(a_k^2+4)) = 8\sum_k(\varepsilon_{k-1}+\varepsilon_k)e_k$. On the right-hand side, the one-arc cost formula of Lemma 4.1, with the factors written in the active margins, gives $8\mathcal{C}/\sqrt{12} = \sum_k \varepsilon_k G(a_k, a_{k+1})$ for the signed action

$$G(a,b) = 3ab - 4 - (a^2+b^2)\log(a+b) + \tfrac{1}{2}a^2\log(a^2+4) + \tfrac{1}{2}b^2\log(b^2+4),$$

and the stationarity identity involves the partial derivatives of G. The Euler-type identity $aG_a(a,b) + bG_b(a,b) = 2G(a,b) + 16/(a^2+4) + 16/(b^2+4)$ turns the sum of the right-hand sides into $2\mathcal{C}/\sqrt{12} + 2\sum_k(\varepsilon_{k-1}+\varepsilon_k)/(a_k^2+4)$. □

In the word $(--+)^m$, $\varepsilon_{k-1}+\varepsilon_k$ is -2 at a centre and 0 at a turn, so the lemma reads $\mathcal{C}/\sqrt{12} = 2\sum_j 1/K_j - 8\sum_j e_{c_j}$.

Proposition 6.11. *A competitive normal vertex period with $N = 6$ does not have the sign word $(--+)^2$.*

Proof. The two recurrences of Lemma 6.9 read $K_1z_1 = -K_0z_{-1} + D_0z_0$ and $K_0z_2 = -K_1z_0 + D_1z_1$, with $z_{-1} = \pm z_1$ and $z_2 = \pm z_0$, and z_0, z_1 are independent. The sign $+$ would give $(K_0 + K_1)z_1 = D_0z_0$, which is impossible. So $z_{-1} = -z_1$, and the recurrences give $K_0 = K_1$ and

$D_0 = D_1 = 0$. Hence both centres have the same active margin a, $K = a^2+4$, and $e_{c_0} = z_{-1}^{\mathsf{T}} S z_0 = -e_{c_1}$. Lemma 6.10 gives the exact cost

$$\mathcal{C} = \sqrt{12} \cdot \frac{4}{K} = \frac{8\sqrt{3}}{a^2+4}.$$

It remains to bound a from above. The equation $D_j = 0$ is $(b_j - a)(b'_j - a) = 2a^2$, and the turn margins exceed a by Proposition 5.6. By Lemma B.5, $b_j b'_j > 5.828\, a^2$ and $(a+b_j)(a+b'_j) = 2b_j b'_j$. The two centre factors exceed one, that is, $(a+b'_1)(a+b_0) < K$ and $(a+b'_0)(a+b_1) < K$. Multiplying, $K^2 > 4\, b_0 b'_0\, b_1 b'_1 > 4 \cdot 5.828^2 a^4$, so $K > 11.656\, a^2$, $a^2 < 4/10.656$ and

$$\mathcal{C} = \frac{8\sqrt{3}}{a^2+4} > \frac{8\sqrt{3}}{4 + 4/10.656} > 3.127 > A_{\mathrm{oct}} = 3.1260\ldots,$$

so the period is not competitive. With the exact bound $b_j b'_j \geq (3+2\sqrt{2})a^2$ from the proof of Lemma B.5, the same computation gives $K > (6+4\sqrt{2})a^2$ and $\mathcal{C} > 2\sqrt{3}(5+4\sqrt{2})/(6+4\sqrt{2}) = 2\sqrt{6} - \sqrt{3} = 3.1669\ldots$. □

Proof of Theorem 6.1. By Lemma 6.2 the word is $(--+)^m$ with $m \equiv 2 \pmod 3$. The period is mixed, so its Casimir is positive (Proposition 4.16), and the competitive cap applies. If $m \geq 5$ the angle count above is contradictory, and if $m = 2$, Proposition 6.11 applies. □

7. Class F: every switch forward

The smoothed octagon's period is entirely forward: its word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ advances the label by one at each of its three switches, and by one more across the twisted seam. This section shows that no other entirely forward period can be competitive once it has six or more pieces.

Theorem 7.1 (Class F)**.** *No competitive normal vertex period with at least six pieces has every switch forward.*

The proof plays two consequences of competitiveness against each other. On one side, the Casimir γ^2 of the normal multiplier (Definition 4.11) is a conserved quantity of the whole period, and a cost at most A_{oct} bounds it from above: the frame has to turn through the angle $\pi/3$ between $g(0) = I$ and $g(T) = R$, and the larger γ^2 is, the more cost each unit of turning requires (Proposition 7.6). On the other side, γ^2 can be read off at any piece from the two switch norms on either side of it and their cross pairing (Lemma 4.15). We read it at a piece of smallest active margin. In an all-forward period the number of pieces satisfies $N \equiv 1 \pmod 3$, so a period with $N \geq 6$ has at least seven pieces, and its cost is a sum of seven or more edge terms that depend only on consecutive active margins. At the octagon's margins each edge costs exactly a quarter of A_{oct}; seven edges fit into the budget only if the smallest margin is markedly smaller than the octagon's. The switching conditions at the five pieces around that small margin (stationarity of the cost, and the requirement that the unused vertex not be preferable at the two

outer switches) then hold the two switch norms close to their least possible value 3/32. With such norms the Gram identity makes γ^2 slightly larger than competitiveness allows. The margin in this contradiction is small: the cap gives $\gamma^2 < 0.18381$, the switching conditions give $\gamma^2 > 23/125 = 0.184$ (Figure 9).

The informal proof is `HAND_FORWARD_PERIOD_EXCLUSION` of [7], which replaced two interval covers of an earlier version (`GLOBAL_FORWARD_PERIOD_EXCLUSION`). We follow its chain of steps; at a few places we use a shorter argument.

7.1. **The count and the cost of a forward period.**

Lemma 7.2. *If every switch of a normal vertex period is forward, then* $N \equiv 1 \pmod 3$. *In particular* $N \geq 6$ *implies* $N \geq 7$.

Proof. The label advances by one at each of the N switches of a period, so $i_N = i_0 + N$, while the twist gives $i_N = i_0 + 1$ in $\mathbb{Z}/3$ (Lemma 4.6). □

At a forward switch from piece $k-1$ to piece k the state has the margins a_{k-1} and a_k for the labels i_{k-1} and i_k, and the relation $p_0p_1 + p_1p_2 + p_2p_0 = 4$ (Lemma 2.8) determines the third margin, $(4 - a_{k-1}a_k)/(a_{k-1} + a_k)$. Its positivity and the one-arc formula (Lemma 4.1) on piece k give the following.

Lemma 7.3. *In a normal vertex period with every switch forward, for every* k,

$$a_{k-1}a_k < 4, \qquad q_k = \frac{(a_{k-1} + a_k)(a_k + a_{k+1})}{a_k^2 + 4} > 1,$$

and the cost is

$$\mathcal{C} = \frac{\sqrt{3}}{4} \sum_{j=0}^{N-1} G(a_j, a_{j+1}), \tag{7.1}$$

where

$$G(a,b) = 3ab - 4 - (a^2 + b^2)\log(a+b) + \tfrac{1}{2}a^2\log(a^2+4) + \tfrac{1}{2}b^2\log(b^2+4).$$

So in a forward period the durations are determined by the active margins, and (7.1) expresses the cost through the margins alone. The function G is the signed action of `MIXED_GLOBAL_COSTATE_ELLIPTIC` (3) of [7] with all signs positive. The arc costs of Lemma 4.1 differ from the terms of (7.1) by a telescoping sum, which vanishes over a period. For the octagon, $N = 4$ and all four active margins equal a with $a^2 = 4/(2\sqrt{2} - 1)$; then $(a^2+4)/(4a^2) = 1/\sqrt{2}$, so $G(a,a) = 3a^2 - 4 + a^2 \log \frac{a^2+4}{4a^2} = 2\delta_{\mathrm{oct}}$ and the four edges cost $\sqrt{3}\,G(a,a) = A_{\mathrm{oct}}$.

Appendix C.1 collects the properties of G that we use: G is symmetric, increasing in each argument as long as $ab \leq 8$, has positive mixed derivative $\partial_a\partial_b G = 2(a^2 + ab + b^2)/(a+b)^2$, and satisfies $G(a,a) \geq 0$ when $a^2 \geq 4/3$.

7.2. **The window at a minimum.** Fix a piece m whose active margin $a = a_m$ is the least of all active margins of the period, and write

$$(p, b, a, c, d) = (a_{m-2}, a_{m-1}, a_m, a_{m+1}, a_{m+2})$$

for the active margins of the five pieces around it; all are at least a. Lowering every active margin outside this window to a lowers each edge term of (7.1), by monotonicity of G, and keeps the two edges $G(a,p)$ and $G(d,a)$ that close the window.

Lemma 7.4 (The seven-entry bound)**.** *If $N \geq 7$, there is an integer $n \geq 0$ with*

$$G(a,p) + G(p,b) + G(b,a) + G(a,c) + G(c,d) + G(d,a) + (n+1)\,G(a,a) \leq \sum_{j=0}^{N-1} G(a_j, a_{j+1}) = \frac{4}{\sqrt{3}}\,\mathcal{C}.$$

The window is described by dimensionless quantities. Put

$$\nu = \frac{4}{a^2}, \qquad z = \frac{b}{a}, \qquad w = \frac{c}{a}, \qquad \hat{p} = \frac{p}{a}, \qquad \hat{d} = \frac{d}{a},$$
$$\hat{s}_- = 1 + z, \quad \hat{s}_+ = 1 + w, \quad \hat{K} = 1 + \nu, \quad \hat{K}_- = z^2 + \nu, \quad \hat{K}_+ = w^2 + \nu,$$

so that $\hat{s}_- = s_{m-1}/a$, $\hat{s}_+ = s_m/a$, and $\hat{K}, \hat{K}_-, \hat{K}_+$ are K_m, K_{m-1}, K_{m+1} divided by a^2. These letters are local to this section and to Appendix C. The factors of the three central pieces are

$$h := q_{m-1} = \frac{(\hat{p}+z)\hat{s}_-}{\hat{K}_-}, \qquad q := q_m = \frac{\hat{s}_-\hat{s}_+}{\hat{K}}, \qquad k := q_{m+1} = \frac{(w+\hat{d})\hat{s}_+}{\hat{K}_+}.$$

Minimality gives $\hat{p}, z, w, \hat{d} \geq 1$, and the products $pb, ab, ac, cd < 4$ of Lemma 7.3 read $\hat{p}z < \nu$, $z < \nu$, $w < \nu$, $w\hat{d} < \nu$. Together these give

$$\frac{\hat{s}_-^2}{\hat{K}_-} \leq h \leq \frac{\hat{s}_-}{z}, \qquad \frac{\hat{s}_+^2}{\hat{K}_+} \leq k \leq \frac{\hat{s}_+}{w}. \tag{7.2}$$

7.3. **The normal multiplier at the window.** The costate enters through three numbers: eight times the switch norms on either side of the central piece, and sixteen times the cross pairing on it (Definition 4.11),

$$u := 8d_{m-1}, \qquad v := 8d_m, \qquad E := 16e_m.$$

(The letter u does not denote the control here.) At a forward switch the switch norm is at least $j/2 \geq 3/32$ (Lemma 4.14), so $u, v \geq 3/4$; the proof will show that u and v exceed $3/4$ only by little.

Three facts about (u, v, E) come from the Pontryagin conditions. First, stationarity (Lemma 4.13) at the piece m gives one linear relation, and at the pieces $m \mp 1$, after the coordinates of the multiplier are transported across the pieces m and $m+1$ (Definition 4.11), two more. Second, at the forward switch from piece $m-2$ into piece $m-1$ the vertices of the two pieces tie, and the third vertex, which neither piece uses, may not do better

(Lemma 3.3): its switching function is $\chi = -4\sqrt{3}\,G_L/(p+b)^2 \le 0$ for an affine expression G_L in the entries of the multiplier. Read in the frame of the piece $m-1$, the inequality $G_L \ge 0$ is a lower bound for u in terms of the exterior factor h and the ratio z; the switch from $m+1$ into $m+2$ gives the same kind of bound for v. Third, the Gram identity (Lemma 4.15) at the doubly forward piece m expresses the Casimir.

Proposition 7.5 (The window data)**.** *In the setting above, put*

$$A = 1 + z + w - \frac{1}{\hat{s}_-} - \frac{1}{\hat{s}_+} + \frac{1}{\hat{K}} - \log q,$$

$$B = \frac{\hat{K}_- h}{\hat{s}_-} + 1 - \frac{z^2 \hat{s}_-}{\hat{K}_- h} - \frac{z^2}{\hat{s}_-} + \frac{z^3}{\hat{K}_-} - z \log h,$$

$$\eta = \frac{\hat{K}(h\hat{s}_- - z)}{\hat{K}_-}, \qquad n_1 = \frac{\hat{s}_-^2(\nu - z)(h-1)}{\hat{K}_-},$$

$$n_2 = \frac{\hat{K}^2}{\hat{s}_- \hat{K}_-}\big(q - 1 + (h-1)(q\hat{s}_- - 1)\big),$$

and let $F_x(r) = 1 + x^2 r - xr \log r/(r-1)$ *for* $r > 1$*. Then:*

(1) (central row) $\hat{s}_- u + \hat{s}_+ v - E = A$*;*

(2) (left row) $n_1 u + n_2 v = \eta A - B$*, and the right row is obtained by exchanging* $(z, \hat{p}, h, u)$ *with* $(w, \hat{d}, k, v)$*;*

(3) (exterior floors) $u \ge F_{z/\hat{s}_-}(h)$ *and* $v \ge F_{w/\hat{s}_+}(k)$*;*

(4) (Gram identity) $\gamma^2 = \hat{K}^2(4uv - E^2)/(16\nu^2)$*, and* $\gamma^2 = \det \Lambda_1(0)/12$*.*

In physical units the central row reads $(a+b)u + (a+c)v - aE = \frac{1}{2}\partial_{a_m} \sum_j G(a_j, a_{j+1})$: it is the stationarity of the cost (7.1) in the central margin. For $x \ge 1/2$ the floor $F_x(r)$ increases in r and tends to $1 - x + x^2 \ge 3/4$ as $r \to 1$. The floor for u is the missing-control inequality `MIXED_GLOBAL_SMALL_FORWARD_MINIMUM` (11)–(12) of [7]: once the stationarity of the piece $m-1$ is used, G_L is a positive multiple of $u - F_{z/\hat{s}_-}(h)$.

7.4. **The competitive cap.** Proposition 4.17 bounds the cost below by a function of the Casimir. Case F uses the following sharper bound; since $(\kappa + \frac{3}{2})(2\kappa + 3) - 12\kappa = 2(\kappa - \frac{3}{2})^2 \ge 0$, it implies $\mathcal{C} \ge 4\pi\kappa/(2\kappa+3)$.

Proposition 7.6 (Spectral cost–Casimir bound)**.** *Let* P *be a normal vertex period with every switch forward. Then* $\det \Lambda_1(0) > 0$*, and with* $\kappa = \sqrt{\det \Lambda_1(0)}$*,*

$$\mathcal{C} \ge \frac{\pi}{3}\Big(\kappa + \frac{3}{2}\Big).$$

Sketch. The Casimir is positive by the projective argument of Proposition 4.16: at each doubly forward piece the projective coordinate of the switch vectors increases strictly, so over a period the increments cannot add up to zero. With one positive switch norm this makes $\Lambda_1(0)$ negative elliptic,

$\Lambda_1(0) = -\kappa\, hJh^{-1}$ for some $h \in \mathrm{SL}_2(\mathbb{R})$. Put $B(t) = h^{-1}g(t)$; then $B' = BX$ and $B(T) = B(0)R$. Write $\rho = -\frac{1}{2}\operatorname{tr}(JX)$ and $\rho_R = -\frac{1}{2}\operatorname{tr}(JBXB^{-1})$, so that the cost density is 3ρ, and let $z_B = \frac{1}{2}(B_{00} + B_{11}) + \frac{i}{2}(B_{10} - B_{01})$ be the rotation coordinate of B. Conservation, $\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)g(t)$, gives $\operatorname{tr}(\Lambda_1 X) = 2\kappa\rho_R$, and the vertex Hamiltonians being nonpositive gives $\operatorname{tr}((\Lambda_1 + \frac{3}{2}J)X) \le 0$, that is, $2\kappa\rho_R \le 3\rho$. The polar angle θ of z_B changes at the rate $\theta' = (\rho + \rho_R)/(2|z_B|^2) > 0$ and turns through at least $\pi/3$, since $B(T) = B(0)R$. The pointwise inequality $(\rho + \rho_R)^2 \le 4|z_B|^2\rho\rho_R$, an explicit sum of squares, gives $\theta' \le 2\rho\rho_R/(\rho + \rho_R) \le 6\rho/(2\kappa + 3)$. Integrating, $\pi/3 \le 2\mathcal{C}/(2\kappa + 3)$. □

Corollary 7.7 (Competitive cap)**.** *A competitive normal vertex period with every switch forward has $\gamma^2 < 23/125$.*

Proof. By Proposition 7.6 and $\mathcal{C} \le A_{\mathrm{oct}} < 3.1268$, $\kappa < 3 \cdot 3.1268/\pi - \frac{3}{2} < 1.4859$, so $\gamma^2 = \kappa^2/12 < 1.4859^2/12 < 23/125$. □

The notes prove the sharper cap $\gamma^2 < 9193/50000$; the weaker cap $23/125$ suffices. The exact bound that Proposition 7.6 allows is $\gamma^2 \le (3A_{\mathrm{oct}}/\pi - \frac{3}{2})^2/12 < 0.18381$.

7.5. **The scalar exclusion.** From now on suppose $N \ge 6$, so $N \ge 7$ by Lemma 7.2. The window data of Lemma 7.4 and Proposition 7.5, the bounds (7.2) and the cap $\gamma^2 \le 23/125$ are finitely many real equations and inequalities. We show that they are inconsistent. The data are symmetric under the exchange of the two sides, which exchanges the left and right rows and the two floors, so we may assume $z \le w$. The steps follow; Appendix C proves the scalar inequalities.

Step 1: the minimum is not small. The first restriction does not use the cost.

Lemma 7.8 (Small forward minimum)**.** *$\nu < 3$, that is, $a^2 > 4/3$.*

If $\nu \ge 3$, the left row and the floors force the residual $\eta A - B - n_1 F_{z/\hat{s}_-}(h) - n_2 V$ to be nonnegative, where V is the floor for v evaluated at the smallest factor that minimality allows. Appendix C.7 shows that this residual is negative: lowering the larger neighbour w does not decrease it, which leaves three faces, and on each face the residual is a function of h with explicit value and slope at $h = 1$ and a controlled second derivative.

Step 2: the minimum is below the octagon's scale. Since $a^2 > 4/3$, $G(a, a) \ge 0$, so the seven-entry bound holds with $n = 0$. If $a^2 \ge 20/11$, each of the seven edges is at least $G(a_0, a_0)$ with $a_0^2 = 20/11$, and $G(a_0, a_0) = \frac{20}{11}(\frac{4}{5} - \log\frac{5}{4}) \ge \frac{23}{22}$, so the cost would be at least $\frac{7\sqrt{3}}{4} \cdot \frac{23}{22} > 3.1268 > A_{\mathrm{oct}}$.

Lemma 7.9. $11/5 < \nu < 3$.

Step 3: the exterior budget. The rectangle inequality $G(x,y) \geq G(x,a) + G(a,y) - G(a,a)$ for $x, y \geq a$, applied to the edges (p,b) and (c,d), turns the seven-entry bound into

$$\frac{\sqrt{3}}{\nu}\big(2g_\nu(\hat{p})+2g_\nu(z)+2g_\nu(w)+2g_\nu(\hat{d})-g_\nu(1)\big) \leq A_{\mathrm{oct}}, \qquad g_\nu(x) = \frac{G(a,ax)}{a^2}.$$

On $[1,\nu]$ the function g_ν is increasing and concave, so for a given sum $\hat{p}+z+w+\hat{d}$ the left side is smallest when the excess over four is carried by one ratio. A one-variable inequality (Appendix C.3) then gives:

Lemma 7.10 (Exterior budget)**.** *$\hat{p}+z+w+\hat{d} < (5\nu+1)/3$. Consequently $z+w < 5(\nu-1)/3$ and $h \leq \hat{s}_-(5\nu-2-3w)/(3\hat{K}_-)$.*

The bound on h follows from $h\hat{K}_-/\hat{s}_- = \hat{p}+z$ and $k\hat{K}_+/\hat{s}_+ = \hat{d}+w \geq \hat{s}_+$.

Step 4: the larger neighbour. The floors and the left row give $\eta A - B = n_1 u + n_2 v \geq n_1 F_{z/\hat{s}_-}(h) + n_2 F_{w/\hat{s}_+}(\hat{s}_+^2/\hat{K}_+)$, since $n_2 \geq n_1 \geq 0$, the floor increases in the factor and $k \geq \hat{s}_+^2/\hat{K}_+$. The difference of the two sides is a residual $\mathcal{E}_N$, and $\mathcal{E}_N < 0$ whenever $w \geq 4/3$ (Appendix C.5).

Lemma 7.11 (Neighbour bound)**.** *If $z \leq w$, then $w < 4/3$.*

Step 5: the switch norms are nearly minimal. Write $u = \frac{3}{4} + \delta u$ and $v = \frac{3}{4} + \delta v$. The floors become $\delta u \geq L(z,h) \geq 0$ and $\delta v \geq L(w, \hat{s}_+^2/\hat{K}_+) \geq 0$ for an explicit function L (Appendix C.4), and the left row becomes the centered identity

$$n_1\,\delta u + n_2\,\delta v = r, \qquad r := \eta A - B - \tfrac{3}{4}(n_1+n_2), \tag{7.3}$$

with $n_2 > n_1 \geq 0$. If $\delta u + \delta v \geq 1/35$, then (7.3), written as $r = n_1(\delta u + \delta v) + (n_2 - n_1)\delta v$, and the floor for δv give

$$\mathcal{E}_\Sigma := r - \frac{n_1}{35} - (n_2 - n_1)\,L\Big(w, \frac{\hat{s}_+^2}{\hat{K}_+}\Big) \geq 0;$$

if $\delta v \geq 1/64$, they give $\mathcal{E}_v := r - n_1 L(z,h) - n_2/64 \geq 0$. Neither residual decreases when w is lowered to z, and on the resulting symmetric domain both are negative (Appendix C.5).

Lemma 7.12 (Norm bounds)**.** *If $z \leq w$, then $\delta u, \delta v \geq 0$, $\delta u + \delta v < 1/35$ and $\delta v < 1/64$.*

Step 6: the Casimir is too large. The central row and the identity $\frac{3}{4}(\hat{s}_- + \hat{s}_+) - A = e_0 - \varphi$, where

$$e_0 = 1 - \frac{\hat{K}}{8} + \frac{1}{\hat{K}}, \quad \varphi = f\Big(\frac{\hat{s}_-}{2}\Big) + f\Big(\frac{\hat{s}_+}{2}\Big) + f\Big(\frac{4}{\hat{K}}\Big), \quad f(r) = \frac{r - r^{-1}}{2} - \log r,$$

give $E = e_0 - \varphi + \hat{s}_-\delta u + \hat{s}_+\delta v$, and $0 \leq \varphi \leq 1/100$ on the domain reached so far. Substituted into the Gram identity, the norm bounds give:

Lemma 7.13 (Conditional Casimir bound)**.** *Let* $11/5 \le \nu < 3$, $1 \le z \le w \le 4/3$, $z + w \le 5(\nu - 1)/3$, $\delta u, \delta v \ge 0$, $\delta u + \delta v \le 1/35$, $\delta v \le 1/64$ *and* $E = e_0 - \varphi + \hat{s}_- \delta u + \hat{s}_+ \delta v$. *Then*

$$\frac{\hat{K}^2(4(\frac{3}{4} + \delta u)(\frac{3}{4} + \delta v) - E^2)}{16\nu^2} > \frac{23}{125}.$$

Proof of Theorem 7.1. Suppose P is competitive and all-forward, with $N \ge 6$. By Lemma 7.2 $N \ge 7$; take the window at a global minimum of the active margins. Its data satisfy Lemma 7.4 and Proposition 7.5 and Corollary 7.7; after exchanging the two sides if necessary, $z \le w$. Lemmas 7.9 to 7.12 give the hypotheses of Lemma 7.13, which together with the Gram identity gives $\gamma^2 > 23/125$, contradicting Corollary 7.7. □

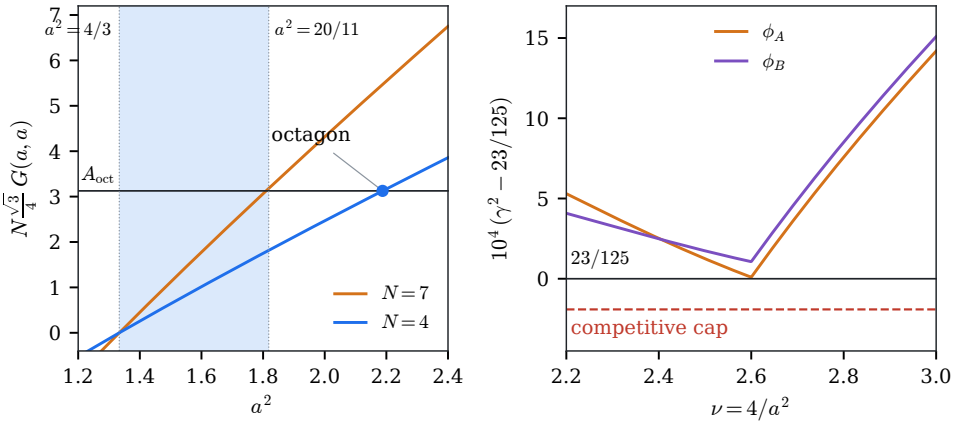


FIGURE 9. The two sides of the case F contradiction. Left: the cost $N\frac{\sqrt{3}}{4}G(a,a)$ of N equal edges against a^2, for $N = 4$ and $N = 7$. Four edges at the octagon's margin cost exactly A_{oct} (dot). Seven edges cost more than A_{oct} once $a^2 \ge 20/11$ (Lemma 7.9), and $a^2 > 4/3$ by Lemma 7.8, so the least active margin of a long competitive forward period lies in the shaded band. Right: on that band, under the bounds of Lemmas 7.10 to 7.12, the Casimir is at least the smaller of the two curves ϕ_A, ϕ_B of Appendix C.6, which stay above 23/125, while competitiveness forces $\gamma^2 < 0.18381$ (dashed). The vertical scale is $10^4(\gamma^2 - 23/125)$.

8. Class M2: the minimum at a reverse centre and the nonshort cases

A period of class M2 has switches of both signs, every maximal reverse run has exactly two switches, and some maximal forward run has at least two (Definition 4.7). Unlike the classes R, F and M1, such a period has no rigid sign word: the forward runs may have any lengths, the number of reverse

runs is arbitrary, and the factors are unrelated. The exclusion is therefore local. We look at a window of at most seven consecutive pieces around the smallest active margin of the period, read the single normal multiplier U of the whole period in that window, and derive a finite list of scalar necessary conditions, which are then shown to be inconsistent. The whole period enters only through three inputs: the compact domain (Theorem 2.14), the cap on the Casimir that competitiveness imposes, and a lower bound on the Casimir obtained from the winding of the switch vectors over the period.

Theorem 8.1 (Class M2)**.** *Let P be a competitive normal vertex period whose state stays in $\mathfrak{h}^{\star\star}$ and which has at least six pieces. Then the sign sequence of P is not of class* M2*.*

The proof has five steps.

(1) A global minimum of the active margins sits at the centre of a reverse run (Proposition 8.5).
(2) Of the two turns beside that centre, choose the one with the smaller active margin; after reflecting the period and reversing time if necessary, it follows the centre (Lemma 8.6).
(3) The switch after the chosen turn is forward or reverse. If it is reverse, the forward run through the turn has a single switch (it is short), and Section 9 excludes the window (Theorem 9.3). If it is forward, the run is nonshort, and the switch after the next piece is again forward or reverse (Propositions 8.7 and 8.8).
(4) A comparison that moves one margin to a boundary value turns the window into one scalar residual N that must be negative (Lemma 8.9), together with a short list of necessary bounds (Lemma 8.11).
(5) A reverse successor contradicts $N < 0$ directly (Theorem 8.13). A forward successor leaves a small region of parameters on which the Casimir of the period exceeds the competitive cap (Theorem 8.15).

This section carries out steps (1), (2), (4), (5) and the nonshort half of (3). The polynomial inequalities are collected in Appendix D. Throughout, γ^2 is the Casimir of Definition 4.11, and the switch norms and turn secants are those of Lemma 4.14.

8.1. **Three inputs from the whole period.** The competitive cap of Corollary 4.18 is not quite sharp enough for class M2. We use the spectral form of the cost–Casimir inequality [7, `GLOBAL_SPECTRAL_COST_CASIMIR`].

Lemma 8.2 (Casimir bounds)**.** *Let P be a normal vertex period with cost $\mathcal{C}$ and primary costate Λ_1.*

(a) If $\det \Lambda_1(0) > 0$ and $\Lambda_1(0)_{10} < 0$, then $\mathcal{C} \geq \frac{\pi}{3}(\kappa + \frac{3}{2})$ with $\kappa = \sqrt{\det \Lambda_1(0)}$.
(b) If P is competitive, then $\gamma^2 < 9193/50000$.
(c) If P is competitive and mixed, then $\gamma^2 > 27/256$.

Sketch. (a) Write $\Lambda_1(0) = -\kappa\, hJh^{-1}$ and $B = h^{-1}g$. The maximum condition gives pointwise $2\kappa m_s \le 3m_b$, where $m_b = -\frac{1}{2}\operatorname{tr}(JX)$ is a third of the cost density and $m_s = -\frac{1}{2}\operatorname{tr}(JBXB^{-1})$. The argument θ of $(B_{00} + B_{11}) + i(B_{10} - B_{01})$ has $\theta' = (m_b + m_s)/(z+1)$ with $z = \frac{1}{2}\|B\|_F^2 \ge 1$, and $g(T) = R$ forces $\Delta\theta \ge \pi/3$. The spectral inequality $(m_b + m_s)^2 \le 4z\, m_b m_s$ combines these into the bound, which is stronger than Proposition 4.17 by the arithmetic–geometric mean inequality. (b) Class R is impossible (Theorem 5.4), so some switch is forward and has positive norm; when $\gamma^2 > 0$, $\Lambda_1(0)$ is then negative elliptic (Proposition 4.16), and (a) with $\mathcal{C} \le A_{\mathrm{oct}} < 3.12606$ gives $\kappa < 1.4853$, so $\gamma^2 = \kappa^2/12 < 9193/50000$. (c) If $\gamma^2 \le 27/256$, the Gram identity (Lemma 4.15) with $d_\pm \ge 3/32$ puts the angle between consecutive switch vectors below $\pi/3$ at a forward piece, and above $2\pi/3$ at a reverse centre, by the angle gap $\cos\varphi < 3/10$ of Section 6; a turn contributes nothing. The increments then add up to more than $\pi/3$, while the winding of the switch vectors over one period is exactly $\pi/3$ (Section 6). □

At a reverse centre (the middle piece of a reverse run of two) the pairing and the Casimir are explicit functions of four quantities. With active margin a, turn neighbours p, n and incident norms $u_{\mathrm{i}}, u_{\mathrm{o}}$ (eight times the switch norms, which are turn secants), put $r = a/(a+p)$, $t = a/(a+n)$ and $\rho = a^2/(a^2+4)$. Stationarity at the centre and the Gram identity read

$$\gamma^2 = \frac{4u_{\mathrm{i}}u_{\mathrm{o}} - E^2}{16(1-\rho)^2}, \qquad E = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1-u_{\mathrm{i}}}{r} - \frac{1-u_{\mathrm{o}}}{t}. \tag{8.1}$$

The strict centre minimum gives $r, t < \frac{1}{2}$, and the positive reverse factor gives $\rho < rt$.

Lemma 8.3 (Scale of a reverse centre)**.** *In a competitive mixed normal vertex period, every reverse centre has active margin a with $a^2 < 12/17$.*

Proof. The bound is $\rho < 3/20$. Lowering both incident norms to their common floor $3/4$, and then raising both ratios r, t to $\frac{1}{2}$, only decreases the right side of (8.1), and gives $\gamma^2 > G(\rho) = (9/4 - B(\rho)^2)/(16(1-\rho)^2)$ with $B(\rho) = 1 - \rho + \log(4\rho)$. The function G increases on $[3/20, 1/4)$ and $G(3/20) > 23/125 > 9193/50000$, which contradicts Lemma 8.2(b) when $\rho \ge 3/20$; and $\rho < rt < 1/4$ always. Figure 10(a) shows G. □

8.2. **The minimum is a reverse centre.** Let a_m be a smallest active margin of the period; the active margins are periodic, so a minimum exists. A minimum cannot be at a turn: at three pieces with labels i, j, i the active margins are strictly log-concave, $a_k a_{k+2} < a_{k+1}^2$ (Lemma 5.5), so the middle one is not the smallest of the three. If both switches at m are reverse, m is the centre of a reverse run. The remaining case, both switches forward, is excluded by the following theorem, which uses only stationarity and the floors that the two missing controls impose.

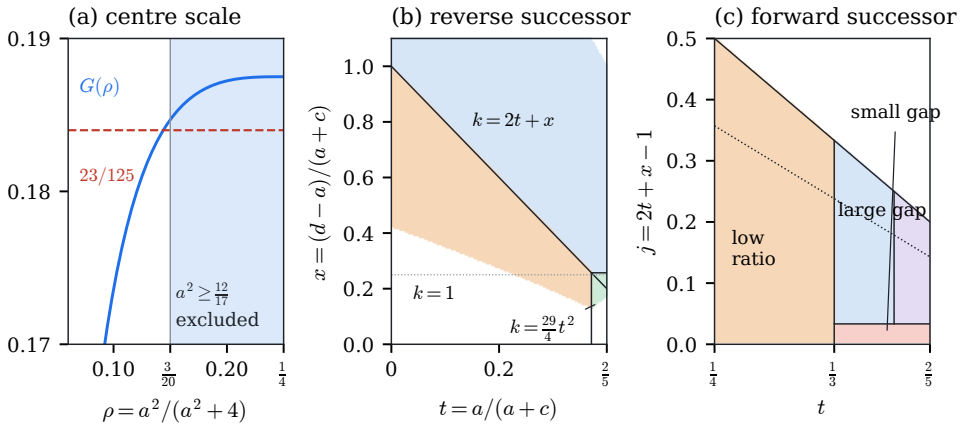


FIGURE 10. What the estimates of class M2 control. (a) The lower bound $G(\rho) = (9/4 - B(\rho)^2)/(16(1-\rho)^2)$ for the Casimir at a reverse centre, with $\rho = a^2/(a^2+4)$. It exceeds the cap $23/125$ (dashed) on $[3/20, 1/4)$, which proves $a^2 < 12/17$ (Lemma 8.3). (b) The parameters (t, x) of the scalar reverse exclusion (Proposition 8.12), below $x = 1.1$, by the face to which k is lowered: the unit shape $k = 1$, the order face $k = 2t + x$, and the scale strip $k = 29t^2/4$, which lies in $13/35 < t < 2/5$, $x \leq 9/35$. Points above $x = 1$ belong to the order face. (c) The parameters (t, j) of a forward successor, in $1/4 < t < 2/5$, $0 < j < 1 - 2t$: the low ratio $t \leq 1/3$, the two large-gap regions $j \geq 1/30$, split at $t = 3/8$, and the small-gap region, where the Casimir exceeds the cap. The dotted line is $j = \frac{5}{7}(1 - 2t)$.

Proposition 8.4 (No small forward minimum)**.** *Let P be a normal vertex period with at least three pieces, and let m be an index such that the switches $m-2, m-1, m, m+1$ are forward, $a_m \leq a_j$ for $m-2 \leq j \leq m+2$, and $a_m^2 \leq 4/3$. This is impossible.*

The proof [7, `SMALL_FORWARD_MINIMUM_HAND_PROOF`] writes the multiplier in the frame of the central piece, eliminates one coordinate with stationarity at the centre, and keeps the stationarity row at the smaller neighbour with the two exterior floors $v \geq F_x(q)$ at the forward switches. The remaining face is a one-variable residual with an exactly controlled second derivative; two integrations and the Padé bound $\frac{1}{2}(h - h^{-1}) - \log h \geq (h-1)^3/(3h(h+1))$ leave a rational expression, negative by an elementary cubic comparison. No Casimir bound is used.

Proposition 8.5 (The minimum is a reverse centre)**.** *Let P be a competitive normal vertex period of class* M2 *with at least six pieces. There is an index m with $a_m \leq a_k$ for every k such that the switches $m-1$ and m are reverse and the switches $m-2$ and $m+1$ are forward.*

Proof. Take a global minimum m. If exactly one of the switches $m-1, m$ is reverse, the pieces $m-1, m, m+1$ have labels i, j, i, and Lemma 5.5 contradicts minimality. If both are forward, the strict centre minimum of the neighbouring reverse runs (Proposition 5.6) forces the switches $m-2$ and $m+1$ to be forward as well: otherwise the centre of the adjacent reverse run would have a smaller active margin. The period is mixed, so it has a reverse centre; by Lemma 8.3 its active margin squared is below $12/17$, hence $a_m^2 < 12/17 < 4/3$, and Proposition 8.4 gives a contradiction. If both switches are reverse, they form a maximal reverse run, of length two in class M2, and m is its centre. □

8.3. **Orientation by reflection.** Let $S = \operatorname{diag}(1, -1)$. Reflecting the plane and reversing time is a symmetry of the control problem: it maps a normal vertex period to another one with the same cost, and reverses the order of the pieces.

Lemma 8.6 (Reflection)**.** *For every normal vertex period P there is a normal vertex period P' with the same number of pieces and the same cost, whose state stays in $\mathfrak{h}^{\star\star}$ when that of P does, with labels $i'_k = c - i_{-1-k}$ for a constant $c \in \mathbb{Z}/3$ and active margins $a'_k = a_{-1-k}$. Its switch k is forward if and only if the switch $-2-k$ of P is forward, and P' is of class* M2 *when P is.*

The construction is $u'(t) = \varsigma u(T-t)$, with ς the permutation of the vertices that reflects the labels, $g'(t) = SR^{-1}g(T-t)S$, $X'(t) = -SX(T-t)S$, and $\Lambda'(t) = -S\Lambda(T-t)S$ for both costates. Since $SJS = -J$, the cost density is preserved, and $\mathfrak{h}^{\star\star}$ is symmetric under $x \mapsto -x$. Reading the signs in reverse order does not negate them: a reverse switch of P is a reverse switch of P'.

By Proposition 8.5 and Lemma 8.6 we may assume that the global minimum $a = a_m$ is a reverse centre and that the turn after it is the smaller one: $a_{m+1} \le a_{m-1}$. (At a tie either choice works.) After relabeling the vertices by a power of R, which changes neither the cost nor the active margins, the pieces $m-2, \dots, m+3$ have labels and active margins

$$0(p), \quad 1(b), \quad 0(a), \quad 2(c), \quad 0(d), \quad 1(r) \tag{8.2}$$

when the switch $m+2$ is forward. These are consecutive pieces of the period, continued through the twisted seam when necessary; they need not be distinct from other pieces named later, and they are not a closed competitor.

8.4. **The nonshort window.** Table 2 lists the local notation. All quantities in the window come from the one normal multiplier of the period. The factors of the same-sign pieces follow from the one-arc map (Lemma 4.1) and the margins $(a_k, a_{k+1}, (4 - a_k a_{k+1})/(a_k + a_{k+1}))$ of the state at a switch: the factor of a piece with incident signs $++$, $--$, $-+$, $+-$ is $s_{k-1}s_k/K_k$, $K_k/(s_{k-1}s_k)$, s_k/s_{k-1}, s_{k-1}/s_k respectively. In the window this gives the central factor $K_a/(ST) > 1$, the factor $q = WZ/Q > 1$ of the piece d, and $d > a$ from the turn c.

Symbol	Meaning (in Section 8 and Appendix D)
p, b, a, c, d, r	active margins of six consecutive pieces (8.2); a is a global minimum, and $c \le b$
$K_a,\ Q,\ K_r$	a^2+4, d^2+4, r^2+4
S, T, W, Z, Y	$a+b$, $a+c$, $c+d$, $d+r$, W^2-T^2 (here T is not the terminal time)
u, v, w	eight times the norms of the switches $b \to a$; $a \to c$ and $c \to d$; $d \to r$
h_a, h_d	the diagonal entries of the multiplier in the frames of the pieces a and d
$U(x),\ g_B(x)$	$1-x+x^2$ and $(B^2+Bx+x^2)/(B+x)^2 = 1-Bx/(B+x)^2$
$\mathrm{Sec}(A, B, P)$	the average of g_B over the interval between A and P
$F_x(q)$	$1+x^2q-xq\log q/(q-1)$, with $F_x(1)=U(x)$
t, x, k, q	a/T, $(d-a)/T$, K_a/T^2, WZ/Q (normalized coordinates)
j	$2t+x-1$, the gap of a forward successor

TABLE 2. Local notation of class M2. The letters p, r, t, u, x have their usual meaning elsewhere in the paper.

Both switches at a turn have the same norm, the turn secant (Lemma 4.14): $u = \mathrm{Sec}(a, b, p)$ and $v = \mathrm{Sec}(a, c, d)$, where

$$\mathrm{Sec}(A, B, P) = 1 + \frac{B^2}{(A+B)(P+B)} - \frac{B}{P-A}\log\frac{P+B}{A+B}.$$

In the frame of the central piece the multiplier is

$$U_a = \begin{pmatrix} h_a & -K_a^2 v/(8S^2) \\ S^2 u/8 & -h_a \end{pmatrix}, \qquad \gamma^2 = \det U_a = \frac{uvK_a^2}{64} - h_a^2, \tag{8.3}$$

and stationarity at the centre (Lemma 4.13) fixes the diagonal:

$$h_a = \frac{K_a}{16a}\big(Su + Tv - C(b, a, c)\big),$$
$$C(p, a, n) = p + a + n - \frac{a^2}{p+a} - \frac{a^2}{a+n} + \frac{a^3}{a^2+4} - a\log\frac{(p+a)(a+n)}{a^2+4}. \tag{8.4}$$

Here $C(p, a, n)$ is four times the partial derivative of the normalized cost $\mathcal{C}/\sqrt{12}$ in the active margin of a same-sign piece. Transport across the centre and the turn to the frame of the piece d gives

$$h_d = h_a + \frac{vY}{8}, \qquad w = \frac{uK_a^2 + 16h_aY + vY^2}{Q^2}, \tag{8.5}$$

and stationarity at d, the first piece of the forward run after the turn, is the root equation

$$R_d := C(c, d, r) - Wv - Zw + \frac{16\,d\,h_d}{Q} = 0. \tag{8.6}$$

Proposition 8.7 (The nonshort window)**.** *Let P be competitive, with state in $\mathfrak{h}^{\star\star}$, of class* M2 *and with at least six pieces; let m be as in Proposition 8.5 with $a_{m+1} \le a_{m-1}$, and let the switch $m{+}2$ be forward. Then $(p, b, a, c, d, r) = (a_{m-2}, \dots, a_{m+3})$, with u, v, h_a, w as above, satisfy:*

(i) the order $a < b$, $a < c \le b$, $a < p$, $a < d$, $a \le r$;

(ii) strict star at the five switches, $pb, ab, ac, cd, dr < 4$; the factors $ST < K_a$ and $Q < WZ$; three-arc log-concavity $ad < c^2$;

(iii) the compact domain: $a \ge 2/15$, $cd + \frac{2}{15}(c+d) \le 4$, $dr + \frac{2}{15}(d+r) \le 4$, $d, r \le 8$, and $a+c$, $c+d$, $d+r \ge \frac{1}{2}$; and the scale bound $a^2 < 12/17$;

(iv) the turn secants $u = \mathrm{Sec}(a, b, p)$, $v = \mathrm{Sec}(a, c, d)$; the switching inequalities $g_b(p) \le u \le g_b(a)$, $g_c(d) \le v \le g_c(a)$, $w \ge g_d(r)$; the missing-control floors $v \ge F_{d/W}(q)$ and $w \ge F_{d/Z}(q)$;

(v) the identities (8.4), (8.5) and (8.6);

(vi) $27/256 < uvK_a^2/64 - h_a^2 < 9193/50000$.

Sketch. Each item is read off the costate cycle at the switches of the window: (i) from minimality, the choice $c \le b$ and Proposition 5.6; (ii) from the margin transports (Lemma 4.1); (iii) from Theorem 2.14 at the switching times (every margin is at least $2/15$, and $y \le 9/2$ on $\mathfrak{h}^{\star\star}$ gives the pair sums and $d, r \le 8$) and Lemma 8.3; (iv) from Lemma 4.14 and, for the floors, from $H_i \le 0$ for the two controls missing at a forward switch, written through stationarity [7, `MIXED_GLOBAL_SMALL_FORWARD_MINIMUM` (11)–(12)], an identity algebraic in x that holds for every x; (v) from exact transports; (vi) from Lemma 8.2. □

8.5. **The successor of the forward piece.** The window does not yet contain a contradiction: the root equation (8.6) has solutions compatible with (i)–(vi). The next actual switch, after the piece r, supplies one more condition. Finiteness of the representation guarantees that it exists, and equal neighbouring labels have been merged, so it is forward or reverse. Let

$$t_{\mathrm{next}} = \frac{Q^2 v - 16 h_d Z^2 + w Z^4}{K_r^2}$$

be the norm of that switch when it is forward, read in the frame of the piece r.

Proposition 8.8 (The continuation alternative)**.** *In the situation of Proposition 8.7:*

(a) if the switch $m{+}3$ is forward, to a piece with active margin s, then $s \ge a$, $rs < 4$, $K_r < Z(r+s)$, $t_{\mathrm{next}} \ge g_r(s) \ge \frac{3}{4}$, and both missing-control floors hold at the piece r;

(b) if the switch $m+3$ is reverse, to a piece with active margin A, then $a \le A < d$, $Ad < r^2$, $w = \mathrm{Sec}(A, r, d)$ and $w \le g_r(A)$.

In (a) the inequality for t_{next} is the forward switching inequality at the piece r, whatever the margin s and the length of the forward run. In (b)

the piece r is a turn, both its switches have the norm $\mathrm{Sec}(A, r, d)$, $A \geq a$ by minimality, and $Ad < r^2$ is three-arc log-concavity. Since g_r decreases on $(0, r)$ and $A < r$, extending the average from $[A, d]$ to $[a, d]$ also gives $w \leq \mathrm{Sec}(a, r, d)$; we use (b) as stated.

8.6. **The boundary comparison and the residual.** The root equation (8.6) is affine in the incident norm u, with slope

$$\partial_u R_d = \frac{K_a}{aQ^2}\big(dSQ - Z(aK_a + SY)\big) < 0, \tag{8.7}$$

negative because $Y > 0$, $Z > Q/W$, and $dSQ - Z(aK_a + SY) < (Q/W)\{a(ST - K_a) + Sc(a-d)\} < 0$. The reverse switch $b \to a$ caps u by the ceiling $U(a/S) = g_b(a)$, so the actual data satisfy $R_d(U(a/S)) \leq R_d(u) = 0$. Stationarity at d thus asks for an incident norm that is not too large. The obstruction we shall exhibit is that, on the necessary domain, it asks for a norm above the ceiling.

Before testing this, we remove the margins b and p. Keep a, c, d, r and the actual v fixed and raise b to $b_* = K_a/T - a$, the value at which the central factor $K_a/(ST)$ equals one. Write $G(b) = R_d(U(a/(a+b)); b)$ for the residual at the ceiling, with h_a given by central stationarity at the margin b.

Lemma 8.9 (Boundary comparison)**.** *For the window of Proposition 8.7, $G(b_*) < 0$; the root u_* of $R_d(\,\cdot\,; b_*) = 0$ satisfies $u_* < U(aT/K_a)$; and the outgoing norm w_* of that boundary root satisfies $w_* < w$.*

Proof. Along $[b, b_*]$ the derivative is $G'(b) = \partial_u R_d \cdot a(b-a)/(a+b)^3 < 0$, and at the actual margin $G(b) = \partial_u R_d\,(U(a/S) - u) \leq 0$, so $G(b_*) < 0$. Along the curve of roots $u(b)$, the outgoing norm has $dw/db = dK_a^3 G(b)/(aQ^3(\partial_u R_d)^2) < 0$, so w decreases from its actual value. □

This is a scalar comparison: the point b_* need not belong to a trajectory, and no switching inequality, floor or Casimir bound is transported to it.

At b_* the logarithm of the centre disappears ($C(b_*, a, c) = 4/T + 4c/K_a$), and homogeneity lets us normalize by $T = a + c$. In the coordinates $t = a/T$, $x = (d-a)/T$, $k = K_a/T^2$, $q = WZ/Q$, write $W = 1 + x$, $p = t + x$ and $B = k + 2tx + x^2$ for the normalized values; the margins are recovered from (t, x, k, q), since $a^2(k - t^2) = 4t^2$. The positive multiple

$$N(t, x, k, q, v, \log q) = t\,B\,q\,W \cdot \frac{G(b_*)}{T}$$

is a polynomial in t, x, k, q, v and $L = \log q$, affine in k, in v and in L; Appendix D.3 gives its coefficients. The normalized incoming secant is $v = \mathrm{Sec}(t, 1-t, t+x) = 1 + (1-t)^2/W - (1-t)\log W/x$.

Lemma 8.10 (Shape monotonicity)**.** *If $0 < t < 1$, $x > 0$, $q \geq 1$, $v \leq U(t)$ and $L < q - 1$, then $N(t, x, k, q, v, L)$ is nondecreasing in k.*

The coefficient of k is at least $q\,t\,x(1-t)\{(q-1)(2x+1) + tx\} > 0$ (Appendix D.3). Since k moves with a at fixed t, the lemma lets every later

argument lower k to the smallest value that the necessary conditions allow, and prove $N > 0$ there.

Lemma 8.11 (Necessary conditions)**.** *Every window of Proposition 8.7 satisfies*

(i) $0 < t < \frac{1}{2}$, $x > 0$, $k > 1$, $1 < q < W/p$, $v = \mathrm{Sec}(t, 1-t, t+x)$, $1-(1-t)p/W^2 \le v \le U(t)$ *and* $F_{p/W}(q) \le v$;

(ii) $N(t, x, k, q, v, \log q) < 0$;

(iii) $t < 2/5$;

(iv) $a < 4/5$, *that is,* $k > 29t^2/4$;

(v) $d^2 + 2cd > 4$, *that is,* $k < 2t + 2x + x^2$;

(vi) $3d < 7c$, *that is,* $x < \frac{7}{3} - \frac{10}{3}t$;

(vii) if the successor is reverse, $d \le b_*$, *that is,* $k \ge 2t + x$.

Item (i) restates the window ($k > 1$ is $ST < K_a$ with $S \ge T$, and $q < W/p$ is $dr < 4$); the others are proved in Appendix D.4.

8.7. **The reverse successor.**

Proposition 8.12 (The scalar reverse exclusion)**.** *Let* $0 < t < 2/5$, $x > 0$, $\max\{1, 2t + x, 29t^2/4\} \le k < 2t + 2x + x^2$, $x < \frac{7}{3} - \frac{10}{3}t$, $1 < q < (1 + x)/(t + x)$, $v = \mathrm{Sec}(t, 1-t, t+x) \le U(t)$ *and* $F_{(t+x)/(1+x)}(q) \le v$. *Then* $N(t, x, k, q, v, \log q) > 0$.

Proof. By Lemma 8.10 it suffices to find a shape $k' \le k$ allowed by the hypotheses with $N(k') > 0$. There are three exhaustive cases (Figure 10(b)).

(1) If $t \le 13/35$ and $x \le 1 - 2t$, lower k to 1, where $N(1) > 0$ (Lemma D.5).

(2) If $2t + x \ge 1$ and $x > 1/4$, lower k to the order face $k = 2t + x$. The floor and the swapped secant order give $q \le (1 + x)/(2t + x)$. On this face N is a positive multiple of a function $\alpha + \beta q + \gamma/q - p \log q$, concave in q, so it suffices to check $q = 1$ and the upper end, where it is positive (Lemma D.6).

(3) Otherwise $13/35 < t < 2/5$ and $x \le 9/35$; lower k to the scale face $k = 29t^2/4$, where $N > 0$ (Lemma D.7). □

On each face the logarithms are first replaced by rational bounds with the correct signs (v by a larger $\bar{v}$, $\log q$ by $L_+(q) \ge \log q$, the floor by a smaller F_ℓ), which lowers N; the result is a cubic in $y = q - 1$ whose positivity for $y \ge 0$ follows from six polynomial signs in (t, x). Appendix D.5 gives the reduction and proves the signs.

Theorem 8.13 (Reverse successor)**.** *No window of Proposition 8.7 has a reverse successor.*

Proof. By Proposition 8.8(b) and Lemma 8.11, the normalized window satisfies every hypothesis of Proposition 8.12, which gives $N > 0$, against Lemma 8.11(ii). □

8.8. **The forward successor.** When the successor is forward the order bound (vii) fails, and the residual alone does not suffice. Two facts replace it. First, the scalar reverse exclusion applies to every window with $k \geq 2t + x$, without any assumption on the successor, so every window has the opposite order

$$k < 2t + x, \qquad \text{equivalently } d > b_* \text{ and } j = 2t + x - 1 > 0. \tag{8.8}$$

Second, the forward successor gives a lower bound for the next norm at the boundary root. Both the actual point and the boundary point satisfy the root equation (8.6), which makes h_d an affine function of w; on that line the next norm is a decreasing affine function of w, and $w_* < w$, so the next norm $t_{\mathrm{next},*}$ at the boundary is at least $t_{\mathrm{next}} \geq \frac{3}{4}$. As a function of the incident norm the next norm is affine, and its value at the root differs from its value $\mathcal{T}_U$ at the ceiling by a multiple of N. For multipliers $\alpha > 0$ and $M \geq 0$ the combined residual

$$\begin{aligned} \mathcal{H} &= \alpha N + C_*(\tfrac{3}{4} - t_{\mathrm{next},*}) + M(F_{p/W}(q) - v) \\ &= (\alpha + 1 - Je^2)N + C_*(\tfrac{3}{4} - \mathcal{T}_U) + M(F_{p/W}(q) - v), \end{aligned} \tag{8.9}$$

with $e = q/W$, $J = t + x(x+2)$ and an explicit $C_* > 0$, is therefore negative at every window with a forward successor. It is affine in v, in $\log q$ and in the floor value, and a concave quadratic in k.

The unit face of the low-scale argument gives $t > 1/4$ (Lemma D.8); with the derivative floor $v \geq g_c(d)$ this gives $x < 1$ and $4t + x < 2$, and a quantitative form of the swapped secant order gives $q < W/(1 + 2j)$. The remaining parameters are cut down in three steps (Figure 10(c)).

Proposition 8.14 (The forward successor in three steps)**.** *Every window of Proposition 8.7 with a forward successor satisfies*

(a) $t > 1/3$*;*

(b) $j = 2t + x - 1 < 1/30$*;*

and then the Casimir of the period is at least $9193/50000$*, against Lemma 8.2(b).*

Outline. (a) If $t \leq 1/3$, take $\alpha = 6$ and $M = 0$. The coefficients of v and of $\log q$ in $\mathcal{H}$ are negative, so raising v to a rational majorant $\hat{v}$ and $\log q$ to $L_+(q)$ keeps $\mathcal{H}$ negative. The result increases in k, so it is negative at $k = 1$ as well; but at $k = 1$ it is positive (Lemma D.9).

(b) Suppose $j \geq 1/30$. Then $1/3 < t < 2/5$ by (a), and the derivative floor in integral form gives $j < \frac{5}{7}(1 - 2t)$. For $1/3 \leq t \leq 3/8$ take $\alpha = \frac{7}{2} - 10(q-1)$ and $M = 8j + 8(q-1) - \frac{1}{6}$, both positive here; after the substitutions of v, $\log q$ and the floor by rational bounds, the residual increases in k and is positive at $k = 1$. For $3/8 \leq t < 2/5$ take $\alpha = 4$ and $M = 1$; the residual increases in k and is positive at the scale face $k = 29t^2/4$ (Lemma D.10).

(c) Now $1/3 < t < 2/5$, $0 < j < 1/30$ and $1 \leq k < 31/30$. With $N < 0$, a rational comparison in q gives $\chi = pq/W > 4/7$ (Lemma D.11). On the

stationarity line of the piece d, the diagonal h_d is an affine function $H_d(w)$ of the outgoing norm, and the Casimir is $\Gamma(w) = (4B^2vw - H_d(w)^2)/(16(k - t^2)^2)$ in normalized form. For $w \geq 3/4$ one has $\Gamma'(w) < 0$ (Lemma D.12). The actual w lies between $3/4$ and the central upper envelope w_U, the transported norm with u at its ceiling and $b = b_*$, so $\gamma^2 = \Gamma(w) > \Gamma(w_U)$. At w_U the event factor q is eliminated with the floor (Lemma D.13), and the remaining function of (t, x, k, v) exceeds $9193/50000$ (Lemma D.14). □

Theorem 8.15 (Forward successor)**.** *No window of Proposition 8.7 has a forward successor.*

This is Proposition 8.14. The obstruction in its last step is the cost itself: the turn data force the Casimir of the period above $9193/50000$, and by Lemma 8.2(a) a period with such a Casimir costs more than A_{oct}.

Remark 8.16 (Small scale). The same methods exclude every window with $a \leq \frac{1}{2}$, whatever the sign of the successor [7, `MIXED_NONSHORT_LOW_SCALE_HAND_EXCLUSION`]: then $k \geq 17t^2$, and N can be lowered to $k = 1$ or to $k = 2t + \frac{1}{2}$. The proof above needs only the unit face of that argument (Lemma D.8), which gives $t > 1/4$.

8.9. **Proof of Theorem 8.1.** Let P be as in Theorem 8.1, and take m from Proposition 8.5. If $a_{m+1} > a_{m-1}$, replace P by its reflection (Lemma 8.6): it is again competitive, of class M2, with state in $\mathfrak{h}^{\star\star}$ and the same number of pieces; the minimum moves to $-1 - m$, the four switches around it keep their signs in reverse order, and the smaller turn now follows the centre. If the switch $m + 2$ after the turn is reverse, the seven pieces $m - 2, \ldots, m + 4$ form a smaller-short window, excluded by Theorem 9.3. If it is forward, Proposition 8.7 gives a nonshort window, and Proposition 8.8 gives a forward or a reverse successor; Theorem 8.15 and Theorem 8.13 exclude them. □

9. Class M2: the smaller-short window

This section completes the exclusion of class M2. By Section 8, a competitive normal period of class M2 with at least six pieces has a global minimum of its active margins at the centre of a reverse run (Proposition 8.5). Of the two turns adjacent to that centre we choose the one with the smaller active margin, and after reflecting and reversing time if necessary (Lemma 8.6) it follows the centre. If the forward run that starts at this turn has at least two switches, Section 8 excludes the period. The remaining case is a forward run with a single switch: a short bridge between two reverse runs. Its data are fixed by six real numbers, and no free continuation of the period is left to compare with. The proof therefore works with a fixed list of necessary conditions on those numbers and shows that the list is inconsistent.

The argument has seven steps, which we call gates, following the notes [7] (`MIXED_SMALLER_SHORT_HAND_PERIOD_EXCLUSION`). Gates S0 to S3 confine the shape of the bridge to a small box, gate S4 gives a lower bound for the Casimir, gate S5 shows that the bridge is sheared in a definite direction, and

gate S6 shows that a bridge sheared in that direction costs more, together with the rest of the period, than the octagon.

9.1. **The window.** Index the pieces of the period so that piece m is the global minimum. The seven pieces $m-2,\dots,m+4$ have active margins

$$p_b,\ b,\ a,\ C,\ B,\ A,\ P_f,$$

and the switches between them and after the last one have signs $+,-,-,+,-,-,+$. Thus a is the centre of a reverse run with the two turns b and C; the turn C, the chosen smaller one, is joined by the single forward switch $C\to B$ to the turn B; and A is the centre of the next reverse run, with far neighbour P_f. At a turn the two incident switch norms are equal (Lemma 4.14). Hence the three switches $a\to C$, $C\to B$, $B\to A$ of the bridge share one norm, and we write u for eight times it. Similarly v is eight times the norm of the switch $b\to a$ and w eight times that of $A\to P_f$; the factor eight is the normalization of the notes. The turn secant of Lemma 4.14 reads, in this normalization,

$$\begin{aligned}\operatorname{Sec}(A_0,B_0,P_0) &= 1+\frac{B_0^2}{(A_0+B_0)(P_0+B_0)}-\frac{B_0}{P_0-A_0}\log\frac{P_0+B_0}{A_0+B_0}\\ &= \frac{1}{P_0-A_0}\int_{A_0}^{P_0} g_{B_0}(s)\,\mathrm{d}s,\end{aligned}\tag{9.1}$$

with $g_{B_0}(s)=1-B_0s/(B_0+s)^2$, and with the continuous value $g_{B_0}(A_0)$ at $A_0=P_0$. The identity

$$g_{B_0}(s)=\frac{3}{4}+\frac{(s-B_0)^2}{4(s+B_0)^2}\tag{9.2}$$

shows that every secant lies in $[3/4,1]$. We also write $\mathcal{U}(s)=1-s+s^2$, so that $g_{B_0}(A_0)=\mathcal{U}(A_0/(A_0+B_0))$; this is four times the quantity j_k of Lemma 4.14, and the switching inequalities of that lemma become $u\ge g$ at a forward switch and $u\le g$ at a reverse one.

Definition 9.1 (Smaller-short window)**.** A *smaller-short window* consists of real numbers p_b,b,a,C,B,A,P_f (margins), u,v,w (norms), E,E_L (centre pairings) and Γ (the Casimir) such that, with $\rho=a^2/(a^2+4)$:

(i) (order) $0<a<b$, $a<C\le b$, $a\le A<C$, $a\le p_b$, $A<B$ and $A<P_f$;
(ii) (strict star at the six switches) p_bb, ab, aC, CB, BA and AP_f are less than 4;
(iii) (reverse centres) $(a+b)(a+C)<a^2+4$ and $(B+A)(A+P_f)<A^2+4$;
(iv) (scales) $2/15\le a$, $a^2<12/17$ and $A^2<12/17$;
(v) (turn secants) $u=\operatorname{Sec}(a,C,B)=\operatorname{Sec}(C,B,A)$ and $v=\operatorname{Sec}(a,b,p_b)$;
(vi) (switching inequalities) $u\le g_C(a)$, $g_B(C)\le u$, $u\le g_B(A)$, $v\le g_b(a)$, $g_b(p_b)\le v$, $w\le g_{P_f}(A)$ and $3/4\le w$;

(vii) (centre stationarity) $E = \mathcal{E}(a, b, C, v, u)$ and $E_L = \mathcal{E}(A, B, P_f, u, w)$, where

$$\mathcal{E}(a_0, p, n, u_i, u_o) = 1 + r' + t' - \rho_0 + \log \frac{\rho_0}{r't'} - \frac{1 - u_i}{r'} - \frac{1 - u_o}{t'},$$

with $r' = a_0/(a_0 + p)$, $t' = a_0/(a_0 + n)$ and $\rho_0 = a_0^2/(a_0^2 + 4)$;

(viii) (Gram determinants and Casimir bounds) $\Gamma = (4uv - E^2)/(16(1-\rho)^2)$, $\Gamma = (4uw - E_L^2)/(16(1 - \rho_A)^2)$ with $\rho_A = A^2/(A^2 + 4)$, and $27/256 < \Gamma < 9193/50000$;

(ix) (same-lift transport) $w = \lambda^2 v + d^2 u - d\lambda E$ and $E_L = 2du - \lambda E$, where $\lambda = (a^2 + 4)/(A^2 + 4)$ and $d = \lambda\rho D$ with the bridge diagonal D of Table 3;

(x) (window defect) the inequality (9.3) below, with $0 < \mathcal{B} < \mathcal{A}$.

The conditions are the facts of Sections 4 and 8 written at the seven pieces. The orders come from the minimum structure (Theorem 5.7) and the choice of the smaller turn; the products in (ii) and (iii) from the strict star condition and the reverse factors (Lemma 4.1); the scales from the compact domain (Theorem 2.14) and the reverse-centre scale bound; and (v) to (ix) from Lemmas 4.13 to 4.15 applied to the one normal multiplier of the period. The Casimir bounds are those of Section 8.

TABLE 3. The normalized variables of a smaller-short window. Lengths are divided by the central margin a.

Symbol	Definition
$x,\ \beta,\ \alpha$	C/a, B/a, A/a (ratio, bridge, outer centre)
$\delta,\ \eta$	$\beta/x - 1 = B/C - 1$ (displacement); $(\alpha - 1)/\delta$
$\rho,\ r,\ t$	$a^2/(a^2 + 4)$; $1/(1 + x) = a/(a + C)$; $a/(a + b)$
$q,\ r_A$	$\rho_A = A^2/(A^2 + 4)$ (outer scale); $\alpha/(\alpha + \beta) = A/(A + B)$
q_c	$\alpha^2/(x(x + 2) + \alpha^2)$, the value of q at $\rho = r^2$
$\tau,\ \tau_{\max}$	$(A + P_f)/A$ (outer duration); r_A/q
$\lambda,\ D$	$(a^2 + 4)/(A^2 + 4)$; $\alpha^2 + 2\alpha\beta + 1 + 2x - 2\beta x$ (bridge diagonal)
$u,\ v,\ w$	bridge, central and outer switch norms (times 8)
$E,\ E_L$	pairings at the centres a and A
$\Gamma,\ \kappa$	the Casimir γ^2; $\sqrt{12\Gamma}$

Proposition 9.2 (Extraction of the window)**.** *Let P be a competitive normal period of class* M2 *in the compact domain with at least six pieces, let m be a piece of least active margin, and suppose that the switches after pieces $m-2, m-1, m, m+1, m+2$ have the signs $+, -, -, +, -$ and that $a_{m+1} \le a_{m-1}$. Then the active margins $a_{m-2}, \dots, a_{m+4}$ are the margins of a smaller-short window.*

The essential new condition is the window defect. Let $\mathcal{C}_{\mathrm{br}}$ be the cost of the two bridge pieces, let θ_{br} be the angle swept over the bridge by the

continuous polar angle of the costate frame, and let $\kappa = \sqrt{\det \Lambda_1} = \sqrt{12\Gamma}$. Over a competitive period the polar angle increases by exactly $\pi/3$. The cost–Casimir inequality (Proposition 4.17) has a pointwise strengthening: the cost density is at least $\kappa + 3/2$ times the rate of the polar angle (`GLOBAL_SPECTRAL_COST_CASIMIR` in [7]). Integrating it over the complement of the bridge gives

$$\mathcal{C}_{\mathrm{br}} + \Big(\kappa + \frac{3}{2}\Big)\Big(\frac{\pi}{3} - \theta_{\mathrm{br}}\Big) \le \mathcal{C}(T, X) \le A_{\mathrm{oct}}. \tag{9.3}$$

Both sides are explicit. Because the two turn secants agree, the logarithms in the two arc costs cancel and

$$\mathcal{C}_{\mathrm{br}} = \sqrt{3}\Big[\frac{\rho u}{1-\rho}(2\beta x - x - \alpha\beta) + \frac{\beta x - \alpha}{(1+x)(\alpha+\beta)}\Big], \tag{9.4}$$

by the rational identity $\frac{x(\beta-1)}{(1+x)(\beta+x)} + \frac{\beta(x-\alpha)}{(\alpha+\beta)(\beta+x)} = \frac{\beta x-\alpha}{(1+x)(\alpha+\beta)}$. The angle is $\theta_{\mathrm{br}} = \arctan(\sqrt{3}\mathcal{B}/\mathcal{A}) \in (0, \pi/3)$, where $\mathcal{A}$ and $\sqrt{3}\mathcal{B}$ are $T_\kappa = 2\kappa(1-\rho)^2$ times the real and imaginary parts of the product of the polar coordinates of the costate frame at the two ends of the bridge. In the coordinates of Table 3 they are rational functions of $x, \alpha, \beta, \rho, u, v, E$ and κ, computed from the relative frame of the bridge and the costate at the central switch. The window defect says that the bridge is cheap compared with the angle it turns, and the gates show that no window satisfies it.

Theorem 9.3 (The smaller-short window does not exist)**.** *There is no smaller-short window. Consequently no period satisfies the hypotheses of Proposition 9.2.*

The second sentence follows from the first by Proposition 9.2. It is the case of a short forward run after the chosen smaller turn, and Section 8 uses it to finish class M2 (Theorem 8.1). The rest of this section proves the first sentence.

9.2. **Normalized data and the shape of the bridge.** The secant is invariant under scaling of its three arguments, so in the variables of Table 3

$$u = \mathrm{Sec}(1, x, \beta) = \mathrm{Sec}(\alpha, \beta, x). \tag{9.5}$$

The first secant averages g_x over $[1, \beta]$ and the second averages g_β over $[\alpha, x]$. Only the conditions of Definition 9.1 are used below. We collect the first consequences.

Lemma 9.4 (First consequences)**.** *Every smaller-short window satisfies:*

(a) $C \le B$, *so that* $1 \le \alpha < x \le \beta$;

(b) $u \le V(x) := \mathrm{Sec}(1, x, x)$, *and* $u \le 13/16$;

(c) $aB/C + \frac{2}{7}(B - C) \le A$, *that is* $\alpha \ge \beta/x + \frac{2}{7}(\beta - x)$, *and in particular* $\beta \le \alpha x$;

(d) *the outer centre is stationary: with* $\tau = (A+P_f)/A$ *one has* $2 < \tau < \tau_{\max}$ *and*

$$E_L = (w-1)\tau + \frac{1}{\tau} + \frac{u-1}{r_A} + r_A + 1 - q + \log\frac{\tau q}{r_A}; \tag{9.6}$$

(e) $q \le q_c$ *and* $q < 1/7$.

Proof sketch. (a) Suppose $B < C$. The cross inequality for secants gives $\mathrm{Sec}(a,B,C) < \mathrm{Sec}(a,C,B) = u$. On the other hand, g_B decreases on $[a,A]$ and $g_B \ge g_B(A) \ge u$ there, so adding $[a,A]$ to the interval of $\mathrm{Sec}(A,B,C) = u$ gives $\mathrm{Sec}(a,B,C) \ge u$. (b) Split the integral of $\mathrm{Sec}(1,x,\beta)$ at x. On $[x,\beta]$ the integrand is at most $g_x(\beta) \le u$, the switching inequality at $C \to B$, so $u \le V(x)$; and $V(x) \le 13/16$ by a logarithm bound. (c) is the strengthened order of the notes, proved by the comparison argument of Appendix E.2. (d) is the stationarity (vii) at A, rewritten: $\tau > 2$ is $A < P_f$, and $\tau < \tau_{\max}$ is the reverse factor at A. (e) $q \le q_c$ is equivalent to $\rho \le r^2$, which follows from $\rho < rt \le r^2$. The bound $q < 1/7$ is the reverse-centre scale bound of the notes, applied at the outer centre: if $q \ge 1/7$, the Gram determinant at A would be at least $9193/50000$ (Appendix E.3). □

Equation (9.6) is the one place where the outer reverse run enters. We write it as $R_\tau(t,v) = 0$, where the *outer residual* $R_\tau(t,v)$ is the right side of (9.6) minus E_L, with w and E_L given by the transport (ix) as functions of the opposite turn (t,v) and with E given by the stationarity at a:

$$E(t,v) = (u-1)(1+x) + \frac{v-1}{t} + 1 + \frac{1}{1+x} + t - \rho + \log\frac{\rho(1+x)}{t}. \tag{9.7}$$

The shape (x,α,β) is constrained by (9.5) alone. The gates use it through explicit inequalities between α and β at fixed x, which we call orders.

Lemma 9.5 (Orders of the bridge)**.** *Let* $1 < x$, $1 \le \alpha < x \le \beta$ *and* $\mathrm{Sec}(1,x,\beta) = \mathrm{Sec}(\alpha,\beta,x)$, *and put* $\delta = \beta/x - 1$.

(a) *If* $x \le 9$, *then* $\alpha \ge \beta/x + k_s(x)(\beta - x)$ *with* $k_s(x) = 4(x+1)/(7x+1)$.
(b) $\alpha \le \beta/x + \frac{5}{4}(\beta - x)$, *and* $\alpha \le \beta/x + \frac{9}{8}(\beta - x)$ *if* $x \le 5$.
(c) *If* $x \le 13/6$ *and* $\delta \le 1/19$, *then* $\alpha \le 1 + 3\delta$.
(d) *If* $2 \le x \le 13/6$, *then* $\alpha \ge 1 + \frac{4x^2+11x+1}{7x+1}\,\delta + \frac{\delta^2}{1+4\delta}$.

In particular, at a window with $x \ge 2$ *one has* $\alpha \ge 1 + (1 + \frac{4x(x+1)}{7x+1})\delta \ge 1 + \frac{13}{5}\delta$.

The last inequality holds because $4x(x+1)/(7x+1)$ increases, with derivative $4(7x^2+2x+1)/(7x+1)^2$, and equals $8/5$ at $x = 2$. In the slope $\eta = (\alpha - 1)/\delta$, parts (a) and (c) say that $\eta \in [13/5, 3]$ on the box $2 \le x \le 13/6$, $0 < \delta \le 1/19$; Figure 11(b) shows the actual slopes.

9.3. **How a gate compares the window with a face.** Gates S0 to S4 share one structure. Each assumes that the window lies outside a region and derives a scalar inequality at the actual window: $G \ge 0$ for a weighted outer residual G, or $\Gamma \ge G$ for a corrected Casimir G. It then moves the opposite

turn, the scale and the outer duration, each time in a direction that keeps the inequality valid, until the point reaches one of two faces where all data are explicit functions of the shape (x, α, β) and of u. There the inequality fails. We describe the moves once.

The opposite turn. The norm v and the ratio t of the turn before the centre are not controlled individually. They satisfy $v \le \mathcal{U}(t)$, the switching inequality at $b \to a$, and $t_0 := \rho(1+x) \le t \le r$, from the central reverse factor and the choice of the smaller turn. For a weight $m \ge 0$ put $G = R_\tau + m(w - 3/4)$. Then

$$\frac{\partial G}{\partial v} = \frac{\lambda}{t} B_{\tau+m}(t), \qquad \frac{\mathrm{d}}{\mathrm{d}t} G(t, \mathcal{U}(t)) = \frac{\lambda(2t-1)}{t} B_{\tau+m}(t),$$

with the linear factor $B_z(s) = 1 + z(\lambda s - \lambda\rho D)$. When $B_{\tau+m} > 0$ on $[t_0, t]$, G increases in v and $G(s, \mathcal{U}(s))$ decreases in s (since $s < 1/2$), so

$$G(t, v) \le G(t_0, \mathcal{U}(t_0)) \tag{9.8}$$

. Without weight, the positivity of B_τ for $\tau \le \tau_{\max}$ reduces to $\alpha(\alpha+\beta) + (1+x) - D = x(\beta - 1) + \beta(x - \alpha) > 0$; with a weight each gate checks it separately. At the point $(t_0, \mathcal{U}(t_0))$ the opposite turn has zero duration and the logarithm in (9.7) vanishes.

The scale. At $(t_0, \mathcal{U}(t_0))$ every quantity is rational in ρ. We reparametrize by the outer scale $q = \rho_A$, through $\rho = q/(\alpha^2 - (\alpha^2 - 1)q)$. Then

$$w_* = 1 - qA_1 + q^2 A_2, \qquad E_{L,*} = -E_0 + qA_3, \qquad E_0 = (1+x)(u-1) + \frac{1}{1+x}, \tag{9.9}$$

with A_1, A_2, A_3 affine in u and rational in x, α, β. At fixed $p = \tau q$ the residual becomes $R^*_{p/q} = \mathcal{C}(p) + q\mathcal{L}(p)$, where $\mathcal{C}(p) = E_0 + (u-1)/r_A + r_A + 1 - pA_1 + \log(p/r_A)$ and $\mathcal{L}(p) = pA_2 + 1/p - 1 - A_3$. So at fixed p every weighted residual is a quadratic polynomial in q. The actual scale satisfies $q \le q_c$ and $q < 1/7$ (Lemma 9.4), and $q \le p/2$ because $\tau \ge 2$.

The faces. The scale is raised at fixed p up to $q_{\max} = \min\{q_c, 1/7, p/2\}$. If $q_{\max} = p/2$, the duration has reached $\tau = 2$, and the scale is raised further along $\tau = 2$. The end point lies on the *central face* $q = q_c \le 1/7$ or on the *outer face* $q = 1/7 \le q_c$. On the central face $\rho = r^2$, that is $(a + C)^2 = a^2 + 4$: the central reverse factor is one when both turns equal C, so the central piece has zero duration. On the outer face the outer centre sits at the reverse-centre scale bound $A^2 = 2/3$. On each face all data are rational in (x, α, β, u), except for the duration $\tau \in [2, \tau_{\max}]$ and the term $\log(\tau q/r_A)$.

The duration. The duration enters only through $(w-1)\tau + 1/\tau + \log\tau$. The identity

$$R_\tau - R_2 - (\tau - 2)\Big(w - \frac{3}{4}\Big) = \log\frac{\tau}{2} - \frac{\tau}{4} + \frac{1}{\tau} \le 0 \qquad (\tau \ge 2) \tag{9.10}$$

bounds every R_τ with $2 \le \tau \le \tau_{\max}$ by $\max\{R_2,\ R_2 + (\tau_{\max} - 2)(w - 3/4)\}$. A variant with two tangent lines to the logarithm reduces the duration to three test values.

The incoming norm. The bridge norm u is the secant (9.5), a transcendental function of (x, β). On each face the tested quantity is monotone in u, and u is replaced by a rational ceiling or floor, for instance $V(x) \le \bar{V}(x) = \frac{3}{4} + \frac{2(x-1)^2}{(x+1)(35x+13)}$ (Appendix E.1). What remains is the sign of an explicit function of (x, α, β), in some cases with one logarithm, on a region bounded by the orders. These are the *face signs*.

9.4. **Gates S0 to S3: the ratio and the displacement.** The ratio $x = C/a$ is the shape parameter of the central reverse run, and the displacement δ measures how far the bridge is from symmetric. Gates S0 to S3 place (x, δ) in the box $[2, 13/6] \times [0, 1/19]$ (Figure 11(a)).

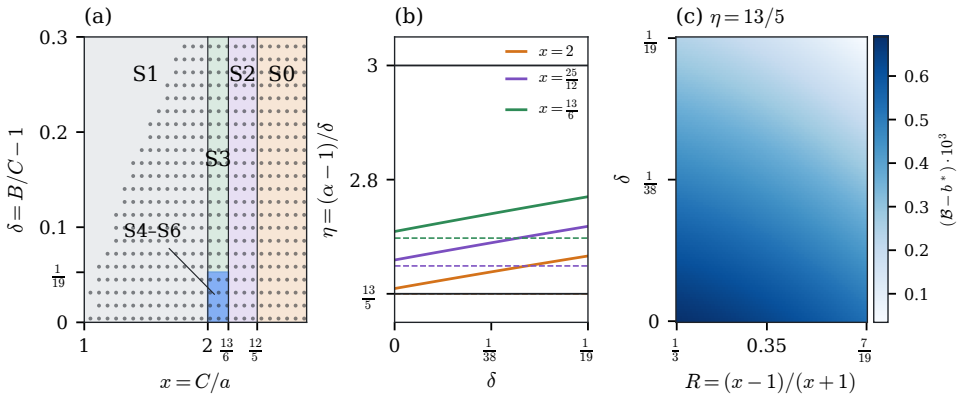


FIGURE 11. What the gates control. (a) The plane of the ratio $x = C/a$ and the displacement $\delta = B/C - 1$. The dots are computed points of the equal-secant locus (9.5) with $1 \le \alpha < x$. Gate S1 removes $x \le 2$, S0 removes $x \ge 12/5$, S2 removes $13/6 \le x < 12/5$ and S3 removes $\delta \ge 1/19$, leaving the box where S4 to S6 work. (b) On that box, the slope $\eta = (\alpha - 1)/\delta$ of the locus for $x = 2, 25/12, 13/6$ (solid), between the lower order $1 + 4x(x+1)/(7x+1)$ (dashed) and the upper order 3 of Lemma 9.5. (c) The polynomial bound $\mathfrak{B} - b^*$ of gate S6 at $\eta = 13/5$: it is positive on the whole box, with minimum $3.4 \cdot 10^{-5}$, small against $b^* = 0.00223$.

Proposition 9.6 (Gate S0)**.** *Every smaller-short window has $x < 12/5$, $D > 0$ and $E_L > 0$.*

Proof sketch. Suppose $x \ge 12/5$. The root gives $R_\tau = 0$ and $w \ge 3/4$, so the floor-weighted residual $G = R_\tau + \frac{1}{2}(w - \frac{3}{4})$ is nonnegative at the actual window. Large ratios force a lopsided outer bridge: the equal secants

and $x \geq 12/5$ give $\alpha/\beta < 3/4$ and then $D < \alpha(\alpha+\beta)$. The weighted envelope (9.8) applies (its factor stays positive for $x > 5/3$), so $G_\tau^* \geq 0$ at the actual scale, and the two shape facts make G_τ^* increase with u. At fixed p its value at the zero scale is negative, and an endpoint argument for the quadratic in q gives a nonnegative value at $q_{\max}$; the move along $\tau = 2$ keeps it nonnegative. On the face reached, u is raised to the ceiling $\hat{u}(x,\beta)$ (central face) or $\bar{u}(x,\beta)$ (outer face), and the duration is reduced to the three tests $F(2,\cdot)$, $F(8\tau_{\max}/9,\cdot)$, $F(\tau_{\max},\cdot)$ of Section 9.3. The three face signs of Proposition E.3 say that each test is negative on the region cut out by the orders. This contradiction gives $x < 12/5$. For $x \leq 12/5$ the identities

$$D - \big[(1+2x)\alpha^2 - 2x^2\alpha + 1 + 2x\big] = 2(\beta - \alpha x)(\alpha - x),$$
$$(1+2x)\alpha^2 - 2x^2\alpha + 1 + 2x - D_0 = (\alpha - 1)\big[(1+2x)(\alpha+1) - 2x^2\big],$$

with $D_0 = 2 + 4x - 2x^2 \geq 2/25$, give $D \geq \alpha D_0 > 0$. Finally $E_L = \lambda(2\rho D u - E) > 0$, because E is at most its envelope value $(1+x)(u-1) + \frac{1}{1+x} + \rho(2x+1)$, which is less than $2\rho D u$. □

Proposition 9.7 (Gate S1)**.** *Every smaller-short window has $x > 2$.*

Proof sketch. Suppose $1 < x \leq 2$. For small ratios the residual alone is not enough, and the gate uses the upper bound $\Gamma < 9193/50000$. Moving the opposite turn changes the Casimir $\Gamma(t,v)$ computed at the central reverse run, but the combination $\Gamma - \frac{7}{96}R_\tau$ does not increase along the two moves of (9.8): in v it is concave with a nonpositive slope at the actual point, and along $v = \mathcal{U}(s)$ its derivative is the derivative of R_τ, which is negative, times a factor that stays below zero. Since $R_\tau = 0$ at the actual window,

$$\Gamma \geq \Gamma_* - \tfrac{7}{96}R_\tau^*$$

at the actual scale, where Γ_* is the Gram expression at $(t_0, \mathcal{U}(t_0))$. At fixed p the right side decreases in q, and along $\tau = 2$ the slope of R_2^* in q exceeds one, so the right side does not increase when the point moves to a face. On the central face $q = q_c \leq 1/7$ and on the outer face $q = 1/7 \leq q_c$, the face signs $\Gamma_* - \frac{7}{96}R_\tau^* > 9193/50000$ for every admissible duration (Proposition E.4) contradict $\Gamma < 9193/50000$. □

Proposition 9.8 (Gate S2)**.** *A smaller-short window with $x < 12/5$ has $x < 13/6$.*

Proof sketch. Suppose $13/6 \leq x < 12/5$. The weight is now variable, $m = 2 - 7\tau/(4\tau_{\max}) \in [1/4, 2)$, and the weighted residual $G_\tau = R_\tau + m(w - 3/4)$ is still nonnegative at the root. The envelope applies with this weight, the scale polynomial is convex in q with a negative value at the zero scale, and the $\tau = 2$ continuation reaches the face $q^* = \min\{q_c, 1/7\}$. After raising u to the ceiling $u_C(x,\beta)$ on the central face, and to $u_s(x,\beta)$ or $\bar{V}(x)$ on the outer face, the face signs $G_\tau < 0$ of Proposition E.5 give the contradiction. □

Proposition 9.9 (Gate S3)**.** *A smaller-short window with $2 \leq x \leq 13/6$ has $\delta < 1/19$.*

Proof sketch. Suppose $\delta \geq 1/19$. The displacement is penalized by the weight $m = 4(4-3\tau/\tau_{\max})/(5\alpha-1)$, with which $G_\tau = R_\tau + m(w-3/4) \geq 0$ at the root. On this range $\beta < 2\alpha$, so $D = \alpha^2 + 1 + 2x + 2\beta(\alpha - x) > 5\alpha^2 - 4\alpha x + 1 + 2x \geq 6 - 2x > 0$. The same three moves reach the face $q^* = \min\{q_c, 1/7\}$. The duration is reduced to two tests, F_0 at $\tau = 2$ and F_1 at $\tau = \tau_{\max}$, the norm is raised to a ceiling, and on the central face a monotonicity in β moves the point to the curve $\beta = x + \frac{25x}{15x+36}(\alpha - 1)$. The face signs $F_0, F_1 < 0$ of Proposition E.6 give the contradiction. □

9.5. **Gate S4: the small box and a floor for the Casimir.** After gates S0 to S3 the shape lies in $2 < x < 13/6$, $0 \leq \delta < 1/19$. Gate S6 needs a two-sided control of α in terms of δ, and also a lower bound for κ, because the completion term in (9.3) is weighted by $\kappa + 3/2$.

Proposition 9.10 (Gate S4)**.** *A smaller-short window with* $2 \leq x \leq 13/6$ *and* $\delta \leq 1/19$ *has* $\alpha \leq 1 + 3\delta$ *and* $\Gamma > 1829/10000$.

Proof sketch. The first inequality is Lemma 9.5(c), proved by the comparison argument with a barrier of slope $2/x$. For the second, the monotonicity used in gate S1, now with the constant $1/7$, gives $\Gamma \geq \Gamma_* - R^*_\tau/7$ at the actual scale. The scale is moved at fixed p, and along $\tau = 2$, up to the central scale q_c, and the duration maximum (9.10) leaves the two functions

$$\Phi = \Gamma_*(q_c) - \tfrac{1}{7}R^*_2(q_c), \qquad \Psi = \Phi - \tfrac{1}{7}(\tau_{\max} - 2)(w_*(q_c) - \tfrac{3}{4}).$$

Both decrease in u, so the norm may be raised to its ceiling $\bar{u}(x, \beta)$ (Appendix E.1). The face sign $\min\{\Phi, \Psi\} > 1829/10000$ on the closed box $2 \leq x \leq 13/6$, $0 \leq \delta \leq 1/19$, $\frac{13}{5}\delta \leq \alpha - 1 \leq 3\delta$ (Proposition E.7) concludes. □

With Lemma 9.5 the window now lies in the *small-shape domain*

$$2 \leq x \leq \tfrac{13}{6}, \qquad 0 \leq \delta \leq \tfrac{1}{19}, \qquad 1 + \tfrac{13}{5}\delta \leq \alpha \leq 1 + 3\delta, \tag{9.11}$$

that is $\alpha = 1 + \eta\delta$ with $13/5 \leq \eta \leq 3$ when $\delta > 0$, and $\alpha = 1$ when $\delta = 0$. No division by δ is used, so the symmetric boundary $\delta = 0$ is included.

9.6. **Gate S5: the root forces a negative shear.** The defect inequality compares the cost of the bridge with the angle it turns, and both are governed by one number. Let M be the relative frame of the bridge, from the central switch to the far switch. After a fixed rotation that depends only on x, its lower row does not depend on ρ, and $\det M = 1$ then determines the upper row up to one real parameter h, the *shear*. Put $R_x = (x-1)/(x+1)$, $T_s = (\beta - \alpha)/(\beta + \alpha)$, $\mathcal{U}_1 = \mathcal{U}(r) = (3 + R_x^2)/4$, $\mathcal{U}_2 = \mathcal{U}(r_A) = (3 + T_s^2)/4$ and the anisotropy $\nu = \mathcal{U}_1/\mathcal{U}_2 - 1 \geq 0$. Then

$$\sqrt{3}\, h = \frac{(\beta x - \alpha)\big((x+2)\alpha + (2x+1)\beta\big)}{(1+x)^2(\alpha^2 + \alpha\beta + \beta^2)} - 2\mathcal{U}_1\,(2\beta x - x - \alpha\beta)\,\frac{\rho}{1-\rho} \tag{9.12}$$

. This is a strictly decreasing function of the central scale ρ, and the bridge cost (9.4) is affine in it:

$$\mathcal{C}_{\mathrm{br}} = H_0 - J_0 h, \qquad J_0 = \frac{3u}{2\mathcal{U}_1}, \qquad H_0 = \frac{\sqrt{3}}{2}\Big[\Big(1+\frac{u}{\mathcal{U}_1}\Big)R_x + \Big(1+\frac{u}{\mathcal{U}_2}\Big)T_s\Big]. \tag{9.13}$$

A more negative shear makes the bridge more expensive. Gate S5 shows that the actual outer root is compatible only with a negative shear.

Proposition 9.11 (Gate S5)**.** *A smaller-short window in the small-shape domain* (9.11) *has* $q > \frac{29}{30}q_c$, $\sqrt{3}hR_x + \frac{13}{3}\nu < 0$, $h < 0$ *and* $-hR_x > \frac{5}{2}\nu$.

Proof sketch. Suppose $q \le q_b := \frac{29}{30}q_c$. With weight zero, (9.8) and (9.9) give $\mathcal{R}(u;p,q) := \mathcal{C}(p) + q\mathcal{L}(p) \ge 0$ at $p = \tau q$, and $2q \le p \le r_A$. The residual $\mathcal{R}$ is affine and increasing in u (Appendix E.6), and $u \le \bar{V}(x) \le u_5 := \frac{755}{996} + \frac{x-2}{100}$, so $\mathcal{R}(u_5;p,q) \ge 0$. But $\mathcal{R}(u_5;p,q) < 0$ on the whole closed region $0 \le q \le q_b$, $2q \le p \le r_A$: it is affine in q at fixed p, negative on $q = 0$ by the tangent bound $\log y \le y - 1$, and negative on the faces $q = q_b$ and $p = 2q$ by two tangent lines to the logarithm and convexity in β. Hence $q > q_b$. Since $\rho/(1-\rho) = q/(\alpha^2(1-q))$,

$$(1-q)\Big(\sqrt{3}hR_x + \frac{13}{3}\nu\Big) = F - Gq,$$

where $F = 2W/(3(1+x)^3 k_A)$ and $G = 2V/(3\alpha^2(1+x)^3 k_A)$, with $k_A = \alpha^2 + \alpha\beta + \beta^2$ and polynomials $W, V > 0$ given in Appendix E.6. So $\sqrt{3}hR_x + \frac{13}{3}\nu < 0$ as soon as $q > F/G$, and $q_b > F/G$ is the polynomial inequality $29V - 30W\big(x(x+2) + \alpha^2\big) > 0$ on the small-shape domain. The last two conclusions follow because $R_x > 0$, $\nu \ge 0$ and $13/(3\sqrt{3}) > 5/2$, that is, $676 > 675$. □

9.7. **Gate S6: the completion is too expensive.**

Proposition 9.12 (Gate S6)**.** *For a smaller-short window in the small-shape domain with* $\Gamma > 1829/10000$, $h < 0$ *and* $-hR_x > \frac{5}{2}\nu$,

$$\mathcal{C}_{\mathrm{br}} + \Big(\kappa + \frac{3}{2}\Big)\Big(\frac{\pi}{3} - \theta_{\mathrm{br}}\Big) > A_{\mathrm{oct}}$$

.

The proof lowers the left side in four steps to an explicit function of the shape.

Step 1: lowering κ. Since $12 \cdot \frac{1829}{10000} - (\frac{40}{27})^2 = \frac{23}{1822500} > 0$, the floor of gate S4 gives $\kappa > \kappa_0 = 40/27$. Since $\theta_{\mathrm{br}} < \pi/3$, we may replace $\kappa + 3/2$ by $c = \kappa_0 + 3/2 = 161/54$.

Step 2: the angle in terms of the shear. Let $\psi = \arctan(R_x/\sqrt{3}) + \arctan(T_s/\sqrt{3})$ be the angle of the fixed lower row of the rotated frame,

let $K = \nu + 2 + 2\kappa_0/J_0$, and

$$\delta(h) = 2\arctan\frac{\sqrt{h^2+\nu^2}-h}{K+\sqrt{K^2-\nu^2}}.$$

In the rotated basis, $\mathcal{A} + i\sqrt{3}\mathcal{B}$ is, up to a positive factor and the rotation by ψ, a quadratic polynomial in one unknown entry of the costate at the central switch. The condition that its values stay on one side of the line through the origin at angle $\psi + \delta$ is a discriminant inequality, which holds at $\delta = \delta(h)$. This gives $\theta_{\mathrm{br}} \le \psi + \delta(h)$ for the actual shear. With (9.13), the left side of the proposition is at least $H_0 + c(\pi/3 - \psi) + \tilde{F}(\delta(h))$, where $\tilde{F}(\delta) = -J_0\tilde{h}(\delta) - c\delta$ and $\tilde{h}$ is the inverse of $h \mapsto \delta(h)$.

Step 3: continuation to the central scale. The shear decreases in ρ, and $\rho < r^2$. Let $h_c < h < 0$ be the shear at the central scale $\rho = r^2$ and $\omega_c = \delta(h_c)$. On $[\delta(h), \omega_c]$ the derivative of $\tilde{F}$ has the sign of a quadratic $Q_c(H^2)$ in $H = -\tilde{h}$. The hypothesis $-hR_x > \frac{5}{2}\nu$, together with $H \le -h_c < 7R_x^2/6$, makes $Q_c(H^2) < 0$ along the whole continuation (the barrier lemma of the notes, applied pointwise). So $\tilde{F}$ decreases, and the left side is at least the central score

$$\mathcal{F}_c = \mathcal{C}_c + c\Big(\frac{\pi}{3} - \psi - \omega_c\Big), \qquad \mathcal{C}_c = \sqrt{3}\Big[\frac{u(2\beta x - x - \alpha\beta)}{x(x+2)} + \frac{R_x + T_s}{2}\Big]$$

. Here $\mathcal{C}_c$ is the bridge cost (9.4) at the central scale.

Step 4: a polynomial bound. The Taylor bound $\arctan y \le y - y^3/3 + y^5/5$ for $y \ge 0$, elementary bounds for ω_c and for the central shear, and the two-sided bound $u_q(R_x, \delta) \le u \le \frac{3}{4} + \frac{R_x^2}{12}$ give

$$\mathcal{F}_c > \frac{c\pi}{3} + \sqrt{3}\,\mathfrak{B}(u_q; R_x, T_s, \delta) - \frac{1}{800000}$$

, with $\mathfrak{B}$ the explicit rational function of Appendix E.7. It is concave in T_s, and T_s is monotone in η, so its minimum over $13/5 \le \eta \le 3$ is attained at an endpoint. At both endpoints $\mathfrak{B} > b^* = \frac{792669623}{356040000000}$ for $\frac{1}{3} \le R_x \le \frac{7}{19}$ and $0 \le \delta \le \frac{1}{19}$ (Proposition E.8, Figure 11(c)). The constant b^* is defined by $\frac{989}{571}b^* = \frac{3126055}{10^6} - \frac{c}{3}\cdot\frac{314159265}{10^8} + \frac{1}{800000}$, so $\pi > 314159265/10^8$, $\sqrt{3} > 989/571$ and $A_{\mathrm{oct}} < 3126055/10^6$ give

$$\mathcal{F}_c > \frac{c}{3}\cdot\frac{314159265}{10^8} + \frac{989}{571}b^* - \frac{1}{800000} = \frac{3126055}{10^6} > A_{\mathrm{oct}}.$$

9.8. **Proof of Theorem 9.3.** Let a smaller-short window be given. Gate S0 gives $x < 12/5$, gate S1 gives $x > 2$, and gate S2 then gives $x < 13/6$. Gate S3 gives $\delta < 1/19$, gate S4 gives $\alpha \le 1 + 3\delta$ and $\Gamma > 1829/10000$, and Lemma 9.5 gives $\alpha \ge 1 + \frac{13}{5}\delta$. So the window lies in the small-shape domain (9.11). Gate S5 gives $h < 0$ and $-hR_x > \frac{5}{2}\nu$, and gate S6 then contradicts the window defect (9.3). □

10. Periods with at most five pieces

By Theorem 4.19, a competitive normal vertex period whose state stays in the compact domain has at most five pieces. This section computes the cost of every normal vertex period with at most five pieces and shows that it is at least A_{oct}, with equality only for the octagon's trajectory. It then completes the proofs of Theorem 1.2 and of part (b) of Theorem 1.1.

For so few pieces nothing has to be estimated abstractly. The tangent closure (Lemma 4.4) is a system of four scalar equations in the factors; with at most five factors it either has no solution or determines the factors up to one or two parameters. The twist $X(T) = R^{-1}X(0)R$ then determines the initial state, and the cost becomes an explicit function of the parameters. The difficulty is that the comparison is sharp. The four-piece family contains the octagon, so its minimum must be located exactly, and the five-piece family approaches the octagon at the edge of its domain, where one of its factors tends to 1 and the period degenerates into the octagon's four pieces. Its strict inequality therefore has no margin near that edge.

10.1. **Closed vertex cycles.** The terminal comparison uses only the data of a period at its switches. We write them in the triangular coordinates $M \mapsto CMC^{-1}$, $C = \left(\begin{smallmatrix} \sqrt{3} & 1 \\ \sqrt{3} & -1 \end{smallmatrix}\right)$, in which the tangent factors of Lemma 4.1 and the rotation R become

$$\widetilde{T}_0(q) = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix}, \qquad \widetilde{T}_1(q) = \begin{pmatrix} q^{-1} & q - q^{-1} \\ 0 & q \end{pmatrix}, \qquad \widetilde{T}_2(q) = \begin{pmatrix} q^{-1} & 0 \\ q^{-1} - q & q \end{pmatrix}, \qquad \widetilde{R} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \tag{10.1}$$

Definition 10.1 (Closed vertex cycle)**.** A *closed vertex cycle* with $N \geq 1$ pieces consists of labels $i_k \in \mathbb{Z}/3$, factors $q_k > 1$ and switch margins $m_k = (m_k(0), m_k(1), m_k(2))$ with positive entries, indexed by $k \in \mathbb{Z}$, such that for every k and j:

(1) (twist) $i_{k+N} = i_k + 1$, $q_{k+N} = q_k$, $m_{k+N}(j) = m_k(j-1)$, and $i_{k+1} \neq i_k$;
(2) (quadric) $m_k(0)m_k(1) + m_k(1)m_k(2) + m_k(2)m_k(0) = 4$;
(3) (one-arc map) with $i = i_k$, $a = m_k(i)$, $b = m_k(i+1)$, $c = m_k(i-1)$,

$$m_{k+1}(i) = a, \qquad m_{k+1}(i+1) = (a+b)q_k - a, \qquad m_{k+1}(i-1) = \frac{a+c}{q_k} - a; \tag{10.2}$$

(4) (tangent closure) $\widetilde{T}_{i_k}(q_k)\,\widetilde{T}_{i_{k+1}}(q_{k+1}) \cdots \widetilde{T}_{i_{k+N-1}}(q_{k+N-1}) = \widetilde{R}$.

Its *raw cost* is $\sum_{k=0}^{N-1} \rho(a_k, b_k, c_k, q_k)$, where a_k, b_k, c_k are the active, growing and decreasing margins of item (3) and

$$\rho(a, b, c, q) = a\big((a+b)(q-1) + (a+c)(1-q^{-1}) - a\log q\big). \tag{10.3}$$

The letter ρ denotes the raw arc cost only in this section and in Appendix F.

Lemma 10.2 (The cycle of a normal period)**.** *Let P be a normal vertex period with N pieces. Its labels, its factors and the margins $m_k(j) = p_j$ of its twisted state at the switching times τ_k form a closed vertex cycle with N pieces, and*

$$\mathcal{C}(P) = \frac{\sqrt{3}}{4}\,(\textit{raw cost}), \qquad \textit{that is,} \qquad \frac{\mathcal{C}(P)}{\sqrt{12}} = \frac{\textit{raw cost}}{8}.$$

Proof. Positivity and the quadric are Lemma 2.8; (10.2) and the cost of a piece, $(\sqrt{3}/4)\,\rho(a,b,c,q)$, are Lemma 4.1. The twist $Y(t+T) = R^{-1}Y(t)R$ of the state shifts the margins, $p_j(R^{-1}XR) = p_{j-1}(X)$, which gives $m_{k+N}(j) = m_k(j-1)$. The tangent closure from the switch τ_k is Lemma 4.4 for the period rebased at τ_k, conjugated by C. □

So a period beats the octagon exactly when the raw cost of its cycle is below $8\delta_{\mathrm{oct}}$. Two operations change a cycle without changing its raw cost: *rebasing*, which shifts all indices by a fixed integer, and *relabeling*, which adds 1 to every label and shifts the margins accordingly; rebasing a cycle with N pieces by r replaces its word by the last $N-r$ letters followed by the first r letters increased by one. Relabeling at most twice makes the first label 0.

10.2. **The words of at most five pieces.** A cycle with first label 0 is described by its word $i_0 i_1 \cdots i_{N-1}$: neighbouring letters differ, and by the twist the letter after i_{N-1} is $i_0 + 1 = 1$, so $i_{N-1} \neq 1$. For $N \leq 5$ there are $1+1+3+5+11 = 21$ such words.

Proposition 10.3 (Words of at most five pieces)**.** *Let c be a closed vertex cycle with $N \leq 5$ pieces.*

(1) $N \geq 4$.

(2) If $N = 4$, every switch is forward: $i_{k+1} = i_k + 1$ for all k.

(3) If $N = 5$, either every switch is reverse, or some rebasing and relabeling of c has the word 0 1 0 2 0 *and the same raw cost.*

Proof. Relabel so that $i_0 = 0$ and compare the closure product of each of the 21 words with $\widetilde{R}$, entry by entry. Each entry of a product of the matrices (10.1) is a rational function of the factors, and for most words a single entry already contradicts $q_k > 1$. For the word 0, the upper left entry of $\widetilde{T}_0(q_0)$ is $q_0 \neq 0$. For 02, 020, 0202 and 02020 the upper right entry of the product vanishes, while that of $\widetilde{R}$ is 1; for 010 and 01010 the lower left entry vanishes instead of being -1; for 0210 and 02010 the upper left entry is a positive monomial in the factors instead of 0. For 012 two entries are needed: the lower row of the product is

$$\Big(-\frac{q_1(q_2^2-1)}{q_0 q_2},\ \frac{q_1 q_2}{q_0}\Big),$$

whose second entry forces $q_0 = q_1 q_2$, after which the first is $-(q_2^2-1)/q_2^2 \neq -1$. The words 0102, 01012 and 01212 fall in the same way, through the two

entries of the lower row: for 01212, the lower right entry gives $q_0q_2q_3 = q_1q_4(q_2^2+q_3^2-1)$, and then the lower left entry is $-1+q_1q_3/(q_0q_2q_4) \neq -1$. For 0212, the upper left and lower left entries give $q_2^2q_3^2 = q_2^2+q_3^2$ and $q_0q_2 = q_1q_3$, and then the lower right entry forces $q_2 = 1$. Table 4 lists the outcome. The surviving words are 0120, whose switches, including the one across the seam to the letter 1, are all forward; the five words 01020, 01202, 01210, 02012, 02120, which are the rebasings of 01020, normalized to start with 0; and 02102, whose five switches are all reverse. □

N	words with $i_0 = 0$, $i_{N-1} \neq 1$	outcome
1, 2, 3	0; 02; 010, 012, 020	excluded by the closure
4	0102, 0202, 0210, 0212	excluded by the closure
	0120 $(+++{+})$	the octagon's word
5	01010, 01012, 01212, 02010, 02020	excluded by the closure
	01020 $(+--++)$, 01202, 01210, 02012, 02120	rebasings of 01020
	02102 $(-----)$	entirely reverse (class R)

TABLE 4. The reduced words of at most five letters and their fate. The signs are those of the N switches of a period, the last one across the seam.

Remark 10.4. Most of these words can also be excluded by the fixed-edge obstruction [7]: a trajectory whose control has one coordinate equal to zero almost everywhere cannot satisfy $g(T) = R$. On a fixed edge of the control simplex the frame can be integrated explicitly, and the terminal condition then forces $x(0) = -1/\sqrt{3}$, on the boundary of the star domain. Rebasing a word that misses a letter in a suitable window produces such a trajectory. The entries of the closure give the same exclusions without integrating along an edge, and neither argument uses optimality.

10.3. **Four pieces.** By Proposition 10.3 and a relabeling, a four-piece cycle has the octagon's word 2 0 1 2. Its closure equations can be solved in closed form.

Lemma 10.5 (Four-piece closure)**.** *If $p, q, r, z > 1$ and*

$$\widetilde{T}_2(p)\widetilde{T}_0(q)\widetilde{T}_1(r)\widetilde{T}_2(z) = \widetilde{R},$$

then $r = p$, $z = q$ and $p^2q^2 = p^2+q^2$.

Proof. The upper row of the product is $\big(-q(r^2z^2-r^2-z^2)/(prz),\ qz(r^2-1)/(pr)\big)$, so $r^2z^2 = r^2+z^2$ and $qz(r^2-1) = pr$; together these give $pz = qr$. The lower right entry then gives $q^2r^2 = q^2+r^2$. Subtracting the two quadratic relations gives $(q^2-z^2)(r^2-1) = 0$, so $z = q$, and $pz = qr$ gives $r = p$. □

Write $a = 1/p$ and $b = 1/q$. The relation $p^2q^2 = p^2+q^2$ becomes $a^2+b^2 = 1$: the four factors are $(1/a, 1/b, 1/a, 1/b)$ for a point (a, b) of the open quarter circle, and every such point satisfies the closure. Put

$$s = a + b \in (1, \sqrt{2}\,], \qquad \mathcal{N} = (s-1)(s+3),$$

and define the *four-piece density*

$$\delta(a,b) = \frac{4s(s-1) + 2(1+a)(1-b)\log a + 2(1-a)(1+b)\log b}{(s-1)(s+3)}. \quad (10.4)$$

Lemma 10.6 (Four-piece cycles)**.** *Let c be a closed vertex cycle with word 2 0 1 2. There are $a, b > 0$ with $a^2 + b^2 = 1$ such that the factors are $(q_0, q_1, q_2, q_3) = (1/a, 1/b, 1/a, 1/b)$, the initial margins are*

$$(m_0(0), m_0(1), m_0(2)) = \frac{1}{\sqrt{\mathcal{N}}}\big(2(s-1),\ 2(1-a+b),\ 2(1+a-b)\big), \quad (10.5)$$

and the raw cost is $8\,\delta(a, b)$.

Proof. The factors come from Lemma 10.5. For the margins, write (A, B, C) for the vector in (10.5) before the factor $\mathcal{N}^{-1/2}$; it satisfies $C + A = 4a$, $C+B = 4$ and $A+B = 4b$. Starting from arbitrary initial margins (x_0, x_1, x_2) and applying (10.2) four times, the twist $m_4(j) = m_0(j-1)$ gives two linear equations, $(1+a-b)x_0 = (s-1)x_2$ and $(1+a-b)x_1 = (1-a+b)x_2$; the quadric fixes the scale, and $x_2 > 0$ fixes the sign. Along the period the margins run through the permutations

$$(A,B,C) \to (B,A,C) \to (B,C,A) \to (A,C,B) \to (C,A,B)$$

of the same three numbers (each divided by $\sqrt{\mathcal{N}}$), so every switch state is strict-star, and the last one is the twist of the first. The active margins are C, B, C, B. For the first piece, (10.3) with $a_0 = C$, $b_0 = A$, $c_0 = B$ (each over $\sqrt{\mathcal{N}}$) and $q_0 = 1/a$ gives, by $C + A = 4a$ and $C + B = 4$,

$$\rho = \frac{C}{\mathcal{N}}\Big(4a\big(\tfrac{1}{a} - 1\big) + 4(1-a) - C\log\tfrac{1}{a}\Big) = \frac{8C(1-a) + C^2\log a}{\mathcal{N}},$$

the third piece costs the same, and the second and fourth cost $(8B(1-b) + B^2\log b)/\mathcal{N}$. On the circle, $C^2 = 4(1+a-b)^2 = 8(1+a)(1-b)$, $B^2 = 8(1-a)(1+b)$, and $(1+a-b)(1-a)+(1-a+b)(1-b) = 1-s+2ab = s(s-1)$. The sum of the four pieces is $8\delta(a, b)$. □

Theorem 10.7 (The four-piece density)**.** *For $a, b > 0$ with $a^2 + b^2 = 1$,*

$$\delta(a,b) \ \geq\ \delta_{\mathrm{oct}} = \frac{8 - 4\sqrt{2} - \log 2}{2\sqrt{2} - 1},$$

with equality only at $a = b = 1/\sqrt{2}$.

The proof is in Appendix F.2. The density is symmetric in a and b, and its value at $a = b = 1/\sqrt{2}$ is δ_{oct}: there $s = \sqrt{2}$, $4s(s-1) = 8 - 4\sqrt{2}$, the logarithmic terms add up to $4(1-a^2)\log a = -\log 2$, and $\mathcal{N} = 2\sqrt{2}-1$. The content of the theorem is that the symmetric point is the minimum. Writing

$b \in (0, 1/\sqrt{2}\,]$ and $a = \sqrt{1-b^2} \geq b$, we show that $\psi(b) = \delta(\sqrt{1-b^2}, b)$ is strictly decreasing. Its derivative has the sign of $(s+1)(w(b)-w(a)) - s(s-1)(a-b)$, where $w(x) = (2x+2+2ab)\log x/(1-x)$, and w is increasing on $(0,1)$ by the elementary bound $\log x \geq (x - x^{-1})/2$. The left panel of Figure 12 shows δ along the quarter circle.

Remark 10.8. Another proof [7] parametrizes the quarter circle by s, with $a \geq b$, writes the derivative as $(2/\mathcal{N}^2)(H(s) - s(s-1)^3/(s+1))$, and proves $H(s) < 0$ through the integral representation $H(s) = (s-1)^2 \int_0^\infty (2t - s^2 - 1)\,\mathrm{d}t/((t+1)(t+a)(t+b))$: the weight $(t+1)^2/((t+a)(t+b))$ is decreasing, so the positive and negative parts of the integrand, separated at $t = (s^2+1)/2$, compare with a negative total.

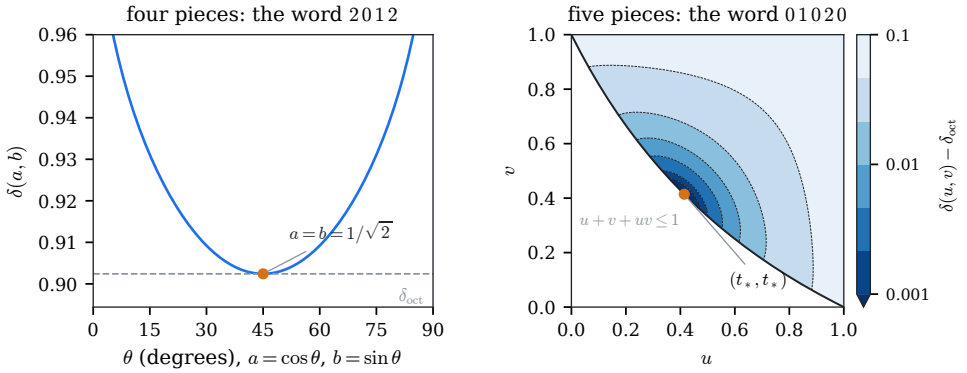


FIGURE 12. Left: the four-piece density $\delta(\cos\theta, \sin\theta)$ of (10.4); its minimum δ_{oct} is attained only at $\theta = 45°$, the octagon. Right: the excess $\delta(u,v) - \delta_{\mathrm{oct}}$ of the five-piece density (10.6) over its domain $u, v \in (0,1)$, $u + v + uv > 1$, on a logarithmic colour scale; it is positive everywhere and tends to 0 only at the boundary point (t_*, t_*), $t_* = \sqrt{2} - 1$, where the middle factor tends to 1. Both are computed from the cycle data; the script also checks the closure, the forced margins and the cost identity at random parameters.

Corollary 10.9. *A closed vertex cycle with four pieces has raw cost at least $8\delta_{\mathrm{oct}}$. If its raw cost is $8\delta_{\mathrm{oct}}$, then every factor is $\sqrt{2}$, every switch is forward, and its initial margins are those of the octagon: the active margin $m_0(i_0)$ and the margin $m_0(i_0+2)$ equal $2/\sqrt{2\sqrt{2}-1}$, and $m_0(i_0+1) = 2(\sqrt{2}-1)/\sqrt{2\sqrt{2}-1}$.*

Proof. Relabel to the word 2012 and apply Lemma 10.6 and Theorem 10.7. At equality $a = b = 1/\sqrt{2}$, so every factor is $\sqrt{2}$, $s = \sqrt{2}$, $\mathcal{N} = 2\sqrt{2} - 1$, and (10.5) gives the stated margins. □

With active margin $a_k = 2/\sqrt{2\sqrt{2}-1}$ and factor $\sqrt{2}$, each piece lasts $(a_k/2)\log q_k = \log 2/(2\sqrt{2\sqrt{2}-1})$, the octagon's switching time t_{sw}.

10.4. **Five pieces.** By Proposition 10.3, a five-piece cycle that is not entirely reverse can be rebased and relabeled to the word 0 1 0 2 0, whose switches have the signs $+--++$. Its closure has a two-parameter family of solutions. Put

$$P(t) = \frac{1+t^2}{2t}, \qquad Q(t) = \frac{1+t^2}{1-t^2} \qquad (0<t<1).$$

Lemma 10.10 (Five-piece closure)**.** *If $p, q, r, s, z > 1$ and*

$$\widetilde{T}_0(p)\,\widetilde{T}_1(q)\,\widetilde{T}_0(r)\,\widetilde{T}_2(s)\,\widetilde{T}_0(z) = \widetilde{R},$$

then $p^2(q^2-1) = q^2$, $z^2(s^2-1) = s^2$ and $r^2 = (q^2-1)(s^2-1)$. Consequently there are $u, v \in (0,1)$ with $u+v+uv > 1$ and

$$(p,q,r,s,z) = \Big(P(u),\ Q(u),\ \frac{4uv}{(1-u^2)(1-v^2)},\ Q(v),\ P(v)\Big).$$

Proof. The three relations come from the entries of the product compared with $\widetilde{R}$. Every $q > 1$ is $Q(u)$ for $u = \sqrt{(q-1)/(q+1)} \in (0,1)$, and then $p^2 = q^2/(q^2-1)$ gives $p = P(u)$; likewise for s, z and v, and $r^2 = (Q(u)^2-1)(Q(v)^2-1)$ gives the middle factor. Finally $r > 1$ is equivalent to $u+v+uv > 1$ by the factorization $4uv-(1-u^2)(1-v^2) = (u+v+uv-1)(u+v+1-uv)$, whose second factor is positive. □

Put $d = (1-u)(1-v)$ and

$$F(t) = \Big(\frac{1+t}{1-t}\Big)^2 \log\frac{1+t^2}{2t} + \frac{1}{t^2}\log\frac{1+t^2}{1-t^2} + \log\frac{2t}{1-t^2},$$

and define the *five-piece density*

$$\delta(u,v) = \frac{2(d+2) - \frac{d}{2}\big(F(u)+F(v)\big)}{4-d}. \tag{10.6}$$

Lemma 10.11 (Five-piece cycles)**.** *Let c be a closed vertex cycle with word 0 1 0 2 0. There are $u, v \in (0,1)$ with $u+v+uv > 1$ such that the factors are those of Lemma 10.10, the initial margins are*

$$(m_0(0), m_0(1), m_0(2)) = \frac{2}{\sqrt{d(4-d)}}\big((1+u)(1-v),\ d,\ (1-u)(1+v)\big),$$

and the raw cost is $8\,\delta(u,v)$.

Proof. As for four pieces, the twist gives two linear equations for the initial margins, which the quadric normalizes. With $A = (1+u)(1-v)$ and $B = (1-u)(1+v)$, the active margins of the five pieces are A, d/u, d, d/v, B, times $2/\sqrt{d(4-d)}$. The algebraic part of the raw cost (all but the terms $-a_k^2\log q_k$) adds up to $16\,d(d+2)/(d(4-d))$, and the logarithmic part is

$$-\frac{4}{d(4-d)}\Big(A^2\log P(u) + \frac{d^2}{u^2}\log Q(u) + d^2\log r + \frac{d^2}{v^2}\log Q(v) + B^2\log P(v)\Big).$$

Since $A/d = (1+u)/(1-u)$, $B/d = (1+v)/(1-v)$ and $\log r = \log \frac{2u}{1-u^2} + \log \frac{2v}{1-v^2}$, the bracket is $d^2(F(u)+F(v))$, and the raw cost is $8\delta(u,v)$. □

Theorem 10.12 (The five-piece density)**.** *For $u, v \in (0,1)$ with $u+v+uv > 1$, $\delta(u,v) > \delta_{\mathrm{oct}}$.*

The proof is in Appendix F.3; here is its structure. The function F enters only through $F(t) + 2\log(1-t)$, and this function of one variable is at most its value at $t_* = \sqrt{2}-1$, namely $(3+2\sqrt{2})\log 2 + 2\log(2-\sqrt{2})$. Summing over $t = u, v$ bounds $F(u)+F(v)$ above by a function of d alone, hence $\delta(u,v)$ below by a function of d. The constraint $u+v+uv > 1$ gives $d < d_* = (2-\sqrt{2})^2$, and at $d = d_*$ the lower bound equals δ_{oct} exactly. A tangent-line estimate for $d\log d$ at d_* shows that the bound exceeds δ_{oct} for $d < d_*$. The infimum of $\delta(u,v)$ is approached only as $u, v \to t_*$, where $d \to d_*$ and the middle factor r tends to 1: the five-piece period degenerates into the octagon's four pieces (right panel of Figure 12).

10.5. **The terminal theorem.**

Theorem 10.13 (Terminal comparison)**.** *Every normal vertex period with at most five pieces costs at least A_{oct}.*

Proof. Let c be the cycle of the period (Lemma 10.2). It is not entirely reverse, by Theorem 5.4, which holds for every number of pieces. By Proposition 10.3, either it has four pieces, and then its raw cost is at least $8\delta_{\mathrm{oct}}$ by Corollary 10.9, or it has five pieces and is a rebasing and relabeling of a cycle with word $0\,1\,0\,2\,0$, whose raw cost exceeds $8\delta_{\mathrm{oct}}$ by Lemma 10.11 and Theorem 10.12. By Lemma 10.2, $\mathcal{C} \geq \frac{\sqrt{3}}{4} \cdot 8\delta_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}} = A_{\mathrm{oct}}$. □

No competitiveness and no compact domain are needed here. The equality case identifies the trajectory itself.

Proposition 10.14 (Terminal equality)**.** *Let P be a normal vertex period with at most five pieces and cost A_{oct}. Then $T = t_{\mathrm{f}}$, and there is $r \in \mathbb{N}$ such that, for $t \in [0,T]$,*

$$R^{-r} g(t) R^r = g_{\mathrm{oct}}(t), \qquad R^{-r} X(t) R^r = X_{\mathrm{oct}}(t),$$

where $(g_{\mathrm{oct}}, X_{\mathrm{oct}})$ is the trajectory of the smoothed octagon (Proposition 2.9).

Proof. A five-piece cycle that is not entirely reverse has raw cost strictly above $8\delta_{\mathrm{oct}}$, and the entirely reverse case is excluded by Theorem 5.4, so $N = 4$. By Corollary 10.9 the period has four forward pieces, all factors $\sqrt{2}$, and the octagon's initial margins. A traceless matrix is determined by its margins, so after relabeling by a power of R the initial state is $\Phi(0, y_0)$ with $y_0^2 = (2\sqrt{2}-1)/3$, and the control agrees almost everywhere with the octagon's control $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ on four pieces of length t_{sw}. Changing a control on a null set does not change a Carathéodory solution, and the octagon's trajectory is the unique solution with $g(0) = I$ and $X(0) = \Phi(0, y_0)$, by uniqueness of solutions of the state equations with a fixed vertex control on each piece. □

10.6. **Proofs of the lower bounds.** The chain of reductions now closes. Recall that a global minimizer is an admissible trajectory whose cost is at most that of every admissible trajectory (Definition 2.7).

Theorem 10.15 (Cost of a minimizer)**.** *Every global minimizer of the control problem costs at least* A_{oct}.

Proof. Let (T, u, g, X) be a global minimizer. The octagon's trajectory is admissible and costs A_{oct} (Proposition 2.9), so $\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$: the minimizer is competitive. By Theorem 3.18, rebased at a switch it is a normal vertex period P with the same terminal time and cost, whose state stays in $\mathfrak{h}^{\star\star}$. By Theorem 4.19, P has at most five pieces, and by Theorem 10.13, $\mathcal{C}(P) \geq A_{\mathrm{oct}}$. □

Proof of Theorem 1.2. Let (T, u, g, X) be admissible, with any terminal time and any measurable control. By Theorem 2.12 a global minimizer (T', u', g', X') exists: the realization of a worst disk. By Theorem 10.15 and minimality, $A_{\mathrm{oct}} \leq \mathcal{C}(T', X') \leq \mathcal{C}(T, X)$, and $A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}$. □

Proof of Theorem 1.1(b). By Theorem 2.12 there is a worst disk K_0, a convex disk symmetric about the origin whose greatest packing density is the least among all such disks, and its realization is a global minimizer of cost $\sqrt{12}\,\delta(K_0)$. By Theorem 10.15, $\sqrt{12}\,\delta(K_0) \geq A_{\mathrm{oct}}$, that is, $\delta(K_0) \geq \delta_{\mathrm{oct}}$. Now let K be a convex disk with a centre c, so that $2c - x \in K$ for $x \in K$. Its translate $K - c$ is a convex disk symmetric about the origin, and translation does not change the greatest packing density, since a translate is a congruent copy (Proposition 11.5). Hence $\delta(K) = \delta(K - c) \geq \delta(K_0) \geq \delta_{\mathrm{oct}}$. □

11. The equality case

Proposition 10.14 identifies a period of cost A_{oct} with the octagon's trajectory, but only after three choices that the proof of the lower bound made along the way: the normalization of the disk to Reinhardt's normal position, the rebasing of its trajectory at a switch, and a relabeling of the vertices of the control simplex by a power of R. To classify the disks of density δ_{oct} these choices must be undone, and the trajectory to which the terminal equality is applied must be the realization of the given disk; another minimizer of the same cost would say nothing about it. This section carries the identification back to the disk (Section 11.1), proves the converse (Section 11.2), identifies Hales's octagon K_{H} with Reinhardt's K_{oct} (Section 11.3), and completes the proof of Theorem 1.1 (Section 11.4).

11.1. **Rigidity.** Let $s_0, \ldots, s_5$ be the sixth roots of unity, so that $Rs_j = s_{j+1}$. The realization of a worst disk (Theorem 2.12) keeps the link between its frame and its boundary.

Lemma 11.1 (Realization with its contact link)**.** *Let K be a worst disk. There are an invertible linear map e of the plane, a balanced pair σ of eK in circle representation and a global minimizer (T, u, g, X) of the control*

problem such that eK is a worst disk, $\delta(K) = \mathcal{C}(T,X)/\sqrt{12}$, and for every j and every $t \in \mathbb{R}$

$$\sigma_j(t) = g(t)\, s_j, \qquad \sigma_j(t+T) = \sigma_{j+1}(t).$$

The first identity is the definition of the frame of the realization, which is defined on all of $\mathbb{R}$; the second is the cyclic shift of the contacts by one sixth of the period. Together they force the frame to be twisted on all of $\mathbb{R}$, beyond the period $[0,T]$ on which the control problem is posed.

Lemma 11.2 (The frame twist)**.** *If $\sigma_j(t) = g(t)s_j$ and $\sigma_j(t+T) = \sigma_{j+1}(t)$ for all j and t, then $g(t+T) = g(t)R$ for every $t \in \mathbb{R}$.*

Proof. $g(t+T)s_j = \sigma_j(t+T) = \sigma_{j+1}(t) = g(t)s_{j+1} = g(t)Rs_j$, and s_0, s_1 span the plane. □

The last ingredient recovers a disk from its boundary.

Lemma 11.3. *Two compact convex sets with nonempty interior, both symmetric about the origin, that have the same frontier are equal.*

Proof. Let A, B be the two sets. The interior of B is connected, contains the origin, and does not meet the common frontier; it is covered by the disjoint open sets $\operatorname{int} A$ and $\mathbb{R}^2 \setminus \overline{A}$, so it lies in $\operatorname{int} A$, which contains the origin. By symmetry the interiors are equal, and a convex body is the closure of its interior. □

Theorem 11.4 (Rigidity)**.** *Every worst disk is the image of the octagon K_{H} under an invertible linear map of the plane. Explicitly, with the notation of the proof, $K = e^{-1}g(s)R^r(K_{\mathrm{H}})$.*

Proof. Let K be a worst disk, and take e, σ and (T, u, g, X) as in Lemma 11.1. By Lemma 11.2, $g(t+T) = g(t)R$ on $\mathbb{R}$.

The cost is A_{oct}. Since K is a worst disk and $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ (Proposition 2.9), $\delta(K) \le \delta_{\mathrm{oct}}$, and part (b) of Theorem 1.1 gives $\delta(K) \ge \delta_{\mathrm{oct}}$. Hence $\mathcal{C}(T,X) = \sqrt{12}\,\delta(K) = A_{\mathrm{oct}}$.

The terminal equality. By Theorem 3.18, the minimizer rebased at some $s \in [0,T)$ is a normal vertex period P with the same terminal time and cost, whose state stays in $\mathfrak{h}^{\star\star}$; its frame is $t \mapsto g(s)^{-1}g(s+t)$, because g is already twisted. By Theorem 4.19, P has at most five pieces, and by Proposition 10.14, $T = t_{\mathrm{f}}$ and there is r with $R^{-r}g(s)^{-1}g(s+t)R^r = g_{\mathrm{oct}}(t)$ for $t \in [0,T]$. With $M = g(s)R^r$ this reads

$$g(s+t) = M\, g_{\mathrm{oct}}(t)\, R^{-r} \qquad (0 \le t < T).$$

Both sides satisfy $F(t+T) = F(t)R$: the left side by the twist of g, the right side because $g_{\mathrm{oct}}(t+t_{\mathrm{f}}) = g_{\mathrm{oct}}(t)R$ and R commutes with R^{-r}. Two curves with the same twist that agree on $[0,T)$ agree everywhere, so the identity holds for all $t \in \mathbb{R}$.

The arcs. Applying it to s_0 and using $R^{-r}s_0 = s_{-r}$,

$$\sigma_0(s+t) = M\, g_{\mathrm{oct}}(t)\, s_{-r} = M\, \sigma^{\mathrm{oct}}_{-r}(t) \qquad (t \in \mathbb{R}),$$

where $\sigma_j^{\mathrm{oct}}(t) = g_{\mathrm{oct}}(t)s_j$ are the contact curves of the octagon. In a balanced pair each contact curve traces the whole frontier once per period, so the frontier of eK is the range of σ_0, and the frontier of K_{H} is the range of $\sigma_{-r}^{\mathrm{oct}}$. Hence $\partial(eK) = M(\partial K_{\mathrm{H}}) = \partial(MK_{\mathrm{H}})$. Both eK and MK_{H} are convex disks symmetric about the origin, so $eK = MK_{\mathrm{H}}$ by Lemma 11.3, and $K = e^{-1}M(K_{\mathrm{H}})$. □

The proof never compares K with a minimizer other than its own realization. It runs the reduction of Sections 3 to 10 a second time, on that realization, and uses the lower bound only to know that its cost is A_{oct}.

11.2. **The converse.** The converse is the affine invariance of the greatest packing density.

Proposition 11.5 (Affine invariance)**.** *Let K be a convex disk symmetric about the origin, A an invertible linear map of the plane and b a vector. Then $\delta(AK) = \delta(K)$ and $\delta(K + b) = \delta(K)$. In particular $\delta(AK_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ for every invertible linear map A.*

Proof. A translate, and more generally a congruent copy, of K has the same packings by congruent copies, so its greatest packing density is the same. By Fejes Tóth's theorem (Theorem 2.2), $\delta(K)$ is the greatest density $\mathrm{area}(K)/\det\Lambda$ of a lattice packing $K + \Lambda$. If $\det A > 0$, the lattice packings of K and of AK correspond through $\Lambda \mapsto A\Lambda$, and area and covolume are both multiplied by $\det A$, so the lattice densities agree. If $\det A < 0$, write $A = (AF)F$ with the reflection $F(x_0, x_1) = (x_0, -x_1)$: then FK is a congruent copy of K, and AF preserves orientation. The last statement follows from $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ (Proposition 2.9). □

11.3. **Reinhardt's construction.** Theorem 1.1 is stated for Reinhardt's octagon K_{oct} (Definition 2.3), while the proof works with K_{H}, the region swept from the origin by the contact curve σ_0^{oct} of the octagon's trajectory over one period $[0, 6t_{\mathrm{f}}]$. By Proposition 2.9, K_{H} is also the convex hull of the six arcs $\sigma_j^{\mathrm{oct}}([0, t_{\mathrm{f}}])$. The two octagons differ by a vertical stretch of about 5.5% (Figure 1).

Proposition 11.6 (Reinhardt's octagon is Hales's octagon)**.** *Let $A = \mathrm{diag}(1, (2\sqrt{2}-1)/\sqrt{3})$. Then $A(K_{\mathrm{H}}) = K_{\mathrm{oct}}$.*

Proof. Write $y_0^2 = (2\sqrt{2}-1)/3$ and $W = \exp(\sqrt{3}\,y_0 s)$ for $s \in [0, t_{\mathrm{sw}}]$, so that W runs over $[1, \sqrt{2}]$; recall that $\sigma_0^{\mathrm{oct}}(0) = s_0 = (1, 0)$.

One eighth of the boundary. On the first switching interval the octagon's frame $g_0(s)$ is the flow of the control $\mathbf{e}_2$ from $\Phi(0, y_0)$; conjugated by A it acts by

$$Ag_0(s)A^{-1} = \begin{pmatrix} 1 & -\frac{W-1}{2\sqrt{2}-1} \\ 1 - W^{-1} & 1 - \frac{W-2+W^{-1}}{2\sqrt{2}-1} \end{pmatrix},$$

and $A\,g_0(t_{\mathrm{sw}})R^{-1}A^{-1}$ is the rotation by $-\pi/4$; both are direct computations from the closed form of the flow of $\mathbf{e}_2$. Since $g_{\mathrm{oct}}(t+t_{\mathrm{sw}}) = g_0(t_{\mathrm{sw}})R^{-1}\,g_{\mathrm{oct}}(t)\,R$, the curve $A\sigma_0^{\mathrm{oct}}$ is, on the first three switching intervals,

$$\begin{aligned}
&(1,\ 1-W^{-1}) && \text{along side 0,}\\
&\Big(\sqrt{2}+1-\tfrac{\sqrt{2}}{2}(W+W^{-1}),\ 1-\tfrac{\sqrt{2}}{2W}\Big) && \text{around corner 0,}\\
&\Big(\sqrt{2}-\tfrac{W}{2},\ \tfrac{W}{2}\Big) && \text{along side 1.}
\end{aligned}$$

The first piece runs from the midpoint $(1,0)$ of side 0 to the point $(1,1-\sqrt{2}/2)$, and the third from $(\sqrt{2}-\frac{1}{2},\frac{1}{2})$ to the midpoint $(\sqrt{2}/2,\sqrt{2}/2)$ of side 1. With d_k the distance to the kth side line, the middle piece has $d_7 = W/2$ and $d_2 = \sqrt{2}/(2W)$, so it lies on the hyperbola $d_7d_2 = \sqrt{2}/4$; it starts at $(1,1-\sqrt{2}/2)$ and ends at $(\sqrt{2}-\frac{1}{2},\frac{1}{2})$, the points where the first and the third pieces end and start. The control on these three intervals is $\mathbf{e}_2$, $\mathbf{e}_0$, $\mathbf{e}_1$, and only the second bends the curve.

The rest. Three switching intervals later the frame relation gives $\sigma_0^{\mathrm{oct}}(t+3t_{\mathrm{sw}}) = -h^3\sigma_0^{\mathrm{oct}}(t)$ with $h = g_0(t_{\mathrm{sw}})R^{-1}$, and $-Ah^3A^{-1}$ is the rotation by $\pi/4$. Over the period $6t_{\mathrm{f}} = 24t_{\mathrm{sw}}$ the curve $A\sigma_0^{\mathrm{oct}}$ therefore traces eight rotated copies of the three pieces, once around.

The two inclusions. Every point q of the three pieces lies in K_{oct}, and μq does not for $\mu > 1$: on the sides this is the inequality $d_k \ge 0$, and on the corner it is the hyperbola condition, which strictly fails past the arc. Since K_{oct} is invariant under the rotation by $\pi/4$, the same holds for every point $A\sigma_0^{\mathrm{oct}}(t)$. The set K_{oct} is star-shaped about 0, so $AK_{\mathrm{H}} \subseteq K_{\mathrm{oct}}$. Conversely, a point $p \neq 0$ of K_{oct} is a positive multiple λq of some $q = A\sigma_0^{\mathrm{oct}}(t)$, by the intermediate value theorem and $\sigma_0^{\mathrm{oct}}(t+3t_{\mathrm{f}}) = -\sigma_0^{\mathrm{oct}}(t)$, and $\lambda \le 1$ because $\mu q \notin K_{\mathrm{oct}}$ for $\mu > 1$. Hence $p \in AK_{\mathrm{H}}$. □

Each side of K_{oct} keeps a straight middle part of length $2-\sqrt{2}$, between the points of tangency $(1,\pm(1-\sqrt{2}/2))$ on side 0. The proposition transfers the facts known for K_{H} to K_{oct}; in particular, the affine images of the two octagons are the same sets.

11.4. **Proofs of parts (a) and (c) of Theorem 1.1.**

Proof of Theorem 1.1(a). By Proposition 11.6, $K_{\mathrm{oct}} = AK_{\mathrm{H}}$. The octagon K_{H} is a compact convex set with nonempty interior, symmetric about the origin, and $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ (Proposition 2.9); these properties pass to its linear image, the density by Proposition 11.5. □

Proof of Theorem 1.1(c). Let K be a convex disk with centre c. Suppose first that $\delta(K) = \delta_{\mathrm{oct}}$. The translate $K-c$ is symmetric about the origin with $\delta(K-c) = \delta_{\mathrm{oct}}$, which by part (b) is the least value; so $K-c$ is a worst disk. By Theorem 11.4, $K-c = LK_{\mathrm{H}}$ for an invertible linear map L, and K is the image of K_{H} under the affine map $x \mapsto Lx+c$. Conversely, an affine image $A'K_{\mathrm{H}}$, with linear part L, is the translate $LK_{\mathrm{H}} + A'(0)$, which has density

δ_{oct} by Proposition 11.5. This proves the equivalence with K_{H} in place of K_{oct}, and the affine images of K_{H} are those of K_{oct} by Proposition 11.6. □

In Hales's form, the equality disks are the affine images of the convex hull of the six arcs $g(t)s_j$, $t \in [0, t_{\mathrm{f}}]$, of the trajectory (g, X) of the octagon's control from $g(0) = I$ and $X(0) = \Phi(0, y_0)$ (Proposition 2.9); this hull is a centrally symmetric convex disk of area $A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}$ and greatest packing density δ_{oct}. The equality data are explicit: four forward pieces with the word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$, every factor equal to $\sqrt{2}$, every duration equal to $\log 2/(2\sqrt{2\sqrt{2}-1})$, and initial state $\Phi(0, y_0)$.

Appendix A. Costate identities for reverse switches

This appendix gives the explicit formulas behind Sections 4 and 5: the one-piece costate map, the two reverse-four identities, the sign of the cost derivative N, and the elementary inequalities of the three-arc lemma, the switch norms and the Casimir cap. Each identity below is an identity of rational functions in the displayed variables and the logarithms that occur; it is verified by clearing denominators and expanding.

A.1. **The one-piece costate map.** On a piece with active margin a, growing margin b and decreasing margin c at its start (Lemma 4.1), the inhomogeneous terms of Lemma 4.10(b) at the factor q are

$$\begin{aligned}
\mathrm{Act}(a,b,c,q) &= \tfrac{\sqrt{3}a}{2}\big((a+c)(1-q^{-1}) - (a+b)(q-1)\big),\\
\mathrm{AG}(a,b,q) &= \tfrac{\sqrt{3}a}{2}\big(a(q^2-1) - (a+b)(q^2-q)\big),\\
\mathrm{AD}(a,c,q) &= \tfrac{\sqrt{3}a}{2}\big((a+c)(q^{-1}-q^{-2}) - a(1-q^{-2})\big),\\
\mathrm{BG}(a,b,q) &= \sqrt{3}a\big((a+b)\,q\log q - 2a(q-1)\big),\\
\mathrm{BD}(a,c,q) &= \sqrt{3}a\,q^{-1}\big(2a(q-1) - (a+c)\log q\big).
\end{aligned}$$

The logarithms come from the cost term $-\frac{3}{2}[J, X]$ of the costate equation, integrated against the exponentially moving margins.

These formulas give the switching functions of the two inactive labels inside the piece. Let i be the active label, and at the start of the piece let $Y = \alpha_i + \alpha_{i+1}$, $Y' = \alpha_i + \alpha_{i-1}$, $B = \beta_i + \beta_{i+1}$, $B' = \beta_i + \beta_{i-1}$. Since $\chi_i = 0$ on the piece, at the intermediate factor $s \in [1, q]$

$$\chi_{i+1}(s) = 2s^2 Y + sB + \sqrt{3}\,a\,s\,F(s), \qquad \chi_{i-1}(s) = \frac{2Y'}{s^2} + \frac{B'}{s} - \frac{\sqrt{3}\,a}{s}\,h(s),$$

where

$$\begin{aligned}
F(s) &= (a+b)\log s - (a+b)(s-1) + a\,\frac{(s-1)^2}{s},\\
h(s) &= (a+c)\log s - a(s-1) - c\,(1-s^{-1}),
\end{aligned}$$

because $2\mathrm{AG} + \mathrm{BG} = \sqrt{3}asF$ and $2\mathrm{AD} + \mathrm{BD} = -\sqrt{3}a\,h/s$. Both F and h vanish to second order at $s = 1$: $F(1) = F'(1) = h(1) = h'(1) = 0$. When the growing label was active on the previous piece, $\chi_{i+1}(1) = 2Y + B = 0$ and

$$\chi_{i+1}(s) = 2s\Big(Y(s-1) + \frac{\sqrt{3}a}{2}F(s)\Big);$$

when the decreasing label was, $B' = -2Y'$ and

$$\chi_{i-1}(s) = -\frac{2}{s}\Big(Y'(1-s^{-1}) + \frac{\sqrt{3}a}{2}h(s)\Big).$$

These are the forms used in Lemma 4.14(a) and in Lemma 5.5. In each case the derivative at $s = 1$ is $\pm 2Y$ or $\pm 2Y'$, which is how the sign condition at the start of a piece fixes the sign of Y or Y'.

A.2. **The reverse-four identities.** In the notation of Section 5.1, let x_j, y_j be the coordinates α, β at the switch τ_{k+1} in the label order $(i, i+1, i-1)$, so that $j = 0, 1, 2$ stands for the labels $i, i+1, i-1$, and let u_j, v_j and r_j, s_j be those at τ_{k+2} and τ_{k+3} in the same order. The transport over piece $k+1$ (active margin Z, growing margin P, decreasing margin Q, factor b) and over piece $k+2$ (active margin A, growing margin Z, decreasing margin B, factor c) gives

$$\begin{aligned}
u_2 &= x_2 + \mathrm{Act}(Z, P, Q, b),\\
u_0 &= b^2(x_2 + x_0) + \mathrm{AG}(Z, P, b) - u_2,\\
u_1 &= b^{-2}(x_2 + x_1) + \mathrm{AD}(Z, Q, b) - u_2,\\
r_1 &= u_1 + \mathrm{Act}(A, Z, B, c),\\
r_0 &= c^{-2}(u_1 + u_0) + \mathrm{AD}(A, B, c) - r_1,\\
r_2 &= c^2(u_1 + u_2) + \mathrm{AG}(A, Z, c) - r_1,
\end{aligned}$$

and the same with $(\alpha, \mathrm{Act}, \mathrm{AG}, \mathrm{AD})$ replaced by $(\beta, -2\mathrm{Act}, \mathrm{BG}, \mathrm{BD})$ and the powers $b^{\pm 2}, c^{\pm 2}$ by $b^{\pm 1}, c^{\pm 1}$. The five relations are

$$h_1 = 2x_0 + y_0, \quad h_2 = 2x_2 + y_2, \quad h_3 = 2u_1 + v_1, \quad h_4 = 2r_0 + s_0,$$
$$h_5 = y_0(Q + Z) + y_1(P + Z) + y_2(P + Q),$$

all equal to 0: the active switching functions at the three switches, and $\langle \Lambda_R, X\rangle = 0$ at τ_{k+1}. Then, with $D = A(c-1) + Z(b-1)$ and as identities in all the variables,

$$\begin{aligned}
&B(c-1)(2x_1 + y_1) + bD(2u_0 + v_0) + \sqrt{3}ZN(A, B, Z, b, c)\\
&\qquad = b^2(c-1)(Q - Zb + Z)\,h_1 + (b^2-1)(c-1)(Pb + Q)\,h_2\\
&\qquad\quad + b(c-1)(Pb^2 + Zb^2 + Zbc - Zc - Z)\,h_3 + Zbc^2(b-1)\,h_4\\
&\qquad\quad - b(b-1)(c-1)\,h_5,
\end{aligned}$$

and

$$\begin{aligned}(b-1)B(2r_2+s_2)+cD(2u_0+v_0)+\sqrt{3}AN(Z,B,A,c,b)\\ =bc(c-1)(Q-Zb+Z)\,h_1\\ +\tfrac{c(b-1)}{b}(Pb^2c+Qbc-Qb+Qc-Q+Zb^2-Zbc+Zc-Z)\,h_2\\ +(b-1)(c^2-1)(Pb+Zb+Zc-Z)\,h_3+Zc^3(b-1)\,h_4\\ -c(b-1)(c-1)\,h_5,\end{aligned}$$

with $A=(Z+Q)/b-Z$ and $B=(Z+P)b-Z$. Setting the five relations to zero gives Lemma 5.2.

A.3. **The sign of the cost derivative.** Let $\eta(t)=t-1-\log t$ and $\zeta(t)=t\log t-t+1$. For $t>1$ both are positive.

Lemma A.1 (Padé bound)**.** *For $t>1$, $3(t^2-1)<(t^2+4t+1)\log t$; equivalently $(2t+1)\,\eta(t)<(t+2)\,\zeta(t)$. In particular $\eta(t)<\zeta(t)$.*

Proof. The function $\psi(t)=\log t-3(t^2-1)/(t^2+4t+1)$ vanishes at 1 and $\psi'(t)=(t-1)^4/(t(t^2+4t+1)^2)>0$ for $t>1$. The second form follows from $(t+2)\zeta(t)-(2t+1)\eta(t)=(t^2+4t+1)\log t-3(t^2-1)$, and the third because $(2t+1)/(t+2)>1$ and $\eta(t)>0$. □

Proof of Lemma 5.1. Recall (5.2) and the hypotheses $b,c>1$, $A,Z>0$, $A\le Z$, $bA+(b-1)Z\le B$ and $(2c-1)A\le B$. In particular $B>cA>0$.

The first bracket of N is negative: $Ac\,\eta(c)<Ac\,\zeta(c)<B\,\zeta(c)$.

The second bracket, $T(B)=bA(B+bZ)\,\eta(b)-B(bA+Z)\,\zeta(b)$, is affine in B with slope $bA\,\eta(b)-(bA+Z)\,\zeta(b)<0$, so $T(B)\le T(B_0)$ with $B_0=bA+(b-1)Z$. At B_0 we have $B_0+bZ=bA+(2b-1)Z$ and $B_0(bA+Z)=b^2A^2+b^2AZ+(b-1)Z^2>0$, and Lemma A.1 at $t=b$ gives

$$\begin{aligned}(b+2)\,T(B_0)&<\eta(b)\big[(b+2)\,bA\big(bA+(2b-1)Z\big)\\ &\qquad-(2b+1)\big(b^2A^2+b^2AZ+(b-1)Z^2\big)\big]\\ &=-(b-1)\,\eta(b)\big(b^2A^2+2bZ(Z-A)+Z^2\big),\end{aligned}$$

which is ≤ 0 because $Z\ge A$. So $T(B)<0$, and $N=b(b-1)A\cdot(\text{first bracket})+(c-1)\,T(B)<0$. □

A.4. **Elementary inequalities.** *Logarithms.* For $q>1$, $2(q-1)<(q+1)\log q$ and $q-1<q\log q$: both differences vanish at $q=1$, with derivatives $\log q+q^{-1}-1>0$ and $\log q>0$. They give the signs in the proof of Lemma 5.5.

The turn secant. With $f(x)=x-x^{-1}-\log x$,

$$f'(x)-\tfrac{3}{4}=\frac{(x-2)^2}{4x^2},\qquad 1-f'(x)=\frac{x-1}{x^2}.$$

So every secant slope of f on $(1,\infty)$ lies in $[3/4,1)$, and the switch norm at a turn, $d=\frac{1}{8}\times(\text{secant slope})$, satisfies $3/32\le d<1/8$. At a forward

switch the lower bound $d_k \geq 3/32$ comes instead from $j_k/2 - 3/32 = (a_k - a_{k+1})^2/(32s_k^2) \geq 0$.

The cross pairing bound. With $a, p, n > 0$ and $j(a,b) = (a^2+ab+b^2)/(4(a+b)^2)$,

$$\frac{1}{2a}\Big((a+p)\,\frac{j(p,a)}{2} + (a+n)\,\frac{j(a,n)}{2} - \frac{G_a(a,p)+G_a(a,n)}{16}\Big) \\ = \frac{1}{16}\Big(-1 + \frac{2a}{a+p} + \frac{2a}{a+n} - \frac{a^2}{a^2+4} + \log\frac{(a+p)(a+n)}{a^2+4}\Big).$$

This is the quantity X_k of Lemma 5.10: by Lemma 4.13, at a same-sign piece e_k is an increasing affine function of d_{k-1} and d_k, and X_k is its value at $d = j/2$.

The Casimir arithmetic. The proof of Corollary 4.18 uses $A_{\mathrm{oct}} < 3.127$ and $\pi > 3.14$. From $4\pi\kappa \leq (2\kappa+3)\mathcal{C}$ and $\mathcal{C} < 3.127$,

$$\kappa < \frac{3 \cdot 3.127}{4 \cdot 3.14 - 2 \cdot 3.127} = 1.48763\ldots < 1.4877,$$

and $1.4877^2/12 = 0.18443\ldots < 97969/529200 = 0.18512\ldots$.

Appendix B. Estimates for class M1

This appendix proves the angle gap, Proposition 6.4, and the arithmetic of the two-centre case. Throughout, $K_* = 97969/529200$ is the competitive cap of Corollary 4.18, $U(s) = 1 - s + s^2$, and $\omega(x) = -x - \log(1-x)$ for $0 \leq x < 1$.

B.1. **Logarithm bounds.**

Lemma B.1. (1) *If $z, z_0 > 0$ and $\log z_0 \leq \ell$, then $\log z \leq \ell + z/z_0 - 1$.*
(2) *For $0 < z \leq 1$ and $\tau = (1-z)/(1+z)$, $\log z \leq -2\tau - \frac{2}{3}\tau^3$.*
(3) *For $0 < x \leq 1$, $\frac{1}{2}(x - x^{-1}) \leq \log x$.*
(4) *If $a, b, c \geq 0$, $c < 1$ and $a^2 + b^2 \leq c^2$, then $\omega(a) + \omega(b) \leq \omega(c)$.*

Proof. (1) is $\log(z/z_0) \leq z/z_0 - 1$. (2) keeps the first two terms of the series $\log z = -2\sum_{n\geq 0} \tau^{2n+1}/(2n+1)$, all of whose terms are nonpositive. (3) is $y \leq \sinh y$ for $y = -\log x \geq 0$. (4) Write $\omega(x) = \sum_{n\geq 0} x^{n+2}/(n+2)$. Since $a, b \leq c$, termwise $a^{n+2} + b^{n+2} \leq (a^2+b^2)c^n \leq c^{n+2}$. □

Bound (2) gives the three rational values used below,

$$\log\tfrac{17}{25} \leq -\tfrac{10712}{27783}, \qquad \log\tfrac{3}{4} \leq -\tfrac{296}{1029}, \qquad \log\tfrac{17}{20} \leq -\tfrac{8232}{50653}, \tag{B.1}$$

for $\tau = 4/21$, $1/7$ and $3/37$.

B.2. **One reverse switch.** The reverse-switch term $\Psi(r,u) = r - \log r - (1-u)/r$ of the stationarity identity increases with u and, at fixed u, is largest when the switch bound $u \le U(r)$ is attained.

Lemma B.2. *Let $0 < r < 1/2$ and $3/4 \le u \le U(r)$, and put $s = \sqrt{4u-3}$. Then $0 \le s < 1$, $2r \le 1-s$, and*

$$r - \log r - \frac{1-u}{r} \le \omega(s) + \log 2.$$

Proof. Put $r' = (1-s)/2$, so that $U(r') = (3+s^2)/4 = u$. Since U decreases on $[0,1/2]$ and $U(r') = u \le U(r)$, we have $r \le r'$. The identity

$$2r' - 1 - \Big(r - \tfrac{1}{2}\big(\tfrac{r}{r'} - \tfrac{r'}{r}\big) - \frac{1-U(r')}{r}\Big) = \frac{(r'-r)^2(1-2r')}{2rr'}$$

and $1 - 2r' = s \ge 0$ give $r - \frac{1}{2}(r/r' - r'/r) - (1-u)/r \le 2r' - 1 = -s$. By Lemma B.1(3) with $x = r/r' \le 1$, $-\log(r/r') \le -\frac{1}{2}(r/r' - r'/r)$, so $r - \log r - (1-u)/r \le -s - \log r'$. Finally $-\log r' = -\log(1-s) + \log 2$. □

B.3. **Reduction to two variables.**

Lemma B.3. *Let $\gamma^2, r, t, \rho, u, v, E$ satisfy* (i)–(iv) *of Lemma 6.3, and suppose that $E \ge \frac{3}{5}\sqrt{uv}$. Let $s = \sqrt{4u-3}$, $s' = \sqrt{4v-3}$, $S = \sqrt{s^2 + s'^2 + s^2 s'^2/3}$ and $z = 4\rho/(1-S)$. Then $0 \le S < 1$, $0 < z \le 1$,*

$$uv = \frac{3(3+S^2)}{16}, \qquad \rho = \frac{(1-S)z}{4},$$

$$E \le (1-S)\Big(1 - \frac{z}{4}\Big) + \log z, \qquad E^2 \ge \frac{9}{25} \cdot \frac{3(3+S^2)}{16}.$$

Proof. The identity (6.2) splits as $E = 1 - \rho + \log\rho + \Psi(r,u) + \Psi(t,v)$, and Lemma B.2 bounds the two last terms by $\omega(s) + \log 2$ and $\omega(s') + \log 2$, with $2r \le 1-s$ and $2t \le 1-s'$. Then $4\rho < 4rt \le (1-s)(1-s')$. Since $u, v \ge 3/4$, $\sqrt{uv} \ge 3/4$ and $E \ge 9/20$, and the bounds just obtained give $9/20 \le E < 1 - \rho - s - s'$, so $s + s' < 11/20$. From $u = (3+s^2)/4$ and $v = (3+s'^2)/4$ one gets $uv = 3(3+S^2)/16$. Next,

$$(s + s' - ss')^2 - S^2 = \tfrac{2}{3}s^2 s'^2 + 2ss'(1 - s - s') \ge 0,$$

so $S \le s + s' - ss'$ and $(1-s)(1-s') \le 1 - S$; in particular $S < 1$ and $4\rho < 1 - S$, so $0 < z < 1$. By Lemma B.1(4), $\omega(s) + \omega(s') \le \omega(S)$, and therefore

$$E \le 1 - \rho + \log(4\rho) + \omega(S) = 1 - S - \rho + \log\frac{4\rho}{1-S} = (1-S)\Big(1 - \frac{z}{4}\Big) + \log z.$$

The last inequality of the lemma is the square of $E \ge \frac{3}{5}\sqrt{uv}$. □

Lemma B.4. *Let $S \ge 0$, $0 < z \le 1$ and $E > 0$ satisfy $E \le (1-S)(1 - z/4) + \log z$ and $E^2 \ge \frac{9}{25}Q$, where $Q = 3(3+S^2)/16$. Then*

$$16K_*\Big(1 - \frac{(1-S)z}{4}\Big)^2 < 4Q - E^2.$$

Proof. The hypotheses give $E \ge \frac{3}{5}\cdot\frac{3}{4} = \frac{9}{20}$.

The domain. By $\log z \le z-1$, $E \le -S + z(3+S)/4 \le \frac{3}{4}(1-S)$, so $S \le 2/5$. If $z < 31/50$, Lemma B.1(1) at $z_0 = 17/25$ with (B.1) gives $E \le -\frac{10712}{27783} + z(\frac{25}{17} - \frac{1}{4}) < \frac{9}{20}$, a contradiction. So $z \in [31/50, 1]$ and $S \in [0, 2/5]$.

A supporting function on each interval. Cover $[31/50, 1]$ by the four intervals of Table 5. On an interval $[z_-, z_+]$ with tangent point z_k, bound ℓ_k and multiplier $\mu = \mu(S) \ge 0$ from the table, let $T(z) = (1-S)(1-z/4) + \ell_k + z/z_k - 1$, so that $0 < E \le T(z)$ by Lemma B.1(1). Since $E^2 - \frac{9}{25}Q \ge 0$,

$$\begin{aligned} 4Q - E^2 &\ge 4Q - E^2 - \mu(E^2 - \tfrac{9}{25}Q) \ge (4 + \tfrac{9}{25}\mu)Q - (1+\mu)T(z)^2 \\ &= G(z) + 16K_*\Big(1 - \frac{(1-S)z}{4}\Big)^2, \end{aligned}$$

where the last equality defines G. It suffices to show $G(z) > 0$ on the interval.

Concavity. G is a constant minus two squares of affine functions of z, and

$$\begin{aligned} (z_+ - z_-)G(z) &= (z_+ - z)G(z_-) + (z - z_-)G(z_+) \\ &\quad + (z_+ - z_-)\Big[(1+\mu)\Big(\frac{1}{z_k} - \frac{1-S}{4}\Big)^2 + K_*(1-S)^2\Big](z - z_-)(z_+ - z). \end{aligned}$$

So $G > 0$ on $[z_-, z_+]$ once $G(z_-) > 0$ and $G(z_+) > 0$.

The eight endpoint values. Each $G(z_\pm)$ is a polynomial $c_0 + c_1S + c_2S^2 + c_3S^3$ in $S \in [0, 2/5]$ with rational coefficients and $c_3 \le 0$. On $[0, 2/5]$, $c_3S^3 \ge \frac{2}{5}c_3S^2$, so $G(z_\pm) \ge c_0 + c_1S + c_2'S^2$ with $c_2' = c_2 + \frac{2}{5}c_3$. Replacing c_0, c_1, c_2' by the smaller numbers $\underline{c}_0$, $\underline{c}_1$, $\underline{c}_2$ of Table 6 only lowers this quadratic for $S \ge 0$, and each of the eight quadratics $\underline{c}_0 + \underline{c}_1S + \underline{c}_2S^2$ is positive on $[0, 2/5]$ by the reason given in the table: all three coefficients are nonnegative and $\underline{c}_0 > 0$; or $\underline{c}_2 > 0$ and the discriminant $\underline{c}_1^2 - 4\underline{c}_0\underline{c}_2$ is negative; or $\underline{c}_1 \ge 0$, $\underline{c}_2 < 0$ and the value $\underline{c}_0 + \frac{4}{25}\underline{c}_2$ at $S = 2/5$ is positive. □

TABLE 5. The tangent lines and multipliers of Lemma B.4.

interval $[z_-, z_+]$	tangent point z_k	bound $\ell_k \ge \log z_k$	multiplier $\mu(S)$
[31/50, 7/10]	17/25	−10712/27783	$41/200 + 9S/20$
[7/10, 4/5]	3/4	−296/1029	$3/50 + 3S/5$
[4/5, 9/10]	17/20	−8232/50653	$1/100 + 3S/20$
[9/10, 1]	1	0	0

Proof of Proposition 6.4. Suppose $E \ge \frac{3}{5}\sqrt{uv}$. Lemma B.3 puts (S, z, E) in the hypotheses of Lemma B.4, whose conclusion, with $Q = uv$ and $(1-S)z/4 = \rho$, reads $16K_*(1-\rho)^2 < 4uv - E^2 = 16(1-\rho)^2\gamma^2$. This contradicts $\gamma^2 \le K_*$. □

TABLE 6. Lower bounds for the coefficients of the quadratic minorants of $G(z_\pm)$ in the proof of Lemma B.4, and the reason why each minorant is positive on $[0, 2/5]$.

interval	end	$\underline{c}_0$	$\underline{c}_1$	$\underline{c}_2$	positivity
[31/50, 7/10]	31/50	0.0105	0.0091	−0.0018	value at 2/5
	7/10	0.0105	0.0691	0.0907	coefficients
[7/10, 4/5]	7/10	0.0112	−0.0435	0.2606	discriminant
	4/5	0.0110	−0.0456	0.3756	discriminant
[4/5, 9/10]	4/5	0.0181	−0.0326	0.0903	discriminant
	9/10	0.0178	−0.0193	0.1181	discriminant
[9/10, 1]	9/10	0.0153	0.0132	−0.0006	value at 2/5
	1	0.0213	0.0142	0.0023	coefficients

The margin is small: on the region of Figure 8 the least Casimir is about 0.1863, against $K_* \approx 0.18513$. This is why the logarithm is replaced by four tangent lines and not by one.

B.4. **The two-centre arithmetic.**

Lemma B.5. *If $0 < a < b$, $a < c$ and $(b-a)(c-a) = 2a^2$, then $bc > 5.828\,a^2$ and $(a+b)(a+c) = 2bc$.*

Proof. With $x = b/a$ and $y = c/a$, $(x-1)(y-1) = 2$, so $xy = x + y + 1$ and, by the inequality between the arithmetic and geometric means, $x + y - 2 \geq 2\sqrt{(x-1)(y-1)} = 2\sqrt{2}$. Hence $xy \geq 3 + 2\sqrt{2} = 5.8284\ldots > 5.828$. Expanding, $(a+b)(a+c) = a^2 + a(b+c) + bc$, and $bc - a(b+c) - a^2 = (b-a)(c-a) - 2a^2 = 0$. □

APPENDIX C. ESTIMATES FOR CLASS F

This appendix proves the scalar inequalities of Section 7, in its notation: $\nu = 4/a^2$, the ratios $z, w, \hat{p}, \hat{d} \geq 1$, $\hat{s}_\pm$, $\hat{K}$, $\hat{K}_\pm$, the factors h, q, k, the row coefficients A, B, η, n_1, n_2 and the floor $F_x(r)$ of Proposition 7.5. Throughout,

$$f(r) = \frac{r - r^{-1}}{2} - \log r, \qquad \beta(y) = \frac{y^2 + y + 1}{(1+y)^2} = 1 - \frac{1}{1+y} + \frac{1}{(1+y)^2},$$

so that $\beta(w)$ is the value $1 - x + x^2$ of the floor F_x at $r = 1$ for $x = w/(1+w)$.

C.1. **The edge function.**

Let G be the edge function of (7.1). Its partial derivatives are

$$\partial_b G(a,b) = 3a - 2b\log(a+b) - \frac{a^2+b^2}{a+b} + b\log(b^2+4) + \frac{b^3}{b^2+4},$$

$$\partial_a\partial_b G(a,b) = \frac{2(a^2+ab+b^2)}{(a+b)^2}, \qquad \partial_a\partial_b^2 G(a,b) = \frac{2a(b-a)}{(a+b)^3}.$$

(1) (Monotonicity) The bound $\log y \geq 1 - 1/y$ at $y = (b^2+4)/(a+b)^2$ gives $\partial_b G(a,b) \geq a(a+2b)(b^2-ab+8)/\big((a+b)(b^2+4)\big)$, which is positive when $ab \leq 8$. So G increases in each argument while the product of its arguments is at most 8. All products of consecutive active margins are below 4.

(2) (Rectangle inequality) Since $\partial_b G(x,\cdot) - \partial_b G(a,\cdot) \geq 0$ for $x \geq a$, integrating in the second variable gives $G(x,y) \geq G(x,a) + G(a,y) - G(a,a)$ for $x, y \geq a > 0$.

(3) (Concavity) If $a^2 \geq 4/3$, then $G(a,\cdot)$ is concave on $[a, 4/a]$. Since $\partial_a \partial_b^2 G \geq 0$ for $a \leq b$, it suffices to check $\partial_b^2 G < 0$ at $a = b$ when $4/3 \leq b^2 \leq 4$, and at $a = 4/b$. With $\varsigma = 4b^2/(b^2+4) \in [1,2]$ one has $\partial_b^2 G(b,b) = -\log\varsigma - \frac{3}{2} + \frac{5}{4}\varsigma - \frac{1}{8}\varsigma^2$, which increases in ς (its derivative is $(\varsigma-1)(4-\varsigma)/(4\varsigma)$) up to the value $\frac{1}{2} - \log 2 < 0$ at $\varsigma = 2$; and $\partial_b^2 G(4/b, b) = \log\varrho + 1 - \varrho < 0$ with $\varrho = b^2/(b^2+4)$.

(4) (Scaling) $G(ax, ay) = a^2 g_\nu(x,y)$ with $g_\nu(x,y) = 3xy - \nu - (x^2+y^2)\log(x+y) + \frac{1}{2}x^2\log(x^2+\nu) + \frac{1}{2}y^2\log(y^2+\nu)$; we write $g_\nu(x) = g_\nu(1,x)$. In particular $g_\nu(1) = 3 - \nu - \log\frac{4}{1+\nu} \geq \nu(3-\nu)/(1+\nu) \geq 0$ for $0 < \nu \leq 3$, by $\log y \geq 1 - 1/y$ again; so $G(a,a) \geq 0$ when $a^2 \geq 4/3$.

C.2. **Logarithm bounds.** All logarithms are bounded by rational functions through the following elementary inequalities, each proved from the sign of a derivative. For $v \geq 1$ and $\tau = (v-1)/(v+1)$,

$$2\tau + \tfrac{2}{3}\tau^3 \leq \log v \leq 2\tau + \frac{2\tau^3}{3(1-\tau^2)},$$

and for $h \geq 1$, $0 \leq y < 1$,

$$\frac{(h-1)^3}{3h(h+1)} \leq f(h) \leq \frac{(h-1)^3}{6}, \qquad \log(1-y) \leq -y - \frac{y^2}{2} - \frac{y^3}{3}.$$

The lower bound for f holds because the difference vanishes at $h = 1$ and has derivative $(h-1)^4/(6h^2(h+1)^2)$; it is the Padé bound, used for every $\log h$ below and, in the form $\log v \leq \frac{v - v^{-1}}{2} - \frac{(v-1)^3}{3v(v+1)}$, for other logarithms. The upper bound for $\log v$ at $v = 5/4$ gives $\log\frac{5}{4} < \frac{9}{40}$.

C.3. **The threshold and the budget.** *The threshold.* With $a_0^2 = 20/11$ and $\nu_0 = 11/5$, $g_{\nu_0}(1) = 3 - \frac{11}{5} - \log\frac{20}{16} = \frac{4}{5} - \log\frac{5}{4}$, so $G(a_0,a_0) = \frac{20}{11}(\frac{4}{5} - \log\frac{5}{4}) > \frac{20}{11}\cdot\frac{23}{40} = \frac{23}{22}$. With $\sqrt{3} > 1.732$ and $A_{\mathrm{oct}} < 3.1268$ this gives $\frac{7\sqrt{3}}{4}G(a_0,a_0) > A_{\mathrm{oct}}$, which proves Lemma 7.9.

The budget. By the rectangle inequality the seven-entry bound becomes $\frac{\sqrt{3}}{\nu}(2g_\nu(\hat{p}) + 2g_\nu(z) + 2g_\nu(w) + 2g_\nu(\hat{d}) - g_\nu(1)) \leq A_{\mathrm{oct}}$ (Section 7, Step 3). The four ratios lie in $[1,\nu]$, where $x \mapsto g_\nu(x)$ is increasing and concave by

Appendix C.1. For an increasing concave function ϕ on $[1, M]$ and four points of $[1, M]$ with sum at least σ, where $1 < \sigma - 3 \le M$,

$$\phi(x_1) + \phi(x_2) + \phi(x_3) + \phi(x_4) \ge 3\phi(1) + \phi(\sigma - 3) :$$

lower the points to sum exactly σ, and use that a concave function lies above the chord from 1 to $\sigma - 3$. If $\hat{p} + z + w + \hat{d} \ge (5\nu + 1)/3$, this with $x_* = (5\nu - 8)/3 \in (1, \nu]$ bounds the bracket below by $5g_\nu(1) + 2g_\nu(x_*)$. Write

$$g_\nu(x) = 3x - \nu - \tfrac{1}{2} \log \frac{(1+x)^2}{1+\nu} - \tfrac{x^2}{2} \log \frac{(1+x)^2}{x^2+\nu},$$

and bound both logarithms above by $L_+(N, M) = 2\frac{N-M}{N+M} + \frac{(N-M)^3}{6NM(N+M)}$, the upper bound of Appendix C.2 at $v = N/M \ge 1$. This gives a rational lower bound $\bar{g}_\nu(x)$, and the remaining claim is

$$5\bar{g}_\nu(1) + 2\bar{g}_\nu(x_*) > \frac{2501}{1384}\,\nu \qquad (11/5 \le \nu \le 3).$$

After multiplication by the positive denominator $934200(\nu - 1)^2(\nu + 1)(\nu + 5)(25\nu^2 - 71\nu + 64)(25\nu^2 - 41\nu + 34)(50\nu^2 - 121\nu + 89)$, this is the positivity on $[11/5, 3]$ of a polynomial of degree 11 in ν, which we verify by expanding it in the Bernstein basis of $[11/5, 3]$: all twelve coefficients are positive (this is the argument of `GLOBAL_FORWARD_LINEAR_BUDGET` in [7]). Since $\sqrt{3} > 173/100$ and $\frac{173}{100} \cdot \frac{2501}{1384} = \frac{2501}{800} > A_{\mathrm{oct}}$, the cost would exceed A_{oct}. This proves Lemma 7.10.

C.4. **The centered system.** The following identities hold for $\nu, z, w, h > 0$, $h \ne 1$; each is an equality of rational expressions in the variables and the logarithms.

(1) (Floors) $F_{z/\hat{s}_-}(h) = \frac{3}{4} + L(z, h)$ with

$$L(z, h) = \frac{(z-1)(z-1+2z(h-1))}{4\hat{s}_-^2} + \frac{zh}{\hat{s}_-} \cdot \frac{f(h)}{h-1},$$

and $L(z, h) \ge 0$ for $z \ge 1$, $h > 1$.

(2) (Central row) $\frac{3}{4}(\hat{s}_- + \hat{s}_+) - A = e_0 - \varphi$.

(3) (Left row) With $\zeta = z - 1$ and $H = h - 1$, the quantity $r = \eta A - B - \frac{3}{4}(n_1 + n_2)$ of (7.3) is given by

$$\begin{aligned} 8\hat{s}_-\hat{K}_-\, r = {} & (3-\nu)\zeta(3-\nu+2\zeta) \\ & + H\big((3-\nu)^2(\zeta^2+4\zeta+2) + 8(3-\nu)\zeta z\big) + 2H\zeta^2(8+4\zeta-\zeta^2) \\ & + \frac{4\hat{s}_- H^2 z(3-\nu+4\zeta+\zeta^2)}{h} \\ & + 8\hat{s}_-\hat{K}(1+\hat{s}_- H)\,\varphi - 8\hat{s}_-\hat{K}_-\, z f(h). \end{aligned}$$

It depends on w only through φ.

(4) (Coefficients) $\hat{s}_-\hat{K}_-(n_2-n_1) = \hat{K}(\hat{s}_-\hat{s}_+ - \hat{K}) + H\big(\hat{s}_-^4 - \hat{K}^2 + \hat{K}\hat{s}_-^2(\hat{s}_+ - \hat{s}_-)\big)$, which is positive for $\nu < 3$, $1 \le z \le w$, $h \ge 1$, because $\hat{s}_-\hat{s}_+ \ge 4 > \hat{K}$ and $\hat{s}_-^4 \ge 16 > \hat{K}^2$.

Two monotonicity facts complete the setup. The floor increases in the factor: for $x \ge 1/2$, $\partial_r F_x(r) = x^2 - x\,(r - 1 - \log r)/(r-1)^2 \ge x^2 - x/2 \ge 0$, since $r - 1 - \log r \le (r-1)^2/2$. And in the variable w, with ν, z, h fixed,

$$n_2(w) - n_2(z) = \eta\,(w - z), \qquad A(w) - A(z) \le (w - z)\,\beta(w) \qquad (z \le w),$$

because $\partial_w A = \beta(w)$ and $\beta'(y) = (y-1)/(1+y)^3 \ge 0$ for $y \ge 1$.

C.5. **The three residual signs.** *Lowering the larger neighbour.* The residuals of Section 7 depend on w through A, n_2, φ and the opposite floor. For $\mathcal{E}_v = r - n_1 L(z,h) - n_2/64$ the increments above give $\mathcal{E}_v(w) - \mathcal{E}_v(z) \le \eta(w-z)\big(\beta(w) - \frac{49}{64}\big) \le 0$, since $\beta(w) \le \beta(4/3) = 37/49 < 49/64$. For $\mathcal{E}_\Sigma = r - n_1/35 - (n_2 - n_1)L(w, \hat{s}_+^2/\hat{K}_+)$ put $V(w) = \frac{3}{4} + L(w, \hat{s}_+^2/\hat{K}_+)$, the floor at the least admissible factor; then $V(w) \ge \beta(w)$, and V is nondecreasing in w (the coefficients of L increase in w, $hf(h)/(h-1)$ increases in h, and $\hat{s}_+^2/\hat{K}_+$ increases in w), so

$$\mathcal{E}_\Sigma(w) - \mathcal{E}_\Sigma(z) \le \eta(w-z)\big(\beta(w) - V(w)\big) - (n_2(z) - n_1)\big(V(w) - V(z)\big) \le 0.$$

The neighbour residual is lowered in the same way to $w = \max(z, 4/3)$. After lowering, $w = z$ for $\mathcal{E}_\Sigma$ and $\mathcal{E}_v$, and the factor ranges over the symmetric domain

$$1 \le z \le \tfrac{4}{3}, \qquad 1 + \tfrac{6}{5}z \le \nu < 3, \qquad \frac{(1+z)^2}{z^2+\nu} \le h \le \frac{(1+z)(5\nu - 2 - 3z)}{3(z^2+\nu)}; \tag{C.1}$$

the lower bound for ν is exactly the condition that the interval for h be nonempty.

Concavity in the factor. As functions of h, with the other variables fixed, the three residuals have the form $c_0 + c_1 h + c_2 h^2 + c_{-1}/h + (d_0 + d_1 h)\log h$, with second derivative $(2c_2 h^3 + 2c_{-1} + d_1 h^2 - d_0 h)/h^3$. For $\mathcal{E}_v$ and $\mathcal{E}_N$ it equals $-zP(h)/(\hat{K}_- h^3)$ with

$$P(h) = (\nu - z)h^2(2zh - \hat{s}_-) + \hat{K}_- h - 2z\hat{s}_-.$$

For $h \ge 1$, $P'(h) = 2(\nu - z)h(3zh - \hat{s}_-) + \hat{K}_- > 0$, and at the least factor $H_0 = \hat{s}_-^2/\hat{K}_-$,

$$P(H_0) = -\frac{\hat{s}_-(2z + 1 - \nu)}{\hat{K}_-^3}\,Q(\nu, z),$$

$$Q(\nu, z) = (1-z)\nu^2 - z(z^3 + 7z^2 + 5z + 3)\nu + z^2(z^4 + 2z^3 + 6z^2 + 5z + 2).$$

On the domains of these residuals, $1 \le z \le 5/3$, $\nu < 3 \le 2z + 1$ and $\nu \ge 1 + 6z/5$. The polynomial Q decreases in ν, and at $\nu = 1 + 6z/5$, with

$x = z - 1 \in [0, \frac{2}{3}]$,

$Q = x^6 + \frac{34}{5}x^5 + \frac{78}{5}x^4 + \frac{124}{25}x^3 - \frac{832}{25}x^2 - \frac{1216}{25}x - \frac{96}{5} \le -\frac{42176}{2025}x^2 - \frac{1216}{25}x - \frac{96}{5} < 0$,

bounding the four positive terms by $((\frac{2}{3})^4 + \frac{34}{5}(\frac{2}{3})^3 + \frac{78}{5}(\frac{2}{3})^2 + \frac{124}{25}\cdot\frac{2}{3})x^2$. So $P \ge 0$ on $[H_0, \infty)$, and $\mathcal{E}_v, \mathcal{E}_N$ are concave in h there. A concave function on an interval lies below its tangent at an endpoint, and the tangent is affine, so the residual is negative on the interval as soon as the tangent is negative at both endpoints. This leaves two values for each residual: the value at one endpoint and the tangent's value at the other. The sum residual is treated in the variable $\tau = (h-1)/h$ instead (below).

The endpoint values. On (C.1) we use the coordinates

$$\alpha = \frac{z-1}{z+1} \in [0, \tfrac{1}{7}], \qquad t = \frac{(1+z)^2}{z^2+\nu} - 1 \in (0, \tfrac{1}{4}],$$

in which $\hat{K}/\hat{K}_- = 1 - \alpha(1+t)$ and $\hat{s}_-^2/\hat{K}_- = 1 + t$; the upper bound for t follows from $5(z^2 + 1 + \frac{6}{5}z) - 4(z+1)^2 = (z-1)^2$. The endpoint values involve, besides rational functions of (α, t), the two logarithms $\log(1+t)$ and $\log(1-\alpha(1+t))$, each with a coefficient that is a polynomial in (α, t) divided by a positive one.

- $\mathcal{E}_v < 0$: the tangent is taken at $h = 1 + t$, so the two values are $\mathcal{E}_v(1+t)$ and the tangent's value at the cost endpoint.
- $\mathcal{E}_\Sigma < 0$: here $\mathcal{E}_\Sigma = A_0 + A_1 H + z(DH^2/(2h) - f(h))$ with $H = h - 1$, $D = \alpha + t + \alpha t$ and coefficients A_0, A_1 free of h. The Padé bound for $f(h)$ gives $\mathcal{E}_\Sigma/H \le \Psi(\tau)$ with $\tau = H/h$, where $\Psi(\tau) = A_0(1/\tau - 1) + A_1 + z(D\tau/2 - \tau^2/(3(2-\tau)))$ is concave once $A_0 \le 0$. The tangent at the largest admissible value $\tau_c = \frac{2-2\alpha^2-t(5+3\alpha+2\alpha^2)}{5-3\alpha-2\alpha^2-t(5+3\alpha+2\alpha^2)}$ gives the two values $\Psi(\tau_c)$ and $\Psi(\tau_c) + \Psi'(\tau_c)(\tau_I - \tau_c)$, where $\tau_I = t/(1+t)$ is the least value. The domain also satisfies $t \ge 39\alpha/40$.
- $\mathcal{E}_N < 0$ for $w \ge 4/3$: after lowering w to $\max(z, 4/3)$ there are two faces. On the symmetric face $w = z \in [\frac{4}{3}, \frac{5}{3}]$ one has $\alpha \in [\frac{1}{7}, \frac{1}{4}]$ and $t \in [\frac{13}{100}, \frac{1}{4}]$, with the same two logarithms. On the asymmetric face $w = 4/3 \ge z$ the budget gives $\nu \ge \frac{9}{5} + \frac{3}{5}z \ge \frac{12}{5}$, the variables are $(\nu, z) \in [\frac{12}{5}, 3] \times [1, \frac{4}{3}]$, and the logarithms are $\log(\hat{s}_-^2/\hat{K}_-)$, $\log q$ and $\log(\hat{s}_+^2/\hat{K}_+)$. On each face the tangent is taken at $h = \hat{s}_-^2/\hat{K}_-$.

Each logarithm is replaced by its Padé upper bound, by the cubic bound for $\log(1-y)$, or by the two-term lower bound of Appendix C.2, according to the sign of its coefficient. Each value then becomes a rational function with positive denominator, and the claim is that its numerator, a polynomial in two variables of degree at most 14, is negative. The polynomial signs used are: for $\mathcal{E}_v$, the two logarithm coefficients and the two numerators on $[0, \frac{1}{7}] \times [0, \frac{1}{4}]$; for $\mathcal{E}_\Sigma$, the coefficients and numerators of A_0 (which is nonpositive) and of the first value on $[0, \frac{1}{7}] \times [0, \frac{1}{4}]$, and those of the second value on the boxes $[0, \frac{1}{10}] \times [0, \frac{1}{4}]$ and $[\frac{1}{10}, \frac{1}{7}] \times [\frac{39}{400}, \frac{1}{4}]$, which cover the part

of the domain with $t \geq 39\alpha/40$; for $\mathcal{E}_N$, the coefficients and numerators of both values on $[\frac{1}{7}, \frac{1}{4}] \times [\frac{13}{100}, \frac{1}{4}]$ (symmetric face) and on $[\frac{12}{5}, 3] \times [1, \frac{4}{3}]$ (asymmetric face). We verify each of these polynomial signs by expanding the polynomial in the Bernstein basis of its box: all coefficients have the required sign. Where a numerator vanishes at a corner of its box and the inequality must be strict, the expansion is applied to the numerator plus a small linear form that is positive away from that corner. Another proof of the three signs, by displayed bounds on the coefficients of the numerators in one variable at a time, an integral bound for the neighbour residual and a split of the sum residual at $D = 38/243$, is given in `HAND_FORWARD_LARGER_NORM_HAND_PROOF`, `HAND_FORWARD_SYMMETRIC_SUM_REDUCTION`, `HAND_FORWARD_SYMMETRIC_SUM_HIGH_D`, `HAND_FORWARD_NEIGHBOR_INTEGRAL` and `HAND_FORWARD_ASYMMETRIC_NEIGHBOR_BOUND` of [7].

C.6. **The conditional Casimir bound.** Assume the hypotheses of Lemma 7.13. Then $0 \leq \varphi \leq 1/100$ and $\frac{3}{4} \leq e_0 \leq \frac{73}{80}$, so $0 \leq E \leq e_0 + \hat{s}_-\delta u + \hat{s}_+\delta v$. The Casimir is $\hat{K}^2(4uv - E^2)/(16\nu^2)$ and decreases in $E \geq 0$, so it is at least $\hat{K}^2 Q(\delta u, \delta v)/(16\nu^2)$ with

$$Q(x, y) = 4\Big(\frac{3}{4} + x\Big)\Big(\frac{3}{4} + y\Big) - \big(e_0 + \hat{s}_- x + \hat{s}_+ y\big)^2.$$

Write $c = 1/35$ and $b = 1/64$. The function Q does not increase in x along the norm polygon: for $x \leq c - y$,

$$Q(c-y, y) - Q(x, y) = (c-y-x)\big(3 + 4y - \hat{s}_-(2e_0 + \hat{s}_-(x+c-y) + 2\hat{s}_+ y)\big) \leq 0,$$

since $2\hat{s}_- e_0 \geq 3$ and $\hat{s}_-\hat{s}_+ \geq 4$. Along the edge $x + y = c$ it is concave, by the exact identity

$$Q(c-y, y) - \big((1 - 64y)\, Q(c, 0) + 64y\, Q(c-b, b)\big) = \big(4 + (\hat{s}_+ - \hat{s}_-)^2\big)\, y\, (b - y),$$

which is nonnegative for $0 \leq y \leq b$. So it suffices to bound $\hat{K}^2 Q/(16\nu^2)$ at the two vertices $A = (c, 0)$ and $B = (c - b, b)$. The ordering and the budget give $\hat{s}_- \leq \min(\frac{7}{3}, \frac{5\nu+1}{6})$ and $\hat{s}_+ \leq \frac{7}{3}$, and $\hat{K} e_0 = \hat{K} - \hat{K}^2/8 + 1$. With these, the two vertex values are bounded below by explicit functions of ν alone:

$$\phi_A(\nu) = \frac{\hat{K}^2}{16\nu^2}\Big(\frac{327}{140} - \big(e_0 + \tfrac{1}{35}\min(\tfrac{7}{3}, \tfrac{5\nu+1}{6})\big)^2\Big),$$

and ϕ_B, defined in the same way at B with $\hat{s}_-(c - b) + \hat{s}_+ b$ bounded by $(c-b)(\hat{s}_- + \hat{s}_+) + (2b - c)\hat{s}_+ \leq (c-b)\big(2 + \frac{5}{3}(\nu - 1)\big) + \frac{7}{3}(2b - c)$ for $\nu \leq 13/5$ and by $\frac{7}{3}c$ for $\nu \geq 13/5$. Multiplied by $16\nu^2$, each of the inequalities $\phi_A > 23/125$ and $\phi_B > 23/125$, on $[\frac{11}{5}, \frac{13}{5}]$ and on $[\frac{13}{5}, 3]$, is the positivity of a quartic polynomial. In the variable $y \in [0, 1]$ of the interval ($\nu = \frac{11}{5} + \frac{2}{5}y$, resp.

$\nu = \frac{13}{5} + \frac{2}{5}y$) the four quartics are

$$\begin{aligned}
&\tfrac{6283}{153125} - \tfrac{92}{3125}y - \tfrac{20162}{1378125}y^2 + \tfrac{34}{7875}y^3 - \tfrac{289}{1102500}y^4 && (\phi_A,\ [\tfrac{11}{5}, \tfrac{13}{5}]),\\
&\tfrac{89}{87500} + \tfrac{11419}{65625}y + \tfrac{10643}{393750}y^2 + \tfrac{1}{375}y^3 - \tfrac{1}{2500}y^4 && (\phi_A,\ [\tfrac{13}{5}, 3]),\\
&\tfrac{9671}{306250} - \tfrac{827}{65625}y - \tfrac{2033761}{176400000}y^2 + \tfrac{21823}{5040000}y^3 - \tfrac{19321}{70560000}y^4 && (\phi_B,\ [\tfrac{11}{5}, \tfrac{13}{5}]),\\
&\tfrac{9203}{800000} + \tfrac{211601}{1200000}y + \tfrac{195547}{7200000}y^2 + \tfrac{1}{375}y^3 - \tfrac{1}{2500}y^4 && (\phi_B,\ [\tfrac{13}{5}, 3]).
\end{aligned}$$

On $[\frac{13}{5}, 3]$ the only negative term $-y^4/2500$ is dominated by $y^3/375$. On $[\frac{11}{5}, \frac{13}{5}]$ the derivative is at most the linear coefficient plus three times the cubic coefficient, which is negative, so the quartic decreases and its minimum is its value at $y = 1$, namely $\frac{89}{87500}$ for ϕ_A and $\frac{9203}{800000}$ for ϕ_B. The tightest case is ϕ_A at $\nu = 13/5$:

$$\phi_A(\tfrac{13}{5}) = \frac{348293}{1892800} = \frac{23}{125} + \frac{89}{9464000}.$$

Another argument, which keeps the exact offset $e_0 - \varphi$ at the vertex B and minimizes ϕ_A by differentiation, is in `HAND_FORWARD_CONDITIONAL_NORM_BOUNDS`, Section 2, of [7].

C.7. **The small forward minimum.** Suppose $\nu \geq 3$ and, after the exchange of sides, $z \leq w$. The left row and the floors give $\mathcal{E}(h) \geq 0$ for the residual

$$\mathcal{E}(h) = \eta A - B - n_1 F_{z/\hat{s}_-}(h) - n_2 V, \qquad V = F_{w/\hat{s}_+}\Big(\max\Big\{1, \frac{\hat{s}_+^2}{\hat{K}_+}\Big\}\Big),$$

the floor for v at the least factor that minimality allows. Lowering w does not decrease $\mathcal{E}$, by the increments of Appendix C.4 and $V \geq \beta(w)$, as long as $q \geq 1$. Lower w until either $q = 1$ or $w = z$. Three faces remain:

(1) $q = 1$, that is, $w = \hat{K}/\hat{s}_- - 1$ (possible when $\hat{s}_-^2 \leq \hat{K}$);

(2) $w = z$ with $\hat{s}_-^2 \geq \hat{K}$ and $\hat{s}_-^2 \leq \hat{K}_-$;

(3) $w = z$ with $\hat{s}_-^2 \geq \hat{K}$ and $\hat{s}_-^2 > \hat{K}_-$, the hard face.

On each face $\mathcal{E}$ has the form in h of Appendix C.5, with second derivative $-zP(h)/(\hat{K}_- h^3)$ and the same P. Since $P(1) = -z\hat{K}_- d$ with $d = \hat{s}_-^2/\hat{K}_- - 1$ and $P'(h) \geq \hat{K}_-$ for $h \geq 1$, integrating twice from $h = 1$ with $\int_1^h (h - r)r^{-3}\,\mathrm{d}r = (h-1)^2/(2h)$ and $\int_1^h (h-r)(r-1)r^{-3}\,\mathrm{d}r = f(h)$ gives

$$\mathcal{E}(h) \leq \mathcal{E}(1) + (h-1)\,\mathcal{E}'(1) + \frac{z^2 d\,(h-1)^2}{2h} - z\,f(h). \tag{C.2}$$

The value and slope at $h = 1$ are explicit: with $\psi(q) = (q-1)/\hat{s}_+ + \frac{w}{\hat{s}_+}(1 - q^{-1}) - \log q$,

$$\begin{aligned}
\mathcal{E}(1) &= \frac{\hat{K}}{\hat{K}_-}\,\psi(q) - \frac{\hat{K}^2(q-1)}{\hat{s}_-\hat{K}_-}(V - \beta(w)),\\
\mathcal{E}'(1) &= \frac{\hat{s}_-\hat{K}}{\hat{K}_-}\Big(\psi(q) + \frac{z(q-1)(q-w)}{\hat{s}_-\hat{s}_+ q}\Big) - \frac{\hat{K}^2(\hat{s}_- q - 1)}{\hat{s}_-\hat{K}_-}(V - \beta(w)).
\end{aligned}$$

Since $\psi'(q) = (q-1)(q-w)/(\hat{s}_+ q^2)$, ψ first decreases and then increases on $[1,\infty)$.

(1) On the face $q=1$: $\psi(1)=0$, and $V=\beta(w)$ because $\hat{s}_+^2 \le \hat{K}_+$ there; so $\mathcal{E}(1)=\mathcal{E}'(1)=0$, $d\le 0$, and (C.2) gives $\mathcal{E}(h) \le -zf(h) < 0$.

(2) On the second face: $V=\beta(z)$, $d\le 0$, and $1\le q\le z$, so $\psi(q)\le 0$ and the extra slope term is nonpositive; again $\mathcal{E}(h)\le -zf(h)<0$.

(3) On the hard face, $\psi(q)\le 0$ still holds; by the shape of ψ it suffices to check the largest admissible q. If $z\le 3$, then $\nu\ge 3$ gives $q\le \hat{s}_-^2/4$, and $\psi(\hat{s}_-^2/4) = \frac{\hat{s}_-}{4} + 1 - \frac{2}{\hat{s}_-} - \frac{4}{\hat{s}_-^2} + \frac{4}{\hat{s}_-^3} - 2\log\frac{\hat{s}_-}{2}$ vanishes at $\hat{s}_- = 2$ and has derivative $(\hat{s}_- - 6)(\hat{s}_- - 2)^2(\hat{s}_- + 2)/(4\hat{s}_-^4) \le 0$ for $2\le \hat{s}_- \le 4$. If $z\ge 3$, then $\nu\ge z$ gives $q\le \hat{s}_-$, and $\psi(\hat{s}_-) = x + x^2 - \log \hat{s}_-$ with $x = z/\hat{s}_-$ decreases in $\hat{s}_-$ (its derivative is $-(\hat{s}_- - 1)(\hat{s}_- - 2)/\hat{s}_-^3$) from the value $\frac{21}{16} - \log 4 < 0$ at $\hat{s}_- = 4$. The opposite floor exceeds $\beta(z)$ by at least $\Delta_0 = x(x-\frac{1}{2})d + xd^2/(3(d+2))$, by the Padé bound, and $f(h) \ge (h-1)^3/(3h(h+1))$. Inserting these into (C.2) gives a rational upper bound $\mathcal{B}$ for $\mathcal{E}(h)$. Put $\mu = (2z+1-\nu)/\hat{s}_- \in (0,1)$, $\lambda = 1-\mu$ and $v = \hat{s}_-(h-1)/h$, so that $\mu \le v \le 1$ and $\hat{K}_- = \hat{s}_-(\hat{s}_- - \mu)$, $\hat{K} = \hat{s}_-(2-\mu)$, $q = \hat{s}_-/(2-\mu)$, $d = \mu/(\hat{s}_- - \mu)$. After the two weakenings $\Delta_0 \ge z\mu(3(z-1)+\mu)/(6\hat{s}_-^2(\hat{s}_- - \mu))$ and $(h-1)^3/(3h(h+1)) \ge (h-1)^3/(6h^2)$, an exact simplification gives $\mathcal{B} \le -zh\,P_5/(6\hat{s}_-\hat{K}_-^2)$ with the cubic

$$\begin{aligned} P_5 = {} & (z+\lambda)^2 v^3 - 3z\mu\hat{s}_-(z+\lambda)v^2 \\ & + \big[(1+\lambda)\mu\hat{s}_- z(3(z-1)+\mu) - 6(z+\lambda)(z-\lambda)(1-z\lambda)\big]v \\ & + (1+\lambda)\mu\hat{s}_-(3(z-1)+\mu)(z-\lambda). \end{aligned}$$

It remains to show $P_5 > 0$.

The cubic P_5. We follow `HAND_H5_INTEGRAL_RATIONAL_BOUND` of [7]. Put $\varpi = (2-\mu)z - (2+\mu) = \nu - 3 \ge 0$. Then $(2-\mu)^3 P_5 = R_0 + R_1\varpi + R_2\varpi^2 + R_3\varpi^3$ with

$$\begin{aligned} R_3 &= 3\big((2-\mu^2)v + \mu(2-\mu-v^2)\big), \\ R_2 &= (2-\mu)v^3 - 3\mu(\mu^2-\mu+10)v^2 \\ &\quad + (\mu^4 - 13\mu^3 - 14\mu^2 + 24\mu + 24)v + E_2, \\ R_1 &= 2(2-\mu)(\mu^2-2\mu+4)v^3 - 3\mu(\mu^3+4\mu^2-4\mu+32)v^2 \\ &\quad + \mu(7\mu^4 - 46\mu^3 + 40\mu^2 - 96\mu + 144)v + C_1, \\ R_0 &= (2-\mu)(\mu^2-2\mu+4)^2 v^3 - 12\mu(\mu^3+8)v^2 \\ &\quad + 2\mu^2(3\mu^4 - 16\mu^3 + 20\mu^2 - 56\mu + 64)v \\ &\quad + 4\mu^3(2-\mu)(4-\mu)(8-\mu), \end{aligned}$$

where $E_2 = 4\mu^4+28\mu^2(1-\mu)+24\mu$ and $C_1 = \mu^2(-\mu^4+14\mu^3-40\mu^2-48\mu+160)$; here $0 \le \mu \le v \le 1$. In R_1 and R_2 the cubic coefficient is nonnegative and the quadratic one nonpositive, so $v^2 \le v$ gives a bound linear in v:

- $R_3 \ge 0$, since $2-\mu-v^2 \ge 1-\mu \ge 0$;
- $R_2 \ge (1-v)E_2 + v(E_2+D_2)$ with $D_2 = \mu^4-16\mu^3-11\mu^2-6\mu+24$, and $E_2+D_2 = 5\mu^4-44\mu^3+17\mu^2+18\mu+24 \ge -27\mu^2+18\mu+24 = 15+9(1-\mu)+27\mu(1-\mu)$;
- $R_1 \ge (1-v)C_1 + v(C_1+D_1)$ with $D_1 = \mu(7\mu^4-49\mu^3+28\mu^2-84\mu+48)$, where $C_1 \ge 71\mu^2$ and $C_1+D_1 = \mu(-\mu^5+21\mu^4-89\mu^3-20\mu^2+76\mu+48) \ge \mu(48-34\mu) \ge 14\mu$.

For R_0 there are two cases. If $\mu \le 1/5$, its three v-coefficients are at least 23, at least -97μ and at least $105\mu^2$, and its constant term is nonnegative, so $R_0 \ge v(23(v-\frac{97\mu}{46})^2+\frac{251}{92}\mu^2) > 0$ for $v > 0$. If $\mu \ge 1/5$, write $v = \mu+(1-\mu)s$ with $s \in [0,1]$; then

$$R_0 = \mu^3 A_0(1-s)^3 + \mu^2 A_1 s(1-s)^2 + \mu^2 A_2 s^2(1-s) + Q_0 s^3$$

with

$$\begin{aligned} A_0 &= 320-384\mu+136\mu^2-68\mu^3+12\mu^4-\mu^5,\\ A_1 &= 9\mu^5-64\mu^4+132\mu^3-736\mu^2+672\mu+32,\\ A_2 &= 6\mu^5-35\mu^4+138\mu^3-776\mu^2+792\mu-80,\\ Q_0 &= 2\mu^6+23\mu^5-190\mu^4+124\mu^3+168\mu^2-144\mu+32. \end{aligned}$$

In $n = 1-\mu$, $A_0 = n^5+7n^4+30n^3-6n^2+273n+15 > 0$. The polynomials A_1, A_2 are concave on $[0,1]$ (their second derivatives are $4(45\mu^3-192\mu^2+198\mu-368) < 0$ and $4(30\mu^3-105\mu^2+207\mu-388) < 0$), with $A_1(0) = 32$, $A_1(1) = 45$, $A_2(\frac{1}{5}) = \frac{151281}{3125}$ and $A_2(1) = 45$. For $\mu \le \frac{1}{2}$, $-190\mu^4 \ge -95\mu^3$ gives $Q_0 \ge 29\mu^3+168(\mu-\frac{3}{7})^2+\frac{8}{7} > 0$; for $\mu \ge \frac{1}{2}$, with $n = 1-\mu \le \frac{1}{2}$ and $r = 2n$,

$$\begin{aligned} Q_0 &= 2n^6-35n^5-45n^4+366n^3-340n^2+69n+15\\ &\ge 15(1-r)^3+\tfrac{159}{2}r(1-r)^2+29r^2(1-r)+\tfrac{203}{32}r^3 > 0, \end{aligned}$$

using $n^5 \le n^3/4$ and $n^4 \le n^3/2$. So $R_0 > 0$, hence $P_5 > 0$, and $\mathcal{E}(h) < 0$ on every face, a contradiction; therefore $\nu < 3$. Another argument for the first two faces, by concavity in h alone, is in `MIXED_GLOBAL_SMALL_FORWARD_MINIMUM`, Sections 4–5, of [7].

Appendix D. Estimates for class M2: the nonshort cases

This appendix proves the scalar inequalities behind Section 8, in the notation of Table 2. Every step concerns explicit functions of at most five real variables. The analytic ingredients are a few rational bounds for logarithms; everything else is polynomial, and every polynomial sign is reduced, one variable at a time, to univariate inequalities on intervals. Several of the polynomials are long; they are determined by the displayed definitions, and

we state the identities and bounds on which each sign rests. The arguments follow [7] and, where indicated, the shorter arguments of [6].

D.1. **Secants and floors.** For $B > 0$ the function $g_B(x) = 1 - Bx/(B+x)^2$ decreases on $(0, B)$, increases on (B, ∞), and has minimum $g_B(B) = 3/4$; also $g_B(x) = U(B/(B+x)) = U(x/(B+x))$. The turn secant $\mathrm{Sec}(A, B, P) = (P - A)^{-1} \int_A^P g_B$ has the antiderivative $s - B\log(B + s) - B^2/(B + s)$.

Lemma D.1 (Secants)**.** *(a) If $0 < a \le c < d$, then $\mathrm{Sec}(a, c, d) < \mathrm{Sec}(a, d, c)$.*

(b) If $0 < c \le d$, then $z \mapsto \mathrm{Sec}(z, d, c)$ strictly decreases on $(0, c]$.

(c) For $P, C > 0$ and $q > 1$, $F_{P/(P+C)}(q) = \mathrm{Sec}((P + C)/q - P, P, C)$. In particular $F_{d/W}(q) = \mathrm{Sec}(B_0, d, c)$ with $B_0 = (4 - dr)/(d + r)$, the third margin at the switch $d \to r$.

(d) If $ad < c^2$, then $\mathrm{Sec}(a, c, d) < \frac{1}{2}(g_c(a) + g_c(d))$.

(e) If $B/(A + B) \ge 3/5$ and $A < P$, then $\mathrm{Sec}(A, B, P) > 94/125$.

Proof. (a) $\mathrm{Sec}(a, d, c) = \int_0^1 g_1(u)\, d\theta$ and $\mathrm{Sec}(a, c, d) = \int_0^1 g_1(v)\, d\theta$ with $u = (a + \theta(c - a))/d$ and $v = (a + \theta(d - a))/c$. Here $v > u$, $uv < 1$, and $g_1(u) - g_1(v) = (v - u)(1 - uv)/((1 + u)^2(1 + v)^2) > 0$. (b) The derivative is $(\mathrm{Sec}(z, d, c) - g_d(z))/(c - z)$, and the average of the decreasing function g_d over $[z, c]$ is below its value at z. (c) Substitute $(P + C)/q - P$ in the closed form of the secant; $W/q - d = Q/Z - d = B_0$. (d) For $a \le x \le d$,

$$\frac{(d-x)g_c(a) + (x-a)g_c(d)}{d-a} - g_c(x) = \frac{c(x-a)(d-x)\,[c^2(a+d+2c) + (c^2 - ad)x]}{(a+c)^2(d+c)^2(x+c)^2} \ge 0;$$

integrate over $[a, d]$. (e) With $\eta = B/(A + B)$ and $q = (P + B)/(A + B) > 1$, the secant is $1 + \eta^2/q - \eta \log q/(q - 1)$. Bound $\log q/(q - 1)$ above by a rational function of q as in Lemma D.2(b); the resulting rational lower bound exceeds $94/125$ for $\eta \ge 3/5$, by a discriminant in q. □

D.2. **Rational bounds for the logarithms.**

Lemma D.2 (Logarithms)**.** *Let $x > 0$, $q > 1$, $c = 1 - t$, $W = 1 + x$, $X = x + 2$, $p = t + x$, $g = 1 - cp/W^2$, and let $v = 1 + c^2/W - c \log W/x$ be the normalized incoming secant.*

(a) $\log(1 + x) > 2x/X + 2x^3/(3X^3)$; hence $v < \bar{v} = 1 + c^2/W - (2c/X)(1 + x^2/(3X^2))$ when $t < 1$.

(b) $\log q < L_+(q) = (q - 1)(q^2 + 10q + 1)/(6q(q + 1))$; hence $F_{p/W}(q) > F_\ell = 1 + p^2 q/W^2 - (p/W)(q^2 + 10q + 1)/(6(q + 1))$.

(c) $\log(1 + x) > x(43x^3 + 252x^2 + 390x + 180)/(12W^2(x^2 + 10x + 15))$; hence $v < v_G = g + (cx/W^2)(T_G(x) - t)$ with $T_G = (90 - 19x^2)/(12(x^2 + 10x + 15))$.

(d) $\log(1+x) > 3x(x+2)/(x^2+6x+6)$*; hence* $v < V = 1 + c^2/W - 3c(x+2)/(x^2+6x+6)$.

(e) $\bar{v} - g = ((1-t)x/W^2)(T(x)-t)$ *with* $T(x) = (-2x^3 - x^2 + 10x + 12)/(3(x+2)^3)$*; so* $v \geq g$ *forces* $t < T(x)$*. The function* T *decreases, and* $T(1) = 19/81 < 1/4$.

Proof. (a) Integrate $1/(1-z^2) > 1 + z^2$ from 0 to x/X. In (b), (c), (d) the difference vanishes at the left end and has derivative $(q-1)^4/(6q^2(q+1)^2)$, $x^6/(W^3(x^2+10x+15)^2)$ and $x^4/(W(x^2+6x+6)^2)$ respectively. (e) Direct subtraction; $T'(x) = -(11x^2+24x+16)/(3(x+2)^4)$. □

D.3. **The residual.** Put $c = 1-t$, $p = t+x$, $W = 1+x$, $Y = x(x+2)$, $B = k + 2tx + x^2$, $Z = qB/W$ (the normalized $d + r$), and

$$H_0 = \frac{k}{t}\,(v - 1 + t^2) + t(2-t), \qquad w_U = \frac{k^2 - kt + t^2 + H_0 Y + vY^2}{B^2},$$

the central diagonal and the transported outgoing norm with the incident norm at the boundary ceiling $U(t/k)$. The root function at the ceiling, divided by T, is

$$R = c + Z - \frac{p^2}{W} - \frac{p^2}{Z} + \frac{p^3}{B} - p\log q - Wv - Zw_U + \frac{p(H_0 + 2vY)}{B},$$

and $N = tBqW\,R$. Clearing denominators cancels every negative power, and N is a polynomial, affine in k, v and $L = \log q$:

- the coefficient of v is $N_v = -xq\,I$ with $I = k + tx + tW(k-2t) + (q-1)(kX + txX^2)$, nonpositive when $k \geq 2t$;
- the coefficient of L is $N_L = -tpqWB \leq 0$;
- the coefficient of k is

$$A = q\{x[(1-v)(1+tW) - t(1-t^2)] \\ + (q-1)[t^2(1+2x-x^2) + 2tx^2 + (1-v)Y] - tpWL\}.$$

In A the coefficient of v is $-qx[(q-1)(x+2) + 1 + tW] < 0$ and that of L is $-qtpW < 0$. Inserting $v \leq U(t) = 1 - t + t^2$ and $L < q - 1$ and collecting gives $A > q\,t\,x(1-t)[(q-1)(2x+1) + tx] > 0$, which is Lemma 8.10.

Lemma D.3 (Rational relaxation)**.** *If* $k \geq 2t$*,* $q > 1$ *and* $F_{p/W}(q) \leq v$*, then*

$$N(v, \log q) \;\geq\; N(\bar{v}, L_+) - (q-1)(\bar{v} - F_\ell),$$

and $P_k := 6(q+1)W^2X^3\{N(\bar{v}, L_+) - (q-1)(\bar{v} - F_\ell)\}$ *is a polynomial in* t*,* x*,* k *and* $y = q - 1$*, cubic in* y.

Proof. $N_v \leq 0$, $N_L \leq 0$, $v < \bar{v}$ and $\log q < L_+$, while $F_\ell < F_{p/W}(q) \leq v < \bar{v}$ makes the subtracted term nonnegative. □

D.4. **The necessary bounds.** We prove Lemma 8.11(iii)–(vii).

The scale. At the centre, (8.1) holds with $u_{\mathrm{i}} = u$, $u_{\mathrm{o}} = v$, $r = a/S$ and $t = a/T$. Lowering both norms to a floor μ and raising both ratios to a cap ζ gives $\gamma^2 \geq (4\mu^2 - B(\rho)^2)/(16(1-\rho)^2)$ with $B(\rho) = 1 - \rho + \log\rho + 2\psi_\mu(\zeta)$, $\psi_\mu(\zeta) = \zeta - \log\zeta - (1-\mu)/\zeta$. The cleared comparison $\varphi(\rho) = 4\mu^2 - B(\rho)^2 - 16\,\Gamma_{\max}(1-\rho)^2$, with $\Gamma_{\max} = 9193/50000$, has derivative $2(1-\rho)(16\Gamma_{\max} - B/\rho)$ and increases, so its positivity at the left end of an interval of ρ excludes that interval by Lemma 8.2(b). With the floor $3/4$ and the cap $1/2$ this gives $a^2 < 2/3$, that is $\rho < 1/7$. When $t \leq 2/5$, both ratios are at most $2/5$ and both turns have middle-margin ratio $B/(A+B) \geq 3/5$, so Lemma D.1(e) gives $\mu = 94/125$; the interval $[4/29, 3/20)$ is then excluded, and $a < 4/5$. In normalized form $a^2(k - t^2) = 4t^2$, so $a^2 < 2/3$ is $k > 7t^2$, and $a < 4/5$ is $k > 29t^2/4$.

Lemma D.4 (The ratio $t < 2/5$)**.** *Every window of Proposition 8.7 has* $t < 2/5$.

Proof. Suppose $t \geq 2/5$. Then $k > 7t^2 \geq 28/25$. By Lemma D.2(e), $t < T(x)$, and the numerator $12 - 22x - 41x^2 - 16x^3$ of $T(x) - 2/5$ is negative from $x = 1/3$ on, so $x < 1/3$. The centre gap (below) gives $x^2 + 2x > 7t^2 - 2t \geq 8/25$, so $x > 1/7$. Lower k to $7t^2$ (Lemma 8.10) and use Lemma D.3: $P_{7t^2} = C_0 + C_1 y + C_2 y^2 + C_3 y^3$ with $C_j = C_j(t, x)$. Each C_j increases in t on $2/5 \leq t \leq 1/2$, $0 < x \leq 1/3$. Indeed $\partial_t C_j$ is a polynomial in x whose coefficients are cubic in t; the bounds $\frac{4}{5}t - \frac{4}{25} \leq t^2 \leq \frac{9}{10}t - \frac{1}{5}$ and $\frac{12}{25}t - \frac{16}{125} \leq t^3 \leq \frac{61}{100}t - \frac{9}{50}$ (with remainders $(t - \frac{2}{5})^2$, $(t - \frac{2}{5})(\frac{1}{2} - t)$, $(t - \frac{2}{5})^2(t + \frac{4}{5})$, $(t - \frac{2}{5})(\frac{1}{2} - t)(t + \frac{9}{10})$) make every coefficient affine in t, and the resulting polynomials in x are positive. At $t = 2/5$,

$$C_0 = -\tfrac{72}{625}x^2(x+1)(15x^4 + 68x^3 + 94x^2 + 20x - 48),$$

$$C_1 = \tfrac{12x}{625}(84 + 2086x - 395x^2 - 8726x^3 - 11540x^4 - 6493x^5 - 1647x^6 - 140x^7),$$

$$C_2 = -\tfrac{6}{625}(144 - 1444x - 3342x^2 + 25127x^3 + 75813x^4 + 88565x^5 + 54334x^6 + 18396x^7 + 3270x^8 + 250x^9),$$

$$C_3 = \tfrac{x+2}{625}(2680 + 17612x + 33046x^2 + 3121x^3 - 42903x^4 - 45902x^5 - 20182x^6 - 3790x^7 - 250x^8).$$

Here $C_0 > 0$ for $x \leq 1/3$. The function C_1 has at most one interior critical point, a maximum, and C_2 at most one, a minimum, because C_1'/x and C_2'/x are monotone; and C_3 increases. Evaluation at the endpoints therefore gives, on the rows $[\frac{1}{7}, \frac{1}{5}]$, $[\frac{1}{5}, \frac{3}{10}]$, $[\frac{3}{10}, \frac{1}{3}]$, the bounds $C_1 > b$, $C_2 > -M$, $C_3 > c$ with $(b, M, c) = (\frac{9}{10}, 1, 20)$, $(\frac{3}{2}, 10, 26)$, $(\frac{77}{50}, \frac{771}{50}, \frac{194}{5})$, and $4bc > M^2$ in each row. Hence $C_1 y + C_2 y^2 + C_3 y^3 \geq y(b - My + cy^2) \geq 0$, so $P_{7t^2} > 0$ and $N > 0$, against $N < 0$. □

The centre gap $d^2 + 2cd > 4$. With u at its ceiling, the diagonal and the transported norm increase in b: $\partial_b h_U = K_a(b-a)/(16S^2)$ and $\partial_b w_U = K_a(b-a)(aK_a+SY)/(S^3Q^2)$. So the actual outgoing norm is below its value $w_U(b_*)$. The incoming secant is below the chord value $V_{\mathrm{ch}} = \frac{1}{2}(g_c(a)+g_c(d))$ (Lemma D.1(d)), and w_U increases in v. If $e = 4-d^2-2cd \geq 0$, a cleared identity with five factored coefficients puts $w_U(b_*)$ at $v = V_{\mathrm{ch}}$ strictly below $g_d(r_0) \leq g_d(r) \leq w$, with $r_0 = Q/W - d$, which is impossible.

The duration $3d < 7c$. Since $v = \mathrm{Sec}(a,c,d) \leq g_c(a)$ and $g_c \geq g_c(a)$ on $[0,a]$, extending the interval to $[0,d]$ raises the average: $\mathrm{Sec}(0,c,d) > v \geq g_c(d)$. With $h = d/c$ this reads $1+1/(1+h)-\log(1+h)/h > (1+h+h^2)/(1+h)^2$, which forces $h < 7/3$; at $h = 7/3$ the two sides are $1.3 - \frac{3}{7}\log\frac{10}{3} < 0.785$ and 0.79.

The reverse order $d \leq b_*$. Suppose $d > b_* \geq c$. By Lemma D.1(a),(c), $\mathrm{Sec}(a,d,c) > \mathrm{Sec}(a,c,d) \geq \mathrm{Sec}(B_0,d,c)$, and Lemma D.1(b) forces $B_0 > a$, that is $(a+d)(a+r) < K_a$; with $d > b_*$ this gives $r < c < d$. The reverse successor gives $w \leq \mathrm{Sec}(a,r,d)$, and the outgoing floor is $\mathrm{Sec}(A_0,d,r)$ with $A_0 = (4-cd)/(c+d)$. The same two steps force $A_0 > a$, but $d > b_*$ is equivalent to $A_0 < a$. In normalized form $b_*/T = k-t$ and $d/T = t+x$, so $d \leq b_*$ is $k \geq 2t+x$.

D.5. **The reverse successor.**

Lemma D.5 (Unit shape)**.** *If* $0 < t \leq 13/35$, $0 < x \leq 1-2t$, $2t+2x+x^2 > 1$, $q > 1$, $v = \mathrm{Sec}(t, 1-t, t+x)$ *and* $F_{p/W}(q) \leq v$, *then* $N(t,x,1,q,v,\log q) > 0$.

Proof. By Lemma D.3 it suffices that $P_1 = C_0 + C_1 y + C_2 y^2 + C_3 y^3 > 0$ for $y \geq 0$. This holds when

$$C_0 > 0, \qquad J_1 = 20xC_1 + 3C_0 > 0, \qquad J_2 = 53C_0 + 20x^2C_2 > 0,$$

$$J_3 = 24x^3C_3 + 12x^2C_2 - C_0 > 0, \qquad J_4 = 40x^3C_3 + 30x^2C_2 + 7C_0 > 0,$$

$$J_* = 24x^3C_3 - C_0 > 0.$$

Indeed, with $z = y/x$ and $B_1 = -x^2C_2/C_0$, these give $xC_1/C_0 \geq -\frac{3}{20}$, $B_1 \leq \frac{53}{20}$ and $x^3C_3/C_0 \geq \max\{\frac{1}{24}, \frac{B_1}{2}+\frac{1}{24}, \frac{3B_1}{4}-\frac{7}{40}\}$. The resulting lower bound for P_1/C_0 is affine in B_1 on each side of $B_1 = 13/15$, so it suffices to take $B_1 \in \{0, \frac{13}{15}, \frac{53}{20}\}$, that is, to check the cubics $1 - \frac{3}{20}z + \frac{1}{24}z^3$,

$$1 - \tfrac{3}{20}z - \tfrac{13}{15}z^2 + \tfrac{19}{40}z^3 = \tfrac{52}{135} + \tfrac{13}{180}(z-\tfrac{4}{3}) + \tfrac{2}{5}(z-\tfrac{4}{3})^2 + \tfrac{19}{40}z(z-\tfrac{4}{3})^2,$$

$$1 - \tfrac{3}{20}z - \tfrac{53}{20}z^2 + \tfrac{29}{16}z^3 = \tfrac{1}{80} - \tfrac{1}{80}(z-1) + \tfrac{39}{40}(z-1)^2 + \tfrac{29}{16}z(z-1)^2,$$

each bounded below by a positive constant on $z \geq 0$.

For the six signs put $s = 1-2t-x \geq 0$. The domain gives $\frac{3}{25} < x < 1$ and $\max\{0, \frac{9}{35}-x\} \leq s \leq \min\{1-x, x+x^2\}$, and $s = \frac{9}{35}-x$ is the edge $t = 13/35$.

First, $C_0 = 12W^2X^3N(\bar v, L_+)|_{q=1,k=1}$. At $q=1$ the residual is affine in k with $\partial_k = x\{(1-\bar v)(1+tW) - t(1-t^2)\} > t^2(1-t)x^2 > 0$, and $2t+x \le 1$, so

$$C_0 \;\ge\; 12W^2X^3\,\frac{(1-t)x^3}{3WX^3}\,P_0(t,x),$$

$P_0 = t(3t-1)x^3 + (18t^2-15t+5)x^2 + (36t^2-40t+14)x + (24t^2-32t+12)$, where the right side is the value at $k = 2t+x$. From $t(3t-1) \ge -\frac{1}{12}$, the completed squares $18(t-\frac{5}{12})^2 + \frac{15}{8}$, $36(t-\frac{5}{9})^2 + \frac{26}{9}$, $24(t-\frac{2}{3})^2 + \frac{4}{3}$, and $x<1$, one gets $P_0 > \frac{43}{24}x^2 + \frac{26}{9}x + \frac{4}{3} > 0$.

Next write $J_i/x^2 = \sum_{m=0}^{3} A_{im}(x)s^m$ for $i = 1,\dots,4$. The functions J_1/x^2 and J_2/x^2 increase in s: their s-derivatives are bounded below, using $s \le x + x^2$, by polynomials in x with positive coefficients. So they are smallest at the lower end of s, where they are positive: on the edge $t = 13/35$ by one-variable inequalities on $[\frac{3}{25}, \frac{9}{35}]$, and at $s=0$ because $A_{i0}/(xW^2)$ increases for $x \ge \frac{9}{35}$ and is positive at $\frac{9}{35}$. The function J_3/x^2 is concave in s ($A_{32} + 3A_{33}s < 0$ on the domain), so it suffices to check the two ends of the s-interval, again one-variable inequalities. The function J_4/x^2 increases in s for $x \ge \frac{1}{8}$: its s-derivative, written in the Bernstein basis of $[0, x+x^2]$, has three coefficients that are polynomials in x with positive non-constant coefficients and positive at $x = \frac{1}{8}$. For $\frac{3}{25} < x < \frac{1}{8}$, one has $A_{40} \ge \frac{7059}{800}$, $A_{41}, A_{43} > 0$, $A_{42} > -217$ and $s \le \frac{9}{64}$, so $J_4/x^2 > 4$. Finally J_*, a cubic in s written in the Bernstein basis of $[0, S]$ with $S = x + x^2$ for $x \le \frac{2}{5}$ and $S = 1 - x$ for $x \ge \frac{2}{5}$, has four coefficients bounded below, after dropping positive terms, by $258xW^2$, $675xW^2$, $570xW^2$, $94xW^2$ in the first case and by $258xW^2$, $100W$, 46, 94 in the second. □

Lemma D.6 (Order face)**.** *If $0 < t < 2/5$, $x > 1/4$, $2t+x \ge 1$, $x < \frac{7}{3} - \frac{10}{3}t$, $v = \mathrm{Sec}(t, 1-t, t+x)$ and $1 < q \le (1+x)/(2t+x)$, then $N(t, x, 2t+x, q, v, \log q) > 0$.*

Proof. Put $k = 2t+x$. Then $B = Wk$, $Z = qk$, $r = Z - p$, and with t, x, k, v fixed, $R'(q) = k(g_p(r) - w_U)$ and $R''(q) = k^2 p(r-p)/(r+p)^3 \le 0$, since $t \le r \le c \le p$ on the interval. So R is concave, and it suffices to check $q = 1$ and $q = W/k$.

At $q=1$, $N = x\{(k+tY)(1-v) - 2t(1-t)(t+x)\}$ with $k + tY = tW^2 + t + x$, and Lemma D.2(a) gives $N > \frac{(1-t)x^3}{3WX^3}P_0(t,x)$ with P_0 as in Lemma D.5, positive (for $t < \frac{1}{3}$ use $t(1-3t) \le \frac{1}{12}$ and $x < \frac{5}{2}$).

At $q = W/k$ replace $v, \log q$ by $\bar v, L_+$, which lowers R. With $h = 1-2t$, $s = k-1$ and $x = h+s$ the domain is $\frac{1}{5} < h < 1$, $0 \le s < \frac{4}{3}$, $h + s > \frac{1}{4}$, and

$$\bar R(W/k) = \frac{P(h,s)}{24tk^2W^2(W+k)X^3}, \qquad P = \sum_{j=0}^{7} C_j(h)s^j,$$

$C_0 = h^3(1+h)^2F(h)$, $\qquad F = -16 + 24h + 120h^2 + 140h^3 + 78h^4 + 23h^5 + 3h^6$,

$C_1 = h^3(1+h)(8 + 396h + 1020h^2 + 1080h^3 + 597h^4 + 179h^5 + 24h^6)$,

and $C_2, \dots, C_7$ are polynomials in h whose only negative monomials are $-70h^2$ in C_5, $-108h^2$ in C_6, and $-4h$, $-20h^2$ in C_7. These are absorbed: $184hs^5 - 70h^2s^5 \ge 0$, and $316hs^4 - 108h^2s^6 - (4h + 20h^2)s^7 \ge hs^4(316 - 108 \cdot \frac{16}{9} - 24 \cdot \frac{64}{27}) > 0$, the term $316hs^4$ being part of C_4s^4. So $P \ge C_0 + C_1 s$. The polynomial F increases and $F(\frac{1}{4}) = \frac{63}{4096} > 0$, which settles $h \ge \frac{1}{4}$. For $\frac{1}{5} \le h \le \frac{1}{4}$ one has $s > \frac{1}{4} - h$ and $C_0 + C_1(\frac{1}{4} - h) = \frac{h^3(1+h)}{4} Q(h)$ with $Q = -56 + 396h + 12h^2 - 1960h^3 - 2851h^4 - 1805h^5 - 588h^6 - 84h^7$, which is concave on $[\frac{1}{5}, \frac{1}{4}]$ ($Q'' < 24 - \frac{11760}{5} < 0$), with $Q(\frac{1}{5}) = \frac{220476}{78125}$ and $Q(\frac{1}{4}) = \frac{315}{4096}$. □

Lemma D.7 (Scale strip)**.** *If $\frac{13}{35} \le t \le \frac{2}{5}$, $0 < x \le \frac{9}{35}$, $\frac{29}{4}t^2 < 2t + 2x + x^2$, $q > 1$, $v = \mathrm{Sec}(t, 1-t, t+x)$ and $F_{p/W}(q) \le v$, then $N(t, x, \frac{29}{4}t^2, q, v, \log q) > 0$.*

Proof. Write $P_{29t^2/4} = \sum_j C_j(t,x) y^j$. Each C_j increases in t on the strip: $\partial_t C_0 \ge 44x^2$, $\partial_t C_1 \ge 99x$, $\partial_t C_2 \ge 48 - \frac{2301}{81}$ and $\partial_t C_3 \ge 64 - \frac{167}{729}$. Each $\partial_t C_j$ is a polynomial in x whose coefficients are cubics $\sum_i a_i t^i$, and on $[\frac{13}{35}, \frac{14}{35}]$ such a cubic is at least $\sum_{a_i \ge 0} a_i (\frac{13}{35})^i + \sum_{a_i < 0} a_i (\frac{14}{35})^i$; one coefficient of $\partial_t C_0$ needs the additional remark that it decreases in t. So $P_{29t^2/4}(t,x,y) \ge P_{29t_0^2/4}(t_0,x,y)$ with $t_0 = \frac{13}{35}$, for every $y \ge 0$. Since $\frac{29}{4}t^2 - 2t$ increases, the strip condition gives $2x + x^2 > \frac{29}{4}t_0^2 - 2t_0 > \frac{9}{35}$, so (t_0, x) satisfies the shape inequalities of Lemma D.5, whose polynomial part gives $P_1(t_0,x,y) > 0$ (no floor is used there). As P_k increases in k at fixed (t_0, x, q) (the coefficient of k is bounded below as in Lemma 8.10, now with $\bar v < U(t_0)$ and $L_+ < q - 1$) and $\frac{29}{4}t_0^2 = \frac{4901}{4900} > 1$, also $P_{29t_0^2/4}(t_0,x,y) > 0$. Hence $P_{29t^2/4} > 0$, and Lemma D.3 at the original (t, x, q), where the floor holds, gives $N > 0$. □

Lemma D.8 (Low-scale unit face)**.** *If $0 < t \le \frac{1}{4}$, $\frac{1}{2} < x \le \frac{7}{3}$, $x \ge 1 - 2t$, $t \le T(x)$, $q > 1$, $v = \mathrm{Sec}(t, 1-t, t+x)$ and $F_{p/W}(q) \le v$, then $N(t, x, 1, q, v, \log q) > 0$.*

Proof. With $I_0 = 1 + tx + tW(1-2t)$, $I_1 = X + txX^2$ and $\mathcal{A} = xq(I_0 + I_1 y) = -N_v$, one has $N > N(\bar v, L_+) - \mathcal{A}(\bar v - F_\ell)$, and $Q = 6(q+1)W^2\{N(\bar v, L_+) - \mathcal{A}(\bar v - F_\ell)\}$ is a quartic $P_0 + P_1 y + \cdots + P_4 y^4$. It is positive for $y \ge 0$ when $P_0 > 0$, $P_1 + 2P_0 > 0$, $P_2 + 48P_0 > 0$, $P_3 + 4P_2 > 0$, $P_3 - P_0 > 0$ and $P_4 - P_0 > 0$; the extreme cases reduce to $1 - 2y - \frac{1}{4}y^2 + y^3 + y^4 = \frac{4}{81} + \frac{5}{27}(y - \frac{2}{3}) + (\frac{29}{12} + \frac{7}{3}y + y^2)(y - \frac{2}{3})^2$ and $1 - 2y - 48y^2 + 192y^3 = \frac{2}{9} - 2(y - \frac{1}{6}) + (16 + 192y)(y - \frac{1}{6})^2$, both positive. The first four signs factor as $4(1-t)x^2 F_0/X^3$, $2(1-t)xF_1/X^3$, F_2/X^3, F_3/X^3 with polynomials $F_i(t,x)$, and $-\partial_t F_i$ is smallest at $t = \frac{1}{4}$ and positive there (in the variables $1 - 4t$ and $2x - 1$ its coefficients are nonnegative after absorbing a few negative terms), so $F_i(t,x) \ge F_i(\frac{1}{4}, x) + (\frac{1}{4} - t)(-\partial_t F_i(\frac{1}{4}, x))$. For $\frac{1}{2} \le x \le 1$ the values $F_i(\frac{1}{4}, x)$ are positive one-variable polynomials. For $1 \le x \le \frac{7}{3}$, where $t < T(x) < \frac{1}{4}$, the tangent value at $t = T(x)$ is a positive multiple of an explicit polynomial $h_i(x)$, positive on $[1, \frac{7}{3}]$ after charging its few negative

high-degree terms to positive lower ones. The comparisons $P_3 > P_0$ and $P_4 > P_0$ are proved in the same way. □

D.6. **The forward successor.** In this subsection write $o = 2t+x$, $j = o-1$, $h = 1-2t$, $y = q-1$, $e = q/W$, $J = t+Y$, $D = eJ-p$ and $E = e^2B+1-2pe$. The next norm at the ceiling is $\mathcal{T}_U = (v - e^2(H_0+2vY) + e^4B^2w_U)/E^2$, and the combined residual (8.9) has $C_* = tW^2DE^2/e$; it is a concave quadratic in k with leading coefficient $-tW^2De^3/4$.

Lemma D.9 (Low ratio)**.** *Let $\frac14 \le t \le \frac13$, $1-2t \le x \le 1$, $1 \le k \le o$, $1 \le q \le W/o$, and let $\widehat{G} = \mathcal{H}(\hat{v}, L_+)$, with $\alpha = 6$, $M = 0$, $\hat{v} = v_G + \rho(v_G - F_\ell)$ and $\rho = \frac34(q+o-1)$. Then $\widehat{G} > 0$. Consequently every window with a forward successor has $t > 1/3$.*

Proof. At fixed (t, x, q) the coefficients of v and $\log q$ in $\mathcal{H}$ are negative: $\mathcal{H}_v < -\frac{17}{12}xq(I_0 + I_1 y)$, from $Je^2 \le \frac{10}{3}$ and $WD/x \le \frac94$, and $\mathcal{H}_{\log q} = (7 - Je^2)N_L < 0$. So a window with a forward successor and $t \le \frac13$, which lies in the domain (with $q < W/(1+2j) \le W/o$), has $\widehat{G} \le \mathcal{H}(v, \log q) < 0$.

The three-term secant majorant $v_3 = 1 + c^2/W - (2c/X)(1 + x^2/(3X^2) + x^4/(5X^4))$ satisfies $v_3 - v_G = cx^6(31x^2+82x+46)/(60W^2X^5(x^2+10x+15)) > 0$, and $v_3 + \rho(v_3 - F_\ell) < U(t)$ (using $F_\ell - g = \eta(\eta - \frac12)y + \eta y^2/(3(y+2)) \ge 0$ with $\eta = p/W \ge \frac12$, and $\rho < 2$); so $\hat{v} < U(t)$. Also $\mathcal{H}_{kv} < 0$ and $\mathcal{H}_{k,\log q} < 0$, so $\widehat{G}_k > \mathcal{H}_k(U(t), y)$, which decreases in k. At $k = o$,

$$\mathcal{H}_k(U(t), y)\big|_{k=o} = \frac{tq}{2W^2}(d_0 + d_1 y + d_2 y^2 + d_3 y^3),$$

with $d_0 = xc\{1 - 2t + x(-2t^2 + 14t - 1) + 24tx^2 + 12tx^3\} > 0$, $d_3 = xWJ > 0$, $d_2 < 0$ (every coefficient in x is negative for $\frac14 \le t \le \frac13$), and $2d_1 + d_2 > 0$; as $y \le \frac12$, $d_1 + d_2 y \ge d_1 + \frac12 d_2 > 0$. Hence $\widehat{G}_k > 0$, and it suffices to show $\widehat{G} > 0$ at $k = 1$.

For $k = 1$ we follow [6, `NONSHORT_LOW_RATIO_LINK`]. With $t = \frac{1-h}{2}$, $x = h + j$ and $q = 1 + y$ on $\frac13 \le h \le \frac12$, $j \ge 0$, $0 \le y \le h$, a region that contains the domain, the polynomial $\Pi = 192q(q+1)W^3(x^2 + 10x + 15)\,\widehat{G}(t, x, q, 1)$ equals $\sum_i \tilde{C}_i(h, y)j^i$. Its part of degree at most two in j is $h^3w^3(\mathcal{A}_0 + \beta\mathcal{A}_1 + w\beta^2\mathcal{A}_2)$ with $w = 1+h$ and $\beta = j/(hw)$; each $\mathcal{A}_i$ increases in h, its h-derivative being a polynomial with nonnegative coefficients after factoring, and at $h = \frac13$ the quadratic in β exceeds 4 by three square identities. The coefficients $\tilde{C}_3, \dots, \tilde{C}_{10}$ are nonnegative from their factored forms. Hence $\Pi > 4h^3w^3 + 700j^3 + 3900j^4 > 0$. □

Lemma D.10 (Large gap)**.** *Every window with a forward successor has $j < 1/30$.*

Proof. Suppose $j \ge \frac{1}{30}$. By Lemma D.9, $t > \frac13$; Lemma D.2(c) with $v \ge g$ gives $t < T_G(x)$, hence $x < \frac35$ and $7x < 12h$, that is, $j < \frac57 h$.

For $\frac{1}{3} \le t \le \frac{3}{8}$ we follow [6, NONSHORT_LOW_T_CALLER]. Take $\alpha = \frac{7}{2} - 10y \ge \frac{11}{16}$ and $M = 8j + 8y - \frac{1}{6} \ge \frac{1}{10}$. The v-coefficient of $\mathcal{H}$ is negative, so $v \mapsto V_L$ (a rational majorant of v), $\log q \mapsto L_+$ and $F \mapsto F_\ell$ keep $\mathcal{H} < 0$. The resulting residual increases in k on $[1, o]$: it is quadratic in k, its shape derivative at the midpoint of two shapes is affine in v, $\log q$ and k with nonpositive slopes, and at the extreme values it equals $q^2 S/(12W)$ with a quartic S in y that is concave in y and positive at both ends. At $k = 1$ a positive multiple S of the residual, a polynomial in (h, j, y), increases in h at fixed (j, y): a cubic Taylor bound from $\partial_h^4 S \ge 500000$ and two derivative signs at $h = \frac{1}{4}$ give $\partial_h S > 0$. So it suffices to check the three faces $h = \frac{1}{4}$, $j = \frac{5}{7}h$ and the largest y, where $S > 0$ by termwise monomial bounds after a shift of the variable.

For $\frac{3}{8} \le t < \frac{2}{5}$ we follow [6, NONSHORT_HIGH_RATIO_SHAPE, NONSHORT_HIGH_RATIO_LOWER_REDUCTION]. Take $\alpha = 4$, $M = 1$ and $v \mapsto v_G$. The residual increases in k on $(\frac{29}{4}t^2, o)$, by the same midpoint argument with endpoint value $2tq^2(d_0' + d_1' y + d_2' y^2 + d_3' y^3)$, a positive cubic. At $k = \frac{29}{4}t^2$ a positive multiple S satisfies $-\partial_h^2 S > 0$, and $-\partial_h S > 0$ at $h = \frac{1}{5}$, which reduces the sign to $h = \frac{1}{4}$; there S is concave in y, and along the remaining line a ninth-derivative bound and exact values at the ends give $S \ge \frac{13}{4}$. In both cases $\mathcal{H} > 0$, a contradiction. □

Lemma D.11 (The ratio χ)**.** *If $\frac{1}{3} \le t \le \frac{19}{50}$, $0 \le j \le \frac{1}{30}$, $k \ge 1$, $q > 1$, $v = \mathrm{Sec}(t, 1-t, t+x) \le U(t)$ and $N(t, x, k, q, v, \log q) < 0$, then $\chi = pq/W > \frac{4}{7}$.*

Proof. Lower k to 1 (Lemma 8.10): $N_1(q) = A_2 q^2 + A_1 q - tp^2 W^2 - Cq \log q$ with $C = tpBW > 0$ and $N_1'' = 2A_2 - C/q$. Suppose $1 < q \le q_\chi = 4W/(7p) \le \frac{8}{7}$. The v-coefficient of A_2 is $-Y - tY^2 < 0$, so with $v > \frac{3}{4}$, $N_1'' < -\frac{1}{8}(7C - 16A_2)|_{v=3/4}$, and $(7C - 16A_2)|_{v=3/4} = \sum_{i=0}^4 b_i(h) j^i$ with

$$b_0 = -\tfrac{h+1}{4}(16h^4 + 23h^3 - 9h^2 - 7h + 1), \qquad b_1 = -\tfrac{h+1}{4}(87h^3 + 12h^2 - 55h - 12),$$

$$b_2 = -\tfrac{1}{2}(66h^3 + 33h^2 - 58h - 33), \qquad b_3 = \tfrac{67}{4}(1 - h^2), \qquad b_4 = \tfrac{3}{2}(1 - h),$$

all positive for $\frac{6}{25} \le h \le \frac{1}{3}$. So N_1 is concave on $[1, q_\chi]$. At $q = 1$ the secant bound $v < V_2 = 1 - cp/W^2 + (cx/W^2)(\frac{1}{2} - \frac{x}{3} + \frac{x^2}{12} - t)$ gives $N_1(1) > 0$ (the cleared value exceeds a positive multiple of $2h - j^3 - 4j^2 - 4j > 0$). At $q = q_\chi$, the bounds V_2 and L_+ give $N_1(q_\chi) > P(h, j)/(294(15c + 11j)p^2)$ with $P = \sum_{i=0}^8 P_i(h) j^i$, where $P_0 > \frac{3}{5}$, $P_1 > -13$, $P_2 > -26$, $P_3, P_4 > 0$, $P_5 > -330$, $P_6 > -585$, $P_7 > -140$ and $P_8 > 0$ on $\frac{6}{25} \le h \le \frac{1}{3}$, each by a one-variable monotonicity or by discarding positive terms; with $j \le \frac{1}{30}$ this gives $P > 0$. So $N_1 > 0$ on $[1, q_\chi]$, against $N < 0$. □

Lemma D.12 (Casimir slope)**.** *Let $\frac{1}{3} \le t \le \frac{2}{5}$, $0 < j < \frac{1}{4}$, $k \ge 1$, $q > 1$, $\chi \ge \frac{4}{7}$, $v \ge \frac{3}{4}$, $F_{p/W}(q) \le v$, and let $r = qB/W - p$ satisfy $t \le r \le p$. With $H_d(w) = BL' + \frac{e}{p}B^2(w-1) + p(1-\chi)/e$ and $L' = \log q + (Wv - c + p^2/W)/p$, one has $2v - \frac{e}{p}H_d(w) < 0$ for every $w \ge \frac{3}{4}$; so $\Gamma(w) = (4B^2 vw - H_d(w)^2)/(16(k - t^2)^2)$ decreases on $[\frac{3}{4}, \infty)$.*

Proof. $\Gamma'(w) = \frac{B^2}{8(k-t^2)^2}(2v - \frac{e}{p}H_d(w))$ and H_d increases in w, so it suffices to take $w = \frac{3}{4}$. With $A = Z/p = 1 + r/p$ the hypotheses give $\frac{15}{11} \le A < 2$ and $A/\chi = B/p^2 > 2$, and $j < \frac{1}{4}$ gives $k - t^2 - p^2 > \frac{7}{144}$. The floor bounds $\log q$ below in terms of v, and $2v - \frac{e}{p}H_d(\frac{3}{4})$ is then bounded above by a convex quadratic in one ratio, negative at both ends of its range [7, `MIXED_NONSHORT_FORWARD_CASIMIR_MONOTONICITY`]. □

Lemma D.13 (Eliminating the event factor)**.** *Let $\frac{1}{3} \le t \le \frac{2}{5}$, $0 \le j \le \frac{1}{30}$, $k \ge 1$, $v \le \min\{1, V\}$, $\eta = p/W$, $\alpha_U = B(w_U - 1)/(pW)$ and $\beta_U = pW/B$. Then for $q > 1$,*

$$\log q + \alpha_U(q-1) + \beta_U(1/q - 1) < F_\eta(q) - U(\eta).$$

Proof. By Lemma D.2(b) the difference of the two sides exceeds

$$\frac{y}{6(y+1)(y+2)}(C_0 + C_1 y + C_2 y^2)$$

with $C_0 = 12\eta^2 - 6\eta - 12\alpha_U + 12\beta_U - 12$, $C_1 = 18\eta^2 - 7\eta - 18\alpha_U + 6\beta_U - 12$ and $C_2 = 6\eta^2 - \eta - 6\alpha_U - 1$. Each C_i increases in k, since $\partial_k \alpha_U < \partial_k \beta_U = -pW/B^2 < 0$, and decreases in v, since $\partial_v \alpha_U = Y(k/t + Y)/(BpW) > 0$; so it suffices to take $k = 1$ and $v = V$. There $w_U < 1$, so $\alpha_U < 0$, and $\eta \ge \frac{1}{2}$ gives $C_2 = (2\eta - 1)(3\eta + 1) - 6\alpha_U > 0$. It remains to show $4C_0C_2 > C_1^2$. With $t = \frac{2}{5} - A$ and $x = \frac{1}{5} + 2A + j$ on $0 \le A \le \frac{1}{15}$, $0 \le j \le \frac{1}{30}$, the cleared discriminant is a polynomial in (A, j), concave in A (the positive part of its second A-derivative is dominated using $A^i j^m \le 15^{-i}30^{-m}$), and positive on the two edges $j = 0$ and $j = \frac{1}{30}$, where it is a univariate polynomial whose negative terms are collected against its positive ones [7, `MIXED_NONSHORT_FORWARD_Q_ELIMINATION`]. □

Lemma D.14 (The envelope Casimir)**.** *On $\frac{1}{3} \le t \le \frac{2}{5}$, $0 \le j \le \frac{1}{30}$, $1 \le k \le \frac{31}{30}$, $\frac{3}{4} \le v \le V$,*

$$\mathcal{G} = \frac{B^2\{4vw_U - (\phi(1) + v - g)^2\}}{16(k - t^2)^2} > \frac{9193}{50000},$$

where $\phi(1) = \alpha_U + \beta_U + (Wv - c + p^2/W)/p - p^2/B$.

Proof. The numerator $\mathcal{N}(v) = B^2\{4vw_U - (\phi(1) + v - g)^2\}$ is a quadratic in v. The numerator of w_U has v-derivative $P_v = Y(k/t + Y)$, and $D_1(v) = B(\phi(1) + v - g)$ has $D_1' = B(1 + W/p) + P_v/(pW)$, so the arithmetic–geometric mean inequality and $B > p^2$ give $D_1'^2 > 4P_v$ and $\mathcal{N}'' = 8P_v - 2D_1'^2 < 0$. Put $t = \frac{2}{5} - A$, $x = \frac{1}{5} + 2A + j$ and $k = 1 + Z$ on the box $0 \le A \le \frac{1}{15}$, $0 \le j, Z \le \frac{1}{30}$. The polynomial $5^{10} \cdot 2t^2p^2W^3(-\mathcal{N}'(\frac{3}{4}))$ has constant coefficient 6465474, while the sum of its negative coefficients, each weighted by the maximum $15^{-i}30^{-m-n}$ of its monomial $A^i j^m Z^n$ on the box, is $787831672153/787320 < 6465474$. So $\mathcal{N}'(\frac{3}{4}) < 0$, $\mathcal{N}$ decreases on $[\frac{3}{4}, V]$, and it remains to show $\mathcal{N}(V) > 16(k - t^2)^2 \cdot \frac{9193}{50000}$. The cleared difference is concave in A by the same domination, and positive on the edges $A = 0$ and $A = \frac{1}{15}$ by collecting

negative coefficients against positive ones [7, `MIXED_NONSHORT_FORWARD_ENVELOPE_CASIMIR_SIGN`]. □

The small gap. By Lemma D.11, the hypotheses of Lemma D.12 hold, so $\gamma^2 = \Gamma(w) > \Gamma(w_U)$. At w_U the diagonal divided by B is $\phi(q) = \log q + \alpha_U q + \beta_U/q + (Wv - c + p^2/W)/p - p^2/B$, and Lemma D.13 with the floor gives $0 < \phi(q) \le \phi(1) + F_\eta(q) - U(\eta) \le \phi(1) + v - g$; positivity comes from Lemma D.12, and $U(\eta) = g$. Hence $\Gamma(w_U) \ge \mathcal{G} > 9193/50000$ by Lemma D.14, which is the last step of Proposition 8.14.

Appendix E. Estimates for the smaller-short window

This appendix gives the scalar inequalities behind Section 9, organized by the gate that uses them. They are identities between rational functions, and inequalities proved by exhibiting a difference as a sum of terms of known sign, by the sign of a derivative on an interval, or by monotonicity or convexity in one variable at a time. The face signs of Appendix E.8 are the longest of them. Their proofs follow the hand arguments of [6] and, for one endpoint, of [7]; they reduce each sign, one variable at a time, to explicit polynomials in one or two variables, mostly of degree at most eight, whose coefficients are printed in those notes. We give the reductions and the final signs. Throughout, Sec, g_B and $\mathcal{U}$ are as in Section 9.1, and the variables are those of Table 3.

E.1. **Secants and the incoming norm.** By (9.1), $(P_0 - A_0)\,\mathrm{Sec}(A_0, B_0, P_0)$ is the integral of g_{B_0} over $[A_0, P_0]$, hence additive over adjacent intervals, and by (9.2) every secant lies in $[3/4, 1]$. The function g_{B_0} decreases on $(0, B_0]$ and increases on $[B_0, \infty)$. For the bridge norm, (9.2) gives

$$u - \frac{3}{4} = \frac{1}{4(\beta - 1)} \int_1^\beta \frac{(s - x)^2}{(s + x)^2}\, \mathrm{d}s. \tag{E.1}$$

The central value. With $z = (x-1)/(x+1)$ one has $2x/(1+x) = 1 + z$ and

$$V(x) - \frac{3}{4} = \mathrm{Sec}(1, x, x) - \frac{3}{4} = \frac{1}{2} + \frac{z}{4} - \frac{(1+z)\log(1+z)}{2z}.$$

A lower bound for $\log(1+z)$ is therefore an upper bound for V. The rational bound

$$\log(1+w) > \frac{w(43w^3 + 252w^2 + 390w + 180)}{12(1+w)^2(w^2 + 10w + 15)} \qquad (w > 0)$$

holds because the difference vanishes at $w = 0$ and has derivative $w^6/((1+w)^3(w^2+10w+15)^2) > 0$. It gives $V(x) < V_G(x) = \frac{3}{4} + \frac{(x-1)^2(71x^2+48x+1)}{96x(x+1)(13x^2+14x+3)}$, and then

$$\bar{V}(x) - V_G(x) = \frac{(x-1)^3(11x^2 + 96x + 13)}{96x(x+1)(35x+13)(13x^2+14x+3)} \ge 0$$

gives $V \le \bar V = \frac34 + \frac{2(x-1)^2}{(x+1)(35x+13)}$. The same bound also follows from $\log(1+z) \ge \frac{z}{1+z}\big(1+\frac{z}{2}-\frac{4z^2}{24+11z}\big)$, whose difference has derivative $z^3(32-33z)/\big(2(1+z)^2(24+11z)^2\big)$.

Ceilings and floors. Splitting (E.1) at x, bounding the integral over $[1,x]$ by $(x-1)(\bar V-\frac34)$ or by a sharper rational bound, and bounding $(s-x)^2/(s+x)^2$ on $[x,\beta]$ by a quadratic or linear function of $s-x$, gives the rational ceilings of the bridge norm used by the gates:

$$\bar u = \tfrac34 + \tfrac{1}{\beta-1}\Big[\tfrac{2(x-1)^3}{(x+1)(35x+13)} + \tfrac{(\beta-x)^3}{48x^2}\Big], \qquad \hat u = \tfrac34 + \tfrac{(x-1)^2(\beta+7x-8)}{4(x+1)(35x+13)(\beta-1)},$$
$$u_\ell = \tfrac34 + \tfrac{(x-1)^2(\beta+15x-16)}{8(x+1)(35x+13)(\beta-1)}, \qquad u_C = \tfrac34 + \tfrac{680(x-1)^3+3(x+1)^2(\beta-x)}{9600(x+1)^2(\beta-1)},$$
$$u_s = \tfrac34 + \tfrac{1}{\beta-1}\Big[\tfrac{17(x-1)^3}{240(x+1)^2} + \tfrac{\beta-x}{1200}\Big], \qquad u_O = \tfrac34 + \tfrac{(x-1)^2(x+\beta-2)}{(\beta-1)(x+1)(35x+13)},$$
$$u_{CB} = \tfrac34 + \tfrac{(x-1)(V_G(x)-3/4)+(\beta-x)/5808}{\beta-1},$$

together with the floor $u_f = \frac34 + \frac{(x-1)^2}{28(x+1)^2}$ and the quadratic floor u_q of gate S6 (Appendix E.7). The ceiling $\bar u$ holds for all $1 < x \le \beta$; each of the others holds on the shapes where its gate uses it.

E.2. **Orders and the comparison argument.** Normalize the shape by β: put $\ell = 1/x$, $p_1 = \alpha/\beta$, $q_1 = x/\beta$ and $\operatorname{avg}(p,q) = \operatorname{Sec}(p,1,q)$, the mean of $g(s) = 1 - s/(1+s)^2$ over $[p,q]$. The equal secants (9.5) read $\operatorname{avg}(\ell, 1/q_1) = \operatorname{avg}(p_1,q_1)$. An order compares p_1 with a *barrier* $P(z)$, a differentiable function with $P(1) = \ell$, through the comparison function

$$\Phi(z) = \operatorname{avg}\big(P(z),z\big) - \operatorname{avg}(\ell, 1/z), \qquad \Phi(1) = 0.$$

Because avg decreases in its left end, the sign of $\Phi(q_1)$ decides whether p_1 lies above or below $P(q_1)$.

Lemma E.1 (First contact)**.** *Let $a < 1$ and let Φ be continuous on $[a,1]$ with $\Phi(1) = 0$. If Φ has a negative derivative at each of its zeros in $(a,1]$, then $\Phi > 0$ on $(a,1)$.*

Proof. Since $\Phi(1) = 0$ and $\Phi'(1) < 0$, $\Phi > 0$ on some interval $(1-\varepsilon, 1)$. If $\Phi(z_0) \le 0$ for some $z_0 \in (a,1)$, then Φ has a zero in $[z_0, 1-\varepsilon]$; let z^* be the largest one. Then $\Phi > 0$ on $(z^*,1)$ and $\Phi(z^*) = 0$, so $\Phi'(z^*) \ge 0$, a contradiction. □

At a zero of Φ both averages equal a common value u, and differentiation gives

$$\Phi'(z)\,\big(z - P(z)\big)\,z\,(x-z)$$
$$\qquad = -P'(z)\big(g(P(z)) - u\big)z(x-z) - \big(u - g(z)\big)\big((2z - P(z))x - z^2\big).$$

So the sign of Φ' at a contact is that of an expression in which u appears only through the two gaps $g(P(z)) - u$ and $u - g(z)$. A bound for the ratio of these gaps turns the contact condition into a polynomial inequality in

(z, x). Two gap ratios are used. The lower one comes from a Padé bound for the logarithm. The upper one comes from a Chebyshev bound: with $y = (1+q)/(1+p)$ and $\theta = 1/(1+p)$, the inequality

$$\log y > \frac{(y-1)\big[3y(y+1) - 2\theta(y-1)^2\big]}{6y^2} \qquad (1 < y \le 2\theta)$$

holds because the difference vanishes at $y = 1$ and has derivative $(y-1)^2(2\theta y + 4\theta - 3y)/(6y^3) \ge 0$. With these, the contact polynomials have Bernstein forms on the contact interval with positive coefficients. For example, for the upper order with slope $5/4$, with $q = q_m + (1-q_m)v$, $q_m = (5x+4)/(9x)$ and $v \in [0,1]$, the contact expression $F_{5/4}$ satisfies

$$\frac{8748x^4}{(x-1)^2} F_{5/4} = \sum_{i=0}^{4} \binom{4}{i} B_i(x)\, v^i (1-v)^{4-i},$$

with $B_0 = 8(5x+4)(7x+2)(9x+4)$, $B_1 = 9(x+4)(142x^2+61x+4)$, $B_2 = 54x(53x^2+102x+24)$, $B_3 = 729x^2(8x+5)$ and $B_4 = 8748x^3$, all positive for $x > 1$. With Lemma E.1 this gives the orders of Lemma 9.5. The strengthened order with slope $2/7$ is simpler: there the comparison function is decreasing on the whole interval, by the gap ratio $9\,\mathrm{avg}(p,q) \ge 7g(q) + 2g(p)$.

E.3. **The outer scale bound.**

Lemma E.2 (Reverse-centre scale bound)**.** *At a reverse centre with neighbour ratios $0 < r, t < 1/2$, scale $0 < \rho < rt$, $\rho < 3/20$, norms $3/4 \le u \le \mathcal{U}(r)$, $3/4 \le v \le \mathcal{U}(t)$ and the stationarity $E = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1-u}{r} - \frac{1-v}{t}$, the inequality $(4uv - E^2)/(16(1-\rho)^2) < 9193/50000$ implies $\rho < 1/7$.*

Proof sketch. Assume $\rho \ge 1/7$. Then $E < -1 + 2r + 2t - rt$; the Gram numerator increases in u and v, which may be lowered to $3/4$; and raising r and t to $1/2$ gives

$$\frac{4uv - E^2}{16(1-\rho)^2} > G(\rho) = \frac{9/4 - B(\rho)^2}{16(1-\rho)^2}, \qquad B(\rho) = 1 - \rho + \log 4\rho.$$

G increases on $[1/7, 1/4)$, which follows from $\log y < 2(y-1)/(y+1)$ and the identity

$$\frac{9}{4} - \hat{B}^2 - \frac{\hat{B}(1-\rho)^2}{\rho} = \frac{(4\rho-1)^2(4\rho^2+\rho+4)}{4\rho(4\rho+1)^2}, \qquad \hat{B} = 1 - \rho + \frac{2(4\rho-1)}{4\rho+1}.$$

Finally $G(1/7) > 9193/50000$ by $\log(7/4) > 450606/805255$; the margin is about $1.6 \cdot 10^{-5}$. □

Applied at the outer centre A, with $r = r_A$, $t = 1/\tau$, $\rho = q$, norms u, w and pairing E_L, the lemma gives $q < 1/7$ (Lemma 9.4(e)).

E.4. **The common moves.** This subsection records the formulas behind the moves of Section 9.3.

The scale coefficients. With $\varepsilon = 1-\alpha^{-2}$, $n = (1+x)/\alpha^2$, $g = D/\alpha^2$ and $E_0 = (1+x)(u-1) + 1/(1+x)$, the coefficients of (9.9) are

$$\begin{aligned} A_1 &= 2\varepsilon + n + gE_0, \\ A_2 &= \varepsilon^2 + \varepsilon n + n^2 + g^2 u + g\varepsilon E_0 - g(2(1+x)-1)/\alpha^2, \\ A_3 &= 2gu + \varepsilon E_0 - (2(1+x)-1)/\alpha^2. \end{aligned}$$

The two scales are related by $\rho/(1-\rho) = q/(\alpha^2(1-q))$, and $q = q_c$ exactly when $\rho = r^2$; also $q_c \le 1/7$ if and only if $6\alpha^2 \le x(x+2)$.

The duration tests. The identity (9.10) is the logarithm bound $\log y \le (y - 1/y)/2$ at $y = \tau/2$. For the three tests of gate S0, the logarithm $\log(\tau/\tau_{\max})$ is bounded by $\log y \le y - 1$ and by the tangent $\log y \le \frac{5}{4}y - \frac{11}{9}$ at $y = 4/5$ (using $\log\frac{5}{4} \ge \frac{2}{9}$), and $1/\tau$ by its chord; an affine function of τ is then at most its larger endpoint value. This gives the values $\tau = 2$, $8\tau_{\max}/9$ and $\tau_{\max}$, with the logarithm replaced by $\frac{5}{2\tau_{\max}} - \frac{11}{9}$, $-\frac{1}{9}$ and 0 respectively.

E.5. **Gates S0 to S4.** Apart from the face signs of Appendix E.8, these gates use the following inequalities.

Gate S0. The positive diagonal is in the proof of Proposition 9.6. For the positive pairing, the envelope bound for E and the replacements $D \ge D_0$ and $u \le u_0 = \frac{3}{4} + \frac{(x-1)^2}{12(x+1)^2}$ give $E - 2\rho D u \le \Lambda(\rho) := \big((1+x) - 2\rho D_0\big)u_0 - (1+x) + \frac{1}{1+x} + \rho(2x+1)$, affine in $\rho \in [0, (1+x)^{-2}]$, with

$$\begin{aligned} \Lambda(0) &= -\frac{(x-1)(x+5)}{6(1+x)}, \\ \Lambda\big((1+x)^{-2}\big) &= -\frac{x^5 - 13x^4 + 6x^3 + 32x^2 + 37x + 9}{6(1+x)^4}, \end{aligned}$$

both negative on $1 < x \le 12/5$. The shape bound $\alpha/\beta < 3/4$ for $x \ge 12/5$ compares two bounds for $u - 3/4$: if $\alpha \ge 3\beta/4$ then $u - \frac{3}{4} \le \frac{(\beta-\alpha)^2}{4(\beta+\alpha)^2} \le \frac{1}{196}$, while (E.1) and the tangent bound $1/y^2 \ge 1/c^2 - 2(y-c)/c^3$ give $u - \frac{3}{4} \ge \frac{4(x-1)^3}{(4x-3)(5x+3)^2}$, which increases in x and exceeds $1/196$ at $x = 12/5$.

Gate S1. The combination $\Gamma - \frac{7}{96}R_\tau$ is monotone along the envelope path because the ratio of the derivatives of Γ and R_τ along that path is at most $7/96$. Along $\tau = 2$ the slope of R_2^* in q exceeds one: after completing the square in Δq with $\Delta = D/\alpha^2$, $q\,\mathrm{d}R_2^*/\mathrm{d}q$ is bounded below by a quadratic in $a = \alpha^{-2} \in (0,1]$ with nonpositive leading coefficient, and its values at $a = 0$ and $a = 1$ exceed $1/7 \ge q$ by explicit rational functions of $z = x - 1$ whose numerators have positive coefficients.

Gate S4. At the central scale, $\rho(q_c) = (1+x)^{-2}$, and the functions Φ and Ψ of Proposition 9.10 have closed forms. On the small box they decrease in u for $3/4 \le u \le u' \le 31/40$, and $\bar{u}(x,\beta) < 31/40$ there.

E.6. **Gate S5.** *Monotonicity in the norm.* At fixed shape, scale and duration the residual $\mathcal{R}(u;p,q)$ is affine in u, with

$$\partial_u \mathcal{R} = (1+x) + \frac{1}{r_A} - pg(1+x) + q\{p(g^2 + g\varepsilon(1+x)) - 2g - \varepsilon(1+x)\}.$$

This is bilinear in (p,q), so it is positive on $0 < q \le 1/7$, $0 < p \le r_A$ as soon as it is positive at the four corners. At each corner it is a polynomial in $\alpha - 1$, $x - \alpha$ and $\beta - x$ with nonnegative coefficients and a positive term.

The ceiling. On the small-shape domain, $u \le \bar{V}(x) \le u_5 = \frac{755}{996} + \frac{x-2}{100}$, since with $t = x - 2$

$$u_5 - \bar{V} = \frac{t(8715t^2 + 4012t + 1)}{24900(t+3)(35t+83)} \ge 0.$$

Convexity of the scale quadratic. With $a = \alpha^{-2}$, $\varepsilon = 1 - a$, $\bar{H} = (1+x)/4 - 1/(1+x)$, $\mathcal{A} = \varepsilon^2 + \varepsilon(1+x)a + (1+x)^2a^2$ and $\bar{K} = \varepsilon\bar{H} + (2x+1)a$, the coefficient A_2 at $u = u_5$ is at least $\mathcal{A} - \bar{K}^2/3$ for every value of g, and

$$\bar{K}^2 - 3\mathcal{A} = \frac{S^4 - 56S^2 + 16 + 2(7S^2 - 4)z + z^2}{16S^2},$$

with $S = 1 + x$ and $z = (S-2)^2 a$. This is negative, because $z \le (7/6)^2 < 2$ and $S \ge 3$. Hence $A_2 > 0$.

The residual is negative below q_b. At fixed p the residual $\mathcal{R}(u_5;p,q)$ is affine in q, so on $\{0 \le q \le \min(q_b, p/2)\}$ its maximum is on $q = 0$, $q = q_b$ or $p = 2q$. On $q = 0$ the tangent bound $\log y \le y - 1$ makes it affine in $p \in [0, r_A]$, with values below $-7/10$ and $-33/190$ at the two ends. On $q = q_b$ the minimum of the two tangent lines $\log y \le y - 1$ and $\log y < \frac{5}{4}y - \frac{1223}{1000}$ is convex in p on each side of their crossing, so three values of p suffice; on $p = 2q$ the bound is a convex quadratic in q. These values are strictly convex in β, so it suffices to take the two extreme slopes $\beta = x(1 + (\alpha-1)/3)$ and $\beta = x(1 + 5(\alpha-1)/13)$. The six resulting signs are polynomials in $t = x - 2 \in [0, 1/6]$ and $c = \alpha - 1 \in [0, 3/19]$ that decrease in c and are positive at $c = 3/19$ by a coefficient comparison [7] (`MIXED_SMALLER_SHORT_SMALL_SHAPE_ROOT_SHEAR`, (9)–(27)).

The shear sign. In Proposition 9.11,

$$m = (8x^2 + 8x - 3)\alpha + (3x^2 - 8x - 8)\beta, \qquad W = (\beta x - \alpha)\, m,$$
$$V = \alpha^2 W + 3(x^2 + x + 1)(\alpha^2 + \alpha\beta + \beta^2)(2\beta x - x - \alpha\beta)(x - 1).$$

On the small-shape domain $1 \le \alpha \le 22/19$, $2 \le \beta \le 130/57$ and $\beta x - \alpha \ge 54/19$. The factorizations $8x^2 + 8x - 48 = 8(x-2)(x+3) \ge 0$ and $3x^2 - 8x + 4 = (x-2)(3x-2) \ge 0$ give $m \ge 45\alpha - 12\beta \ge 335/19$, so $W > 0$; and $2\beta x - x - \alpha\beta - x(2x - \alpha - 1) = (\beta - x)(2x - \alpha) \ge 0$

gives $2\beta x - x - \alpha\beta \geq 70/19$, so $V > 0$. For the final inequality write $x = 2 + t$, $\beta = x(1 + \delta)$ and $\alpha = 1 + (3 - k)\delta$ with $t \in [0, 1/6]$, $\delta \in [0, 1/19]$, $k \in [0, 2/5]$. Then $\Pi = 29V - 30W(x(x + 2) + \alpha^2)$ is a polynomial of 98 monomials whose part free of δ is at least $1869 + 9378t$, and whose 28 negative monomials all contain δ. Bounding these, grouped by powers of δ, by linear functions of t on the box gives $\Pi > 69 + 5578t > 0$ [7] (`MIXED_SMALLER_SHORT_SMALL_SHAPE_ROOT_SHEAR`, (32)–(34)).

E.7. **Gate S6.** *The tangent curve.* Fix $\nu \geq 0$ and $K > \nu$, and put $\mathcal{D} = \sqrt{K^2 - \nu^2}$ and $\tilde{h}(\delta) = (K \cos\delta - \mathcal{D})/\sin\delta$ for $0 < \delta < \pi$. Then $K \sin\delta + \tilde{h}\cos\delta = \sqrt{\tilde{h}^2 + \nu^2}$, because $(K - \mathcal{D}\cos\delta)^2 - (K\cos\delta - \mathcal{D})^2 = \nu^2 \sin^2\delta$; moreover $\tilde{h}'(\delta) = (\mathcal{D}\cos\delta - K)/\sin^2\delta < 0$, and $\tilde{h}$ inverts the function $\delta(h)$ of Section 9.7. Along the curve,

$$\frac{\mathrm{d}}{\mathrm{d}\delta}\bigl(-J_0\tilde{h}(\delta) - c\delta\bigr) = \frac{J_0\sqrt{\tilde{h}^2 + \nu^2} - c\sin\delta}{\sin\delta}.$$

On the small-shape domain this is negative while $-\tilde{h}R_x \geq \frac{5}{2}\nu$ and $0 < -\tilde{h} < 7R_x^2/6$ (the barrier lemma of the notes): its numerator has the sign of a quadratic $Q_c(H^2)$ in $H = -\tilde{h}$, which decreases when the norm deficit $1 - u/\mathcal{U}_1$ is lowered to its least value, increases with ν up to $\frac{2}{5}HR_x$, and is then an upward quadratic in H whose value at $H = 7R_x^2/6$ is a negative polynomial in R_x.

The polynomial bound. With $L(z) = \frac{z}{162} + \frac{161z^3}{1458} - \frac{161z^5}{7290}$, $b_i = 1 - u/\mathcal{U}_i$ and $c = 161/54$,

$$\mathfrak{B}(u; R, T, \delta) = L(R) + L(T) - \frac{b_1R + b_2T}{2} - \frac{(b_1 + b_2)Z_U}{8} - \frac{c\,\nu^2}{8Z_L},$$

where $Z_U = \frac{2R^2(1+R)}{3-R} + \frac{1+2R}{3(1-R)}\delta - \frac{3}{2}\delta^2$ and $Z_L = \frac{2R^2(1+R)}{3-R}(1 + \frac{11}{2}\delta)$ bound the central shear $-\sqrt{3}h_c$ from above and below, and

$$u_q(R, \delta) = \frac{3}{4} + \frac{3R^2 + 10R + 10}{15(2 + R)^3}\Bigl(R^2 - \frac{R(1 + R)\delta}{2} + \frac{19(1 + R)^2\delta^2}{84}\Bigr)$$

is the quadratic floor of the bridge norm. The coefficients of L come from the Taylor bound $\arctan y \leq y - y^3/3 + y^5/5$ $(y \geq 0)$, whose difference has derivative $y^6/(1 + y^2)$, combined with $1 - c/3 = 1/162$, $c/27 = 161/1458$ and $c/135 = 161/7290$. Since $\mathfrak{B}$ is affine in u with positive slope, u may be lowered to u_q. Writing $\alpha = 1 + e\delta$, $\beta = x(1 + \delta)$ and $x = (1 + R)/(1 - R)$ gives

$$T_s = T(R, \delta, e) = \frac{2R + ((e + 1)R - (e - 1))\delta}{2 + (e + 1 - (e - 1)R)\delta},$$

and the second derivative of $\mathfrak{B}$ in T_s is negative on $9/31 \leq T_s \leq 7/19$, $u \geq 3/4$. Proposition E.8 bounds $\mathfrak{B}$ from below.

E.8. **The face signs.** The face values are explicit functions of the shape (x, α, β), the norm u, the scale q and the duration τ, built from the scale form (9.9): w_*, $E_{L,*}$ and R_τ^* are evaluated at the given u and q, Γ_* is the Gram expression at $(t_0, \mathcal{U}(t_0))$, and $T = r_A/q$ is the upper end of the duration interval. Write $c_0(x) = \frac{x(7x+1)}{4x^2+11x+1}$ and call

$$x + \frac{8x(\alpha-1)}{9x+8} \le \beta \le x + c_0(x)(\alpha-1)$$

the *strip*; by Lemma 9.5 it contains the actual shapes for $1 < x \le 5$. In every case the proof runs in the same order: the norm is raised to its ceiling (the face value is monotone in u), the duration is removed by (9.10) or by a tangent bound for the logarithm, then β, then the shape α or $1/\alpha$, then x, each by the sign of a first or second derivative, and the remaining polynomials in one variable are checked on their intervals. Numerically, the smallest margins are about $3 \cdot 10^{-3}$ for gate S0, $6 \cdot 10^{-6}$ for gate S1, $4 \cdot 10^{-4}$ for gate S2, $2 \cdot 10^{-4}$ for gate S3, $3 \cdot 10^{-6}$ for gate S4 and $3 \cdot 10^{-5}$ for gate S6.

Proposition E.3 (Faces of gate S0)**.** *Let* $F(\tau, \ell) = (w_* - 1)\tau + \frac{1}{\tau} + \frac{u-1}{r_A} + r_A + 1 - q - E_{L,*} + \ell + \frac{1}{2}(w_* - \frac{3}{4})$.

(a) On $12/5 \le x \le 9$, $1 \le \alpha \le (5x+4)/12$, $x \le \beta \le x + c_0(x)(\alpha-1)$, *with* $q = q_c$ *and* $u = \hat{u}$, *the values* $F(2, \frac{5}{2T} - \frac{11}{9})$, $F(\frac{8T}{9}, -\frac{1}{9})$ *and* $F(T, 0)$ *are negative.*

(b) On $12/5 \le x \le 9$, $(22x+20)/55 \le \alpha \le x$, $x + \frac{4x(\alpha-1)}{5x+4} \le \beta \le x + c_0(x)(\alpha-1)$, *with* $q = 1/7$ *and* $u = \bar{u}$, *the same three values are negative.*

Proof sketch. (a) Each test is convex in β: as a function of β it is a quadratic with positive leading coefficient plus simple poles with positive residues. So it suffices to take $\beta = x$ and an enlarged upper face $\beta = x + c_j(\alpha - 1)$, with constant slopes $c_j \in \{7/8, 1, 5/4, 3/2\}$ on the four ranges $[12/5, 5/2]$, $[5/2, 3]$, $[3, 5]$, $[5, 9]$ of x; these slopes dominate c_0 because c_0 increases, with $c_0' = (73x^2 + 14x + 1)/(4x^2 + 11x + 1)^2$. On $\beta = x$ the cleared numerators increase in α, and at $\alpha = 1$ they factor through three polynomials in $x - 2$ that are positive for $x \ge 12/5$. On each upper face the cleared numerator $N_j(x, \alpha)$ has $\partial_x^2 N_j > 0$: its expansion in $X = x - z_j$, $A = \alpha - 1$ has degree eight, and bounding each coefficient of X^k below by discarding positive terms and evaluating negative ones at the largest A leaves a polynomial in X that is positive on the range. The values and first x-derivatives at the lower end of each range are polynomials of degree seven in α, positive by a second-order Taylor bound around the midpoint of the α-interval. Where the α-range ends on a slanted edge $x = 2\alpha - o_j$, a third derivative in α is positive, and the values on the edge follow.

(b) Here $\bar{u} \le \bar{V}(x) \le u_b := \frac{x+74}{100}$, by $96(4x^2 + 11x + 1)^2 - (7x + 1)^2(x + 1)(35x + 13) = -179x^4 + 5606x^3 + 11040x^2 + 1882x + 83 > 0$ on $[0, 9]$ and $u_b - \bar{V} = (x - 1)(35x^2 - 152x + 213)/(100(x + 1)(35x + 13)) > 0$. The

shape is split at $\alpha = m(x) = (2x+3)/5$. For $m \le \alpha \le x$ use u_b; with $\beta = x + c(\alpha - 1)$ and $3/5 \le c \le 3/2$, a cubic comparison in β leaves twelve polynomials of degree eight whose signs follow from one sign change in the shape variable. For $(22x+20)/55 \le \alpha \le m$ use the sharper ceiling $\bar{u} \le \hat{u}$, which follows from $75(4x^2+11x+1)^2 - (7x+1)^2(x+1)(35x+13) = -515x^4 + 3758x^3 + 8331x^2 + 1420x + 62 > 0$ on $[0,9]$ (its derivative divided by x^2 decreases, so its minimum is at an endpoint). Each test is then convex in β, and two shape derivatives on five fixed β-faces reduce every sign to a single rational evaluation [6]. □

Proposition E.4 (Faces of gate S1). *Let* $\Phi_\tau = \Gamma_*(q) - \frac{7}{96} R^*_\tau(q)$.

(a) On $1 < x \le 2$, $1 \le \alpha$, $6\alpha^2 \le x(x+2)$, β *in the strip and* $\beta \le \alpha x$, *with* $q = q_c$ *and* $3/4 \le u \le u_\ell$: $\Phi_\tau > 9193/50000$ *for* $2 \le \tau \le T$.

(b) On $1 < x \le 2$, $1 \le \alpha \le x$, $x(x+2) \le 6\alpha^2$, β *in the strip and* $\beta \le \alpha x$, *with* $q = 1/7$, $u_f \le u \le \bar{V}(x)$, *and* $u \le u_\ell$ *when* $x \ge 3/2$ *and* $\alpha \le 6/5$: $\Phi_\tau > 9193/50000$ *for* $2 \le \tau \le T$.

Proof sketch. By (9.10) and the bound $\log y \le 2z + \frac{2}{3}z^3$ ($z = \frac{y-1}{y+1}$, $0 < y \le 1$) at $y = 2/T$, it suffices to show $F_0 > 0$ and $F_1 > 0$, where $F_0 = \Gamma_* - \frac{7}{96}\bar{R}_2 - \frac{9193}{50000}$, $F_1 = F_0 - \frac{7}{96}(T-2)(w_* - \frac{3}{4})$ and $\bar{R}_2$ is R^*_2 with that bound for the logarithm.

(a) The cap $6\alpha^2 \le x(x+2)$ gives $x \ge \sqrt{7} - 1 > 8/5$ and $\alpha \le 2/\sqrt{3} < 7/6$, and the strip lies in $x + \frac{1}{2}(\alpha - 1) \le \beta \le x + \frac{10}{13}(\alpha - 1)$, by $13x(7x+1) - 10(4x^2+11x+1) = (x-2)(51x+5) \le 0$. On $u \in [3/4, 19/25]$ one has $-\frac{1}{5} < \partial_u F_0 \le \partial_u F_1 < 0$, and along the ceiling $u = u_\ell(x,\beta)$ the total β-derivative of each F_j is less than $-101/318720$; so u may be raised to u_ℓ and β to $\beta_+ = x + \frac{10}{13}(\alpha - 1)$. For F_0 the total α-derivative is less than $-4553/2589600$ below the cap, so α may be raised to $\sqrt{x(x+2)/6}$, where the central and outer formulas coincide and the argument of (b) gives $F_0 \ge 167/144000000$. For F_1, raise u to $19/25$; a third derivative shows that the cost part of F_1, which enters with a negative sign, has no interior maximum in $\alpha \in [1, 7/6]$, so F_1 is least at $\alpha = 1$ or $\alpha = 7/6$, where it decreases in x and exceeds $1/50000$ at $x = 2$.

(b) Three branches cover the domain. For $1 < x \le 3/2$ both tests exceed $499/884000$: monotonicity in u and in β, then concavity in $1/\alpha$, reduce them to $\alpha = 1$ and $\alpha = x$, where derivative signs in one variable conclude. For $3/2 \le x \le 2$ and $6/5 \le \alpha \le x$, concavity in $1/\alpha$ reduces to $\alpha = 6/5$ and $\alpha = x$, with margins $1/2000$, $3/500$ and $1/15000$. For $3/2 \le x \le 2$ and $1 \le \alpha \le 6/5$, $F_0 \ge 167/144000000$ after one split at $x = 5/3$; for F_1, concavity in β and in $u \in [3/4, 19/25]$ reduce to the four constant pairs $(\frac{\beta - x}{\alpha - 1}, u) \in \{\frac{1}{2}, \frac{10}{13}\} \times \{\frac{3}{4}, \frac{19}{25}\}$, the second derivative in $1/\alpha$ is less than $-11/800$, and the two shape ends give $F_1 \ge 1597/144000000$ [6]. □

Proposition E.5 (Faces of gate S2). *Let* $G_\tau = R^*_\tau(q) + (2 - \frac{7\tau}{47})(w_* - \frac{3}{4})$.

(a) *On* $13/6 \le x \le 12/5$, $1 \le \alpha$, $6\alpha^2 \le x(x+2)$, β *in the strip, with* $q = q_c$ *and* $u = u_C$*:* $G_\tau < -1/6000$ *for* $2 \le \tau \le T$.

(b) *On* $13/6 \le x \le 12/5$, $\alpha \le x$, $x(x+2) \le 6\alpha^2$, β *in the strip, with* $q = 1/7$*, and* $u = u_s$ *if* $\alpha \le 3/2$, $u = \bar{V}(x)$ *if* $\alpha \ge 3/2$*:* $G_\tau < 0$ *for* $2 \le \tau \le T$.

Proof sketch. (a) The domain lies in $1 \le \alpha \le 4/3$, $x + \frac{5}{8}(\alpha - 1) \le \beta \le \beta_+ = x + \frac{24x+51}{128}(\alpha - 1)$. The function $\log\tau + 1/\tau$ is concave on $[2, \infty)$, its second derivative being $(2-\tau)/\tau^3$; its tangent at $t(\alpha) = \frac{20}{3} - 4\alpha$ ($\alpha \le 7/6$), $t = 2$ ($\alpha \ge 7/6$) makes the bound for G_τ affine in τ, so $G_\tau \le \max\{F_0, F_1\}$ for two explicit tests. Both increase in β, so $\beta = \beta_+$; both have negative total x-derivative (below $-1/250$ and $-1/2000$), so $x = 13/6$. There the norm is raised to an affine upper bound in α, the logarithm is bounded by a quadratic, and four polynomials in one variable with nine terms each give $F_0, F_1 < -1/6000$.

(b) Here $G'_\tau \le (1 - \frac{7}{4T})(w_* - \frac{3}{4})$, so $G_\tau \le \max\{H_0, H_1\}$, the values at $\tau = 2$ and at the end of the affine bound. The domain gives $\alpha \ge \frac{36x+25}{84} > \frac{6}{5}$, by $84^2 x(x+2) - 6(36x+25)^2 = -720x^2 + 3312x - 3750 < 0$, and $\beta = x + c(\alpha - 1)$ with $\frac{5}{8} \le c \le \frac{24x+51}{128} < \frac{7}{8}$. For $\alpha \le 3/2$, each H_j is convex in $1/\alpha$ (second derivative greater than $2/5$). At $\alpha = \frac{36x+25}{84}$ it increases in β and is negative at the largest slope, with margins $1/5000$ and $1/1000$ on $[13/6, 9/4]$ and $[9/4, 12/5]$; at $\alpha = 3/2$, $u_s \le \bar{V}$, H_j increases in u, and its value at $\bar{V}$ is negative. For $\alpha \ge 3/2$ the second derivative in $1/\alpha$ exceeds $1/2$, and the values at $\alpha = 3/2$ and $\alpha = x$ are negative for every slope [6]. □

Proposition E.6 (Faces of gate S3)**.** *Put* $y = 2q/r_A$, $k = 5\alpha - 1$, $J = \frac{u-1}{r_A} + r_A + 1 - q - E_{L,*} + \log y$, $F_0 = J + (2 + \frac{4(4-3y)}{k})(w_* - \frac{3}{4})$ *and* $F_1 = J + (\frac{r_A}{q} + \frac{4}{k})(w_* - \frac{3}{4})$.

(a) *On* $2 \le x \le 13/6$, $\frac{15x+511}{475} \le \alpha \le \frac{89}{72}$, $\beta = x + \frac{25x}{15x+36}(\alpha - 1)$*, with* $q = q_c$ *and* $u = u_{CB}$*:* $F_0, F_1 < 0$.

(b) *On* $2 \le x \le 13/6$, $\alpha \le x$, $x(x+2) \le 6\alpha^2$, $\beta = x + C(\alpha - 1)$ *with* $\frac{8x}{9x+8} \le C \le \frac{115x}{304}$*, with* $q = 1/7$ *and* $u = u_O$*:* $F_0, F_1 < 0$.

Proof sketch. (a) Along the curve both tests decrease in α. Grouping three terms of the α-derivative, and bounding the norm and the transported quantities by their ranges, reduces its sign to the single quadratic inequality

$$\frac{65452}{825}q^2 - \frac{68203}{3750}q + \frac{29299}{30000} < 0 \qquad (63/500 \le q \le 1/7),$$

which holds because the quadratic is convex and negative at both ends [6]. So both tests are at most their values at $\alpha = \frac{15x+511}{475}$, $\beta = \frac{20x}{19}$. There $\log y \le -\frac{40153}{180000} + \frac{2(5y-4)}{5y+4}$ for $0 < y \le 6/7$, and the tests become $-N_i(x - 2)$ divided by a positive polynomial, where N_0 and N_1 have degrees 17 and 15 and all their coefficients are positive except the linear one [7] (`MIXED_SMALLER_SHORT_DISPLACEMENT_CENTRAL_FACE`, (12)). For N_1 the

quadratic part has negative discriminant. For N_0, write c_j for its coefficients and $h = x - 2 \in [0, 1/6]$: the quadratic part decreases on $[0, 1/6]$, since its vertex $-c_1/(2c_2)$ exceeds $1/6$, and it is positive at $h = 1/10$; on $[1/10, 1/6]$ the cubic term $c_3 h^3 \geq c_3/1000$ exceeds the absolute value of the quadratic part at $h = 1/6$. So $N_0, N_1 > 0$.

(b) The multipliers satisfy $n_0, n_1 \in [25/8, 7/2]$ for $\alpha \leq 5/4$ and $n_0, n_1 \in [5/2, 7/2]$ for $\alpha \geq 5/4$, by elementary quadratic bounds, and the domain gives $\alpha \geq \alpha_* = \frac{5(x+1)}{13}$, since $\frac{x(x+2)}{6} - \alpha_*^2 = \frac{19x^2+38x-150}{1014} > 0$. For a constant n the function $R_n = J + n(w_* - \frac{3}{4})$ is affine in n and has second derivative greater than $1/4$ in $1/\alpha$. Its values $R_{25/8}, R_{7/2}$ at α_*, $R_{5/2}$ at $\alpha = 5/4$ and R_n at $\alpha = x$ are negative for every slope C, by polynomial signs in one and two variables of degree at most eight [6]. Convexity in $1/\alpha$ and affinity in n conclude. □

Proposition E.7 (Face of gate S4)**.** *On $2 \leq x \leq 13/6$, $0 \leq \delta \leq 1/19$, $\alpha = 1 + e\delta$ with $13/5 \leq e \leq 3$, $\beta = x(1+\delta)$, and $u = \bar{u}(x, \beta)$, the functions Φ and Ψ of Proposition 9.10 exceed* $1829/10000$.

Proof sketch. Six derivative signs reduce Φ and Ψ to three points [6]: $-\Phi_\delta > 27/5000$ on the whole box; $\Psi_e/(x\delta) > 0$, extended continuously to $\delta = 0$ (note $\Psi_e = \delta\Psi_\alpha$); $-\Psi_x > 11/14000$ at $e = 13/5$; $-\Phi_x > 1/1750$ at $\delta = 1/19$; $-\Phi_e > 0$ at $x = 13/6$, $\delta = 1/19$; and $-\Psi_{\delta\delta} > 81/1000$ at $x = 13/6$, $e = 13/5$. Hence $\Phi \geq \Phi(\frac{13}{6}, \frac{1}{19}, 3)$ and $\Psi \geq \min\{\Psi(\frac{13}{6}, 0, \frac{13}{5}), \Psi(\frac{13}{6}, \frac{1}{19}, \frac{13}{5})\}$. Each of the three values is a rational number minus $\frac{1}{7}\log y$, with y equal to $\frac{103488}{134749}$, $\frac{12}{19}$ and $\frac{2524608}{3353029}$, and the bound $\log y < 2(z + z^3/3 + z^5/5)$, $z = \frac{y-1}{y+1} < 0$, shows that they exceed $1829/10000$ by more than $2053/25000000$, $4017/20000000$ and $303/100000000$. □

Proposition E.8 (Face of gate S6)**.** *For $1/3 \leq R \leq 7/19$, $0 \leq \delta \leq 1/19$ and $13/5 \leq e \leq 3$,*

$$\mathfrak{B}\big(u_q(R, \delta); R, T(R, \delta, e), \delta\big) > \frac{113}{50000} > b^*.$$

Proof sketch. By concavity in T_s it suffices to take $e \in \{13/5, 3\}$. Put $v = 4u_q - 3$ and $k(a) = \frac{a}{162} + \frac{161a^3}{1458} - \frac{161a^5}{7290} - \frac{a^3}{2(3+a^2)}$, $g(a) = \frac{a}{2(3+a^2)}$, $p(a) = \frac{a^2}{3+a^2}$, $\bar{q}(a) = \frac{1}{3+a^2}$. Since $b_i = (a_i^2 - v)/(3 + a_i^2)$ for $a_1 = R$ and $a_2 = T_s$, the bound is exactly

$$\mathfrak{B} = K + vG - \frac{Z_U P}{8} + \frac{v Z_U \bar{Q}}{8} - \frac{c\,\nu^2}{8 Z_L},$$

with $K = k(R) + k(T_s)$ and similarly $G, P, \bar{Q}$. In $s = a^2$,

$$\frac{k''(a)}{a} = -\frac{322s^4 + 2415s^3 + 4347s^2 - 6534s + 6642}{729(s+3)^3},$$

which with $k' < 0$ gives $k'' < -2/25$ on $[9/31, 7/19]$. The derivatives of $g, p, \bar{q}$, of v (a quadratic in δ with explicit coefficients in R), of T_s (fractional

linear in R) and of Z_U have simple explicit signs and bounds. Adding the bounds term by term gives $\mathfrak{B}_{RR} < -\frac{17}{100} + \frac{19}{100} - \frac{1}{10} + \frac{3}{50} + \frac{1}{100} = -\frac{1}{100}$ on the whole box, so the minimum in R is at $R = 1/3$ or $R = 7/19$. On these four curves $\mathfrak{B}$ decreases in δ: an upper bound for the derivative, assembled from the same term bounds at the ends of $[0, 1/19]$ (split at $1/38$ when $e = 3$), is below $-1/1000$. So the minimum is at $\delta = 1/19$, where the four values exceed $2260000 \cdot 10^{-9} = 113/50000$, while $b^* < 223/100000$ [6]. □

Appendix F. The four- and five-piece estimates

This appendix proves Theorems 10.7 and 10.12. Both reduce to functions of one variable, and both rest on the same elementary bound for the logarithm.

F.1. **A logarithm bound.**

Lemma F.1. *For* $0 < x \le 1$, $\dfrac{x - x^{-1}}{2} \le \log x$.

Proof. With $y = -\log x \ge 0$ the inequality reads $y \le \sinh y$. □

The two sides agree to second order at $x = 1$: both are $(x-1) - \frac{1}{2}(x-1)^2 + O((x-1)^3)$. This is what makes the bound useful near the octagon, where the parameters below approach the points at which the estimates are sharp.

F.2. **The four-piece density.**

Recall $\delta(a, b)$ from (10.4), on the open quarter circle $a, b > 0$, $a^2 + b^2 = 1$, with $s = a + b$.

Lemma F.2 (A monotone weight)**.** *For* $c \ge 2$ *the function* $w_c(x) = (2x + c)\log x/(1-x)$ *is nondecreasing on* $(0, 1)$. *Consequently, for* $0 < b \le a < 1$,

$$(2b + c)(1 - a)\log b \;\le\; (2a + c)(1 - b)\log a.$$

Proof. The derivative is $w_c'(x) = \big((2 + c)\log x + (2x + c)(1 - x)/x\big)/(1 - x)^2$. By Lemma F.1 and $2 + c > 0$, the numerator is at least

$$(2 + c)\frac{x - x^{-1}}{2} + \frac{(2x + c)(1 - x)}{x} = \frac{(c - 2)(1 - x)^2}{2x} \;\ge\; 0.$$

The second statement is $w_c(b) \le w_c(a)$ multiplied by $(1 - a)(1 - b) > 0$. □

Proof of Theorem 10.7. Since $\delta(a, b) = \delta(b, a)$, we may assume $b \le a$, that is $b \in (0, 1/\sqrt{2}\,]$ and $a = \sqrt{1 - b^2}$. Put $\psi(b) = \delta(\sqrt{1 - b^2}, b)$. For $0 < b < 1/\sqrt{2}$, differentiating (10.4) with $a' = -b/a$ and simplifying on the circle gives, with $c = 2 + 2ab$,

$$\psi'(b) = \frac{2\Big((s + 1)\big(w_c(b) - w_c(a)\big) - s(s - 1)(a - b)\Big)}{a\,(s + 1)\,(s + 3)^2}. \tag{F.1}$$

Here $c \ge 2$ and $b < a$, so $w_c(b) \le w_c(a)$ by Lemma F.2, while $s > 1$ and $a > b$ make the last term negative. Hence $\psi' < 0$ on $(0, 1/\sqrt{2})$, and ψ, which is continuous at $1/\sqrt{2}$, is strictly decreasing on $(0, 1/\sqrt{2}\,]$. For

$b < 1/\sqrt{2}$ this gives $\delta(a,b) = \psi(b) > \psi(1/\sqrt{2}) = \delta_{\mathrm{oct}}$, the value computed after Theorem 10.7. So $\delta(a,b) \geq \delta_{\mathrm{oct}}$ with equality only at $a = b$. □

The computation behind (F.1) is mechanical. Write $\delta = \mathcal{D}/\mathcal{N}$ with $\mathcal{D}$ the numerator of (10.4) and $\mathcal{N} = (s-1)(s+3)$, and let a prime denote $\mathrm{d}/\mathrm{d}b$ along the circle, so that $a' = -b/a$ and $s' = (a-b)/a$. By the quotient rule, $\psi'(b) = \kappa/(a^2 b \mathcal{N}^2)$ with $\kappa = a^2 b\,(\mathcal{D}'\mathcal{N} - \mathcal{D}\mathcal{N}')$, a polynomial in a, b, $\log a$ and $\log b$, and modulo $a^2 + b^2 - 1$ it satisfies

$$\kappa\,(1-a)(1-b)(s+1) = 2ab(s-1)^2\Big((s+1)\,\Delta - (a-b)\,s\,(s-1)(1-a)(1-b)\Big),$$

where $\Delta = (2b+c)(1-a)\log b - (2a+c)(1-b)\log a$ is the quantity of Lemma F.2. Since $\Delta = (1-a)(1-b)\big(w_c(b) - w_c(a)\big)$ and $\mathcal{N}^2 = (s-1)^2(s+3)^2$, this is (F.1).

F.3. **The five-piece density.** Recall F, $d = (1-u)(1-v)$ and $\delta(u,v)$ from (10.6), and put $t_* = \sqrt{2} - 1$, $d_* = (2-\sqrt{2})^2$ and $\mathcal{K} = (3 + 2\sqrt{2})\log 2$. Since $(3+2\sqrt{2})(2-\sqrt{2})^2 = 2$, we have $\mathcal{K} d_* = 2\log 2$.

The function F is not concave (`FIVE_ARC_PROGRESS`, §7), so its maximum cannot be read off from a second derivative. It becomes tractable after a change of variables that splits it into two values of one concave function at two points of the unit circle.

Lemma F.3 (A concave function)**.** *Let $h(t) = -2t\log t/(1-t) - \log(1+t)$ for $0 < t < 1$. Then $x \mapsto h(\sqrt{x})$ is concave on $(0,1)$. Consequently $h(\alpha) + h(\beta) \leq 2h(1/\sqrt{2})$ whenever $\alpha, \beta > 0$ and $\alpha^2 + \beta^2 = 1$.*

Proof. Put $E(t) = t - 1 - \log t \geq 0$. Then $h'(t) = 2E/(1-t)^2 - 1/(1+t)$, and the derivative of $h(\sqrt{x})$ is $k(\sqrt{x})/2$ with $k(t) = h'(t)/t$, so it suffices that k is nonincreasing. Now $k'(t) = (th''(t) - h'(t))/t^2$ and

$$th'' - h' = -\frac{2}{1-t} + \frac{2(3t-1)E}{(1-t)^3} + \frac{1+2t}{(1+t)^2} = \frac{\mathcal{M}(t)}{(1-t)^3(1+t)^2},$$

$$\mathcal{M}(t) = -2(1-t)^2(1+t)^2 + 2(3t-1)E(1+t)^2 + (1+2t)(1-t)^3.$$

If $t \leq 1/3$, the middle term of $\mathcal{M}$ is at most 0, and the other two add up to $-(1-t)^2(1+3t+4t^2) < 0$. If $t > 1/3$, Lemma F.1 gives $E \leq t - 1 - \frac{1}{2}(t - t^{-1}) = (1-t)^2/(2t)$, so $2tE \leq (1-t)^2$ and

$$\begin{aligned} t\,\mathcal{M}(t) &\leq -2t(1-t)^2(1+t)^2 + (3t-1)(1+t)^2(1-t)^2 + t(1+2t)(1-t)^3 \\ &= (1-t)^3(t^2 - t - 1) < 0. \end{aligned}$$

For the second statement apply concavity at $x = \alpha^2$ and $x = \beta^2$ with weights $\frac{1}{2}$: $\frac{1}{2}h(\alpha) + \frac{1}{2}h(\beta) \leq h\big(\sqrt{(\alpha^2+\beta^2)/2}\big) = h(1/\sqrt{2})$. □

Lemma F.4 (The envelope)**.** *For $0 < t < 1$,*

$$F(t) + 2\log(1-t) \;\leq\; \mathcal{K} + 2\log(2-\sqrt{2}) = \mathcal{K} + \log d_*.$$

The bound is attained at $t = t_*$: there $(1+t^2)/(2t) = (1+t^2)/(1-t^2) = \sqrt{2}$, $2t/(1-t^2) = 1$ and $(1+t)/(1-t) = 1/t = \sqrt{2}+1$, so $F(t_*) = (3+2\sqrt{2})\log 2 = \mathcal{K}$, and $1 - t_* = 2 - \sqrt{2}$.

Proof. With $\alpha = 2t/(1+t^2)$ and $\beta = (1-t^2)/(1+t^2)$, so that $\alpha^2 + \beta^2 = 1$, expanding the logarithms of the products $1-\alpha = (1-t)^2/(1+t^2)$, $1+\alpha = (1+t)^2/(1+t^2)$, $1-\beta = 2t^2/(1+t^2)$ and $1+\beta = 2/(1+t^2)$ gives the identity

$$F(t) + 2\log(1-t) = \log 2 + h(\alpha) + h(\beta).$$

By Lemma F.3 this is at most $\log 2 + 2h(1/\sqrt{2})$. Since $\log(1/\sqrt{2}) = -\frac{1}{2}\log 2$ and $1 + 1/\sqrt{2} = (2+\sqrt{2})/2 = 1/(2-\sqrt{2})$,

$$h(1/\sqrt{2}) = \frac{\sqrt{2}\cdot\frac{1}{2}\log 2}{1 - 1/\sqrt{2}} - \log\frac{1}{2-\sqrt{2}} = (1+\sqrt{2})\log 2 + \log(2-\sqrt{2}),$$

so $\log 2 + 2h(1/\sqrt{2}) = (3+2\sqrt{2})\log 2 + 2\log(2-\sqrt{2})$. □

Lemma F.5. *If $u, v \in (0,1)$ and $u+v+uv > 1$, then $d = (1-u)(1-v) < d_*$.*

Proof. Put $x = 1-u$, $y = 1-v$ and $\omega = \sqrt{xy}$. The hypothesis is $(2-x)(2-y) > 2$, that is, $2 - 2(x+y) + xy > 0$. Since $x + y \geq 2\omega$, also $\omega^2 - 4\omega + 2 > 0$, so $\omega < 2 - \sqrt{2}$ or $\omega > 2 + \sqrt{2}$; the second is impossible because $\omega < 1$. Hence $d = \omega^2 < d_*$. □

Proof of Theorem 10.12. Adding Lemma F.4 at $t = u$ and $t = v$, and using $\log d = \log(1-u) + \log(1-v)$,

$$F(u) + F(v) \leq 2\mathcal{K} + 2\log d_* - 2\log d.$$

Multiplying by $d/2 > 0$,

$$(4-d)\,\delta(u,v) = 2d + 4 - \tfrac{d}{2}\big(F(u)+F(v)\big) \;\geq\; 2d + 4 - \mathcal{K}d - d\log d_* + d\log d.$$

The tangent line of $d \mapsto d\log d$ at d_* gives $d\log d \geq d\log d_* + d - d_*$ (this is $\log(d/d_*) \geq 1 - d_*/d$), so

$$(4-d)\,\delta(u,v) \;\geq\; 4 - d_* + (3-\mathcal{K})\,d.$$

At $d = d_*$ the right side equals $\delta_{\mathrm{oct}}(4-d_*)$: indeed $2d_* + 4 - \mathcal{K}d_* = 16 - 8\sqrt{2} - 2\log 2$ and $4 - d_* = 2(2\sqrt{2}-1)$. Subtracting $\delta_{\mathrm{oct}}(4-d) = \delta_{\mathrm{oct}}(4-d_*) + \delta_{\mathrm{oct}}(d_*-d)$,

$$(4-d)\big(\delta(u,v) - \delta_{\mathrm{oct}}\big) \;\geq\; (\mathcal{K} - 3 - \delta_{\mathrm{oct}})(d_* - d).$$

Both factors on the right are positive: $d < d_*$ by Lemma F.5, and $\mathcal{K} = 4.0400\ldots > 3 + \delta_{\mathrm{oct}} = 3.9024\ldots$; indeed $\sqrt{2} > 1.4142$ and $\log 2 > 0.6931$ give $\mathcal{K} > 4.039$ and $\delta_{\mathrm{oct}} < 0.9025$. Since $4 - d > 0$, $\delta(u,v) > \delta_{\mathrm{oct}}$. □

Remark F.6. Another way to finish [7]: with $M = \mathcal{K} + \log d_*$, the bound obtained from Lemma F.4 is $\delta(u,v) \geq B(d) = \big(2(d+2) - d(M - \log d)\big)/(4-d)$, and

$$B'(d) = \frac{4(4 - M + \log d) - d}{(4-d)^2} < 0 \qquad (0 < d \leq d_*),$$

because $4 - M + \log d \le 4 - \mathcal{K} < 0$ there. So B decreases on $(0, d_*]$ to $B(d_*) = \delta_{\mathrm{oct}}$. The inequality $\mathcal{K} > 4$ also follows from the rational bounds $\sqrt{2} > 7/5$ and $\log 2 > 56/81$.

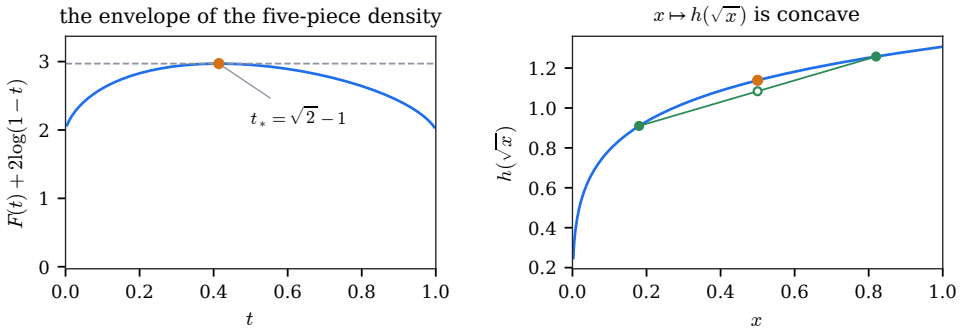


FIGURE 13. Left: $F(t) + 2\log(1-t)$ on $(0,1)$; its maximum $(3+2\sqrt{2})\log 2 + 2\log(2-\sqrt{2}) = 2.970\ldots$ is attained only at $t_* = \sqrt{2} - 1$ (Lemma F.4). Right: the concave function $x \mapsto h(\sqrt{x})$ of Lemma F.3; a chord (green) lies below the graph, and the midpoint of the chord between α^2 and $\beta^2 = 1-\alpha^2$ lies below the value at $x = 1/2$ (orange). Computed; the script checks the identity $F(t) + 2\log(1-t) = \log 2 + h(\alpha) + h(\beta)$ and the concavity on a grid.

APPENDIX G. THE FORMAL VERIFICATION

The proof of this paper is formalized in Lean 4 [8] with Mathlib [33], in the repository [37]. This appendix says what the formalization states, how it is checked, on what it depends, and where its arguments differ from the text, and it lists the Lean declarations that prove the numbered results of the paper.

G.1. **The statements.** The file `Palomar/Challenge.lean` imports only Mathlib and states the three parts of Theorem 1.1 as the theorems `octagon_density`, `reinhardt` and `reinhardt_equality` of the namespace `ReinhardtChallenge`. Its definitions are those of Definition 2.1, which is Definition 2.1.1 of [15]. Points of the plane are maps $\{0,1\} \to \mathbb{R}$ and areas are Lebesgue measure. A convex disk is a compact convex set with nonempty interior, and a congruent copy of K is its image under a map $x \mapsto Ax + b$ with A orthogonal. A packing is a family of convex disks whose members have pairwise disjoint interiors. Its density exists when the upper and lower limits, as $r \to \infty$, of the fraction of the disk of radius r about the origin that it covers are finite and equal, and $\delta(K)$ is the supremum of the densities that exist over the packings by congruent copies of K. A disk is centrally symmetric when there is a point c with $2c - x \in K$ for every $x \in K$. The

smoothed octagon is given in Reinhardt's description (Definition 2.3): with $d_k(x)$ the signed distance from x to the line of the k-th side of the regular octagon of inradius 1, it is the octagon $d_k \geq 0$ with the corner between sides k and $k+1$ cut off along the hyperbola $d_{k-1}d_{k+2} = \sqrt{2}/4$. These definitions are copied word for word from the library, so the theorems the library proves are the theorems the file states.

G.2. **The check.** The root module `SmoothedOctagon.lean` restates the three theorems and proves them with no hypothesis. Lean's kernel checks every step, and every theorem depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`; an audit script checks this for each of the 8,294 declarations of the library. The Comparator configuration `Palomar/comparator.json` checks that the root module proves the statements of `Palomar/Challenge.lean` with the same definitions. The proof is registered in the Palomar registry [28] as `PALOMAR-2026-11-09-000008`, at trust level high: Palomar's verifier checked it with the independent kernels nanoda and con-ron at commit `9ba6bf8` of [37]. Later commits rename and reorganize the library and leave the three statements unchanged; the names below are those of the current library. Lean and Mathlib are pinned at version `v4.36.0-rc2`. The library has about 110,000 lines in 344 modules, and a full build takes about 3.5 CPU-hours.

The library depends on two others. The first [38] formalizes the parts of the book of Hales and Vajjha [15] that the proof uses, together with the classical results behind them: among them Fejes Tóth's theorem (Theorem 2.2), the theorems of Minkowski and Mahler on critical lattices and circumscribed hexagons, the compact domain (Theorem 2.14) and the lift of a global minimizer to a Pontryagin extremal (Theorem 3.2). The second [36] proves the Pontryagin maximum principle [29] with free terminal time, an integral cost and mixed boundary conditions, by the argument of Agrachev and Sachkov [1], and Filippov's theorem [13]; Brouwer's fixed-point theorem comes from [35].

G.3. **The scalar estimates.** Most scalar inequalities of the paper are proved in Lean as in the text: polynomial identities by ring normalization, polynomial inequalities by nonlinear arithmetic with explicit products as hints, monotonicity and concavity in one variable by derivatives, and the logarithm by the tangent and Padé bounds of the text and Mathlib's bounds for $\log 2$ and π. Some polynomial signs on boxes, in class F (Section 7) and in the nonshort windows of class M2 (Section 8), are checked by expansion in the Bernstein basis of the box: scripts in `scripts/certificates/` of [37] compute the coefficients, `ring` checks the identity between the polynomial and its Bernstein form, and normalization checks the signs of the rational coefficients. Neither interval arithmetic nor floating-point computation enters the proof.

G.4. **Where the formal proof argues differently.** The statements of the paper are those that Lean proves, sometimes in a simpler notation. In a few places the text proves an intermediate inequality by a different argument from the Lean proof, usually because the hand argument is shorter to read and the formal one shorter to check:

- In class F, the four quartic signs of the conditional Casimir bound (Lemma 7.13) have hand arguments in the text; Lean proves them by nonlinear arithmetic with products of Bernstein basis polynomials as hints.
- In the nonshort windows of class M2 (Appendix D), the text proves the scale strip by monotonicity in t and a comparison at $t_0 = 13/35$, the low ratio by shape monotonicity, and the bound $\chi > 4/7$, the elimination of q and the envelope by concavity arguments; Lean proves the corresponding signs by Bernstein expansion, and the low ratio by concavity in k with Bernstein expansions at both ends.
- In class M1 the text proves the two-centre bound in the exact form $\mathcal{C} > 2\sqrt{6} - \sqrt{3}$ (Proposition 6.11); Lean proves the weaker numerical bound that the argument needs.
- In the smaller-short window (Section 9), the text proves the outer face sign of gate S3 (Proposition E.6) on a different domain from the Lean statement; both domains contain the cases that occur.

The full small-scale exclusion of Remark 8.16 is not formalized; the proof uses only its unit face (Lemma D.8).

G.5. **Dictionary.** Table 7 lists, for each numbered result of the paper, the Lean declarations that prove it. Names are relative to the namespace `SmoothedOctagon`; the prefixes `ReinhardtChallenge` and `SmoothedPolygons` mark the statements of the challenge file and the declarations of [38]. Several names in a row are the pieces of one result.

Table 7: The Lean declarations behind the numbered results.

Result	Lean declarations
Sections 1 and 2	
Theorem 1.2	`reinhardt_cost`, `controlLowerBound`
Theorem 1.1(a)	`ReinhardtChallenge.octagon_density`, `smoothedOctagon_spec`
Theorem 1.1(b)	`ReinhardtChallenge.reinhardt`, `reinhardtConjecture`
Theorem 1.1(c)	`ReinhardtChallenge.reinhardt_equality`, `reinhardt_equality_trajectory`
Definition 2.1	`ReinhardtChallenge.greatestPackingDensity`
Definition 2.3	`ReinhardtChallenge.smoothedOctagon`
Definition 2.7	`SmoothedPolygons.IsReinhardtMinimizer`
Theorem 2.14	`SmoothedPolygons.theorem5_0_2`
Section 3	

Result	Lean declarations
Definition 3.1	`SmoothedPolygons.IsPontryaginExtremal`
Theorem 3.2	`SmoothedPolygons.exists_isPontryaginExtremal_of_isReinhardtMinimizer`
Lemma 3.3	`IsPontryaginExtremal.continuousOn_vertexHamiltonian`, `IsPontryaginExtremal.vertexHamiltonian_max_eq_zero`
Lemma 3.4	`hamiltonian_vertices_equal_iff_reducedCostate_eq_zero`, `hamiltonian_maximizers_in_proper_edge`
Lemma 3.5	`Reduction.clr_winding`, `Reduction.clr_angular`
Lemma 3.6	`Reduction.clr_vertex_bounds`
Proposition 3.7	`Reduction.clr_budget`, `Reduction.clr_lam_sq_le`
Proposition 3.8	`Reduction.covectorNormSq_le_singularGapSq`, `Reduction.switch_singularGapSq_pos`
Proposition 3.10	`Reduction.edgeBound_of_isReinhardtMinimizer`, `Reduction.Edge.edge_bangBang`
Lemma 3.11	`Reduction.switch_exists_excluded_vertex`
Proposition 3.12	`Reduction.exists_isPiecewiseVertex`
Definition 3.13	`VertexRepresentation`
Lemma 3.14	`Reduction.exists_rebase_vertexRepresentation`, `Reduction.isReinhardtMinimizer_rebase`
Definition 3.15	`NormalPeriod`, `NormalPeriod.IsCompetitive`, `NormalPeriod.InCompactDomain`, `NormalPeriod.activeMargin`
Theorem 3.16	`Reduction.lam_lt_zero_of_vertexRepresentation`, `Reduction.exists_normal_lift`
Lemma 3.17	`Reduction.gn_piece`
Theorem 3.18	`finiteNormalRepresentation`, `Reduction.cost_le_octagonCost`
Sections 4 and 5, Appendix A	
Lemma 4.1	`vertexPiece_eq`, `vertexPiece_margin`, `vertexPiece_cost`, `NormalPeriod.piece_margin`, `NormalPeriod.cost_eq_sum`
Lemma 4.2	`CostateCycle.factor_pp`, `CostateCycle.factor_mp`, `CostateCycle.factor_pm`, `CostateCycle.centre_factor`
Lemma 4.3	`CostateCycle.rawCost_eq`, `CostateCycle.arc_eq`, `NormalPeriod.cost_eq_rawCost`
Lemma 4.4	`NormalPeriod.tangentProduct_N`, `Term.triC_conj_tangentFactor`, `Term.triC_conj_R`
Definition 4.5	`VertexRepresentation.IsForward`, `VertexRepresentation.IsReverse`, `VertexRepresentation.IsReverseRun`, `VertexRepresentation.IsForwardRun`
Lemma 4.6	`NormalPeriod.ff_N_mod_three`

Result	Lean declarations
Definition 4.7	`VertexRepresentation.AllReverse`, `VertexRepresentation.AllForward`, `VertexRepresentation.ClassM3`, `VertexRepresentation.ClassM1`, `VertexRepresentation.ClassM2`
Proposition 4.8	`VertexRepresentation.partition`
Definition 4.9	`costateL`, `costateA`, `costateB`, `costateAlpha`, `costateBeta`, `CostateCycle.sig`
Lemma 4.10	`margin_mul_hamiltonian`, `costateB_perp`, `vertexPiece_costate`, `NormalPeriod.cut_transport`, `NormalPeriod.sig_le_of_mem`, `NormalPeriod.cut_sig_grow_le`, `NormalPeriod.cut_sig_decr_le`, `CostateCycle.sig_act_succ`, `NormalPeriod.costateCycle`
Definition 4.11	`NormalPeriod.Ucal`, `CostateCycle.mu`, `NormalPeriod.mu_zero`, `CostateCycle.gam2`, `NormalPeriod.det_Ucal`, `triY`, `CostateCycle.Mloc`
Lemma 4.12	`CostateCycle.yv_norm`, `CostateCycle.cross_pp`, `CostateCycle.cross_mm`, `CostateCycle.yv_pair`, `CostateCycle.outv_turn`, `CostateCycle.yv_add_N`, `CostateCycle.pre_N`
Lemma 4.13	`CostateCycle.stationarity`, `CostateCycle.centre_stat`
Lemma 4.14	`CostateCycle.swd_fwd`, `CostateCycle.swd_rev`, `CostateCycle.turn_secant`, `CostateCycle.turn_swd_ge`
Lemma 4.15	`CostateCycle.gram_pp`, `CostateCycle.gram_mm`
Proposition 4.16	`NormalPeriod.gam2_pos_of_mixed`, `NormalPeriod.wire_gam2_pos_of_allForward`
Proposition 4.17	`NormalPeriod.cost_casimir`, `el_winding`
Corollary 4.18	`NormalPeriod.gam2_le_cap`
Theorem 4.19	`period_le_five`
Lemma 5.1	`cstRevN_neg`
Lemma 5.2	`cst_rev4_third_identity`, `cst_rev4_second_identity`
Lemma 5.3	`CostateCycle.reverse_four_third`, `CostateCycle.reverse_four_second`
Theorem 5.4	`reverse_exclusion`
Lemma 5.5	`CostateCycle.three_arc`, `CostateCycle.three_arc_not_min`, `CostateCycle.turn_up`, `CostateCycle.turn_down`, `cst_three_grow`, `cst_three_decr`
Proposition 5.6	`reverseRun_two_and_centre_min`
Theorem 5.7	`minimumStructure_exclusion`
Corollary 5.8	`NormalPeriod.rev_run`, `NormalPeriod.centre_min`, `NormalPeriod.swd_ge_of_mixed`

Result	Lean declarations
Lemma 5.9	`el_three_gram`
Lemma 5.10	`CostateCycle.cre_pos_pp`, `CostateCycle.cre_neg_mm`, `el_cross_bound`
Lemma A.1	`cst_log_pade`, `cst_zeta_gt`, `cst_eta_lt_zeta`
Section 6, Appendix B	
Theorem 6.1	`shortRuns_exclusion`
Lemma 6.2	`VertexRepresentation.m1_pattern`
Lemma 6.3	`NormalPeriod.m1_centreData`, `m1_CentreData`
Proposition 6.4	`m1_angle_gap`
Corollary 6.5	`m1_arccos_gt`, `m1_cos_two_pi_div_five`
Proposition 6.7	`m1_winding`, `wind_winding_aux`, `wind_polar`, `wind_centre`, `wind_track_range`, `wind_seam`
Lemma 6.9	`CostateCycle.m1_next_centre`, `CostateCycle.next_centre_local`, `CostateCycle.cross_mm`, `CostateCycle.pre_N`
Lemma 6.10	`CostateCycle.euler_sum`, `CostateCycle.rawCost_eq`, `elGa_euler`
Proposition 6.11	`m1_two_centre`, `m1_two_centre_numeric`
Lemma B.1	`ag_log_le_tangent`, `ag_log_le_atanh`, `ag_half_sub_inv_le_log`, `ag_sigma_superadd`
Lemma B.2	`ag_phi_le`
Lemma B.3	steps of the proof of `m1_angle_gap`
Lemma B.4	`ag_core`
Lemma B.5	`m1_bc_bound`
Section 7, Appendix C	
Theorem 7.1	`forward_exclusion`
Lemma 7.2	`NormalPeriod.ff_N_mod_three`
Lemma 7.3	`CostateCycle.ff_act_mul_lt`, `CostateCycle.ff_q_eq`, `CostateCycle.ff_rawCost`
Lemma 7.4	`ff_seven_le`
Proposition 7.5	`CostateCycle.ff_rows`, `CostateCycle.ff_floors`, `CostateCycle.ff_gram`, `NormalPeriod.ff_gam2_eq`, `ff_Win.toDim`
Proposition 7.6	`NormalPeriod.ff_cost_casimir`, `NormalPeriod.wire_gam2_pos_of_allForward`, `NormalPeriod.wire_spectral`
Corollary 7.7	`ff_gam2_cap`
Lemma 7.8	`ff_Dim.U_lt_three`, `fs_U_lt_three`
Lemma 7.9	`ff_Dim.eleven_fifths_lt`
Lemma 7.10	`ff_Dim.budget`, `ff_Dim.h_le_c`, `ff_budget_g`
Lemma 7.11	`ff_Dim.neighbor`, `ff_resN_neg`
Lemma 7.12	`ff_Dim.norms`, `ff_Dim.norms_nonneg`, `ff_resS_neg`, `ff_resV_neg`
Lemma 7.13	`ff_casimir_lower`
Section 8, Appendix D	

Result	Lean declarations
Theorem 8.1	`nonshort_exclusion`
Lemma 8.2	`NormalPeriod.wire_spectral`, `NormalPeriod.wire_cas_lt`, `NormalPeriod.dens_cas_gt`
Lemma 8.3	`m2_centre`, `m2_scale`
Proposition 8.4	`m2_small_forward_minimum`
Proposition 8.5	`m2_min_centre`
Lemma 8.6	`m2_reflect`, `m2_reflect_fwd_iff`, `m2_reflect_classM2`
Proposition 8.7	`m2_window`
Proposition 8.8	`m2_window`, `M2Window.ForwardNext`, `M2Window.ReverseNext`
Lemma 8.9	`m2_comparison`
Lemma 8.10	`nsrN_mono_k`, `nsrA_pos`
Lemma 8.11	`nsr_window_basic`, `nsr_window_N_neg`, `nsr_high_t`, `nsr_scale_lt`, `nsr_scale_k`, `nsr_centre_gap`, `nsr_centre_gap_k`, `nsr_window_dur`, `nsr_window_dur_x`, `nsr_reverse_order`, `nsr_reverse_order_k`
Proposition 8.12	`nsr_reverse_core`
Theorem 8.13	`m2_reverse_exclusion`
Proposition 8.14	`nsf_low_ratio`, `nsf_small_gap_j`, `nsf_small_gap`
Theorem 8.15	`m2_forward_exclusion`
Lemma D.1	`nsr_sec_swap`, `nsr_sec_start`, `nsr_m2F_eq_sec`, `nsr_sec_lt_chord`, `nsr_turn_floor`
Lemma D.2	`nsr_log_one_add_gt`, `nsr_secN_lt_vbar`, `nsr_log_lt_Lp`, `nsr_m2F_gt_Fl`, `nsf_log_gt_G`, `nsf_secN_lt_VG`, `nsf_secN_lt_Vq`, `nsr_t_lt_T`
Lemma D.3	`nsrNv_nonpos`, `nsrNL_nonpos`, `nsr_N_pos_of_relax`
Lemma D.4	`nsr_high_t'`, `nsr_row_cubic`
Lemma D.5	`nsr_unit_shape`, `nsr_cubic_pos`, `nsrU_signs`
Lemma D.6	`nsr_order_face`, `nsr_q_le_order`, `nsr_concave_min`
Lemma D.7	`nsr_strip`
Lemma D.8	`nsr_low_unit`, `nsr_quartic_pos`
Lemma D.9	`nsf_lowRatio_core`, `nlr_unit_pos`
Lemma D.10	`nsf_LGN_core`, `nsf_LGB_core`, `nlt_unit_pos`, `nlt_shape_ge`, `hrl_lower_pos`, `hrs_shape_lt`
Lemma D.11	`nsf_chi_core`
Lemma D.12	`nsf_cas_slope`
Lemma D.13	`nsf_qelim`
Lemma D.14	`nsf_envelope`
Section 9, Appendix E	
Definition 9.1	`M2Short`
Proposition 9.2	`m2_short_window`
(9.3)	`NormalPeriod.df_window`, `NormalPeriod.df_defect`
Theorem 9.3	`m2_smallerShort_exclusion`

Result	Lean declarations
Lemma 9.4	`M2Short.ss_C_le_B`, `SmallerShortData.u_le_V`, `M2Short.ss_u_le`, `ss_order`, `SmallerShortData.root_zero`, `SmallerShortData.q_le_qc`, `SmallerShortData.q_lt_seventh`
Lemma 9.5	`ss_strong_order`, `ss_upper_order_54`, `ss_upper_order_98`, `ss_small_upper_order`, `ss_quad_order`, `SmallerShortData.strong_box`
Proposition 9.6	`M2Short.ratio_lt_twelve_fifths`, `SmallerShortData.ratio_lt_twelve_fifths`, `lgr_D_pos`, `lgr_pairing`
Proposition 9.7	`M2Short.two_lt_ratio`, `SmallerShortData.two_lt_ratio`, `SmallerShortData.lr_cap_env`
Proposition 9.8	`M2Short.ratio_lt_thirteen_sixths`, `SmallerShortData.ratio_lt_thirteen_sixths`
Proposition 9.9	`M2Short.displacement_lt`, `SmallerShortData.displacement_lt`
Proposition 9.10	`M2Short.upper_order_casimir_floor`, `SmallerShortData.upper_order_casimir_floor`, `SmallerShortData.cas_cap_env`
Proposition 9.11	`M2Short.shear_neg`, `SmallerShortData.shear_neg`, `shr_residual_neg`, `shr_shear_sign`
Proposition 9.12	`M2Short.octagonCost_lt_completion`, `SmallerShortData.octagonCost_lt_completion`, `SmallerShortData.cost_angle`, `SmallerShortData.cost_continuation`, `cost_Fc_gt`, `cost_constants`
Lemma E.1	`ss_first_contact`
Lemma E.2	`ss_rev_centre_scale`
Proposition E.3	`lgr_central_tau_two`, `lgr_central_duration`, `lgr_outer_tests`
Proposition E.4	`lr_central_face`, `lr_outer_face`
Proposition E.5	`ur_central_sign`, `ur_outer_sign_small`, `ur_outer_sign_large`
Proposition E.6	`dsp_central_alpha_proof`, `dsp_outer_face_proof`
Proposition E.7	`cas_floor_box`
Proposition E.8	`cost_B_concave`, `cost_B_endpoint`
Sections 10 and 11, Appendix F	
Definition 10.1	`Term.Cycle`, `Term.Cycle.rawCost`, `Term.rawArc`
Lemma 10.2	`NormalPeriod.termCycle`, `NormalPeriod.cost_eq_rawCost`

Result	Lean declarations
Proposition 10.3	`Term.Cycle.cycle_one_false`, `Term.Cycle.cycle_two_false`, `Term.Cycle.cycle_three_false`, `Term.Cycle.cycle_four_forward`, `Term.Cycle.cycle_five`
Lemma 10.5	`Term.four_closure`
Lemma 10.6	`Term.Cycle.four_param`
Theorem 10.7	`Term.fourPiece_density_ge`, `Term.fourPiece_density_eq`, `ra_four_density`
Corollary 10.9	`Term.Cycle.four_rawCost_ge`, `Term.four_equality_data`
Lemma 10.10	`Term.five_closure`, `Term.Cycle.five_param`
Lemma 10.11	`Term.Cycle.five_param`
Theorem 10.12	`Term.fivePiece_density_gt`
Theorem 10.13	`terminal_cost`, `Term.rawCost_ge_of_le_five`
Proposition 10.14	`terminal_equality`, `Term.four_of_rawCost_eq`, `Term.octagon_identification`
Theorem 10.15	`minimizer_cost_ge`
Lemma 11.1	`eq_worstDisk_minimizer_realization`
Lemma 11.2	`eq_frame_twist_of_contact`
Lemma 11.3	`eq_disk_eq_of_frontier_eq`
Theorem 11.4	`worstDisk_eq_linear_image`
Proposition 11.5	`bridge_greatestPackingDensity_linear_image`, `bridge_greatestPackingDensity_translate`, `greatestPackingDensity_linear_image_octagon`
Proposition 11.6	`cls_image_octagon`, `cls_exists_affine_iff`
Lemma F.1	`ra_half_sub_inv_le_log`
Lemma F.2	`ra_weight_monotoneOn`, `ra_weight_le`
Lemma F.3	`ra_fiveH_concaveOn`, `ra_fiveH_circle_le`
Lemma F.4	`ra_five_envelope`, `Term.fiveEnvelope`
Lemma F.5	a private lemma of `Term.fivePiece_density_gt`

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